"The purpose of learning mathematics is not merely to find answers, but to understand the ideas behind them."



POSTGRADUATE LECTURE NOTES

Linear Algebra

Selected Topics in Linear Algebra

Vector Spaces • Matrices • Linear Map • Bilinear Maps

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Study Materials for UG Mathematics, PG Mathematics and Entrance Examinations

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CHAPTER

1

Foundations of Vector Spaces and Linear Maps

This unit establishes the foundational theory of vector spaces and subspaces, exploring structural properties such as linear independence, bases, and dimensions. We investigate the mechanics of linear transformations, rigorously defining their behavior through the study of kernel, range, and the fundamental rank-nullity theorem. By examining isomorphism theorems and Sylvester's Law, we build the essential framework necessary for understanding the algebraic relationships between different linear systems.

1.1 Compositions and Vector Spaces

Before defining a vector space, it is crucial to understand the mathematical operations that give the space its structure. These operations are formally known as compositions.

Definition 1.1.1 (Internal Composition)**.** Let V be a non-empty set. A mapping $f : V \times V \to V$ is called an **internal composition** (or binary operation) on V. If we denote the image of an ordered pair $(u, v) \in V \times V$ under this mapping by $u * v$, then the internal composition means that for every $u, v \in V$, the element $u * v$ uniquely belongs to V. In the context of vector spaces, this operation is denoted by "$+$" and is called *vector addition*.

Definition 1.1.2 (External Composition)**.** Let V and $\mathbb{K}$ be two non-empty sets. A mapping $g : \mathbb{K} \times V \to V$ is called an **external composition** in V over $\mathbb{K}$. If we denote the image of an ordered pair $(\alpha, v) \in \mathbb{K} \times V$ by $\alpha \cdot v$, then this means that for every scalar $\alpha \in \mathbb{K}$ and every vector $v \in V$, the element $\alpha \cdot v$ uniquely belongs to V. In linear algebra, the set $\mathbb{K}$ is usually a field (like the real numbers $\mathbb{R}$ or complex numbers $\mathbb{C}$), and this operation is called *scalar multiplication*.

With these compositions established, we can now formally define a vector space.

Definition 1.1.3 (Vector Space)**.** Let $\mathbb{K}$ be a field (whose elements are called scalars) and V be a non-empty set (whose elements are called vectors). The set V is called a **vector space** over the field $\mathbb{K}$ if there are two compositions defined on V:

- An internal composition called **vector addition**, denoted by $+$, such that $u + v \in V$ for all $u, v \in V$.
- An external composition called **scalar multiplication**, denoted by $\cdot$, such that $\alpha \cdot v \in V$ for

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all $\alpha \in \mathbb{K}$ and $v \in V$.

Furthermore, these operations must satisfy the following ten axioms:

I. Axioms for Vector Addition (V is an abelian group under $+$):

A1. Closure under addition: For all $u, v \in V$, $u + v \in V$.

A2. Associativity of addition: For all $u, v, w \in V$, $(u + v) + w = u + (v + w)$.

A3. Existence of zero vector: There exists an element $0 \in V$ such that $v + 0 = 0 + v = v$ for all $v \in V$.

A4. Existence of additive inverse: For each $v \in V$, there exists an element $-v \in V$ such that $v + (-v) = 0$.

A5. Commutativity of addition: For all $u, v \in V$, $u + v = v + u$.

II. Axioms for Scalar Multiplication:

M1. Closure under scalar multiplication: For all $\alpha \in \mathbb{K}$ and $v \in V$, $\alpha \cdot v \in V$.

M2. Distributivity of scalar multiplication over vector addition: For all $\alpha \in \mathbb{K}$ and $u, v \in V$, $\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$.

M3. Distributivity of scalar multiplication over field addition: For all $\alpha, \beta \in \mathbb{K}$ and $v \in V$, $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$.

M4. Associativity of scalar multiplication: For all $\alpha, \beta \in \mathbb{K}$ and $v \in V$, $\alpha \cdot (\beta \cdot v) = (\alpha\beta) \cdot v$.

M5. Identity element of scalar multiplication: For the multiplicative identity $1 \in \mathbb{K}$, $1 \cdot v = v$ for all $v \in V$.

A vector space V over a field $\mathbb{K}$ is often denoted as $V(\mathbb{K})$.

Example 1.1.4 (The n-tuple space $\mathbb{R}^n$ over $\mathbb{R}$)**.** Let $V = \mathbb{R}^n = \{(x_1, x_2, \ldots, x_n) \mid x_i \in \mathbb{R}\}$ be the set of all ordered n-tuples of real numbers, and let the field be $\mathbb{K} = \mathbb{R}$. We define the internal composition (vector addition) component-wise:

$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n)$$

We define the external composition (scalar multiplication) component-wise for $\alpha \in \mathbb{R}$:

$$\alpha \cdot (x_1, \ldots, x_n) = (\alpha x_1, \ldots, \alpha x_n)$$

Since real numbers are closed under addition and multiplication, the resulting vectors are clearly in $\mathbb{R}^n$. The zero vector is $(0, 0, \ldots, 0)$. The additive inverse of $(x_1, \ldots, x_n)$ is $(-x_1, \ldots, -x_n)$. Because the operations are defined using the field operations of $\mathbb{R}$, the associative, commutative, and distributive properties naturally inherit from the real numbers, satisfying all ten axioms. Thus, $\mathbb{R}^n(\mathbb{R})$ is a vector space.

Example 1.1.5 (The space of Matrices $M_{m\times n}(\mathbb{K})$)**.** Let $V = M_{m\times n}(\mathbb{K})$ be the set of all $m \times n$ matrices with entries from a field $\mathbb{K}$. Let the field of scalars also be $\mathbb{K}$. We define vector addition

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as standard matrix addition. If $A = [a_{ij}]$ and $B = [b_{ij}]$, then:

$$A + B = [a_{ij} + b_{ij}]$$

We define scalar multiplication as multiplying every entry of the matrix by the scalar $\alpha \in \mathbb{K}$:

$$\alpha \cdot A = [\alpha a_{ij}]$$

The sum of two $m \times n$ matrices is another $m \times n$ matrix, satisfying closure. The zero vector is the $m \times n$ zero matrix $O = [0]$, where all entries are the zero element of $\mathbb{K}$. The additive inverse of $A = [a_{ij}]$ is $-A = [-a_{ij}]$. The laws of matrix addition and scalar multiplication inherently satisfy the vector space axioms. Therefore, $M_{m\times n}(\mathbb{K})$ forms a vector space over $\mathbb{K}$.

Example 1.1.6. Let $V = \mathbb{K}[x]$ be the set of all polynomials in the variable x with coefficients from a field $\mathbb{K}$. Let $p(x) = a_0 + a_1x + \cdots + a_nx^n$ and $q(x) = b_0 + b_1x + \cdots + b_mx^m$. Vector addition is defined by adding corresponding coefficients:

$$p(x) + q(x) = (a_0 + b_0) + (a_1 + b_1)x + \dots$$

Scalar multiplication by $\alpha \in \mathbb{K}$ is defined by multiplying each coefficient by α:

$$\alpha \cdot p(x) = (\alpha a_0) + (\alpha a_1)x + \cdots + (\alpha a_n)x^n$$

The sum of two polynomials is a polynomial, and scaling a polynomial yields a polynomial, verifying closure. The zero vector is the zero polynomial 0 (all coefficients are 0). The additive inverse of $p(x)$ is $-p(x) = -a_0 - a_1x - \cdots - a_nx^n$. Since polynomial arithmetic follows the ring properties of $\mathbb{K}$, all distributive and associative vector space axioms hold true. Thus, $\mathbb{K}[x]$ is a vector space over $\mathbb{K}$.

Example 1.1.7 (The space of Polynomials of bounded degree $P_n(\mathbb{K})$)**.** Let $V = P_n(\mathbb{K})$ be the set of all polynomials of degree less than or equal to a fixed integer n, including the zero polynomial, over the field $\mathbb{K}$. The operations of addition and scalar multiplication are defined as:

$$(a_0 + a_1x + \cdots + a_nx^n) + (b_0 + b_1x + \cdots + b_mx^m) = (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n$$

and

$$\alpha(a_0 + a_1x + \cdots + a_nx^n) = \alpha a_0 + \alpha a_1x + \cdots + \alpha a_nx^n$$

To verify this is a vector space, the critical check is closure. If $p(x)$ and $q(x)$ have degree $\leq n$, their sum $p(x) + q(x)$ cannot have a degree exceeding n (though it may be less if leading coefficients cancel). Thus, addition is closed. Similarly, multiplying a polynomial of degree $\leq n$ by a scalar α yields a polynomial of degree $\leq n$. The zero polynomial is in $P_n(\mathbb{K})$, and the additive inverse of a polynomial of degree $\leq n$ also has degree $\leq n$. The remaining axioms are inherited from the general polynomial space $\mathbb{K}[x]$. Hence, $P_n(\mathbb{K})$ is a vector space.

Example 1.1.8. Let $V = C[a, b]$ be the set of all real-valued continuous functions defined on the closed interval $[a, b]$. The field of scalars is $\mathbb{K} = \mathbb{R}$. For $f, g \in C[a, b]$ and $\alpha \in \mathbb{R}$, we define the

compositions point-wise:

$$(f+g)(x) = f(x) + g(x) \quad \text{for all } x \in [a,b]$$
$$(\alpha \cdot f)(x) = \alpha f(x) \quad \text{for all } x \in [a,b]$$

From real analysis, we know that the sum of two continuous functions is continuous, and a scalar multiple of a continuous function is continuous. This guarantees closure under both operations. The zero vector is the constant zero function $z(x) = 0$ for all x. The additive inverse of f is the function $-f$ defined by $(-f)(x) = -f(x)$. The algebraic axioms (associativity, distributivity, etc.) follow directly from the properties of real numbers at each point x. Thus, $C[a,b]$ is a vector space over $\mathbb{R}$.

Example 1.1.9 (The Complex Numbers $\mathbb{C}$ over the Real Field $\mathbb{R}$)**.** Let $V = \mathbb{C} = \{a + ib \mid a, b \in \mathbb{R}, i^2 = -1\}$ be the set of complex numbers. Let the underlying field of scalars be $\mathbb{K} = \mathbb{R}$. Vector addition is standard complex addition:

$$(a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$$

Scalar multiplication by a real scalar $\alpha \in \mathbb{R}$ is defined as:

$$\alpha \cdot (a + ib) = (\alpha a) + i(\alpha b)$$

Adding two complex numbers yields a complex number. Multiplying a complex number by a real scalar α yields another complex number. The zero vector is $0 + i0$. The inverse of $a + ib$ is $-a - ib$. Because $\mathbb{C}$ is itself a field containing $\mathbb{R}$ as a subfield, the arithmetic operations automatically satisfy all abelian group requirements and distributive properties required by the vector space axioms. Therefore, $\mathbb{C}$ acts as a vector space over $\mathbb{R}$ (notably, its dimension is 2, with a standard basis $\{1, i\}$).

1.2 Subspaces of a Vector Space

In the study of vector spaces, we frequently encounter sets that are contained within a larger vector space and possess the structure of a vector space themselves under the same operations. These "spaces within a space" are called subspaces.

Definition 1.2.1 (Subspace)**.** Let V be a vector space over a field $\mathbb{K}$. A non-empty subset W of V is called a **subspace** of V if W is itself a vector space over the field $\mathbb{K}$ under the identical operations of vector addition and scalar multiplication defined on V.

Remark 1.2.2. Every vector space V has at least two trivial subspaces (provided V contains more than just the zero vector):

1. The singleton set $\{0\}$ containing only the zero vector is a subspace, called the **zero subspace** or **trivial subspace**.
2. The entire space V itself is a subspace of V, sometimes called the **improper subspace**.

Any subspace of V other than $\{0\}$ and V itself is called a **proper subspace**.

To verify whether a given subset W is a subspace, it is tedious to check all ten vector space axioms.

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Since $W \subset V$, the vectors in W are already vectors in V. Therefore, properties like associativity, commutativity, and distributivity hold automatically for elements in W because they hold for all elements in V. The only properties that might fail are the closure properties and the existence of the zero vector and additive inverses within W. This leads to the fundamental criteria for subspaces.

Theorem 1.2.3 (Two-Step Subspace Test). *Let V be a vector space over a field $\mathbb{K}$. A non-empty subset W of V is a subspace of V if and only if the following two conditions hold:*

1. ***Closure under addition:*** *If $u, v \in W$, then $u + v \in W$.*
2. ***Closure under scalar multiplication:*** *If $u \in W$ and $\alpha \in \mathbb{K}$, then $\alpha u \in W$.*

Proof. The proof requires showing that the two conditions are both necessary and sufficient for W to be a subspace of V.

($\Rightarrow$) Suppose W is a subspace of V. By definition, a subspace is itself a vector space under the operations of addition and scalar multiplication inherited from V. One of the fundamental requirements for any vector space is that the set must be closed under its operations. Thus, if $u, v \in W$, then $u + v$ must be in W, and if $u \in W$ and $\alpha \in \mathbb{K}$, then αu must be in W. Therefore, conditions 1 and 2 hold.

($\Leftarrow$) Suppose W is a non-empty subset of V that satisfies the two closure conditions. To prove that W is a subspace, we must show that W satisfies all the axioms of a vector space. Since $W \subseteq V$ and V is a vector space, many properties are inherited automatically, such as the commutativity and associativity of addition, the distributive laws, and the associativity of scalar multiplication.

We only need to verify that the additive identity (zero vector) and additive inverses exist within W:

1. **Zero Vector:** Since W is non-empty, there exists some vector $u \in W$. By condition 2, we can multiply u by the scalar $0 \in \mathbb{K}$. Thus, $0u \in W$. Since $0u = \mathbf{0}$ for any vector in a vector space, it follows that $\mathbf{0} \in W$.

2. **Additive Inverses:** Let $u \in W$. By condition 2, we can multiply u by the scalar $-1 \in \mathbb{K}$. Thus, $(-1)u \in W$. Since $(-1)u = -u$ for any vector in a vector space, it follows that $-u \in W$.

Since W is closed under the operations and contains the zero vector and additive inverses, W satisfies all axioms of a vector space. Thus, W is a subspace of V. ■

Theorem 1.2.4 (One-Step Subspace Test). *Let V be a vector space over a field $\mathbb{K}$. A non-empty subset W of V is a subspace of V if and only if for all $\alpha, \beta \in \mathbb{K}$ and all $u, v \in W$, we have $\alpha u + \beta v \in W$. (Equivalently, for all $\alpha \in \mathbb{K}$ and $u, v \in W$, $\alpha u + v \in W$).*

Proof. If W is a subspace, it is closed under scalar multiplication (so $\alpha u \in W$ and $\beta v \in W$) and closed under addition. Therefore, their sum $\alpha u + \beta v \in W$.

Assume $\alpha u + \beta v \in W$ for all $\alpha, \beta \in \mathbb{K}$ and $u, v \in W$. Since W is non-empty, let $u \in W$.

- Let $\alpha = 0$ and $\beta = 0$. Then $0 \cdot u + 0 \cdot u = 0 \in W$. So, the zero vector is in W.
- Let $\alpha = -1$ and $\beta = 0$. Then $(-1) \cdot u + 0 \cdot u = -u \in W$. Additive inverses exist in W.
- Let $\alpha = 1$ and $\beta = 1$. Then $1 \cdot u + 1 \cdot v = u + v \in W$. W is closed under addition.
- Let $\beta = 0$. Then $\alpha u + 0 \cdot v = \alpha u \in W$. W is closed under scalar multiplication.

By Theorem 1.2.3, W is a subspace of V. ■

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Example 1.2.5. Let $V = \mathbb{R}^3$ over the field $\mathbb{R}$. Determine whether the subset $W = \{(x, y, z) \in \mathbb{R}^3 \mid x + 2y - z = 0\}$ is a subspace of $\mathbb{R}^3$.

Solution. We will use the two-step subspace test. First, we check if W is non-empty. Letting $x = 0, y = 0, z = 0$, we see that $0 + 2(0) - 0 = 0$. So the zero vector $(0, 0, 0) \in W$, meaning W is non-empty.

1. Closure under addition: Let $u = (x_1, y_1, z_1)$ and $v = (x_2, y_2, z_2)$ be any two vectors in W. This means:

$$x_1 + 2y_1 - z_1 = 0 \quad \text{and} \quad x_2 + 2y_2 - z_2 = 0$$

We consider their sum $u + v = (x_1 + x_2, y_1 + y_2, z_1 + z_2)$. To see if $u + v \in W$, we substitute its components into the defining equation of W:

$$\begin{aligned}(x_1 + x_2) + 2(y_1 + y_2) - (z_1 + z_2) &= (x_1 + 2y_1 - z_1) + (x_2 + 2y_2 - z_2) \\ &= 0 + 0 = 0\end{aligned}$$

Since the condition is satisfied, $u + v \in W$.

2. Closure under scalar multiplication: Let $u = (x_1, y_1, z_1) \in W$ and let $\alpha \in \mathbb{R}$. The scalar multiple is $\alpha u = (\alpha x_1, \alpha y_1, \alpha z_1)$. Checking the condition:

$$(\alpha x_1) + 2(\alpha y_1) - (\alpha z_1) = \alpha(x_1 + 2y_1 - z_1) = \alpha(0) = 0$$

Since the condition is satisfied, $\alpha u \in W$.

Since both closure properties hold, W is a subspace of $\mathbb{R}^3$ (geometrically, it represents a plane passing through the origin). ■

Example 1.2.6. Let $V = \mathbb{R}^2$ over $\mathbb{R}$. Show that the set $W = \{(x, y) \in \mathbb{R}^2 \mid x \geq 0 \text{ and } y \geq 0\}$ (the first quadrant) is **not** a subspace of $\mathbb{R}^2$.

Solution. To show that a subset is not a subspace, it is sufficient to find a single counterexample where one of the subspace axioms fails. The set W is clearly non-empty (e.g., $(0, 0) \in W$) and is closed under addition (the sum of two non-negative numbers is non-negative).

However, let us test closure under scalar multiplication. Let $u = (1, 2) \in W$ (since $1 \geq 0$ and $2 \geq 0$). Let $\alpha = -3 \in \mathbb{R}$. Then $\alpha u = -3(1, 2) = (-3, -6)$. Since the coordinates -3 and -6 are not ≥ 0, the vector $(-3, -6) \notin W$. Because W is not closed under scalar multiplication by negative scalars, W is not a subspace of $\mathbb{R}^2$. ■

Example 1.2.7. Let $V = M_{n\times n}(\mathbb{R})$ be the vector space of all $n \times n$ matrices over $\mathbb{R}$. Let W be the set of all symmetric matrices in V. That is, $W = \{A \in M_{n\times n}(\mathbb{R}) \mid A^T = A\}$. Show that W is a subspace of V.

Solution. First, observe that the zero matrix $O_{n\times n}$ satisfies $O^T = O$. Thus, $O \in W$, so W is non-empty. Let us use the one-step test. Let $A, B \in W$ and $\alpha, \beta \in \mathbb{R}$. Since $A, B \in W$, we know $A^T = A$ and $B^T = B$. Consider the matrix $C = \alpha A + \beta B$. We must check if $C^T = C$. Taking the transpose of C:

$$\begin{aligned}C^T &= (\alpha A + \beta B)^T \\ &= (\alpha A)^T + (\beta B)^T \quad \text{(Properties of transpose)} \\ &= \alpha(A^T) + \beta(B^T) \\ &= \alpha A + \beta B \quad \text{(Since } A^T = A \text{ and } B^T = B\text{)} \\ &= C\end{aligned}$$

Since $C^T = C$, the linear combination $\alpha A + \beta B \in W$. Therefore, by the one-step test, the set of symmetric matrices W is a subspace of $M_{n\times n}(\mathbb{R})$. ■

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Example 1.2.8. Let $P_n(\mathbb{R})$ be the vector space of all polynomials of degree $\leq n$. Let $W = \{p(x) \in P_n(\mathbb{R}) \mid p(1) = 0\}$ be the set of polynomials that have a root at $x = 1$. Prove that W is a subspace.

Solution. The zero polynomial $z(x) = 0$ satisfies $z(1) = 0$, so the zero polynomial is in W, making W non-empty. Let $p(x), q(x) \in W$ and $\alpha \in \mathbb{R}$. This means $p(1) = 0$ and $q(1) = 0$. We need to test if the polynomial $r(x) = \alpha p(x) + q(x)$ is in W. We evaluate $r(x)$ at $x = 1$:

$$\begin{aligned} r(1) &= (\alpha p + q)(1) \\ &= \alpha p(1) + q(1) \\ &= \alpha(0) + 0 \\ &= 0 \end{aligned}$$

Since $r(1) = 0$, the linear combination $\alpha p(x) + q(x)$ belongs to W. By the one-step subspace test, W is a subspace of $P_n(\mathbb{R})$. ■

Example 1.2.9 (Solution Space of Homogeneous Systems)**.** Let A be an $m \times n$ matrix over $\mathbb{R}$. Let $V = \mathbb{R}^n$ (represented as column vectors). Show that the set of all solutions to the homogeneous linear system $Ax = 0$, denoted by $W = \{x \in \mathbb{R}^n \mid Ax = 0\}$, is a subspace of $\mathbb{R}^n$.

Solution. The set W is known as the null space of the matrix A. First, note that the zero vector $0 \in \mathbb{R}^n$ satisfies the equation $A(0) = 0$. Therefore, $0 \in W$, so W is non-empty.

We apply the two-step test: **1. Closure under addition:** Let $x_1, x_2 \in W$. Then $Ax_1 = 0$ and $Ax_2 = 0$. We evaluate the matrix multiplication on their sum $x_1 + x_2$:

$$A(x_1 + x_2) = Ax_1 + Ax_2 = 0 + 0 = 0$$

Since $A(x_1 + x_2) = 0$, it follows that $x_1 + x_2 \in W$.

2. Closure under scalar multiplication: Let $x \in W$ and let $\alpha \in \mathbb{R}$. Then $Ax = 0$. We evaluate the action of A on αx:

$$A(\alpha x) = \alpha(Ax) = \alpha(0) = 0$$

Since $A(\alpha x) = 0$, it follows that $\alpha x \in W$.

Because both closure conditions are satisfied, the solution set W to the homogeneous system $Ax = 0$ is a subspace of $\mathbb{R}^n$. ■

Theorem 1.2.10 (Algebra of Subspaces)**.** *Let W_1 and W_2 be subspaces of a vector space V. Then:*

1. *$W_1 \cap W_2$ is a subspace of V.*
2. *$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}$ is a subspace of V.*
3. *$W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.*

Proof. We will use the subspace test: a non-empty subset is a subspace if it is closed under linear combinations. Let $\mathbb{K}$ be the underlying field. Since W_1 and W_2 are subspaces, both contain the zero vector 0, making their intersection, sum, and union non-empty.

1. Intersection ($W_1 \cap W_2$):
Let $x, y \in W_1 \cap W_2$ and $c \in \mathbb{K}$. By definition, x, y belong to both W_1 and W_2. Since W_1 and W_2 are subspaces, the linear combination $cx + y$ must belong to W_1 and to W_2. Therefore, $cx + y \in W_1 \cap W_2$, proving it is a subspace.

2. Sum ($W_1 + W_2$):
Let $u, v \in W_1 + W_2$ and $c \in \mathbb{K}$. We can write $u = w_1 + w_2$ and $v = w_1' + w_2'$ for some $w_1, w_1' \in W_1$ and

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$w_2, w_2' \in W_2$. Then,

$$cu + v = c(w_1 + w_2) + (w_1' + w_2') = (cw_1 + w_1') + (cw_2 + w_2')$$

Since W_1 and W_2 are subspaces, $(cw_1 + w_1') \in W_1$ and $(cw_2 + w_2') \in W_2$. Thus, $cu + v \in W_1 + W_2$, proving it is a subspace.

3. Union ($W_1 \cup W_2$):

($\Longleftarrow$) If $W_1 \subseteq W_2$, then $W_1 \cup W_2 = W_2$, which is a subspace. The same logic holds if $W_2 \subseteq W_1$.

($\Longrightarrow$) Assume $W_1 \cup W_2$ is a subspace, but suppose for contradiction that neither is a subset of the other. Then there exist elements $x \in W_1 \setminus W_2$ and $y \in W_2 \setminus W_1$. Since $x, y \in W_1 \cup W_2$ and the union is a subspace, their sum $x + y$ must also be in $W_1 \cup W_2$. This gives two cases:

- If $x + y \in W_1$, then $y = (x + y) - x \in W_1$ (since W_1 is a subspace). This contradicts $y \notin W_1$.
- If $x + y \in W_2$, then $x = (x + y) - y \in W_2$ (since W_2 is a subspace). This contradicts $x \notin W_2$.

Both cases yield a contradiction. Hence, we must have either $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$. ■

1.3 Product Spaces and Quotient Spaces

In addition to finding vector spaces within existing ones (subspaces), we can construct entirely new vector spaces from given ones. Two fundamental constructions in linear algebra are the product space (combining spaces together) and the quotient space (collapsing spaces along a subspace).

Definition 1.3.1 (Product Space)**.** Let V and W be two vector spaces over the same field $\mathbb{K}$. The **Cartesian product** of V and W, denoted by $V \times W$, is the set of all ordered pairs (v, w) where $v \in V$ and $w \in W$:

$$V \times W = \{(v, w) \mid v \in V, w \in W\}$$

We define vector addition and scalar multiplication component-wise:

1. **Addition:** $(v_1, w_1) + (v_2, w_2) = (v_1 + v_2, w_1 + w_2)$
2. **Scalar Multiplication:** $\alpha(v, w) = (\alpha v, \alpha w)$ for any $\alpha \in \mathbb{K}$.

Theorem 1.3.2. *Let V and W be vector spaces over $\mathbb{K}$. Under the component-wise operations defined above, the product space $V \times W$ is a vector space over $\mathbb{K}$.*

Proof. The proof consists of verifying the vector space axioms. Let $(v_1, w_1), (v_2, w_2), (v_3, w_3) \in V \times W$ and $\alpha, \beta \in \mathbb{K}$.

- **Closure:** Since V and W are vector spaces, $v_1 + v_2 \in V$ and $w_1 + w_2 \in W$. Thus $(v_1 + v_2, w_1 + w_2) \in V \times W$. Similarly, $\alpha v_1 \in V$ and $\alpha w_1 \in W$, so $(\alpha v_1, \alpha w_1) \in V \times W$.
- **Zero Vector:** The element $(0_V, 0_W) \in V \times W$ acts as the additive identity, because $(v, w) + (0_V, 0_W) = (v + 0_V, w + 0_W) = (v, w)$.
- **Additive Inverse:** For any (v, w), the element $(-v, -w) \in V \times W$ is its inverse, since $(v, w) + (-v, -w) = (v - v, w - w) = (0_V, 0_W)$.
- **Associativity/Commutativity/Distributivity:** These follow naturally from the corresponding properties in V and W. For example, $\alpha((v_1, w_1) + (v_2, w_2)) = \alpha(v_1 + v_2, w_1 + w_2) = (\alpha(v_1 + v_2), \alpha(w_1 + w_2)) = (\alpha v_1 + \alpha v_2, \alpha w_1 + \alpha w_2) = (\alpha v_1, \alpha w_1) + (\alpha v_2, \alpha w_2) = \alpha(v_1, w_1) + \alpha(v_2, w_2)$.

Thus, $V \times W$ satisfies all axioms and is a vector space. ■

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While a product space builds a larger space from smaller ones, a quotient space effectively "compresses" or "collapses" a space by treating a subspace as if it were the zero vector.

Definition 1.3.3 (Coset)**.** Let W be a subspace of a vector space V. For any vector $v \in V$, the **coset** of W containing v is the set:

$$v + W = \{v + w \mid w \in W\}$$

The vector v is called the *representative* of the coset.

Remark 1.3.4. Two cosets $v_1 + W$ and $v_2 + W$ are equal if and only if $v_1 - v_2 \in W$. If $v_1 - v_2 \notin W$, the cosets are completely disjoint. Thus, the cosets partition the vector space V.

Definition 1.3.5 (Quotient Space)**.** Let W be a subspace of V. The **quotient space** of V modulo W, denoted by V/W (read as "V mod W"), is the set of all cosets of W in V:

$$V/W = \{v + W \mid v \in V\}$$

We define addition and scalar multiplication on V/W as follows:

1. **Addition:** $(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$
2. **Scalar Multiplication:** $\alpha(v + W) = \alpha v + W$

Theorem 1.3.6 (Well-Definedness of Operations)**.** *The operations defined on V/W are well-defined, meaning they do not depend on the choice of the representative vectors for the cosets.*

Proof. Let $v_1 + W = u_1 + W$ and $v_2 + W = u_2 + W$. This means $v_1 - u_1 = w_1 \in W$ and $v_2 - u_2 = w_2 \in W$.

1. Addition: We must show $(v_1 + v_2) + W = (u_1 + u_2) + W$. Consider the difference of the representatives:

$$(v_1 + v_2) - (u_1 + u_2) = (v_1 - u_1) + (v_2 - u_2) = w_1 + w_2$$

Since W is a subspace, it is closed under addition, so $w_1 + w_2 \in W$. Because the difference is in W, the cosets are equal: $(v_1 + v_2) + W = (u_1 + u_2) + W$.

2. Scalar Multiplication: We must show $\alpha v_1 + W = \alpha u_1 + W$. Consider the difference: $\alpha v_1 - \alpha u_1 = \alpha(v_1 - u_1) = \alpha w_1$. Since W is a subspace, it is closed under scalar multiplication, so $\alpha w_1 \in W$. Thus, $\alpha v_1 + W = \alpha u_1 + W$. ■

Theorem 1.3.7. *Let W be a subspace of V. The quotient set V/W is a vector space over $\mathbb{K}$ under the operations defined above.*

Proof. The proof is straightforward verifying axioms. The zero vector of V/W is $0_V + W = W$. Let's check: $(v + W) + (0_V + W) = (v + 0_V) + W = v + W$. The additive inverse of $v + W$ is $(-v) + W$, since $(v + W) + ((-v) + W) = (v - v) + W = 0_V + W = W$. Distributivity and associativity follow directly from the properties of V. For instance: $\alpha((v_1 + W) + (v_2 + W)) = \alpha((v_1 + v_2) + W) = \alpha(v_1 + v_2) + W = (\alpha v_1 + \alpha v_2) + W = (\alpha v_1 + W) + (\alpha v_2 + W) = \alpha(v_1 + W) + \alpha(v_2 + W)$. Thus, V/W is a vector space. ■

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1.4 Linear Combinations and Span

Before discussing the basis and dimension of a vector space, we must understand how a set of vectors can "generate" other vectors in the space. This idea is captured by the concepts of linear combinations and the linear span.

1.4.1 Linear Combinations

Definition 1.4.1 (Linear Combination)**.** Let V be a vector space over a field $\mathbb{K}$. A vector $v \in V$ is said to be a **linear combination** of a finite set of vectors $\{v_1, v_2, \ldots, v_n\}$ in V if there exist scalars $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{K}$ such that:

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n = \sum_{i=1}^{n} \alpha_i v_i$$

Remark 1.4.2. In vector spaces, we only ever consider sums of a *finite* number of vectors. Infinite sums require a concept of limits and convergence, which belongs to analysis (topological vector spaces) rather than pure linear algebra.

1.4.2 Linear Span

Definition 1.4.3 (Linear Span)**.** Let S be a non-empty subset of a vector space $V(\mathbb{K})$. The **linear span** of S, denoted by $\text{span}(S)$ or $L(S)$, is the set of all possible finite linear combinations of the vectors in S. Formally:

$$L(S) = \left\{ \sum_{i=1}^{k} \alpha_i v_i \;\middle|\; k \in \mathbb{N},\ \alpha_i \in \mathbb{K},\ v_i \in S \right\}$$

If S is the empty set $\emptyset$, we define $L(\emptyset) = \{0_V\}$.

If $L(S) = V$, we say that the set S **spans** V, or that S is a **spanning set** (or generating set) for V.

The most important property of the linear span is that it always creates a subspace, regardless of whether the original set S was a subspace or not.

Theorem 1.4.4. *Let S be a non-empty subset of a vector space $V(\mathbb{K})$. Then the linear span $L(S)$ is a subspace of V.*

Proof. Since S is non-empty, there exists some vector $v \in S$. Then $0 \cdot v = 0_V$ is a linear combination of elements in S, so $0_V \in L(S)$. Thus, $L(S)$ is non-empty.

Let $u, w \in L(S)$ and let $c \in \mathbb{K}$. We must show that $cu + w \in L(S)$ (using the one-step subspace test). Because u and w belong to $L(S)$, they can be written as finite linear combinations of elements in S:

$$u = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_m x_m \quad \text{for some } x_i \in S, \alpha_i \in \mathbb{K}$$

$$w = \beta_1 y_1 + \beta_2 y_2 + \cdots + \beta_n y_n \quad \text{for some } y_j \in S, \beta_j \in \mathbb{K}$$

Now consider the vector $cu + w$:

$$cu + w = c(\alpha_1 x_1 + \cdots + \alpha_m x_m) + (\beta_1 y_1 + \cdots + \beta_n y_n)$$

$$cu + w = (c\alpha_1)x_1 + \cdots + (c\alpha_m)x_m + \beta_1 y_1 + \cdots + \beta_n y_n$$

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This expression is simply another finite linear combination of vectors (x_i and y_j) from S. Therefore, by definition, $cu + w \in L(S)$. Hence, $L(S)$ is a subspace of V. ■

Theorem 1.4.5 (Smallest Subspace Property). *Let S be a subset of a vector space V. Then $L(S)$ is the **smallest subspace** of V that contains S. In other words, if W is any other subspace of V containing S, then $L(S) \subseteq W$.*

Proof. First, it is clear that $S \subseteq L(S)$, because for any $v \in S$, we can write $v = 1 \cdot v$, which is a trivial linear combination of elements in S. We also know from Theorem 1.4.4 that $L(S)$ is a subspace.

Now, let W be any subspace of V such that $S \subseteq W$. We must show that $L(S) \subseteq W$. Let x be an arbitrary element in $L(S)$. By definition, x is a finite linear combination of vectors in S:

$$x = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_k v_k$$

where $v_i \in S$ and $\alpha_i \in \mathbb{K}$. Since $S \subseteq W$, all the vectors $v_1, v_2, \ldots, v_k$ are also in W. Because W is a subspace, it is closed under scalar multiplication and vector addition. Therefore, any linear combination of vectors in W must also lie in W. Consequently, $x = \sum_{i=1}^{k} \alpha_i v_i \in W$. Since every element $x \in L(S)$ is also in W, we conclude that $L(S) \subseteq W$. Thus, $L(S)$ is indeed the smallest subspace containing S. ■

Theorem 1.4.6. *Let V be a vector space over $\mathbb{K}$. A subset $W \subseteq V$ is a subspace of V if and only if $L(W) = W$.*

Proof. Assume W is a subspace. We know that $W \subseteq L(W)$ is always true. We need to show $L(W) \subseteq W$. Let $x \in L(W)$. Then $x = \sum_{i=1}^{k} \alpha_i w_i$ for some $w_i \in W$. Because W is a subspace, it is closed under linear combinations, meaning $x \in W$. Thus $L(W) \subseteq W$. Combining the two inclusions gives $L(W) = W$. (Alternatively, by Theorem 1.4.5, $L(W)$ is the smallest subspace containing W. Since W is already a subspace containing W, $L(W)$ must be exactly W).

Assume $L(W) = W$. By Theorem 1.4.4, the span of any set is a subspace. Since W is equal to its span, W itself must be a subspace. ■

Corollary 1.4.7 (Idempotence of Span). *For any subset S of a vector space V, $L(L(S)) = L(S)$.*

Proof. By Theorem 1.4.4, $L(S)$ is a subspace of V. By the preceding theorem, if a set W is a subspace, then its span is itself. Letting $W = L(S)$, we immediately get $L(L(S)) = L(S)$. ■

The linear span operation interacts beautifully with set operations like subsets and unions. The following theorems establish the algebraic rules for dealing with spans.

Theorem 1.4.8 (Monotonicity of Span). *Let V be a vector space over a field $\mathbb{K}$, and let S_1 and S_2 be subsets of V. If $S_1 \subseteq S_2$, then $L(S_1) \subseteq L(S_2)$.*

Proof. Let $v \in L(S_1)$ be an arbitrary vector. By definition, v can be expressed as a finite linear combination of vectors in S_1:

$$v = \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n$$

where $\alpha_i \in \mathbb{K}$ and $x_i \in S_1$ for all $i = 1, \ldots, n$. Since we are given that $S_1 \subseteq S_2$, every vector x_i also belongs to S_2. Therefore, the exact same expression represents v as a finite linear combination of vectors in S_2. By definition, this means $v \in L(S_2)$. Hence, $L(S_1) \subseteq L(S_2)$. ■

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Theorem 1.4.9 (Span of a Union). *Let S_1 and S_2 be subsets of a vector space V. Then the span of their union is equal to the sum of their individual spans. That is:*

$$L(S_1 \cup S_2) = L(S_1) + L(S_2)$$

Proof. We will prove this by showing double inclusion.

We show $L(S_1) + L(S_2) \subseteq L(S_1 \cup S_2)$. Let $v \in L(S_1) + L(S_2)$. By definition of the sum of subspaces, $v = u + w$ for some $u \in L(S_1)$ and $w \in L(S_2)$. Since $S_1 \subseteq S_1 \cup S_2$, by Theorem 1.4.8, $L(S_1) \subseteq L(S_1 \cup S_2)$. Thus $u \in L(S_1 \cup S_2)$. Similarly, $S_2 \subseteq S_1 \cup S_2 \implies L(S_2) \subseteq L(S_1 \cup S_2)$. Thus $w \in L(S_1 \cup S_2)$. Since $L(S_1 \cup S_2)$ is a subspace, it is closed under addition. Therefore, $v = u + w \in L(S_1 \cup S_2)$. This proves the first inclusion.

Now, we show $L(S_1 \cup S_2) \subseteq L(S_1) + L(S_2)$. Let $x \in L(S_1 \cup S_2)$. Then x is a finite linear combination of vectors from $S_1 \cup S_2$. We can group the terms originating from S_1 and the terms originating from S_2 (if a vector is in both, we can arbitrarily assign it to one group). Thus, we can write $x = \sum \alpha_i u_i + \sum \beta_j w_j$ where $u_i \in S_1$ and $w_j \in S_2$. The first sum, $\sum \alpha_i u_i$, is a linear combination of elements in S_1, so it belongs to $L(S_1)$. The second sum, $\sum \beta_j w_j$, is a linear combination of elements in S_2, so it belongs to $L(S_2)$. Therefore, $x \in L(S_1) + L(S_2)$, proving the second inclusion.

Combining both parts, we get $L(S_1 \cup S_2) = L(S_1) + L(S_2)$. ■

Theorem 1.4.10 (Redundancy Theorem). *Let S be a subset of a vector space V. If a vector v is already in the span of S (i.e., $v \in L(S)$), then adding v to the set S does not enlarge the span. That is:*

$$L(S \cup \{v\}) = L(S)$$

Proof. Since $S \subseteq S \cup \{v\}$, by monotonicity (Theorem 1.4.8), we have $L(S) \subseteq L(S \cup \{v\})$.

To show the reverse inclusion, let $x \in L(S \cup \{v\})$. Then x can be written as:

$$x = c_0 v + c_1 u_1 + c_2 u_2 + \cdots + c_k u_k$$

where $c_i \in \mathbb{K}$ and $u_i \in S$. We are given that $v \in L(S)$, which means v can be written as a linear combination of elements in S:

$$v = d_1 w_1 + d_2 w_2 + \cdots + d_m w_m \quad (\text{where } d_j \in \mathbb{K}, w_j \in S)$$

Substitute this expression for v into the equation for x:

$$x = c_0(d_1 w_1 + \cdots + d_m w_m) + c_1 u_1 + \cdots + c_k u_k$$

Distributing c_0, we see that x is now expressed entirely as a finite linear combination of vectors (w_j and u_i) that are exclusively in S. Therefore, $x \in L(S)$. This proves $L(S \cup \{v\}) \subseteq L(S)$. Thus, $L(S \cup \{v\}) = L(S)$. ■

Example 1.4.11. Let $V = \mathbb{R}^2$. Show that the set $S = \{(1,2), (3,4)\}$ spans $\mathbb{R}^2$.

Solution. To show that S spans $\mathbb{R}^2$, we must demonstrate that *any* arbitrary vector $(a, b) \in \mathbb{R}^2$ can be written as a linear combination of the vectors in S. We set up the linear combination equation with unknown scalars $c_1, c_2 \in \mathbb{R}$:

$$c_1(1,2) + c_2(3,4) = (a, b)$$

This yields the vector equation $(c_1 + 3c_2, 2c_1 + 4c_2) = (a, b)$, which leads to a system of linear equations:

$$\begin{aligned} c_1 + 3c_2 &= a \\ 2c_1 + 4c_2 &= b \end{aligned}$$

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We need to show this system always has a solution for c_1 and c_2 regardless of the values of a and b. We can solve this using substitution or elimination. Multiply the first equation by 2:

$$2c_1 + 6c_2 = 2a$$
$$2c_1 + 4c_2 = b$$

Subtract the second equation from the first:

$$2c_2 = 2a - b \implies c_2 = a - \frac{b}{2}$$

Substitute c_2 back into the first equation:

$$c_1 + 3\left(a - \frac{b}{2}\right) = a \implies c_1 = a - 3a + \frac{3b}{2} = -2a + \frac{3b}{2}$$

Since we can uniquely determine the real scalars c_1 and c_2 for any real numbers a and b, every vector in $\mathbb{R}^2$ is in the span of S. Thus, $L(S) = \mathbb{R}^2$. ■

Example 1.4.12. Find the linear span of the set $S = \{(1,1,1), (0,1,1)\}$ in the vector space $\mathbb{R}^3$, and describe it geometrically.

Solution. The span of S consists of all vectors $(x, y, z) \in \mathbb{R}^3$ that can be written as:

$$c_1(1,1,1) + c_2(0,1,1) = (x, y, z)$$

for some scalars $c_1, c_2 \in \mathbb{R}$. Combining the left side, we get:

$$(c_1, c_1 + c_2, c_1 + c_2) = (x, y, z)$$

Equating the corresponding components gives:

$$x = c_1$$
$$y = c_1 + c_2$$
$$z = c_1 + c_2$$

From the second and third equations, we can immediately observe a restriction on the coordinates: $y = z$. There are no restrictions on x (it can be any real number c_1). Therefore, the span of S is the set:

$$L(S) = \{(x, y, z) \in \mathbb{R}^3 \mid y = z\}$$

Geometrically, this represents a **plane** in 3-dimensional space that passes through the origin and contains the x-axis (since setting $y = 0$ forces $z = 0$, allowing x to be anything). ■

Example 1.4.13. Let V be the vector space of all 2×2 symmetric matrices over $\mathbb{R}$. Let S be the set of matrices:

$$S = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

Show that $L(S) = V$.

Solution. A matrix A in $M_{2\times 2}(\mathbb{R})$ is symmetric if $A^T = A$. The general form of any 2×2 symmetric matrix is:

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

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where a, b, c are arbitrary real numbers. We need to check if A can be written as a linear combination of the matrices in S. Let M_1, M_2, M_3 denote the three matrices in S respectively. We look for scalars c_1, c_2, c_3 such that:

$$c_1M_1 + c_2M_2 + c_3M_3 = A$$

Substituting the matrices:

$$c_1\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} + c_2\begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix} + c_3\begin{pmatrix}0 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}a & b\\ b & c\end{pmatrix}$$

Performing the scalar multiplication and matrix addition on the left side yields:

$$\begin{pmatrix}c_1 & c_2\\ c_2 & c_3\end{pmatrix} = \begin{pmatrix}a & b\\ b & c\end{pmatrix}$$

By equating corresponding entries, it is obvious that $c_1 = a$, $c_2 = b$, and $c_3 = c$. Since for any symmetric matrix, we can trivially find these scalars, every symmetric matrix lies in the span of S. Therefore, $L(S) = V$. ■

Example 1.4.14 (Polynomial Space $P_2(\mathbb{R})$)**.** Let $V = P_2(\mathbb{R})$, the space of all polynomials of degree ≤ 2. Show that the set $S = \{1, x, x^2\}$ spans $P_2(\mathbb{R})$.

Solution. An arbitrary element in $P_2(\mathbb{R})$ is a polynomial of the form $p(x) = a + bx + cx^2$, where $a, b, c \in \mathbb{R}$. This polynomial is already written explicitly as a linear combination of the vectors (functions) $1, x$, and x^2 with scalars a, b, and c. Therefore, every polynomial in $P_2(\mathbb{R})$ is in the span of S. Hence, $L(S) = P_2(\mathbb{R})$. ■

1.5 Linear Dependence and Independence

The concept of the linear span allows us to describe all vectors that can be generated by a specific set S. However, a spanning set might contain "redundant" vectors—vectors that don't actually contribute to generating any new parts of the space because they can already be formed by the other vectors in the set. To Formalize the idea of a set of vectors having no redundancies, we introduce linear independence.

Consider the homogeneous vector equation:

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0_V \tag{1.1}$$

where $v_1, \ldots, v_n \in V$ and $c_1, \ldots, c_n \in \mathbb{K}$. Notice that this equation *always* has at least one solution: setting all scalars $c_1 = c_2 = \cdots = c_n = 0$. This is called the **trivial solution**. The critical question is whether there are any *other* solutions.

Definition 1.5.1 (Linear Independence)**.** A finite set of vectors $S = \{v_1, v_2, \ldots, v_n\}$ in a vector space V is said to be **linearly independent** (often abbreviated as LI) if the equation

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0_V$$

has *only* the trivial solution. That is, the relation implies $c_1 = c_2 = \cdots = c_n = 0$.

Definition 1.5.2 (Linear Dependence)**.** A finite set of vectors $S = \{v_1, v_2, \ldots, v_n\}$ in a vector space V is said to be **linearly dependent** (abbreviated as LD) if it is not linearly independent.

Equivalently, S is linearly dependent if there exist scalars $c_1, c_2, \ldots, c_n \in \mathbb{K}$, **not all zero**, such that:

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0_V$$

Such a combination is called a **non-trivial** linear combination resulting in the zero vector.

Remark 1.5.3 (Infinite Sets). An infinite set of vectors is said to be linearly independent if every finite subset of it is linearly independent. If there exists even one finite subset that is linearly dependent, the entire infinite set is considered linearly dependent.

1.5.1 Basic Properties

Several properties of linear dependence and independence follow immediately from the definitions:

1. **Zero Vector:** Any set containing the zero vector 0_V is linearly dependent. *(Proof: If $0_V \in S$, we can assign the scalar 1 to 0_V and 0 to all other vectors. Then $1 \cdot 0_V + 0 \cdot v_2 + \cdots = 0_V$, which is a non-trivial solution).*
2. **Singleton Set:** A set containing a single non-zero vector $\{v\}$ is linearly independent. *(Proof: $c_1v = 0_V$ with $v \neq 0_V$ implies $c_1 = 0$ since we are in a field).*
3. **Subsets:** Every non-empty subset of a linearly independent set is linearly independent.
4. **Supersets:** Every superset of a linearly dependent set is linearly dependent.
5. **Two Vectors:** Two vectors v_1 and v_2 are linearly dependent if and only if one is a scalar multiple of the other (i.e., they are collinear).

The following theorem formalizes the idea of "redundancy." It states that a set is dependent exactly when at least one vector is redundant (it can be built from the others).

Theorem 1.5.4 (Linear Dependence Criterion). *Let V be a vector space over $\mathbb{K}$. A set of vectors $S = \{v_1, v_2, \ldots, v_n\}$ (where $n \geq 2$) is linearly dependent if and only if at least one vector in S can be expressed as a linear combination of the remaining vectors.*

Proof. Assume S is linearly dependent. By definition, there exist scalars $c_1, c_2, \ldots, c_n \in \mathbb{K}$, not all zero, such that:

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0_V$$

Since not all scalars are zero, suppose $c_k \neq 0$ for some index k ($1 \leq k \leq n$). We can subtract c_kv_k from both sides:

$$c_1v_1 + \cdots + c_{k-1}v_{k-1} + c_{k+1}v_{k+1} + \cdots + c_nv_n = -c_kv_k$$

Since $c_k \neq 0$ and $\mathbb{K}$ is a field, the inverse c_k^{-1} exists. Multiplying both sides by $-c_k^{-1}$:

$$v_k = \left(-\frac{c_1}{c_k}\right)v_1 + \cdots + \left(-\frac{c_{k-1}}{c_k}\right)v_{k-1} + \left(-\frac{c_{k+1}}{c_k}\right)v_{k+1} + \cdots + \left(-\frac{c_n}{c_k}\right)v_n$$

Thus, v_k is a linear combination of the remaining vectors.

Assume that one vector in S, say v_k, can be written as a linear combination of the remaining vectors. Thus, there exist scalars a_i such that:

$$v_k = a_1v_1 + \cdots + a_{k-1}v_{k-1} + a_{k+1}v_{k+1} + \cdots + a_nv_n$$

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We can bring v_k to the right side of the equation:

$$0_V = a_1v_1 + \cdots + a_{k-1}v_{k-1} + (-1)v_k + a_{k+1}v_{k+1} + \cdots + a_nv_n$$

This is a linear combination of the vectors in S that equals the zero vector. Notice that the coefficient of v_k is -1, which is non-zero. Since we have found a non-trivial linear combination that equals zero, the set S is linearly dependent. ■

Theorem 1.5.5 (Extension Lemma / Independence and Span). *Let $S = \{v_1, \ldots, v_n\}$ be a linearly independent set of vectors in V. Let $v \in V$ be a vector not in S. Then the enlarged set $S \cup \{v\} = \{v_1, \ldots, v_n, v\}$ is linearly dependent if and only if $v \in \text{span}(S)$.*

Proof. Assume $S \cup \{v\}$ is linearly dependent. Then there exist scalars $c_1, \ldots, c_n, c \in \mathbb{K}$, not all zero, such that:

$$c_1v_1 + \cdots + c_nv_n + cv = 0_V$$

We claim that $c \neq 0$. Suppose, for the sake of contradiction, that $c = 0$. Then the equation becomes:

$$c_1v_1 + \cdots + c_nv_n = 0_V$$

Since S is linearly independent, this forces $c_1 = c_2 = \cdots = c_n = 0$. But this means all scalars (including c) are zero, contradicting our initial assumption. Thus, $c \neq 0$. Since $c \neq 0$, we can solve for v:

$$v = \left(-\frac{c_1}{c}\right) v_1 + \cdots + \left(-\frac{c_n}{c}\right) v_n$$

This shows that v is a linear combination of vectors in S, so $v \in \text{span}(S)$.

Assume $v \in \text{span}(S)$. Then there exist scalars $a_1, \ldots, a_n$ such that:

$$v = a_1v_1 + \cdots + a_nv_n$$

Rearranging this equation gives:

$$a_1v_1 + \cdots + a_nv_n + (-1)v = 0_V$$

This is a linear combination of vectors in $S \cup \{v\}$ equating to zero, where the coefficient of v is $-1 \neq 0$. Thus, $S \cup \{v\}$ is linearly dependent. ■

Example 1.5.6. Determine whether the set of vectors $S = \{(1,2,3), (2,-1,4), (0,5,2)\}$ in $\mathbb{R}^3$ is linearly dependent or independent.

Solution. We set up the homogeneous vector equation:

$$c_1(1,2,3) + c_2(2,-1,4) + c_3(0,5,2) = (0,0,0)$$

This yields a system of three linear equations:

$$\begin{aligned} c_1 + 2c_2 + 0c_3 &= 0 \quad \Rightarrow \quad c_1 = -2c_2 \\ 2c_1 - c_2 + 5c_3 &= 0 \\ 3c_1 + 4c_2 + 2c_3 &= 0 \end{aligned}$$

Substitute $c_1 = -2c_2$ into the second equation:

$$2(-2c_2) - c_2 + 5c_3 = 0 \implies -5c_2 + 5c_3 = 0 \implies c_2 = c_3$$

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Now substitute $c_1 = -2c_2$ and $c_3 = c_2$ into the third equation:

$$3(-2c_2) + 4c_2 + 2(c_2) = 0 \implies -6c_2 + 4c_2 + 2c_2 = 0 \implies 0 = 0$$

This means the system has infinitely many solutions. For instance, let $c_2 = 1$. Then $c_3 = 1$ and $c_1 = -2$. Because we found a non-trivial solution $(c_1, c_2, c_3) = (-2, 1, 1)$ such that:

$$-2(1, 2, 3) + 1(2, -1, 4) + 1(0, 5, 2) = (0, 0, 0)$$

The set S is **linearly dependent**. ■

Example 1.5.7. Let $V = P_2(\mathbb{R})$. Test the linear independence of the set of polynomials $S = \{1 - x, 1 + x, x^2\}$.

Solution. Set up the homogeneous equation for polynomials:

$$c_1(1 - x) + c_2(1 + x) + c_3(x^2) = 0 \quad \text{for all } x \in \mathbb{R}$$

Distribute and group the terms by powers of x:

$$(c_1 + c_2) + (-c_1 + c_2)x + c_3x^2 = 0 + 0x + 0x^2$$

By equating the corresponding coefficients of both sides, we get:

$$\begin{aligned} c_1 + c_2 &= 0 \\ -c_1 + c_2 &= 0 \\ c_3 &= 0 \end{aligned}$$

Adding the first two equations gives $2c_2 = 0 \implies c_2 = 0$. Substituting $c_2 = 0$ into the first equation gives $c_1 = 0$. Since the only solution is $c_1 = 0, c_2 = 0, c_3 = 0$, the set S is **linearly independent**. ■

Example 1.5.8. Let $V = M_{2\times2}(\mathbb{R})$. Determine if the following set of matrices is linearly independent:

$$S = \left\{ \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

Solution. We write the linear combination equating to the zero matrix:

$$c_1 \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} + c_4 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Adding the matrices on the left gives:

$$\begin{pmatrix} c_1 + c_4 & c_1 + c_2 \\ c_2 + c_3 & c_3 + c_4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

This yields the system:

$$c_1 + c_4 = 0 \implies c_4 = -c_1 \tag{1.2}$$
$$c_1 + c_2 = 0 \implies c_2 = -c_1 \tag{1.3}$$
$$c_2 + c_3 = 0 \implies c_3 = -c_2 = c_1 \tag{1.4}$$
$$c_3 + c_4 = 0 \tag{1.5}$$

Substitute the results of (1) and (3) into equation (4):

$$c_3 + c_4 = (c_1) + (-c_1) = 0$$

Equation (4) is automatically satisfied without forcing c_1 to be zero. We can choose $c_1 = 1$. This implies $c_2 = -1, c_3 = 1, c_4 = -1$. Since we found a non-zero solution $(1, -1, 1, -1)$, the matrices are **linearly dependent**. Indeed, notice that $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. ■

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1.6 Basis and Dimension of Vector Spaces

We have seen that a spanning set generates the entire vector space, while a linearly independent set contains no redundant vectors. When a set possesses both of these properties simultaneously, it forms the most efficient possible "skeleton" or foundation for the vector space. Such a set is called a basis.

Definition 1.6.1 (Basis)**.** Let V be a vector space over a field $\mathbb{K}$. A subset $\mathcal{B}$ of V is called a **basis** for V if it satisfies two conditions:

1. $\mathcal{B}$ is linearly independent.
2. $\mathcal{B}$ spans V (i.e., $L(\mathcal{B}) = V$).

Definition 1.6.2 (Finite-Dimensional Vector Space)**.** A vector space V is said to be **finite-dimensional** if it has a basis consisting of a finite number of vectors. If V cannot be spanned by any finite set of vectors, it is called infinite-dimensional (e.g., the space of all polynomials $\mathbb{K}[x]$). The trivial vector space $\{0_V\}$ is defined to be finite-dimensional with an empty basis $\emptyset$.

Definition 1.6.3 (Dimension)**.** The **dimension** of a finite-dimensional vector space V, denoted as $\dim(V)$, is defined as the exact number of elements in a basis for V.

Theorem 1.6.4 (Unique Representation Theorem)**.** *Let V be a vector space over $\mathbb{K}$. A set $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ is a basis for V if and only if every vector $v \in V$ can be uniquely expressed as a linear combination of the vectors in $\mathcal{B}$.*

Proof. Assume $\mathcal{B}$ is a basis for V. By definition, $\mathcal{B}$ spans V, so every vector $v \in V$ can be written as *at least one* linear combination:

$$v = c_1v_1 + c_2v_2 + \cdots + c_nv_n$$

We must show this representation is unique. Suppose there is another way to write v:

$$v = d_1v_1 + d_2v_2 + \cdots + d_nv_n$$

Subtracting the second equation from the first yields:

$$0_V = (c_1 - d_1)v_1 + (c_2 - d_2)v_2 + \cdots + (c_n - d_n)v_n$$

Since $\mathcal{B}$ is a basis, it is linearly independent. By definition of linear independence, the coefficients of the zero vector must all be zero:

$$c_1 - d_1 = 0, \quad c_2 - d_2 = 0, \quad \ldots, \quad c_n - d_n = 0$$

Thus, $c_i = d_i$ for all $i = 1, 2, \ldots, n$. The representation is indeed unique.

Assume every vector $v \in V$ has a unique representation as a linear combination of vectors in $\mathcal{B}$. Since every vector can be represented, $\mathcal{B}$ clearly spans V. To show $\mathcal{B}$ is linearly independent, consider the zero vector. We know 0_V can be written with all zero coefficients: $0_V = 0v_1 + \cdots + 0v_n$. Because we assumed the representation of *every* vector (including 0_V) is unique, this must be the *only* way to write 0_V. Therefore, the only solution to $c_1v_1 + \cdots + c_nv_n = 0_V$ is the trivial solution $c_i = 0$. Thus, $\mathcal{B}$ is linearly independent. Since it spans V and is linearly independent, $\mathcal{B}$ is a basis. ■

To prove that dimension is a well-defined concept, we must prove that two different bases cannot have different numbers of elements. We first need a crucial lemma.

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Lemma 1.6.5 (Size of LI set vs Spanning set). *Let V be a vector space over $\mathbb{K}$. If V is spanned by a finite set S of m vectors, then any linearly independent set L in V must contain at most m vectors.*

Proof. Let $S = \{v_1, v_2, \ldots, v_m\}$ be a spanning set for V. Suppose, for the sake of contradiction, that there exists a linearly independent set $L = \{w_1, w_2, \ldots, w_n\}$ in V with $n > m$. Since S spans V, every vector in L can be written as a linear combination of vectors in S:

$$\begin{aligned} w_1 &= a_{11}v_1 + a_{21}v_2 + \cdots + a_{m1}v_m \\ w_2 &= a_{12}v_1 + a_{22}v_2 + \cdots + a_{m2}v_m \\ &\vdots \\ w_n &= a_{1n}v_1 + a_{2n}v_2 + \cdots + a_{mn}v_m \end{aligned}$$

Consider the linear combination equation for testing the independence of L:

$$c_1w_1 + c_2w_2 + \cdots + c_nw_n = 0_V$$

Substitute the expressions for w_j into this equation and regroup the terms by v_i:

$$\left(\sum_{j=1}^{n} a_{1j}c_j\right) v_1 + \left(\sum_{j=1}^{n} a_{2j}c_j\right) v_2 + \cdots + \left(\sum_{j=1}^{n} a_{mj}c_j\right) v_m = 0_V$$

This gives us a homogeneous system of m linear equations in n unknowns $(c_1, \ldots, c_n)$:

$$\sum_{j=1}^{n} a_{ij}c_j = 0 \quad \text{for } i = 1, 2, \ldots, m$$

From the theory of linear equations, a homogeneous system with more unknowns than equations ($n > m$) *always* has non-trivial solutions. Thus, there exist scalars $c_1, \ldots, c_n$, not all zero, satisfying the system. But this means $c_1w_1 + \cdots + c_nw_n = 0_V$ has a non-trivial solution, implying L is linearly dependent. This contradicts our assumption that L is linearly independent. Therefore, our assumption $n > m$ must be false. Hence, $n \leq m$. ■

Theorem 1.6.6 (Invariance of Dimension). *If a vector space V has a basis of n vectors, then every basis of V contains exactly n vectors.*

Proof. Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two bases for V, containing n and m vectors respectively.

- Because $\mathcal{B}_1$ is a spanning set (size n) and $\mathcal{B}_2$ is a linearly independent set (size m), Lemma 1.6.5 implies that $m \leq n$.
- Because $\mathcal{B}_2$ is a spanning set (size m) and $\mathcal{B}_1$ is a linearly independent set (size n), Lemma 1.6.5 implies that $n \leq m$.

Combining $m \leq n$ and $n \leq m$ yields $m = n$. Thus, the number of basis vectors is an invariant property of the space. ■

Theorem 1.6.7 (Basis Extension Theorem). *Let V be a vector space of dimension n. Any linearly independent set $S = \{v_1, \ldots, v_k\}$ in V (where $k < n$) can be extended to form a basis for V.*

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Proof. Let $\mathcal{B} = \{e_1, e_2, \ldots, e_n\}$ be a known basis for V. Consider the set $S' = S \cup \mathcal{B} = \{v_1, \ldots, v_k, e_1, \ldots, e_n\}$. Clearly, S' spans V because it contains $\mathcal{B}$, but it is linearly dependent because $k + n > n$. We build a basis by adding vectors from $\mathcal{B}$ to S one by one, only if they do not destroy linear independence. Examine e_1. If $e_1 \in L(S)$, discard it. If $e_1 \notin L(S)$, add it to S. By the Extension Lemma, $S \cup \{e_1\}$ is linearly independent. Continue this process for $e_2, \ldots, e_n$. At each step, add e_i only if it is not in the span of the previously accepted vectors. When the list is exhausted, the resulting set will still be linearly independent. Furthermore, since every discarded vector e_j was in the span of the accepted vectors, the span of our new set contains all of $\mathcal{B}$, meaning it spans V. Thus, we have successfully extended S to a basis for V. (Since $\dim(V) = n$, we will have added exactly $n - k$ vectors). ■

Example 1.6.8. Show that the set of vectors $\mathcal{B} = \{(1, 1), (1, -1)\}$ is a basis for $\mathbb{R}^2$.

Solution. Since we know $\dim(\mathbb{R}^2) = 2$ and the set $\mathcal{B}$ has exactly 2 vectors, we only need to show that they are linearly independent (any linearly independent set of size n in an n-dimensional space automatically spans the space and is a basis). We set up the homogeneous equation:

$$c_1(1, 1) + c_2(1, -1) = (0, 0)$$

This gives the system of equations:

$$\begin{aligned} c_1 + c_2 &= 0 \\ c_1 - c_2 &= 0 \end{aligned}$$

Adding the two equations yields $2c_1 = 0 \implies c_1 = 0$. Substituting $c_1 = 0$ into the first equation yields $c_2 = 0$. Since the only solution is the trivial one, the vectors are linearly independent. Because there are 2 linearly independent vectors in a 2-dimensional space, $\mathcal{B}$ forms a basis for $\mathbb{R}^2$. ■

Example 1.6.9. Find a basis and the dimension for the subspace W of $\mathbb{R}^3$ defined by:

$$W = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\}$$

Solution. The subspace W consists of all vectors (x, y, z) that satisfy the condition $x + y + z = 0$. We can express one variable in terms of the others. Let $z = -x - y$. Now, rewrite an arbitrary vector in W using this substitution:

$$\begin{aligned} (x, y, z) &= (x, y, -x - y) \\ &= (x, 0, -x) + (0, y, -y) \\ &= x(1, 0, -1) + y(0, 1, -1) \end{aligned}$$

This shows that every vector in W can be written as a linear combination of the vectors $v_1 = (1, 0, -1)$ and $v_2 = (0, 1, -1)$. Thus, the set $\mathcal{B} = \{v_1, v_2\}$ spans W. Next, we check if $\mathcal{B}$ is linearly independent. Set up the homogeneous equation:

$$c_1(1, 0, -1) + c_2(0, 1, -1) = (0, 0, 0)$$

This yields the vector $(c_1, c_2, -c_1 - c_2) = (0, 0, 0)$. Equating components gives $c_1 = 0$ and $c_2 = 0$. Since $\mathcal{B}$ is linearly independent and spans W, $\mathcal{B} = \{(1, 0, -1), (0, 1, -1)\}$ is a basis for W. Because the basis contains exactly 2 vectors, the dimension of the subspace is $\dim(W) = 2$ (geometrically, it is a plane in $\mathbb{R}^3$). ■

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Example 1.6.10. Find a basis and the dimension for the vector space of all 2×2 symmetric matrices over $\mathbb{R}$.

Solution. Let V be the space of 2×2 symmetric matrices. A matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is symmetric if $A = A^T$, which means $b = c$. Thus, any matrix in V has the form:

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$$

We can decompose this generic matrix into a linear combination by factoring out the variables a, b, and d:

$$\begin{aligned} \begin{pmatrix} a & b \\ b & d \end{pmatrix} &= \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & b \\ b & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix} \\ &= a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Let $M_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $M_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, and $M_3 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. The set $\mathcal{B} = \{M_1, M_2, M_3\}$ clearly spans V. To check linear independence, set $c_1M_1 + c_2M_2 + c_3M_3 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$. This gives:

$$\begin{pmatrix} c_1 & c_2 \\ c_2 & c_3 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

This immediately implies $c_1 = 0$, $c_2 = 0$, and $c_3 = 0$. Thus, $\mathcal{B}$ is linearly independent. Therefore, $\mathcal{B} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ is a basis for V. Since the basis contains 3 matrices, the dimension of the space of 2×2 symmetric matrices is $\dim(V) = 3$. ■

Theorem 1.6.11 (Dimension of Product Space). *If V and W are finite-dimensional vector spaces over $\mathbb{K}$, then $V \times W$ is finite-dimensional, and:*

$$\dim(V \times W) = \dim(V) + \dim(W)$$

Proof. Let $\dim(V) = n$ and $\dim(W) = m$. Let $\mathcal{B}_V = \{v_1, v_2, \ldots, v_n\}$ be a basis for V and $\mathcal{B}_W = \{w_1, w_2, \ldots, w_m\}$ be a basis for W. Consider the set $S \subset V \times W$ defined by:

$$S = \{(v_1, 0_W), \ldots, (v_n, 0_W), (0_V, w_1), \ldots, (0_V, w_m)\}$$

We claim that S is a basis for $V \times W$. Note that S contains exactly $n + m$ elements.

1. S **spans** $V \times W$**:** Let (v, w) be any element in $V \times W$. Since $\mathcal{B}_V$ spans V and $\mathcal{B}_W$ spans W, we can write $v = \sum_{i=1}^{n} \alpha_i v_i$ and $w = \sum_{j=1}^{m} \beta_j w_j$. Then:

$$\begin{aligned} (v, w) &= (v, 0_W) + (0_V, w) \\ &= \left(\sum_{i=1}^{n} \alpha_i v_i, 0_W \right) + \left(0_V, \sum_{j=1}^{m} \beta_j w_j \right) \\ &= \sum_{i=1}^{n} \alpha_i (v_i, 0_W) + \sum_{j=1}^{m} \beta_j (0_V, w_j) \end{aligned}$$

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This shows (v, w) is a linear combination of elements in S. Thus, S spans $V \times W$.

2. S is linearly independent: Suppose $\sum_{i=1}^{n} \alpha_i(v_i, 0_W) + \sum_{j=1}^{m} \beta_j(0_V, w_j) = (0_V, 0_W)$. Combining the terms on the left gives:

$$\left(\sum_{i=1}^{n} \alpha_i v_i, \sum_{j=1}^{m} \beta_j w_j\right) = (0_V, 0_W)$$

This implies $\sum_{i=1}^{n} \alpha_i v_i = 0_V$ and $\sum_{j=1}^{m} \beta_j w_j = 0_W$. Because $\mathcal{B}_V$ and $\mathcal{B}_W$ are bases (and therefore linearly independent), all scalars must be zero: $\alpha_i = 0$ for all i and $\beta_j = 0$ for all j. Therefore, S is linearly independent. Since S is a basis with $n + m$ elements, $\dim(V \times W) = n + m = \dim(V) + \dim(W)$. ■

Example 1.6.12 (Basis of a Product Space)**.** Let $V = \mathbb{R}^2$ and $W = P_1(\mathbb{R})$ (polynomials of degree ≤ 1). Find the dimension and a basis for the product space $V \times W$.

Solution. We know $\dim(V) = 2$ with standard basis $\mathcal{B}_V = \{e_1 = (1, 0), e_2 = (0, 1)\}$. We know $\dim(W) = 2$ with standard basis $\mathcal{B}_W = \{1, x\}$. By the dimension theorem for product spaces, $\dim(V \times W) = \dim(V) + \dim(W) = 2 + 2 = 4$. To form the basis for $V \times W$, we pair the basis vectors of V with the zero vector of W (which is the zero polynomial 0), and pair the zero vector of V (which is $(0, 0)$) with the basis vectors of W. The basis is:

$$\mathcal{B}_{V \times W} = \{((1, 0), 0), ((0, 1), 0), ((0, 0), 1), ((0, 0), x)\}$$

■

Theorem 1.6.13 (Dimension of Quotient Space)**.** *Let V be a finite-dimensional vector space over $\mathbb{K}$, and let W be a subspace of V. Then V/W is finite-dimensional, and:*

$$\dim(V/W) = \dim(V) - \dim(W)$$

Proof. Let $\dim(V) = n$ and $\dim(W) = m$. Since W is a subspace of V, $m \leq n$. Let $\mathcal{B}_W = \{w_1, w_2, \ldots, w_m\}$ be a basis for W. By the basis extension theorem, we can extend this to a basis for the entire space V:

$$\mathcal{B}_V = \{w_1, w_2, \ldots, w_m, v_1, v_2, \ldots, v_{n-m}\}$$

We claim that the set of cosets $S = \{v_1 + W, v_2 + W, \ldots, v_{n-m} + W\}$ forms a basis for V/W.

1. S spans V/W: Let $x + W$ be an arbitrary element in V/W. Since $x \in V$, we can write it using the basis $\mathcal{B}_V$:

$$x = \alpha_1 w_1 + \cdots + \alpha_m w_m + \beta_1 v_1 + \cdots + \beta_{n-m} v_{n-m}$$

Notice that $\sum_{i=1}^{m} \alpha_i w_i \in W$. Let's call this sum w. Then $x = w + \sum_{j=1}^{n-m} \beta_j v_j$. Now, consider the coset $x + W$:

$$\begin{aligned} x + W &= \left(w + \sum_{j=1}^{n-m} \beta_j v_j\right) + W \\ &= \left(\sum_{j=1}^{n-m} \beta_j v_j\right) + (w + W) \\ &= \sum_{j=1}^{n-m} \beta_j (v_j + W) \quad (\text{since } w \in W \implies w + W = W) \end{aligned}$$

This shows $x + W$ is a linear combination of elements in S. Thus S spans V/W.

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2. S is linearly independent: Suppose $\sum_{j=1}^{n-m} c_j(v_j + W) = W$ (the zero vector in V/W). Then $\left(\sum_{j=1}^{n-m} c_j v_j\right) + W = W$. By definition of coset equality, this means the representative vector must belong to W:

$$\sum_{j=1}^{n-m} c_j v_j \in W$$

If this sum is in W, it can be written as a linear combination of the basis vectors of W:

$$\sum_{j=1}^{n-m} c_j v_j = \sum_{i=1}^{m} d_i w_i$$

Rearranging gives: $\sum_{j=1}^{n-m} c_j v_j - \sum_{i=1}^{m} d_i w_i = 0_V$. Because the entire set $\mathcal{B}_V = \{w_1, \ldots, w_m, v_1, \ldots, v_{n-m}\}$ is linearly independent, all coefficients in this sum must be zero. In particular, $c_1 = c_2 = \cdots = c_{n-m} = 0$. Therefore, S is linearly independent. Since S is a basis for V/W and contains exactly $n - m$ elements, $\dim(V/W) = n - m = \dim(V) - \dim(W)$. ■

Example 1.6.14 (Geometric interpretation of a Quotient Space)**.** Let $V = \mathbb{R}^2$ and let $W = \{(x, y) \in \mathbb{R}^2 \mid y = 2x\}$. Describe the elements of the quotient space $\mathbb{R}^2/W$ geometrically, and determine its dimension.

Solution. The space $V = \mathbb{R}^2$ is the standard Cartesian plane. The subspace W is a line passing through the origin with a slope of 2. An element of V/W is a coset $v + W$, where $v = (a, b) \in \mathbb{R}^2$. Geometrically, adding a fixed vector $v = (a, b)$ to every vector in the line W translates the entire line. The new set $v + W$ is a line parallel to W that passes through the point (a, b). Therefore, the elements of the quotient space $\mathbb{R}^2/W$ are **all the lines in $\mathbb{R}^2$ that are parallel to the line $y = 2x$**. To find the dimension, we use Theorem 1.6.13:

$$\dim(V/W) = \dim(\mathbb{R}^2) - \dim(W)$$

We know $\dim(\mathbb{R}^2) = 2$. The line W is spanned by the single vector $(1, 2)$, so $\dim(W) = 1$.

$$\dim(V/W) = 2 - 1 = 1$$

Thus, the quotient space is 1-dimensional. This makes geometric sense: we can uniquely identify every parallel line by its y-intercept, which is a single real number (a 1-dimensional parameter). ■

Example 1.6.15 (Quotient space of Matrices)**.** Let $V = M_{2\times 2}(\mathbb{R})$ be the space of 2×2 matrices. Let W be the subspace of all scalar matrices, i.e., $W = \left\{ \begin{pmatrix} c & 0 \\ 0 & c \end{pmatrix} \;\middle|\; c \in \mathbb{R} \right\}$. Find the dimension of V/W. Give an example of two distinct matrices in V that belong to the same coset in V/W.

Solution. First, we find the dimensions. The space V has $\dim(V) = 4$. The subspace W is spanned by the identity matrix $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, since every element is cI. Thus, $\dim(W) = 1$. By the quotient dimension theorem:

$$\dim(V/W) = \dim(V) - \dim(W) = 4 - 1 = 3$$

Next, we want to find two matrices $A, B \in V$ that belong to the same coset. Recall that $A + W = B + W$ if and only if $A - B \in W$. So, we need to pick two matrices whose difference is a scalar matrix. Let $A = \begin{pmatrix} 5 & 2 \\ 3 & 4 \end{pmatrix}$. Let us choose a scalar matrix from W, say $S = \begin{pmatrix} 10 & 0 \\ 0 & 10 \end{pmatrix}$. Let $B = A - S =$

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$\begin{pmatrix} 5-10 & 2-0 \\ 3-0 & 4-10 \end{pmatrix} = \begin{pmatrix} -5 & 2 \\ 3 & -6 \end{pmatrix}$. Because $A - B = S \in W$, the matrices A and B define the exact same coset in the quotient space: $A + W = B + W$. ■

1.7 Linear Transformations

Having studied the static structure of vector spaces, we now turn our attention to the dynamic aspect of linear algebra: the mappings or functions between vector spaces that preserve this algebraic structure. These structure-preserving maps are called linear transformations.

Definition 1.7.1 (Linear Transformation)**.** Let V and W be vector spaces over the same field $\mathbb{K}$. A mapping (or function) $T : V \to W$ is called a **linear transformation** (or vector space homomorphism) if it satisfies the following two conditions for all vectors $u, v \in V$ and all scalars $c \in \mathbb{K}$:

1. **Additivity:** $T(u + v) = T(u) + T(v)$
2. **Homogeneity:** $T(cu) = cT(u)$

Remark 1.7.2 (One-Step Condition). The two conditions above can be combined into a single, highly useful equivalent condition. A map $T : V \to W$ is a linear transformation if and only if:

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all $u, v \in V$ and all $\alpha, \beta \in \mathbb{K}$. This trivially extends by induction to any finite linear combination: $T(\sum_{i=1}^{n} c_i v_i) = \sum_{i=1}^{n} c_i T(v_i)$.

Example 1.7.3. Let $T : \mathbb{R} \to \mathbb{R}$ be defined by $T(x) = ax$, where $a \in \mathbb{R}$. Then T is a linear transformation.

Solution. To verify that T is a linear transformation, we check the two defining properties:

1. **Additivity:** For any $x, y \in \mathbb{R}$:

$$T(x + y) = a(x + y) = ax + ay = T(x) + T(y)$$

2. **Homogeneity:** For any $x \in \mathbb{R}$ and scalar $c \in \mathbb{R}$:

$$T(cx) = a(cx) = c(ax) = cT(x)$$

Since both properties hold, T is a linear transformation. ■

Example 1.7.4. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by $T(x, z, y) = (a_1x + b_1y + c_1z, a_2x + b_2y + c_2z)$, where $a_i, b_i \in \mathbb{R}$, $1 \leq i \leq 3$. Then T is a linear transformation.

Solution. To show that T is a linear transformation, we must verify that it satisfies the properties of additivity and homogeneity. Let $\mathbf{u} = (x_1, z_1, y_1)$ and $\mathbf{v} = (x_2, z_2, y_2)$ be vectors in $\mathbb{R}^3$, and let $\alpha \in \mathbb{R}$ be a scalar.

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1. **Additivity:** We check if $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$. The sum of the vectors is $\mathbf{u}+\mathbf{v} = (x_1+x_2, z_1+z_2, y_1+y_2)$. Applying T:

$$\begin{aligned}
T(\mathbf{u}+\mathbf{v}) &= T(x_1+x_2, z_1+z_2, y_1+y_2) \\
&= \big(a_1(x_1+x_2)+b_1(y_1+y_2)+c_1(z_1+z_2), a_2(x_1+x_2)+b_2(y_1+y_2)+c_2(z_1+z_2)\big) \\
&= \big((a_1x_1+b_1y_1+c_1z_1)+(a_1x_2+b_1y_2+c_1z_2), \\
&\quad (a_2x_1+b_2y_1+c_2z_1)+(a_2x_2+b_2y_2+c_2z_2)\big) \\
&= (a_1x_1+b_1y_1+c_1z_1, a_2x_1+b_2y_1+c_2z_1)+(a_1x_2+b_1y_2+c_1z_2, a_2x_2+b_2y_2+c_2z_2) \\
&= T(x_1, z_1, y_1) + T(x_2, z_2, y_2) \\
&= T(\mathbf{u}) + T(\mathbf{v})
\end{aligned}$$

2. **Homogeneity:** We check if $T(\alpha\mathbf{u}) = \alpha T(\mathbf{u})$. The scalar multiple is $\alpha\mathbf{u} = (\alpha x_1, \alpha z_1, \alpha y_1)$. Applying T:

$$\begin{aligned}
T(\alpha\mathbf{u}) &= T(\alpha x_1, \alpha z_1, \alpha y_1) \\
&= \big(a_1(\alpha x_1)+b_1(\alpha y_1)+c_1(\alpha z_1), a_2(\alpha x_1)+b_2(\alpha y_1)+c_2(\alpha z_1)\big) \\
&= \big(\alpha(a_1x_1+b_1y_1+c_1z_1), \alpha(a_2x_1+b_2y_1+c_2z_1)\big) \\
&= \alpha(a_1x_1+b_1y_1+c_1z_1, a_2x_1+b_2y_1+c_2z_1) \\
&= \alpha T(x_1, z_1, y_1) \\
&= \alpha T(\mathbf{u})
\end{aligned}$$

Since both conditions hold, T is a linear transformation. ■

Example 1.7.5 (The Most General Linear Transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$)**.** Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be any linear transformation. Then T is completely and uniquely determined by an $m \times n$ matrix A. Specifically, if we represent vectors as column vectors, then for every input vector $x = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$, the output vector $y = (y_1, y_2, \ldots, y_m)^T \in \mathbb{R}^m$ is given by the matrix multiplication:

$$T(x) = Ax$$

Explicitly, the transformation can be written as a system of linear equations where every output coordinate is a linear combination of the input coordinates:

$$\begin{aligned}
y_1 &= a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\
y_2 &= a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\
&\vdots \\
y_m &= a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n
\end{aligned}$$

where $a_{ij} \in \mathbb{R}$ are constants. *(No constant terms like $+5$ or non-linear terms like x_1^2 or $\sin(x_1)$ are allowed).*

Solution. Let $\{e_1, e_2, \ldots, e_n\}$ be the standard basis for $\mathbb{R}^n$. Any vector $x = (x_1, x_2, \ldots, x_n)^T \in \mathbb{R}^n$ can be expressed as $x = x_1e_1 + x_2e_2 + \cdots + x_ne_n$.

By the linearity of T, we can distribute the transformation over the sum and pull out the scalars:

$$T(x) = T(x_1e_1 + x_2e_2 + \cdots + x_ne_n) = x_1T(e_1) + x_2T(e_2) + \cdots + x_nT(e_n)$$

Let A be the $m \times n$ matrix whose columns are the fixed image vectors $T(e_1), T(e_2), \ldots, T(e_n)$. By the standard definition of matrix-vector multiplication, a linear combination of the columns of a matrix A with weights x_j is exactly the product Ax.

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Thus, we can write:

$$T(x) = \begin{bmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = Ax$$

This proves that every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is simply matrix multiplication, where the j-th column of A is the image of the j-th standard basis vector. This linear transformation can also be expressed as

$$T(x_1, x_2, \dots, x_n) = \left(\sum_{i=1}^{n} a_{1i}x_i, \sum_{i=1}^{n} a_{2i}x_i, \dots, \sum_{i=1}^{n} a_{mi}x_i \right)$$

as a mapping from $\mathbb{R}^n \to \mathbb{R}^m$

■

Example 1.7.6. Let $C^1(a,b)$ be the vector space of functions that are continuously differentiable on the interval (a,b), and let $F(a,b)$ be the vector space of all real-valued functions on (a,b). The derivative operator $D : C^1(a,b) \to F(a,b)$ defined by

$$D(f) = f'$$

is a linear transformation.

Solution. To prove that D is a linear transformation, we must show that it satisfies the two defining properties of linearity: additivity and homogeneity. Let $f, g \in C^1(a,b)$ and let $c \in \mathbb{R}$ be a scalar.

1. **Additivity:** We must show that $D(f+g) = D(f) + D(g)$. By the definition of the derivative:

   $$D(f+g) = \frac{d}{dx}[f(x) + g(x)]$$

   Using the sum rule for derivatives from calculus:

   $$\frac{d}{dx}[f(x) + g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$$

   Substituting the operator notation back in, we get:

   $$D(f+g) = D(f) + D(g)$$

   Thus, D is additive.

2. **Homogeneity:** We must show that $D(cf) = cD(f)$. By the definition of the derivative:

   $$D(cf) = \frac{d}{dx}[cf(x)]$$

   Using the constant multiple rule for derivatives:

   $$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

   Substituting the operator notation back in, we get:

   $$D(cf) = cD(f)$$

   Thus, D is homogeneous.

Since the operator D satisfies both additivity and homogeneity, it is a linear transformation. ■

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Example 1.7.7 (Finding a Transformation from its action on a Basis)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation. Suppose we know how T acts on two specific vectors:

$$T(1,1) = (1,0,2) \quad \text{and} \quad T(2,3) = (1,-1,4)$$

Find the explicit formula for $T(x,y)$ for any arbitrary vector $(x,y) \in \mathbb{R}^2$.

Solution. A fundamental property of linear transformations is that they are completely determined by their action on a basis. First, we observe that the set $S = \{(1,1),(2,3)\}$ is linearly independent and therefore forms a basis for $\mathbb{R}^2$. We need to express an arbitrary vector (x,y) as a linear combination of these basis vectors:

$$(x,y) = a(1,1) + b(2,3)$$

This leads to the system of equations:

$$\begin{aligned} a + 2b &= x \\ a + 3b &= y \end{aligned}$$

Subtracting the first equation from the second gives $b = y - x$. Substituting b back into the first equation gives $a = x - 2(y-x) = 3x - 2y$. So, we can uniquely write:

$$(x,y) = (3x-2y)(1,1) + (y-x)(2,3)$$

Now, we apply the linear transformation T to both sides and use the properties of linearity (additivity and homogeneity):

$$\begin{aligned} T(x,y) &= T\big((3x-2y)(1,1) + (y-x)(2,3)\big) \\ &= (3x-2y)T(1,1) + (y-x)T(2,3) \end{aligned}$$

Substitute the given output vectors:

$$\begin{aligned} T(x,y) &= (3x-2y)(1,0,2) + (y-x)(1,-1,4) \\ &= (3x-2y,\ 0,\ 6x-4y) + (y-x,\ -(y-x),\ 4y-4x) \\ &= (3x-2y+y-x,\ x-y,\ 6x-4y+4y-4x) \\ &= (2x-y,\ x-y,\ 2x) \end{aligned}$$

Thus, the explicit formula is $T(x,y) = (2x-y, x-y, 2x)$. ■

Example 1.7.8 (The Definite Integral as a Linear Transformation)**.** Let $V = P_2(\mathbb{R})$ be the vector space of polynomials of degree ≤ 2, and let $W = P_3(\mathbb{R})$ be the vector space of polynomials of degree ≤ 3. Define the mapping $T : V \to W$ by:

$$T(p(x)) = \int_0^x p(t)\,dt$$

Prove that T is a linear transformation.

Solution. We must verify the one-step linearity condition. Let $p(x)$ and $q(x)$ be any two polynomials in $P_2(\mathbb{R})$, and let $\alpha, \beta \in \mathbb{R}$ be any scalars. We apply T to the linear combination $\alpha p(x) + \beta q(x)$:

$$T(\alpha p(x) + \beta q(x)) = \int_0^x (\alpha p(t) + \beta q(t))\,dt$$

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By the fundamental properties of definite integrals in calculus (the integral of a sum is the sum of integrals, and constants can be factored out), we have:

$$\begin{aligned}\int_0^x (\alpha p(t) + \beta q(t))\,dt &= \int_0^x \alpha p(t)\,dt + \int_0^x \beta q(t)\,dt \\ &= \alpha \int_0^x p(t)\,dt + \beta \int_0^x q(t)\,dt \\ &= \alpha T(p(x)) + \beta T(q(x))\end{aligned}$$

Since $T(\alpha p(x) + \beta q(x)) = \alpha T(p(x)) + \beta T(q(x))$, the integral mapping T is a linear transformation. (Note: Integration always increases the degree of a polynomial by 1, which perfectly maps $P_2(\mathbb{R})$ into $P_3(\mathbb{R})$). ■

Example 1.7.9 (The Trace of a Matrix)**.** Let $V = M_{n\times n}(\mathbb{R})$ be the vector space of all $n \times n$ matrices over $\mathbb{R}$. Let $W = \mathbb{R}$ be the vector space of real numbers (a 1-dimensional space over itself). The **trace** of a square matrix $A = [a_{ij}]$, denoted $\text{tr}(A)$, is defined as the sum of its main diagonal entries: $\text{tr}(A) = \sum_{i=1}^{n} a_{ii}$. Define $T : M_{n\times n}(\mathbb{R}) \to \mathbb{R}$ by $T(A) = \text{tr}(A)$. Prove T is a linear transformation.

Solution. Notice that this transformation maps a matrix to a single scalar. Such a linear transformation is formally called a **linear functional**. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two $n \times n$ matrices, and let $c \in \mathbb{R}$. We want to evaluate $T(cA + B)$. First, the matrix $cA + B$ has entries of the form $[ca_{ij} + b_{ij}]$. The trace is the sum of the diagonal entries (where row index equals column index, $i = j$):

$$\begin{aligned}T(cA + B) &= \text{tr}(cA + B) \\ &= \sum_{i=1}^{n} (ca_{ii} + b_{ii})\end{aligned}$$

Using the standard properties of finite summations:

$$\begin{aligned}\sum_{i=1}^{n} (ca_{ii} + b_{ii}) &= \sum_{i=1}^{n} ca_{ii} + \sum_{i=1}^{n} b_{ii} \\ &= c\sum_{i=1}^{n} a_{ii} + \sum_{i=1}^{n} b_{ii} \\ &= c\,\text{tr}(A) + \text{tr}(B) \\ &= cT(A) + T(B)\end{aligned}$$

Therefore, the trace operator is a linear transformation. ■

Theorem 1.7.10 (Basic Properties of Linear Transformations)**.** *If $T : V \to W$ is a linear transformation, then:*

1. $T(0_V) = 0_W$ *(It maps the zero vector of V to the zero vector of W).*
2. $T(-v) = -T(v)$ *for all $v \in V$.*
3. $T(u - v) = T(u) - T(v)$ *for all $u, v \in V$.*

Proof. For (1), we use the scalar $0 \in \mathbb{K}$. We have $T(0_V) = T(0 \cdot 0_V) = 0 \cdot T(0_V) = 0_W$. For (2), we use the scalar $-1 \in \mathbb{K}$. We have $T(-v) = T((-1)v) = (-1)T(v) = -T(v)$. For (3), $T(u - v) = T(u + (-v)) = T(u) + T(-v) = T(u) - T(v)$. ■

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Theorem 1.7.11. *Let V be a finite-dimensional vector space over the field F and let $\{v_1, \ldots, v_n\}$ be an ordered basis for V. Let W be a vector space over the same field F and let $w_1, \ldots, w_n$ be any vectors in W. Then there is precisely one linear transformation T from V into W such that*

$$Tv_j = w_j, \quad j = 1, \ldots, n.$$

Proof. To prove there is some linear transformation T with $Tv_j = w_j$ we proceed as follows. Given v in V, there is a unique n-tuple $(\alpha_1, \ldots, \alpha_n)$ such that

$$v = \alpha_1 v_1 + \cdots + \alpha_n v_n.$$

For this vector v we define

$$Tv = \alpha_1 w_1 + \cdots + \alpha_n w_n.$$

Then T is a well-defined rule for associating with each vector v in V a vector Tv in W. From the definition it is clear that $Tv_j = w_j$ for each j. To see that T is linear, let

$$u = \beta_1 v_1 + \cdots + \beta_n v_n$$

be in V and let c be any scalar. Now

$$cv + u = (c\alpha_1 + \beta_1)v_1 + \cdots + (c\alpha_n + \beta_n)v_n$$

and so by definition

$$T(cv + w) = (c\alpha_1 + \beta_1)w_1 + \cdots + (c\alpha_n + \beta_n)w_n.$$

On the other hand,

$$\begin{aligned} c(Tv) + Tu &= c\sum_{i=1}^{n} \alpha_i w_i + \sum_{i=1}^{n} \beta_i w_i \\ &= \sum_{i=1}^{n} (c\alpha_i + \beta_i) w_i \end{aligned}$$

and thus

$$T(cv + u) = c(Tv) + Tu.$$

If U is a linear transformation from V into W with $Uv_j = w_j, j = 1, \ldots, n$, then for the vector $v = \sum_{i=1}^{n} \alpha_i v_i$ we have

$$\begin{aligned} Uv &= U\left(\sum_{i=1}^{n} \alpha_i v_i\right) \\ &= \sum_{i=1}^{n} \alpha_i (Uv_i) \\ &= \sum_{i=1}^{n} \alpha_i w_i \end{aligned}$$

which results in $Uv = Tv$, confirming the uniqueness of T. ■

Example 1.7.12. Find the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ given the basis $B = \{v_1 = (1, 1, -1), v_2 = (1, -1, 1), v_3 = (0, 1, 1)\}$ and images $w_1 = (1, 2), w_2 = (2, 3)$. (Note: For a unique transformation on $\mathbb{R}^3$, we assume $w_3 = (0, 0)$ if not provided). Find the image of $T(-1, 1, 0)$.

Coordinate Representation. We express a general vector $v = (x, y, z)$ as a linear combination of the basis B:

$$(x, y, z) = \alpha_1(1, 1, -1) + \alpha_2(1, -1, 1) + \alpha_3(0, 1, 1)$$

This results in the following system:

$$\alpha_1 + \alpha_2 = x$$

$$\alpha_1 - \alpha_2 + \alpha_3 = y$$

$$-\alpha_1 + \alpha_2 + \alpha_3 = z$$

Adding the second and third equations gives $2\alpha_3 = y + z$, so $\alpha_3 = \dfrac{y+z}{2}$. Using the first equation $\alpha_2 = x - \alpha_1$ and substituting into the second:

$$\alpha_1 - (x - \alpha_1) + \frac{y+z}{2} = y \implies 2\alpha_1 = x + \frac{y-z}{2} \implies \alpha_1 = \frac{2x+y-z}{4}$$

Then $\alpha_2 = x - \dfrac{2x+y-z}{4} = \dfrac{2x-y+z}{4}$.

Defining the Transformation. Following the provided proof, $Tv = \alpha_1 w_1 + \alpha_2 w_2 + \alpha_3 w_3$. Setting $w_3 = (0,0)$:

$$T(x,y,z) = \left(\frac{2x+y-z}{4}\right)(1,2) + \left(\frac{2x-y+z}{4}\right)(2,3) + \left(\frac{y+z}{2}\right)(0,0)$$

Simplifying the components:

$$T(x,y,z) = \left(\frac{2x+y-z+4x-2y+2z}{4}, \frac{4x+2y-2z+6x-3y+3z}{4}\right)$$

$$T(x,y,z) = \left(\frac{6x-y+z}{4}, \frac{10x-y+z}{4}\right)$$

Calculating the Image. To find the image of $T(-1,1,0)$, we substitute $x = -1, y = 1, z = 0$:

$$T(-1,1,0) = \left(\frac{6(-1)-1+0}{4}, \frac{10(-1)-1+0}{4}\right)$$

$$T(-1,1,0) = \left(-\frac{7}{4}, -\frac{11}{4}\right)$$

The linear transformation is $T(x,y,z) = \left(\dfrac{6x-y+z}{4}, \dfrac{10x-y+z}{4}\right)$ and the image of the requested vector is $T(-1,1,0) = (-1.75, -2.75)$.

A special and highly important case of a linear transformation occurs when the domain and codomain are the exact same vector space.

Definition 1.7.13 (Linear Operator)**.** A linear transformation from a vector space V into itself is called a **linear operator** on V. That is, a mapping $T : V \to V$ is a linear operator if $T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$ for all $u, v \in V$ and $\alpha, \beta \in \mathbb{K}$.

Two fundamental linear operators exist on every vector space:

- **The Identity Operator (I):** Defined by $I(v) = v$ for all $v \in V$.
- **The Zero Operator (0):** Defined by $0(v) = 0_V$ for all $v \in V$.

Just as we can add vectors and multiply them by scalars, we can perform these exact operations on linear transformations themselves.

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Definition 1.7.14. Let V and W be vector spaces over a field $\mathbb{K}$. Let $T_1 : V \to W$ and $T_2 : V \to W$ be linear transformations. We define their **sum** $T_1 + T_2$ and the **scalar multiplication** cT_1 (for $c \in \mathbb{K}$) point-wise as follows:

$$(T_1 + T_2)(v) = T_1(v) + T_2(v) \quad \text{for all } v \in V$$
$$(cT_1)(v) = c(T_1(v)) \quad \text{for all } v \in V$$

Theorem 1.7.15. *Let $L(V, W)$ (sometimes denoted $\mathrm{Hom}(V, W)$) be the set of all linear transformations from V to W. Under the point-wise operations of addition and scalar multiplication defined above, $L(V, W)$ forms a vector space over the field $\mathbb{K}$.*

Composition of Transformations:
If we have three vector spaces U, V, W over $\mathbb{K}$ and linear transformations $T_1 : U \to V$ and $T_2 : V \to W$, we can define their **composition** (or product) $T_2 \circ T_1 : U \to W$ by:

$$(T_2 \circ T_1)(u) = T_2(T_1(u)) \quad \text{for all } u \in U$$

The composition of two linear transformations is always a linear transformation. When dealing with linear operators on a single space V (i.e., $T_1, T_2 \in L(V)$), we can always compose them. We often write the composition simply as multiplication: T_2T_1.

Theorem 1.7.16 (Algebra of Linear Operators)**.** *The set of all linear operators on V, denoted $L(V)$ or $\mathrm{End}(V)$, forms a vector space under addition and scalar multiplication. Furthermore, under the operation of composition (multiplication), it satisfies:*

1. ***Associativity:*** $T_1(T_2T_3) = (T_1T_2)T_3$
2. ***Distributivity:*** $T_1(T_2 + T_3) = T_1T_2 + T_1T_3$ *and* $(T_1 + T_2)T_3 = T_1T_3 + T_2T_3$
3. ***Identity:*** $TI = IT = T$

*Because $L(V)$ is a vector space that is also a ring with an identity under composition, it is called a **Linear Algebra** over $\mathbb{K}$. (Note: Matrix multiplication, which represents operator composition, is generally non-commutative, so $T_1T_2 \neq T_2T_1$ in general).*

Proof. The fact that $L(V)$ forms a vector space follows naturally from point-wise definitions; the closure and axioms are inherited directly from the target space V. We prove the algebraic properties of composition by applying the operators to an arbitrary vector $v \in V$.

1. Associativity: Function composition is inherently associative. For any $v \in V$:

$$(T_1(T_2T_3))(v) = T_1((T_2T_3)(v)) = T_1(T_2(T_3(v))) = (T_1T_2)(T_3(v)) = ((T_1T_2)T_3)(v)$$

2. Distributivity: For left distributivity, we crucially use the fact that T_1 is a *linear* transformation:

$$\begin{aligned}(T_1(T_2 + T_3))(v) &= T_1((T_2 + T_3)(v)) \\ &= T_1(T_2(v) + T_3(v)) \quad \text{(definition of operator addition)} \\ &= T_1(T_2(v)) + T_1(T_3(v)) \quad \text{(by linearity of } T_1\text{)} \\ &= (T_1T_2)(v) + (T_1T_3)(v) = (T_1T_2 + T_1T_3)(v)\end{aligned}$$

For right distributivity, we only need the definition of operator addition:

$$\begin{aligned}((T_1+T_2)T_3)(v) &= (T_1+T_2)(T_3(v)) \\ &= T_1(T_3(v)) + T_2(T_3(v)) \quad \text{(definition of operator addition)} \\ &= (T_1T_3)(v) + (T_2T_3)(v) = (T_1T_3+T_2T_3)(v)\end{aligned}$$

3. Identity: Using the definition of the identity operator $I(v) = v$:

$$(TI)(v) = T(I(v)) = T(v) \quad \text{and} \quad (IT)(v) = I(T(v)) = T(v)$$

Thus, $(TI)(v) = (IT)(v) = T(v)$.

Since these equalities hold for every vector $v \in V$, the operator equations $T_1(T_2T_3) = (T_1T_2)T_3$, $T_1(T_2+T_3) = T_1T_2 + T_1T_3$, and $TI = IT = T$ are universally true. ■

Example 1.7.17. Determine whether the mapping $T : \mathbb{R}^2 \to \mathbb{R}^3$ defined by $T(x,y) = (x+y, x-y, 3x)$ is a linear transformation.

Solution. We must verify that T preserves vector addition and scalar multiplication. Let $u = (x_1, y_1)$ and $v = (x_2, y_2)$ be vectors in $\mathbb{R}^2$, and let $c \in \mathbb{R}$ be a scalar.

1. Additivity:

$$\begin{aligned}T(u+v) &= T(x_1+x_2, y_1+y_2) \\ &= ((x_1+x_2)+(y_1+y_2), (x_1+x_2)-(y_1+y_2), 3(x_1+x_2)) \\ &= (x_1+y_1+x_2+y_2, x_1-y_1+x_2-y_2, 3x_1+3x_2) \\ &= (x_1+y_1, x_1-y_1, 3x_1) + (x_2+y_2, x_2-y_2, 3x_2) \\ &= T(x_1,y_1) + T(x_2,y_2) \\ &= T(u) + T(v)\end{aligned}$$

2. Homogeneity:

$$\begin{aligned}T(cu) &= T(cx_1, cy_1) \\ &= (cx_1+cy_1, cx_1-cy_1, 3(cx_1)) \\ &= c(x_1+y_1, x_1-y_1, 3x_1) \\ &= cT(x_1,y_1) \\ &= cT(u)\end{aligned}$$

Since both conditions are satisfied, T is indeed a linear transformation. ■

Example 1.7.18. Show that the mapping $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x,y) = (x^2, y)$ is **not** a linear transformation.

Solution. To show a map is not a linear transformation, we only need to find one counterexample where either additivity or homogeneity fails. Let us test homogeneity. Let $u = (2,3)$ and let the scalar $c = 2$. First, we find $T(cu)$:

$$T(2\cdot(2,3)) = T(4,6) = (4^2, 6) = (16,6)$$

Now, we find $cT(u)$:

$$2\cdot T(2,3) = 2\cdot(2^2,3) = 2\cdot(4,3) = (8,6)$$

Since $(16,6) \neq (8,6)$, we have $T(cu) \neq cT(u)$. Therefore, T fails to preserve scalar multiplication and is not a linear transformation. ■

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Example 1.7.19 (The Derivative Operator)**.** Let $V = P_n(\mathbb{R})$ be the vector space of all polynomials of degree $\leq n$ over $\mathbb{R}$. Define the mapping $D : V \to V$ by $D(p(x)) = p'(x)$, where $p'(x)$ is the formal derivative of $p(x)$ with respect to x. Show that D is a linear operator on V.

Solution. First, note that if $p(x)$ has degree $\leq n$, its derivative $p'(x)$ has degree $\leq n-1$, which is still in V. Thus D maps V into V. Let $p(x)$ and $q(x)$ be polynomials in V, and let $\alpha, \beta \in \mathbb{R}$. We use the one-step condition to verify linearity:

$$\begin{aligned} D(\alpha p(x) + \beta q(x)) &= \frac{d}{dx}[\alpha p(x) + \beta q(x)] \\ &= \frac{d}{dx}[\alpha p(x)] + \frac{d}{dx}[\beta q(x)] \quad \text{(Sum rule of differentiation)} \\ &= \alpha \frac{d}{dx}[p(x)] + \beta \frac{d}{dx}[q(x)] \quad \text{(Constant multiple rule)} \\ &= \alpha D(p(x)) + \beta D(q(x)) \end{aligned}$$

Because differentiation is inherently a linear process, the derivative map D is a linear operator. ■

Example 1.7.20 (Algebra of Transformations)**.** Let $T_1, T_2 : \mathbb{R}^2 \to \mathbb{R}^2$ be linear operators defined by $T_1(x, y) = (x - y, y)$ and $T_2(x, y) = (2x, x + y)$. Find the explicit formulas for the operators $(T_1 + T_2)$ and $(T_1 T_2)$.

Solution. By the definition of the algebra of linear transformations:

1. Finding $(T_1 + T_2)$**:** We add the mappings point-wise:

$$\begin{aligned} (T_1 + T_2)(x, y) &= T_1(x, y) + T_2(x, y) \\ &= (x - y, y) + (2x, x + y) \\ &= (x - y + 2x, y + x + y) \\ &= (3x - y, x + 2y) \end{aligned}$$

2. Finding $(T_1 T_2)$ **(Composition):** Recall that $T_1 T_2$ means $T_1 \circ T_2$, which is applying T_2 first, then T_1.

$$\begin{aligned} (T_1 T_2)(x, y) &= T_1(T_2(x, y)) \\ &= T_1(2x, x + y) \end{aligned}$$

Now, apply the rule for $T_1(a, b) = (a - b, b)$, where $a = 2x$ and $b = x + y$:

$$\begin{aligned} T_1(2x, x + y) &= ((2x) - (x + y), x + y) \\ &= (2x - x - y, x + y) \\ &= (x - y, x + y) \end{aligned}$$

Thus, the composed operator formula is $(T_1 T_2)(x, y) = (x - y, x + y)$. ■

Example 1.7.21. Let V be the vector space of all infinite sequences of real numbers, $V = \{(x_1, x_2, x_3, \dots) \mid x_i \in \mathbb{R}\}$. Define two linear operators on V:

- The **Right Shift Operator** $R(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$
- The **Left Shift Operator** $L(x_1, x_2, x_3, \dots) = (x_2, x_3, x_4, \dots)$

Compute the compositions LR and RL, and show that they are not equal.

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Solution. Let $x = (x_1, x_2, x_3, \dots)$ be an arbitrary sequence in V.

First, we compute the composition LR (applying R then L):

$$\begin{aligned}(LR)(x) &= L(R(x_1, x_2, \dots)) \\ &= L(0, x_1, x_2, \dots)\end{aligned}$$

The operator L takes a sequence and deletes the first term. Therefore:

$$L(0, x_1, x_2, \dots) = (x_1, x_2, x_3, \dots) = x$$

Since $(LR)(x) = x$ for all x, we have $LR = I$ (the identity operator).

Next, we compute the composition RL (applying L then R):

$$\begin{aligned}(RL)(x) &= R(L(x_1, x_2, x_3, \dots)) \\ &= R(x_2, x_3, x_4, \dots)\end{aligned}$$

The operator R takes a sequence and inserts a 0 at the beginning. Therefore:

$$R(x_2, x_3, x_4, \dots) = (0, x_2, x_3, x_4, \dots)$$

Clearly, $(RL)(x) \neq x$ in general (it replaces the first term with 0). Thus, $RL \neq I$.

This elegantly demonstrates that while the operator algebra $L(V)$ has an associative multiplication, it is strictly non-commutative. Here, $LR = I$, but $RL \neq I$. ■

Theorem 1.7.22 (Dimension of the Space of Linear Transformations)**.** *Let V and W be finite-dimensional vector spaces over a field F. Let $\dim(V) = n$ and $\dim(W) = m$. The vector space $\mathcal{L}(V, W)$, consisting of all linear transformations from V to W, is finite-dimensional and:*

$$\dim(\mathcal{L}(V, W)) = \dim(V) \cdot \dim(W) = nm$$

Proof. To prove the theorem, we construct a basis for $\mathcal{L}(V, W)$ and show it contains nm elements. Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis for V and $\mathcal{C} = \{w_1, \dots, w_m\}$ be a basis for W.

For each pair of integers (i, j) such that $1 \leq i \leq m$ and $1 \leq j \leq n$, define a linear transformation $E_{ij} : V \to W$ by its effect on the basis $\mathcal{B}$:

$$E_{ij}(v_k) = \begin{cases} w_i & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases}$$

There are exactly $m \times n$ such transformations. Let $\mathcal{E} = \{E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$.

1. Linear Independence: Suppose $\sum_{i,j} c_{ij} E_{ij} = \mathbf{0}$ for some scalars $c_{ij} \in F$. Applying this sum to an arbitrary basis vector v_k:

$$\left(\sum_{j=1}^{n} \sum_{i=1}^{m} c_{ij} E_{ij} \right) (v_k) = \mathbf{0}(v_k) \implies \sum_{i=1}^{m} c_{ik} w_i = 0$$

Since $\{w_1, \dots, w_m\}$ is a basis for W, it is linearly independent. Thus, $c_{ik} = 0$ for all i. Since this holds for every $k \in \{1, \dots, n\}$, all coefficients c_{ij} must be zero. Hence, $\mathcal{E}$ is linearly independent.

2. Spanning Property: Let $T \in \mathcal{L}(V, W)$. For each $v_j \in \mathcal{B}$, $T(v_j)$ is a vector in W, so it can be written as

$$\begin{aligned}T(v_j) &= \sum_{i=1}^{m} a_{ij} w_i \\ &= \sum_{j=1}^{n} a_{ij} E_{ij}(v_j) \\ &= \sum_{j=1}^{n} \sum_{i=1}^{m} a_{ij} E_{ij}(v_k)\end{aligned}$$

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Thus, $\mathcal{E}$ spans $\mathcal{L}(V, W)$. Since $\mathcal{E}$ is a basis with nm elements, the dimension is nm. ■

Definition 1.7.23 (Sum of Vector Spaces)**.** Let V and W be subspaces of a vector space U over a field F. The **sum** of V and W, denoted by $V + W$, is the set of all vectors that can be written as the sum of a vector from V and a vector from W. Formally, this is defined as:

$$V + W = \{v + w : v \in V, w \in W\}$$

The sum $V + W$ is the smallest subspace of U that contains both V and W.

Theorem 1.7.24 (Dimension of the Sum of Subspaces)**.** *Let V and W be finite-dimensional subspaces of a vector space U. The dimension of their sum $V + W$ is given by:*

$$\dim(V + W) = \dim(V) + \dim(W) - \dim(V \cap W)$$

Proof. Let $V \cap W$ have dimension k, and let $\{u_1, \ldots, u_k\}$ be a basis for $V \cap W$. By the basis extension theorem, we can extend this to:

1. A basis for V: $\{u_1, \ldots, u_k, v_1, \ldots, v_n\}$, so $\dim(V) = k + n$.
2. A basis for W: $\{u_1, \ldots, u_k, w_1, \ldots, w_m\}$, so $\dim(W) = k + m$.

We claim that $\mathcal{B} = \{u_1, \ldots, u_k, v_1, \ldots, v_n, w_1, \ldots, w_m\}$ is a basis for $V + W$.

1. Spanning Property: Any $x \in V + W$ is $x = v + w$. Since $v \in \text{span}\{u_i, v_j\}$ and $w \in \text{span}\{u_i, w_l\}$, their sum x is clearly in $\text{span}(\mathcal{B})$.

2. Linear Independence: Suppose $\sum a_i u_i + \sum b_j v_j + \sum c_l w_l = 0$. Then:

$$\sum a_i u_i + \sum b_j v_j = -\sum c_l w_l$$

The left side is a vector in V and the right side is a vector in W. Thus, the vector $y = -\sum c_l w_l$ is in $V \cap W$. We can write y in terms of the basis for $V \cap W$ as $y = \sum d_i u_i$. Substituting this: $\sum d_i u_i + \sum c_l w_l = 0$. Since $\{u_i, w_l\}$ is a basis for W, all $c_l = 0$. The original equation then becomes $\sum a_i u_i + \sum b_j v_j = 0$. Since $\{u_i, v_j\}$ is a basis for V, all $a_i = 0$ and $b_j = 0$. Thus, $\mathcal{B}$ is linearly independent.

Conclusion: The number of vectors in $\mathcal{B}$ is $k + n + m$. Note that $(k + n) + (k + m) - k = k + n + m$. Therefore, $\dim(V + W) = \dim(V) + \dim(W) - \dim(V \cap W)$. ■

Definition 1.7.25 (Null Space and Range)**.** Let $T : V \to W$ be a linear transformation.

1. The **null space** (or kernel) of T is $N(T) = \{v \in V \mid T(v) = 0_W\}$.
2. The **range** (or image) of T is $R(T) = \{T(v) \mid v \in V\}$.

Theorem 1.7.26. *If $T : V \to W$ is a linear transformation, then $N(T)$ is a subspace of V and $R(T)$ is a subspace of W.*

Proof of Theorem (Subspaces $N(T)$ and $R(T)$). Since $T(0_V) = 0_W$, both $0_V \in N(T)$ and $0_W \in R(T)$, meaning the sets are non-empty. We use the subspace test.

- **Null Space $N(T)$:** Let $x, y \in N(T)$ and $c \in \mathbb{K}$. Then $T(x) = 0_W$ and $T(y) = 0_W$. By linearity, $T(cx + y) = cT(x) + T(y) = c(0_W) + 0_W = 0_W$. Thus, $cx + y \in N(T)$, proving $N(T)$ is a subspace of V.

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- **Range $R(T)$:** Let $u, v \in R(T)$ and $c \in \mathbb{K}$. By definition, there exist $x, y \in V$ such that $T(x) = u$ and $T(y) = v$. By linearity, $cu + v = cT(x) + T(y) = T(cx + y)$. Since $cx + y \in V$, $cu + v \in R(T)$, proving $R(T)$ is a subspace of W.

■

Definition 1.7.27 (Rank and Nullity)**.** Let $T : V \to W$ be a linear transformation.

1. The **nullity** of T is the dimension of the null space, $\dim(N(T))$.
2. The **rank** of T is the dimension of the range, $\dim(R(T))$.

Theorem 1.7.28 (Sylvester's Law of Nullity / Rank-Nullity Theorem)**.** *Let V and W be vector spaces over a field $\mathbb{K}$, and let $T : V \to W$ be a linear transformation. If V is finite-dimensional, then:*

$$\dim(V) = rank(T) + nullity(T)$$

Proof of Theorem (Sylvester's Law of Nullity). Let $\dim(N(T)) = k$ and $\dim(V) = n$. Let $\{u_1, u_2, \ldots, u_k\}$ be a basis for $N(T)$. Since $N(T)$ is a subspace of V, we can extend this to a basis for the entire space V: $\{u_1, \ldots, u_k, v_1, \ldots, v_{n-k}\}$. We claim that the set $S = \{T(v_1), \ldots, T(v_{n-k})\}$ is a basis for $R(T)$.

1. S spans $R(T)$: Let $y \in R(T)$. Then $y = T(x)$ for some $x \in V$. We can write $x = \sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j$.

Applying T and using linearity gives:

$$y = T(x) = \sum_{i=1}^{k} a_i T(u_i) + \sum_{j=1}^{n-k} b_j T(v_j)$$

Since $u_i \in N(T)$, $T(u_i) = 0$. Thus $y = \sum_{j=1}^{n-k} b_j T(v_j)$, meaning S spans $R(T)$.

2. S is linearly independent: Suppose $\sum_{j=1}^{n-k} c_j T(v_j) = 0_W$. By linearity, $T\left(\sum_{j=1}^{n-k} c_j v_j\right) = 0_W$. This implies the vector $\sum_{j=1}^{n-k} c_j v_j$ lies in $N(T)$. Therefore, it can be written as a linear combination of the basis vectors of $N(T)$:

$$\sum_{j=1}^{n-k} c_j v_j = \sum_{i=1}^{k} d_i u_i \implies \sum_{j=1}^{n-k} c_j v_j - \sum_{i=1}^{k} d_i u_i = 0_V$$

Because $\{u_1, \ldots, u_k, v_1, \ldots, v_{n-k}\}$ is a basis for V (and thus linearly independent), all coefficients must be zero. In particular, $c_j = 0$ for all j. Hence, S is linearly independent. Since S is a basis for $R(T)$ with $n - k$ elements, $\text{rank}(T) = n - k$. Therefore, $\dim(V) = n = k + (n - k) = \text{nullity}(T) + \text{rank}(T)$. ■

Example 1.7.29. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation defined by:

$$T(x, y, z) = (x - y, y + z)$$

Find the null space, range, nullity, and rank of T. Verify the Rank-Nullity Theorem.

Solution. **1. Null Space and Nullity:** The null space $N(T)$ consists of all vectors $(x, y, z) \in \mathbb{R}^3$ such that

$T(x, y, z) = (0, 0)$. Setting the output to the zero vector gives a system of linear equations:

$$x - y = 0 \implies x = y$$
$$y + z = 0 \implies z = -y$$

Let $y = t$, where $t \in \mathbb{R}$ is a free parameter. Then $x = t$ and $z = -t$. Thus, any vector in the null space has the form:

$$(x, y, z) = (t, t, -t) = t(1, 1, -1)$$

This shows that $N(T)$ is spanned by the single vector $(1, 1, -1)$. Since this single non-zero vector is linearly independent, it forms a basis for $N(T)$. Therefore, the **null space** is $N(T) = \text{span}\{(1, 1, -1)\}$, and the **nullity** is $\dim(N(T)) = 1$.

2. Range and Rank: The range $R(T)$ is the span of the images of the standard basis vectors of the domain $\mathbb{R}^3$. Let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, and $e_3 = (0, 0, 1)$.

$$T(e_1) = T(1, 0, 0) = (1 - 0, 0 + 0) = (1, 0)$$
$$T(e_2) = T(0, 1, 0) = (0 - 1, 1 + 0) = (-1, 1)$$
$$T(e_3) = T(0, 0, 1) = (0 - 0, 0 + 1) = (0, 1)$$

So, $R(T) = \text{span}\{(1, 0), (-1, 1), (0, 1)\}$. To find a basis for the range, we need to extract the linearly independent vectors from this set. Notice that $(1, 0)$ and $(0, 1)$ are the standard basis vectors for $\mathbb{R}^2$, and they are clearly linearly independent. Since they span all of $\mathbb{R}^2$, the range is the entire codomain. Therefore, the **range** is $R(T) = \mathbb{R}^2$, and the **rank** is $\dim(R(T)) = 2$.

3. Verification of Rank-Nullity Theorem: The domain is $V = \mathbb{R}^3$, so $\dim(V) = 3$.

$$\text{Rank}(T) + \text{Nullity}(T) = 2 + 1 = 3 = \dim(\mathbb{R}^3)$$

The theorem holds perfectly. ■

Example 1.7.30 (Constructing a Transformation with a specific Null Space)**.** Construct a linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^2$ such that its null space is exactly the line generated by the vector $(1, 2, 3)$.

Solution. We want the null space $N(T) = \text{span}\{(1, 2, 3)\}$. This means we require:

$$T(1, 2, 3) = (0, 0)$$

To completely define T, we must specify its action on a basis of $\mathbb{R}^3$. We start with $(1, 2, 3)$ and extend it to a full basis for $\mathbb{R}^3$. Let's add the standard basis vectors $e_2 = (0, 1, 0)$ and $e_3 = (0, 0, 1)$. The set $\mathcal{B} = \{(1, 2, 3), (0, 1, 0), (0, 0, 1)\}$ is easily verified to be linearly independent, hence a basis.

To ensure the null space doesn't contain any *more* vectors than the span of $(1, 2, 3)$, the other two basis vectors must map to linearly independent vectors in $\mathbb{R}^2$. Let's map them to the standard basis of $\mathbb{R}^2$:

$$T(0, 1, 0) = (1, 0)$$
$$T(0, 0, 1) = (0, 1)$$

Now, we find the general formula for $T(x, y, z)$. We express (x, y, z) in terms of our basis $\mathcal{B}$:

$$(x, y, z) = a(1, 2, 3) + b(0, 1, 0) + c(0, 0, 1)$$

$$(x, y, z) = (a, 2a + b, 3a + c)$$

Equating components: $a = x$
$2a + b = y \implies b = y - 2x$
$3a + c = z \implies c = z - 3x$

So, $(x,y,z) = x(1,2,3) + (y-2x)(0,1,0) + (z-3x)(0,0,1)$. Applying the linear transformation T:

$$\begin{aligned} T(x,y,z) &= xT(1,2,3) + (y-2x)T(0,1,0) + (z-3x)T(0,0,1) \\ &= x(0,0) + (y-2x)(1,0) + (z-3x)(0,1) \\ &= (0,0) + (y-2x,0) + (0,z-3x) \\ &= (y-2x,\, z-3x) \end{aligned}$$

Thus, the constructed linear transformation is $T(x,y,z) = (y-2x, z-3x)$. *(Quick check:* $T(1,2,3) = (2-2(1), 3-3(1)) = (0,0)$*, as required!)* ■

Example 1.7.31 (Polynomial Space)**.** Let $P_2(\mathbb{R})$ be the vector space of polynomials of degree ≤ 2. Let $T : P_2(\mathbb{R}) \to P_2(\mathbb{R})$ be the linear operator defined by taking the derivative and multiplying by x:

$$T(p(x)) = x \cdot p'(x)$$

Find a basis for the null space and the range, and determine the rank and nullity.

Solution. An arbitrary polynomial in $P_2(\mathbb{R})$ has the form $p(x) = a + bx + cx^2$. Let's apply the transformation T to this arbitrary polynomial:

$$p'(x) = b + 2cx$$
$$T(p(x)) = x(b + 2cx) = bx + 2cx^2$$

1. Null Space and Nullity: To find $N(T)$, we set $T(p(x)) = 0$ (the zero polynomial).

$$bx + 2cx^2 = 0 + 0x + 0x^2$$

Equating coefficients, we get $b = 0$ and $2c = 0 \implies c = 0$. There is no restriction on a. Thus, if $p(x) \in N(T)$, it must be of the form $p(x) = a$. This means the null space consists exclusively of constant polynomials. A basis for $N(T)$ is the constant polynomial $\{1\}$. Therefore, the **nullity** is $\dim(N(T)) = 1$.

2. Range and Rank: From our general formula, the output of the transformation is always of the form:

$$T(p(x)) = bx + 2cx^2 = b(x) + c(2x^2)$$

This explicitly shows that any polynomial in the range is a linear combination of the polynomials x and $2x^2$. Thus, the set $\{x, 2x^2\}$ spans $R(T)$. Since these two polynomials are not scalar multiples of each other, they are linearly independent and form a basis for $R(T)$. (Equivalently, you could use $\{x, x^2\}$ as the basis). Therefore, the **rank** is $\dim(R(T)) = 2$.

3. Verification: The dimension of the domain $P_2(\mathbb{R})$ is 3 (with standard basis $\{1, x, x^2\}$).

$$\text{Rank} + \text{Nullity} = 2 + 1 = 3 = \dim(P_2(\mathbb{R}))$$

■

Example 1.7.32 (Matrix Space)**.** Let $V = M_{2\times 2}(\mathbb{R})$ be the vector space of all 2×2 matrices over $\mathbb{R}$. Define a linear transformation $T : V \to V$ by:

$$T(A) = A - A^T$$

Find the rank and nullity of T.

Solution. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be an arbitrary matrix in V. Then its transpose is $A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$. Applying the transformation:

$$T(A) = \begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} 0 & b-c \\ c-b & 0 \end{pmatrix}$$

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1. Null Space and Nullity: To find $N(T)$, we set $T(A)$ equal to the 2×2 zero matrix:

$$\begin{pmatrix} 0 & b-c \\ c-b & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

This yields the single independent equation $b - c = 0 \implies b = c$. There are no restrictions on a and d. Thus, a matrix in the null space must look like:

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$$

This is the definition of a symmetric matrix. Thus, $N(T)$ is the subspace of all 2×2 symmetric matrices. We can decompose this generic matrix to find a basis:

$$\begin{pmatrix} a & b \\ b & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

These three matrices are linearly independent and span $N(T)$. Therefore, the **nullity** is 3.

2. Range and Rank: Looking at the output formula, any matrix in the range has the form:

$$\begin{pmatrix} 0 & b-c \\ c-b & \end{pmatrix} = \begin{pmatrix} 0 & (b-c) \\ -(b-c) & 0 \end{pmatrix}$$

Let $k = b - c$, where k is an arbitrary real number. The range consists of all matrices of the form:

$$k \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

This is the subspace of all 2×2 skew-symmetric matrices. It is spanned by the single non-zero matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, which forms its basis. Therefore, the **rank** is 1.

3. Verification: The dimension of the domain $M_{2\times2}(\mathbb{R})$ is 4.

$$\text{Rank} + \text{Nullity} = 1 + 3 = 4 = \dim(M_{2\times2}(\mathbb{R}))$$

■

Theorem 1.7.33 (First Isomorphism Theorem). *Let $T : V \to W$ be a linear transformation. Then the quotient space $V/N(T)$ is isomorphic to the range of T, i.e.,*

$$V/N(T) \cong R(T)$$

Proof. Define a map $\bar{T} : V/N(T) \to R(T)$ by $\bar{T}(v + N(T)) = T(v)$.

- **Well-defined and Injective:** $v + N(T) = u + N(T) \iff v - u \in N(T) \iff T(v - u) = 0_W \iff T(v) = T(u) \iff \bar{T}(v + N(T)) = \bar{T}(u + N(T))$. Reading forward, $\bar{T}$ is well-defined; reading backward, $\bar{T}$ is injective.
- **Surjective:** For any $y \in R(T)$, there exists $v \in V$ such that $T(v) = y$. Thus, $\bar{T}(v + N(T)) = y$, making $\bar{T}$ surjective.
- **Linearity:** $\bar{T}(\alpha(u + N(T)) + (v + N(T))) = \bar{T}((\alpha u + v) + N(T)) = T(\alpha u + v) = \alpha T(u) + T(v) = \alpha\bar{T}(u + N(T)) + \bar{T}(v + N(T))$.

Since $\bar{T}$ is a bijective linear transformation, it is an isomorphism. ■

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Theorem 1.7.34 (Second and Third Isomorphism Theorems)**.** *Let U and W be subspaces of a vector space V. Then:*

1. *(Second)* $(U+W)/W \cong U/(U\cap W)$
2. *(Third) If* $W \subseteq U$*, then* $(V/W)/(U/W) \cong V/U$

Proof. Both theorems elegantly follow from applying the First Isomorphism Theorem to carefully chosen linear transformations.

1. Second Isomorphism Theorem: Define a map $\phi : U \to (U+W)/W$ by $\phi(u) = u + W$. Clearly, ϕ is linear. To show it is surjective, let $x + W \in (U+W)/W$. By definition, $x = u + w$ for some $u \in U, w \in W$. Then $x + W = (u+w) + W = u + W = \phi(u)$. Thus $R(\phi) = (U+W)/W$. Now, find the null space: $u \in N(\phi) \iff \phi(u) = W \iff u + W = W \iff u \in W$. Since the domain is U, $N(\phi) = U \cap W$. By the First Isomorphism Theorem, $U/N(\phi) \cong R(\phi)$, which yields $U/(U\cap W) \cong (U+W)/W$.

2. Third Isomorphism Theorem: Assume $W \subseteq U \subseteq V$. Define a map $\psi : V/W \to V/U$ by $\psi(v+W) = v + U$. This is well-defined because if $v_1 + W = v_2 + W$, then $v_1 - v_2 \in W \subseteq U$, meaning $v_1 + U = v_2 + U$. Linearity is straightforward. ψ is clearly surjective because for any $v + U \in V/U$, we have $\psi(v+W) = v + U$. Thus $R(\psi) = V/U$. Now, find the null space: $v + W \in N(\psi) \iff \psi(v+W) = U \iff v + U = U \iff v \in U$. So, $N(\psi)$ consists of all cosets $v + W$ where $v \in U$. This is exactly the definition of the quotient space U/W. Thus, $N(\psi) = U/W$. By the First Isomorphism Theorem, $(V/W)/N(\psi) \cong R(\psi)$, which yields $(V/W)/(U/W) \cong V/U$. ■

Exercise 1.7.35. Solve the following exercises based on Chapter 1.

1. Let $V = \mathbb{R}^+$ be the set of strictly positive real numbers. Define vector addition as $x \oplus y = xy$ (standard real multiplication) and scalar multiplication for any $\alpha \in \mathbb{R}$ as $\alpha \odot x = x^\alpha$. Prove that V is a vector space over $\mathbb{R}$ under these operations. What is the zero vector in this space?
2. Determine which of the following subsets of $M_{n\times n}(\mathbb{R})$ are subspaces:
   (a) The set of all upper triangular matrices.
   (b) The set of all matrices A such that $\det(A) = 0$.
   (c) The set of all matrices A such that the trace of A is zero (i.e., $\operatorname{tr}(A) = 0$).
3. Let W_1 and W_2 be subspaces of a vector space V. Prove that the sum $W_1 + W_2$ is the smallest subspace of V containing both W_1 and W_2.
4. Let $V = \mathbb{R}^3$ and let $W = \{(x,y,z) \in \mathbb{R}^3 \mid x = y = z\}$ be a subspace of V. Find the dimension and a basis for the quotient space V/W.
5. Let $U = P_1(\mathbb{R})$ (polynomials of degree ≤ 1) and $W = \mathbb{R}^2$. Consider the product space $U \times W$. Find its dimension and write down a standard basis for $U \times W$.
6. Let $V = C(\mathbb{R})$ be the vector space of continuous functions from $\mathbb{R}$ to $\mathbb{R}$. Determine whether the set of functions $S = \{1, \sin^2(x), \cos^2(x)\}$ is linearly independent or dependent. Justify your answer.
7. Let $\{u, v, w\}$ be a linearly independent set of vectors in a vector space V over $\mathbb{R}$. Prove that the set $\{u+v, v+w, w+u\}$ is also linearly independent.
8. In the vector space $P_3(\mathbb{R})$, consider the polynomials $p_1(x) = 1 + x$, $p_2(x) = x + x^2$, $p_3(x) = x^2 + x^3$, and $p_4(x) = 1 + x^3$. Are these polynomials linearly dependent or independent? If dependent, express one as a linear combination of the others.

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9. Let $W = \text{span}\{(1,2,-1,3),(2,4,-2,6),(0,1,1,-1)\} \subset \mathbb{R}^4$. Find the dimension of W and extract a basis for W from the given spanning set.

10. Let $S = \{(1,0,1,0),(0,1,0,1)\}$. Extend the set S to form a basis for $\mathbb{R}^4$.

11. Find the dimension and a basis for the subspace W of $M_{3\times3}(\mathbb{R})$ consisting of all 3×3 skew-symmetric matrices (matrices where $A^T = -A$).

12. Let $V = P_n(\mathbb{R})$. Let $a \in \mathbb{R}$ be a fixed scalar. Consider the set $W = \{p(x) \in V \mid p(a) = 0\}$. Prove that W is a subspace of V and find its dimension.

13. Determine whether the following mappings are linear transformations:

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$ defined by $T(x,y) = (x+1, y)$.

(b) $T : M_{2\times2}(\mathbb{R}) \to \mathbb{R}$ defined by $T(A) = \det(A)$.

(c) $T : C[0,1] \to \mathbb{R}$ defined by $T(f) = \int_0^1 f(x)\,dx$.

14. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be a linear transformation. Suppose we know that $T(1,1) = (1,0,2)$ and $T(2,3) = (1,-1,4)$. Find the explicit formula for $T(x,y)$ for any arbitrary vector $(x,y) \in \mathbb{R}^2$.

15. Let $T : P_2(\mathbb{R}) \to P_3(\mathbb{R})$ be a linear transformation defined by $T(p(x)) = xp(x) + p'(x)$. Find a basis for the null space $N(T)$ and the range $R(T)$. State the rank and nullity of T.

16. Define a linear transformation $T : M_{2\times2}(\mathbb{R}) \to M_{2\times2}(\mathbb{R})$ by $T(A) = AB - BA$, where $B = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Find the nullity and rank of T. Verify Sylvester's theorem for this transformation.

17. Use Sylvester's Law of Nullity to prove the following: If $T : V \to W$ is a linear transformation between finite-dimensional vector spaces and $\dim(V) > \dim(W)$, then T cannot be an injective (one-to-one) mapping.

18. Let $T : \mathbb{R}^4 \to \mathbb{R}^3$ be defined by $T(x,y,z,w) = (x+y-z, y+z+w, x-2z-w)$. Find the dimension of the null space and the dimension of the range of T.

19. Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V such that $T^2 = T \circ T = 0$ (the zero operator).

(a) Prove that $R(T) \subseteq N(T)$.

(b) Using Sylvester's theorem, deduce that $\text{rank}(T) \leq \frac{1}{2}\dim(V)$.

20. Let V be a vector space, and let W_1 and W_2 be subspaces such that $V = W_1 \oplus W_2$ (that is, $V = W_1 + W_2$ and $W_1 \cap W_2 = \{0\}$). Use the Isomorphism Theorems to prove that $V/W_1 \cong W_2$.

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CHAPTER

2

Matrix Representations, Dual Spaces, and Spectral Theory

This unit explores the fundamental relationship between linear transformations and their matrix representations, emphasizing how these maps behave under changes of coordinate systems. We introduce the concepts of dual spaces, dual bases, and annihilators to deepen our understanding of the underlying structure of vector spaces and linear functionals. The study further extends into spectral theory, utilizing characteristic and minimal polynomials to analyze the intrinsic properties of linear operators. Ultimately, these concepts culminate in the study of eigenvalues and diagonalizability, providing essential tools for simplifying and solving complex systems in linear algebra.

2.1 Matrix Representation of a Linear Transformation

We have seen that a linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented by a matrix multiplication. A profound result in linear algebra is that *every* linear transformation between *any* finite-dimensional vector spaces (polynomials, matrices, etc.) can be perfectly represented by a matrix, provided we fix a basis for both spaces. This allows us to translate abstract linear algebra problems into concrete matrix calculations.

2.1.1 Coordinates and Ordered Bases

To bridge the gap between abstract vectors and matrices (which are arrays of numbers), we need the concept of coordinates.

Definition 2.1.1 (Ordered Basis)**.** An **ordered basis** for a finite-dimensional vector space V is a basis for V endowed with a specific, fixed order. If V has dimension n, an ordered basis is written as a sequence: $\alpha = (v_1, v_2, \ldots, v_n)$.

Definition 2.1.2 (Coordinate Vector)**.** Let V be a vector space over $\mathbb{K}$ with an ordered basis $\alpha = (v_1, v_2, \ldots, v_n)$. By the Unique Representation Theorem, any vector $v \in V$ can be uniquely written as:

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$$

The **coordinate vector** of v relative to the basis α, denoted by $[v]_\alpha$, is the column vector in $\mathbb{K}^n$

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whose components are the unique coefficients c_i:

$$[v]_\alpha = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$$

Let V be an n-dimensional vector space with ordered basis $\alpha = (v_1, v_2, \ldots, v_n)$. Let W be an m-dimensional vector space with ordered basis $\beta = (w_1, w_2, \ldots, w_m)$. Let $T : V \to W$ be a linear transformation.

To define the transformation entirely, we only need to know how it acts on the basis vectors of V. We apply T to each basis vector $v_j \in \alpha$, resulting in a vector $T(v_j) \in W$. We then write each image vector $T(v_j)$ as a linear combination of the basis vectors in β:

$$\begin{aligned} T(v_1) &= a_{11}w_1 + a_{21}w_2 + \cdots + a_{m1}w_m \\ T(v_2) &= a_{12}w_1 + a_{22}w_2 + \cdots + a_{m2}w_m \\ &\vdots \\ T(v_n) &= a_{1n}w_1 + a_{2n}w_2 + \cdots + a_{mn}w_m \end{aligned}$$

Definition 2.1.3 (Matrix Representation)**.** The **matrix representation of T relative to the bases α and β**, denoted by $[T]_\alpha^\beta$, is the $m \times n$ matrix whose j-th column is the coordinate vector of $T(v_j)$ relative to the basis β. That is:

$$[T]_\alpha^\beta = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix} = \begin{bmatrix} [T(v_1)]_\beta & [T(v_2)]_\beta & \ldots & [T(v_n)]_\beta \end{bmatrix}$$

(Note: If T is a linear operator on V (so $V = W$), we typically use the same basis for both domain and codomain, writing $[T]_\alpha^\alpha$ simply as $[T]_\alpha$).

The following theorem is arguably the most important bridge between linear transformations and matrices. It states that applying the transformation to a vector is algebraically identical to multiplying their matrix representations.

Theorem 2.1.4 (Matrix Action Theorem)**.** *Let $T : V \to W$ be a linear transformation, and let α and β be ordered bases for V and W respectively. For any vector $v \in V$:*

$$[T(v)]_\beta = [T]_\alpha^\beta [v]_\alpha$$

Proof. Let $\alpha = (v_1, \ldots, v_n)$ and $\beta = (w_1, \ldots, w_m)$. Let $A = [T]_\alpha^\beta$. By definition, the entry a_{ij} is the coefficient of w_i in the expansion of $T(v_j)$, so $T(v_j) = \sum_{i=1}^{m} a_{ij} w_i$.

Let $v \in V$. We can write $v = \sum_{j=1}^{n} x_j v_j$. Therefore, its coordinate vector is $[v]_\alpha = (x_1, x_2, \ldots, x_n)^T$.

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Applying the linear transformation T to v:

$$\begin{aligned} T(v) &= T\left(\sum_{j=1}^{n} x_j v_j\right) \\ &= \sum_{j=1}^{n} x_j T(v_j) \quad \text{(by linearity of } T\text{)} \\ &= \sum_{j=1}^{n} x_j \left(\sum_{i=1}^{m} a_{ij} w_i\right) \quad \text{(substituting the image of basis vectors)} \\ &= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} x_j\right) w_i \quad \text{(rearranging the finite sums)} \end{aligned}$$

This final equation expresses $T(v)$ explicitly as a linear combination of the basis vectors w_i in β. The coefficient of w_i (which is the i-th entry of the coordinate vector $[T(v)]_\beta$) is exactly $\sum_{j=1}^{n} a_{ij} x_j$. By the definition of matrix multiplication, this sum is precisely the i-th entry of the product of the matrix A and the column vector $[v]_\alpha$. Thus, $[T(v)]_\beta = A[v]_\alpha = [T]_\alpha^\beta [v]_\alpha$. ■

Theorem 2.1.5 (Composition Theorem). *Let U, V, and W be finite-dimensional vector spaces with ordered bases α, β, and γ respectively. Let $T : U \to V$ and $S : V \to W$ be linear transformations. Then the matrix representation of the composition $S \circ T$ is the product of their matrix representations:*

$$[S \circ T]_\alpha^\gamma = [S]_\beta^\gamma [T]_\alpha^\beta$$

Proof. The map $S \circ T : U \to W$ is linear. Let $v \in V$, then using applying the Matrix Action Theorem (Theorem 2.1.4) $\left[T(u)_\beta\right] = [T]_\alpha^\beta [v]_\alpha^\beta$, we have

$$\begin{aligned} [S \circ T]_\gamma^\alpha [v]_\alpha &= [(S \circ T)(v)]_\gamma \\ &= [S(T(v))]_\gamma \\ &= [S]_\gamma^\beta [T(v)]_\beta \\ &= [S]_\gamma^\beta [T]_\beta^\alpha [v]_\alpha \end{aligned}$$

Hence,

$$[S \circ T]_\gamma^\alpha = [S]_\gamma^\beta [T]_\beta^\alpha.$$

■

Example 2.1.6 (Standard Basis in $\mathbb{R}^n$). Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be defined by $T(x, y) = (2x - y, x + 3y, -x)$. Find the matrix of T relative to the standard bases of $\mathbb{R}^2$ and $\mathbb{R}^3$.

Solution. Let $\alpha = \{e_1 = (1, 0), e_2 = (0, 1)\}$ be the standard basis for $\mathbb{R}^2$. Let $\beta = \{f_1 = (1, 0, 0), f_2 = (0, 1, 0), f_3 = (0, 0, 1)\}$ be the standard basis for $\mathbb{R}^3$. We evaluate T on the basis vectors of the domain α:

$$\begin{aligned} T(1, 0) &= (2(1) - 0, 1 + 3(0), -1) = (2, 1, -1) \\ T(0, 1) &= (2(0) - 1, 0 + 3(1), -0) = (-1, 3, 0) \end{aligned}$$

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Next, we write these images as coordinate vectors in the codomain basis β. Since β is the standard basis, the coordinates are simply the components of the vectors:

$$[T(e_1)]_\beta = 2f_1 + 1f_2 - 1f_3 = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$$
$$[T(e_2)]_\beta = -1f_1 + 3f_2 + 0f_3 = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}$$

We assemble these coordinate vectors as the columns of the matrix:

$$[T]_\alpha^\beta = \begin{pmatrix} 2 & -1 \\ 1 & 3 \\ -1 & 0 \end{pmatrix}$$

Observe that the coefficients of x form the first column, and the coefficients of y form the second column. This shortcut works exclusively when using standard bases. ■

Example 2.1.7 (Non-Standard Basis)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear operator defined by $T(x, y) = (x + y, x - y)$. Find the matrix representation $[T]_\alpha^\alpha$ relative to the non-standard ordered basis $\alpha = (v_1, v_2)$ where $v_1 = (1, 1)$ and $v_2 = (1, -1)$.

Solution. First, evaluate the transformation on the given basis vectors v_1 and v_2:

$$T(v_1) = T(1, 1) = (1 + 1, 1 - 1) = (2, 0)$$
$$T(v_2) = T(1, -1) = (1 + (-1), 1 - (-1)) = (0, 2)$$

Now, we must express these output vectors as linear combinations of the *same basis* $\alpha = \{(1, 1), (1, -1)\}$ to find their coordinate vectors. We need scalars such that:

$$(2, 0) = c_1(1, 1) + c_2(1, -1)$$
$$(0, 2) = d_1(1, 1) + d_2(1, -1)$$

By inspection (or solving the simple systems):

$$(2, 0) = 1(1, 1) + 1(1, -1) \implies [T(v_1)]_\alpha = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
$$(0, 2) = 1(1, 1) - 1(1, -1) \implies [T(v_2)]_\alpha = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

The matrix representation is formed by placing these coordinate vectors as columns:

$$[T]_\alpha^\alpha = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

■

Example 2.1.8 (Matrix of a Derivative Operator)**.** Let $V = P_3(\mathbb{R})$ be the space of polynomials of degree ≤ 3, and let $W = P_2(\mathbb{R})$ be the space of polynomials of degree ≤ 2. Let $D : V \to W$ be the derivative transformation defined by $D(p(x)) = p'(x)$. Find $[D]_\alpha^\beta$ relative to the standard bases $\alpha = (1, x, x^2, x^3)$ for V and $\beta = (1, x, x^2)$ for W.

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Solution. We apply the derivative operator to each vector in the domain basis α:

$$\begin{aligned} D(1) &= 0 \\ D(x) &= 1 \\ D(x^2) &= 2x \\ D(x^3) &= 3x^2 \end{aligned}$$

Now, we write these results as coordinate vectors relative to the codomain basis $\beta = (1, x, x^2)$:

$$D(1) = 0(1) + 0(x) + 0(x^2) \implies [D(1)]_\beta = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$D(x) = 1(1) + 0(x) + 0(x^2) \implies [D(x)]_\beta = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$D(x^2) = 0(1) + 2(x) + 0(x^2) \implies [D(x^2)]_\beta = \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$$

$$D(x^3) = 0(1) + 0(x) + 3(x^2) \implies [D(x^3)]_\beta = \begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$$

We arrange these column vectors left to right to form the matrix representation:

$$[D]_\alpha^\beta = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

To see Theorem 2.1.4 in action, consider taking the derivative of $p(x) = 4 + 2x - 5x^3$. Its coordinate vector is $[p]_\alpha = (4, 2, 0, -5)^T$. Multiplying our matrix by this vector perfectly yields $(2, 0, -15)^T$, which corresponds to $2 - 15x^2$, the correct derivative! ■

2.2 Change of Basis and Coordinate Matrices

In vector spaces, the choice of basis is arbitrary. Some bases simplify problems significantly (for example, making a matrix representation diagonal). Therefore, it is essential to understand how the coordinates of a vector—and the matrix representation of a linear transformation—change when we switch from one basis to another.

2.2.1 The Transition Matrix

Consider an n-dimensional vector space V. Let $\alpha = (u_1, u_2, \ldots, u_n)$ be an "old" ordered basis, and let $\beta = (v_1, v_2, \ldots, v_n)$ be a "new" ordered basis. Any vector $x \in V$ has a coordinate vector $[x]_\alpha$ relative to the old basis, and a different coordinate vector $[x]_\beta$ relative to the new basis. We seek an algebraic mechanism to directly convert $[x]_\beta$ into $[x]_\alpha$.

Recall the Identity transformation $I : V \to V$ defined by $I(x) = x$. We can represent this transformation using β as the domain basis and α as the codomain basis.

Definition 2.2.1 (Change of Coordinate Matrix)**.** Let α and β be ordered bases for a finite-dimensional vector space V. The **change of coordinate matrix** (or transition matrix) from β to α, denoted by Q, is defined as the matrix representation of the identity operator I relative to the

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bases β and α:

$$Q = [I]_{\beta}^{\alpha}$$

The j-th column of Q is simply the j-th basis vector of β written in the coordinates of α. That is:

$$Q = \begin{bmatrix} [v_1]_\alpha & [v_2]_\alpha & \dots & [v_n]_\alpha \end{bmatrix}$$

Theorem 2.2.2 (Change of Coordinates Theorem). *Let Q be the change of coordinate matrix from basis β to basis α in a vector space V. Then for any vector $x \in V$:*

$$[x]_\alpha = Q[x]_\beta$$

Proof. This is a direct application of the Matrix Action Theorem (Theorem 2.1.4). We apply it to the identity transformation I:

$$[x]_\alpha = [I(x)]_\alpha = [I]_\beta^\alpha [x]_\beta = Q[x]_\beta$$

■

Theorem 2.2.3 (Invertibility of the Transition Matrix). *Let α and β be ordered bases for V. The change of coordinate matrix $Q = [I]_\beta^\alpha$ is always invertible, and its inverse is the change of coordinate matrix from α to β:*

$$Q^{-1} = [I]_\alpha^\beta$$

Proof. Consider the composition of the identity transformation with itself: $I \circ I = I$. Using the Composition Theorem (Theorem 2.1.5) to transition from α to β and back to α:

$$[I]_\beta^\alpha [I]_\alpha^\beta = [I \circ I]_\alpha^\alpha = [I]_\alpha^\alpha$$

The matrix $[I]_\alpha^\alpha$ is simply the $n \times n$ identity matrix I_n, because the coordinates of a basis vector relative to its own basis is just a column of the identity matrix. Thus, $([I]_\beta^\alpha)([I]_\alpha^\beta) = I_n$. This implies that $[I]_\beta^\alpha$ is invertible, and its inverse is exactly $[I]_\alpha^\beta$. ■

How does changing the basis affect the matrix representation of a linear operator $T : V \to V$? If we know the matrix of T in the old basis α, how do we find its matrix in the new basis β?

Theorem 2.2.4 (Similarity Transformation). *Let T be a linear operator on a finite-dimensional vector space V. Let α and β be ordered bases for V, and let Q be the change of coordinate matrix from β to α (so $Q = [I]_\beta^\alpha$). Then:*

$$[T]_\beta = Q^{-1}[T]_\alpha Q$$

Proof. We can view the transformation T mapping from (V, β) to (V, β) as a composition of three transformations: (The identity mapping from (V, β) to (V, α), the transformation T mapping from (V, α) to (V, α), the identity mapping from (V, α) back to (V, β).) Algebraically, $T = I \circ T \circ I$. By applying the Composition Theorem:

$$\begin{aligned} [T]_\beta &= [T]_\beta^\beta \\ &= [I \circ T \circ I]_\beta^\beta \\ &= [I]_\alpha^\beta [T]_\alpha^\alpha [I]_\beta^\alpha \end{aligned}$$

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By our definitions, $[I]_{\beta}^{\alpha} = Q$, $[T]_{\alpha}^{\alpha} = [T]_{\alpha}$, and $[I]_{\alpha}^{\beta} = Q^{-1}$. Substituting these in yields:

$$[T]_{\beta} = Q^{-1}[T]_{\alpha}Q$$

■

Definition 2.2.5 (Similar Matrices)**.** Two $n \times n$ matrices A and B are said to be **similar** if there exists an invertible $n \times n$ matrix Q such that $B = Q^{-1}AQ$. Theorem 2.2.4 tells us that similar matrices are simply different matrix representations of the *exact same linear operator* viewed under different bases.

Example 2.2.6 (Finding Coordinates in a New Basis)**.** In $\mathbb{R}^2$, let $\alpha = \{e_1, e_2\}$ be the standard ordered basis. Let $\beta = \{v_1, v_2\}$ be a new ordered basis where $v_1 = (1, 2)$ and $v_2 = (3, 4)$. Let $x = (5, 6)$. Find the transition matrix Q from β to α, and use it to find $[x]_{\beta}$.

Solution. The transition matrix $Q = [I]_{\beta}^{\alpha}$ has columns consisting of the β vectors written in α coordinates. Since α is the standard basis, the coordinates of a vector are just its components:

$$[v_1]_{\alpha} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad [v_2]_{\alpha} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

Thus, the transition matrix is:

$$Q = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

We are given the vector $x = (5, 6)$, which means its standard coordinates are $[x]_{\alpha} = \begin{pmatrix} 5 \\ 6 \end{pmatrix}$. By Theorem 2.2.2, $[x]_{\alpha} = Q[x]_{\beta}$, which implies $[x]_{\beta} = Q^{-1}[x]_{\alpha}$. First, we find Q^{-1}. The determinant is $(1)(4) - (3)(2) = 4 - 6 = -2$.

$$Q^{-1} = -\frac{1}{2}\begin{pmatrix} 4 & -3 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 3/2 \\ 1 & -1/2 \end{pmatrix}$$

Now, multiply Q^{-1} by $[x]_{\alpha}$:

$$[x]_{\beta} = \begin{pmatrix} -2 & 3/2 \\ 1 & -1/2 \end{pmatrix}\begin{pmatrix} 5 \\ 6 \end{pmatrix} = \begin{pmatrix} -10 + 9 \\ 5 - 3 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Thus, the coordinates of x in the new basis β are $(-1, 2)$. *(Verification:* $-1(1, 2) + 2(3, 4) = (-1, -2) + (6, 8) = (5, 6) = x$*. It works!)* ■

Example 2.2.7 (Transition Matrix between Non-Standard Bases)**.** In the vector space $P_1(\mathbb{R})$ (polynomials of degree ≤ 1), consider two bases: $\alpha = \{1, x\}$ and $\beta = \{1 + x, 1 - x\}$. Find the change of coordinate matrix from β to α, and the matrix from α to β.

Solution. **1. From β to α ($Q = [I]_{\beta}^{\alpha}$):** We must express the vectors of β as linear combinations of the vectors in α. Let $v_1 = 1 + x$ and $v_2 = 1 - x$. The basis α is the standard basis $\{1, x\}$.

$$1 + x = 1(1) + 1(x) \implies [1 + x]_{\alpha} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$1 - x = 1(1) - 1(x) \implies [1 - x]_{\alpha} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

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Placing these as columns, the transition matrix from β to α is:

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

2. From α to β ($P = [I]_{\alpha}^{\beta}$): This matrix is simply Q^{-1}. The determinant of Q is $(1)(-1) - (1)(1) = -2$.

$$P = Q^{-1} = -\frac{1}{2}\begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$$

(Verification: Does $1 = \frac{1}{2}(1+x) + \frac{1}{2}(1-x)$? Yes. Does $x = \frac{1}{2}(1+x) - \frac{1}{2}(1-x)$? Yes. The columns are perfectly correct.) ■

Example 2.2.8 (Changing the Matrix Representation of an Operator)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear operator defined by $T(x, y) = (2x + y, x - 3y)$. Let $\alpha = \{e_1, e_2\}$ be the standard basis, and let $\beta = \{(1, 1), (1, -1)\}$ be a new basis. Find $[T]_{\alpha}$, and then use the similarity transformation theorem to compute $[T]_{\beta}$.

Solution. **Find $[T]_{\alpha}$.** Since α is the standard basis, we can read the matrix directly from the coefficients of the transformation:

$$[T]_{\alpha} = A = \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}$$

Find the transition matrix $Q = [I]_{\beta}^{\alpha}$. The columns of Q are the vectors of β written in the standard basis α:

$$Q = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

From the previous example, we already know the inverse of this exact matrix:

$$Q^{-1} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}$$

Apply the similarity transformation $[T]_{\beta} = Q^{-1}AQ$. First, multiply AQ:

$$AQ = \begin{pmatrix} 2 & 1 \\ 1 & -3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} (2+1) & (2-1) \\ (1-3) & (1+3) \end{pmatrix} = \begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix}$$

Next, multiply $Q^{-1}(AQ)$:

$$\begin{aligned} [T]_{\beta} &= \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix}\begin{pmatrix} 3 & 1 \\ -2 & 4 \end{pmatrix} \\ &= \begin{pmatrix} (3/2 - 2/2) & (1/2 + 4/2) \\ (3/2 + 2/2) & (1/2 - 4/2) \end{pmatrix} \\ &= \begin{pmatrix} 1/2 & 5/2 \\ 5/2 & -3/2 \end{pmatrix} \end{aligned}$$

Thus, the matrix representation of T in the basis β is $\begin{pmatrix} 1/2 & 5/2 \\ 5/2 & -3/2 \end{pmatrix}$.

(Verification via direct computation: Let $v_1 = (1, 1)$. $T(1, 1) = (3, -2)$. We express $(3, -2)$ in basis β: $c_1(1, 1) + c_2(1, -1) = (3, -2)$. Solving gives $c_1 = 1/2, c_2 = 5/2$, which matches the first column of our result perfectly!) ■

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2.3 Dual Spaces, Dual Basis, and Annihilators

We have previously established that the set of all linear transformations from a vector space V to a vector space W, denoted $L(V, W)$, forms a vector space itself. A particularly profound and useful structure arises when we choose the codomain W to be the underlying scalar field $\mathbb{K}$ itself.

2.3.1 Linear Functionals and the Dual Space

Definition 2.3.1 (Linear Functional)**.** Let V be a vector space over a field $\mathbb{K}$. A **linear functional** on V is a linear transformation from V into the scalar field $\mathbb{K}$ (viewed as a 1-dimensional vector space over itself). That is, a mapping $f : V \to \mathbb{K}$ is a linear functional if for all $u, v \in V$ and $\alpha, \beta \in \mathbb{K}$:

$$f(\alpha u + \beta v) = \alpha f(u) + \beta f(v)$$

Definition 2.3.2 (Dual Space)**.** Let V be a vector space over $\mathbb{K}$. The set of all linear functionals on V, denoted by V^* (or V'), is called the **dual space** of V. Symbolically, $V^* = L(V, \mathbb{K})$. Under the standard pointwise addition and scalar multiplication of functions, V^* is a vector space over $\mathbb{K}$.

Theorem 2.3.3 (Dimension of the Dual Space)**.** *If V is a finite-dimensional vector space over $\mathbb{K}$ of dimension n, then its dual space V^* is also finite-dimensional and has the exact same dimension:*

$$\dim(V^*) = \dim(V) = n$$

Proof. We know that for finite-dimensional spaces, $\dim(L(V, W)) = \dim(V) \times \dim(W)$. Since $V^* = L(V, \mathbb{K})$ and the field $\mathbb{K}$ is a 1-dimensional space over itself, $\dim(V^*) = \dim(V) \times 1 = \dim(V)$. ■

2.3.2 The Dual Basis

Since V and V^* have the same dimension, they are isomorphic. However, there is a very natural and specific way to link a basis in V to a corresponding basis in V^*.

Theorem 2.3.4 (Existence and Uniqueness of the Dual Basis)**.** *Let V be an n-dimensional vector space with a basis $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$. Then there exists a uniquely determined basis $\mathcal{B}^* = \{f_1, f_2, \ldots, f_n\}$ for V^* such that:*

$$f_i(v_j) = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

*where δ_{ij} is the Kronecker delta. The set $\mathcal{B}^*$ is called the **dual basis** corresponding to $\mathcal{B}$.*

Proof. **Existence of the functionals:** A linear transformation is completely determined by its action on a basis. For each $i \in \{1, \ldots, n\}$, we define a linear functional $f_i \in V^*$ by declaring its values on the basis vectors of V: $f_i(v_j) = \delta_{ij}$. By the linear extension theorem, this uniquely defines n linear functionals $f_1, \ldots, f_n$.

Linear Independence of $\mathcal{B}^*$: Suppose a linear combination of these functionals yields the zero functional:

$$c_1 f_1 + c_2 f_2 + \cdots + c_n f_n = 0_{V^*}$$

This means that for *any* vector $x \in V$, $(c_1 f_1 + \cdots + c_n f_n)(x) = 0$. Let us evaluate this expression at the specific basis vector $v_j \in \mathcal{B}$:

$$(c_1 f_1 + \cdots + c_n f_n)(v_j) = c_1 f_1(v_j) + \cdots + c_j f_j(v_j) + \cdots + c_n f_n(v_j) = 0$$

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Because $f_i(v_j) = 0$ for all $i \neq j$ and $f_j(v_j) = 1$, the entire sum collapses to:

$$c_j(1) = 0 \implies c_j = 0$$

Since this holds for every $j = 1, \ldots, n$, all coefficients c_j must be zero. Thus, $\{f_1, \ldots, f_n\}$ is a linearly independent set in V^*.

Spanning: Let $f \in V^*$ be an arbitrary linear functional. We claim that f can be written as the linear combination:

$$f = \sum_{i=1}^{n} f(v_i) f_i$$

To verify this, we evaluate both sides on an arbitrary basis vector v_j:

$$\left(\sum_{i=1}^{n} f(v_i) f_i \right)(v_j) = \sum_{i=1}^{n} f(v_i) f_i(v_j) = \sum_{i=1}^{n} f(v_i) \delta_{ij} = f(v_j)$$

Since the functional f and the linear combination $\sum f(v_i) f_i$ agree on every vector in a basis of V, they are equal as functions. Thus, $f \in \operatorname{span}\{f_1, \ldots, f_n\}$, proving the set spans V^*.

Spanning: (*2nd Proof*) We have found n linearly independent vectors in V^*. Since we proved earlier that $\dim(V^*) = n$, any set of n linearly independent vectors automatically forms a basis for V^*. ■

Theorem 2.3.5 (Coordinate Representation via Dual Basis). *Let $\mathcal{B} = \{v_1, \ldots, v_n\}$ be a basis for V and $\mathcal{B}^* = \{f_1, \ldots, f_n\}$ be its dual basis.*

1. *For any vector $v \in V$:* $\quad v = \sum_{i=1}^{n} f_i(v) v_i$, *that is,* $[v]_{\mathcal{B}} = [f_1(v) \quad f_2(v) \quad \ldots \quad f_n(v)]^t$
2. *For any functional $g \in V^*$:* $\quad g = \sum_{i=1}^{n} g(v_i) f_i$

Proof. For (1), since $\mathcal{B}$ is a basis, $v = c_1 v_1 + \cdots + c_n v_n$. Applying the functional f_j to both sides gives $f_j(v) = f_j(\sum c_i v_i) = \sum c_i f_j(v_i) = c_j(1) = c_j$. Thus the coordinates c_j are exactly the values $f_j(v)$. For (2), since $\mathcal{B}^*$ is a basis for V^*, $g = k_1 f_1 + \cdots + k_n f_n$. Evaluating both sides at the vector v_j gives $g(v_j) = \sum k_i f_i(v_j) = k_j(1) = k_j$. ■

Theorem 2.3.6. *Let V and W be finite-dimensional vector spaces over a field $\mathbb{K}$ with ordered bases $\alpha = \{v_1, \ldots, v_n\}$ and $\beta = \{w_1, \ldots, w_m\}$ respectively. Let $T : V \to W$ be a linear transformation. Define the dual map $T^* : W^* \to V^*$ by $T^*(f) = f \circ T$ for all $f \in W^*$. Then:*

1. *T^* is a linear transformation.*
2. *If $A = [T]_{\alpha}^{\beta}$ is the matrix of T relative to α and β, then the matrix of T^* relative to the dual bases β^* and α^* is the transpose of A. That is, $[T^*]_{\beta^*}^{\alpha^*} = A^t$.*

Proof. **1. Linearity of T^*:** Let $f, g \in W^*$ and $c \in \mathbb{K}$. We must show that $T^*(cf + g) = cT^*(f) + T^*(g)$. By the definition of T^*:

$$T^*(cf + g) = (cf + g) \circ T$$

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Applying this to any vector $v \in V$:

$$\begin{aligned}(T^*(cf+g))(v) &= ((cf+g)\circ T)(v)\\ &= (cf+g)(T(v))\\ &= cf(T(v)) + g(T(v)) \quad \text{(by definition of addition and scalar mult. in } W^*)\\ &= c(f\circ T)(v) + (g\circ T)(v)\\ &= c(T^*(f))(v) + (T^*(g))(v)\\ &= (cT^*(f) + T^*(g))(v)\end{aligned}$$

Since this holds for all $v \in V$, we have $T^*(cf+g) = cT^*(f) + T^*(g)$. Thus, T^* is linear.

2. Matrix Representation: Let $\alpha^* = \{f_1, \ldots, f_n\}$ be the dual basis of V^* and $\beta^* = \{g_1, \ldots, g_m\}$ be the dual basis of W^*. Recall that $f_i(v_j) = \delta_{ij}$ and $g_i(w_j) = \delta_{ij}$.

Let $A = [T]_\alpha^\beta$. By definition, the j-th column of A consists of the coordinates of $T(v_j)$ relative to β:

$$T(v_j) = \sum_{i=1}^{m} A_{ij} w_i \quad \text{for } j = 1, \ldots, n$$

Let $B = [T^*]_{\beta^*}^{\alpha^*}$. The k-th column of B consists of the coordinates of $T^*(g_k)$ relative to α^*:

$$T^*(g_k) = \sum_{j=1}^{n} B_{jk} f_j \quad \text{for } k = 1, \ldots, m$$

To find the entry B_{rk}, we evaluate the functional $T^*(g_k)$ at the basis vector v_r:

$$(T^*(g_k))(v_r) = \left(\sum_{j=1}^{n} B_{jk} f_j\right)(v_r) = \sum_{j=1}^{n} B_{jk} f_j(v_r) = \sum_{j=1}^{n} B_{jk}\delta_{jr} = B_{rk}$$

On the other hand, using the definition $T^*(g_k) = g_k \circ T$:

$$(T^*(g_k))(v_r) = g_k(T(v_r)) = g_k\left(\sum_{i=1}^{m} A_{ir} w_i\right) = \sum_{i=1}^{m} A_{ir} g_k(w_i) = \sum_{i=1}^{m} A_{ir}\delta_{ki} = A_{kr}$$

Comparing the two expressions for $(T^*(g_k))(v_r)$, we get:

$$B_{rk} = A_{kr}$$

Since the (r,k) entry of B is the (k,r) entry of A, we conclude that $B = A^t$. Thus:

$$[T^*]_{\beta^*}^{\alpha^*} = \left([T]_\alpha^\beta\right)^t$$

■

Example 2.3.7 (Finding the Dual Basis from a Given Basis)**.** Consider the basis $\mathcal{B} = \{v_1, v_2\}$ for $\mathbb{R}^2$, where:

$$v_1 = (2,1), \quad v_2 = (3,1)$$

Find the dual basis $\mathcal{B}^* = \{f_1, f_2\}$ for $(\mathbb{R}^2)^*$.

Solution. The dual basis $\{f_1, f_2\}$ consists of linear functionals such that $f_i(v_j) = \delta_{ij}$. Let $f_1(x,y) = a_1x + b_1y$. To find f_1, we impose the conditions:

$$\begin{aligned}f_1(v_1) = 1 &\implies 2a_1 + b_1 = 1\\ f_1(v_2) = 0 &\implies 3a_1 + b_1 = 0 \implies b_1 = -3a_1\end{aligned}$$

Substituting $b_1 = -3a_1$ into the first equation gives $2a_1 - 3a_1 = 1$, which implies $a_1 = -1$. Then $b_1 = -3(-1) = 3$. Thus, the first functional is:

$$f_1(x, y) = -x + 3y$$

To find $f_2(x, y) = a_2x + b_2y$, we impose the conditions:

$$\begin{aligned} f_2(v_1) = 0 &\implies 2a_2 + b_2 = 0 \implies b_2 = -2a_2 \\ f_2(v_2) = 1 &\implies 3a_2 + b_2 = 1 \end{aligned}$$

Substituting $b_2 = -2a_2$ into the second equation gives $3a_2 - 2a_2 = 1$, which implies $a_2 = 1$. Then $b_2 = -2(1) = -2$. Thus, the second functional is:

$$f_2(x, y) = x - 2y$$

The dual basis is $\mathcal{B}^* = \{-x + 3y, x - 2y\}$. ■

Example 2.3.8 (Finding the Basis from a Given Dual Basis)**.** Let $\{f_1, f_2\}$ be the dual basis of $(\mathbb{R}^2)^*$ defined by:

$$f_1(x, y) = x - y, \quad f_2(x, y) = y$$

Find the corresponding basis $\mathcal{B} = \{v_1, v_2\}$ of $\mathbb{R}^2$.

Solution. We seek vectors $v_1 = (x_1, y_1)$ and $v_2 = (x_2, y_2)$ such that $f_i(v_j) = \delta_{ij}$. To find v_1, we require $f_1(v_1) = 1$ and $f_2(v_1) = 0$:

$$\begin{aligned} x_1 - y_1 &= 1 \\ y_1 &= 0 \end{aligned}$$

From $y_1 = 0$, the first equation becomes $x_1 - 0 = 1$, so $x_1 = 1$. Thus, the first basis vector is:

$$v_1 = (1, 0)$$

To find v_2, we require $f_1(v_2) = 0$ and $f_2(v_2) = 1$:

$$\begin{aligned} x_2 - y_2 = 0 &\implies x_2 = y_2 \\ y_2 &= 1 \end{aligned}$$

Since $y_2 = 1$ and $x_2 = y_2$, we have $x_2 = 1$. Thus, the second basis vector is:

$$v_2 = (1, 1)$$

The basis for $\mathbb{R}^2$ is $\mathcal{B} = \{(1, 0), (1, 1)\}$. ■

Example 2.3.9 (Finding the Dual Basis in $\mathbb{R}^3$)**.** Consider the basis $\mathcal{B} = \{v_1, v_2, v_3\}$ for $\mathbb{R}^3$ given by:

$$v_1 = (1, 0, 0), \quad v_2 = (1, 1, 0), \quad v_3 = (1, 1, 1)$$

Find the dual basis $\mathcal{B}^* = \{f_1, f_2, f_3\}$.

Solution. We seek linear functionals $f_i(x, y, z) = a_ix + b_iy + c_iz$ such that $f_i(v_j) = \delta_{ij}$. To find f_1, we solve the following system:

$$\begin{aligned} f_1(1, 0, 0) = 1 &\implies a_1 = 1 \\ f_1(1, 1, 0) = 0 &\implies a_1 + b_1 = 0 \implies 1 + b_1 = 0 \implies b_1 = -1 \\ f_1(1, 1, 1) = 0 &\implies a_1 + b_1 + c_1 = 0 \implies 1 - 1 + c_1 = 0 \implies c_1 = 0 \end{aligned}$$

Thus, $f_1(x,y,z) = x - y$. To find f_2, we solve:

$$\begin{aligned} f_2(1,0,0) = 0 &\implies a_2 = 0 \\ f_2(1,1,0) = 1 &\implies a_2 + b_2 = 1 \implies 0 + b_2 = 1 \implies b_2 = 1 \\ f_2(1,1,1) = 0 &\implies a_2 + b_2 + c_2 = 0 \implies 0 + 1 + c_2 = 0 \implies c_2 = -1 \end{aligned}$$

Thus, $f_2(x,y,z) = y - z$. To find f_3, we solve:

$$\begin{aligned} f_3(1,0,0) = 0 &\implies a_3 = 0 \\ f_3(1,1,0) = 0 &\implies a_3 + b_3 = 0 \implies 0 + b_3 = 0 \implies b_3 = 0 \\ f_3(1,1,1) = 1 &\implies a_3 + b_3 + c_3 = 1 \implies 0 + 0 + c_3 = 1 \implies c_3 = 1 \end{aligned}$$

Thus, $f_3(x,y,z) = z$. The dual basis is $\mathcal{B}^* = \{x - y, y - z, z\}$. ■

Example 2.3.10 (Finding the Basis from a Dual Basis in $\mathbb{R}^3$)**.** Let the dual basis $\mathcal{B}^* = \{f_1, f_2, f_3\}$ of $(\mathbb{R}^3)^*$ be defined by:

$$f_1(x,y,z) = x, \quad f_2(x,y,z) = x + y, \quad f_3(x,y,z) = x + y + z$$

Find the corresponding basis $\mathcal{B} = \{v_1, v_2, v_3\}$ of $\mathbb{R}^3$.

Solution. We determine vectors $v_j = (x_j, y_j, z_j)$ such that $f_i(v_j) = \delta_{ij}$. To find v_1, we solve:

$$\begin{aligned} f_1(v_1) = 1 &\implies x_1 = 1 \\ f_2(v_1) = 0 &\implies x_1 + y_1 = 0 \implies 1 + y_1 = 0 \implies y_1 = -1 \\ f_3(v_1) = 0 &\implies x_1 + y_1 + z_1 = 0 \implies 1 - 1 + z_1 = 0 \implies z_1 = 0 \end{aligned}$$

So, $v_1 = (1, -1, 0)$. To find v_2, we solve:

$$\begin{aligned} f_1(v_2) = 0 &\implies x_2 = 0 \\ f_2(v_2) = 1 &\implies x_2 + y_2 = 1 \implies 0 + y_2 = 1 \implies y_2 = 1 \\ f_3(v_2) = 0 &\implies x_2 + y_2 + z_2 = 0 \implies 0 + 1 + z_2 = 0 \implies z_2 = -1 \end{aligned}$$

So, $v_2 = (0, 1, -1)$. To find v_3, we solve:

$$\begin{aligned} f_1(v_3) = 0 &\implies x_3 = 0 \\ f_2(v_3) = 0 &\implies x_3 + y_3 = 0 \implies 0 + y_3 = 0 \implies y_3 = 0 \\ f_3(v_3) = 1 &\implies x_3 + y_3 + z_3 = 1 \implies 0 + 0 + z_3 = 1 \implies z_3 = 1 \end{aligned}$$

So, $v_3 = (0, 0, 1)$. The corresponding basis is $\mathcal{B} = \{(1, -1, 0), (0, 1, -1), (0, 0, 1)\}$.

■

2.3.3 Annihilators

The concept of an annihilator provides a dual perspective on subspaces. Instead of describing a subspace by the vectors that span it, we can describe it by the linear equations (functionals) that vanish (evaluate to zero) on it.

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Definition 2.3.11 (Annihilator)**.** Let S be a non-empty subset of a vector space V. The **annihilator** of S, denoted by S^0 (or $S^\perp$), is the set of all linear functionals in the dual space V^* that map every element of S to zero.

$$S^0 = \{f \in V^* \mid f(s) = 0 \text{ for all } s \in S\}$$

Theorem 2.3.12. *For any subset $S \subseteq V$, the annihilator S^0 is a subspace of V^*.*

Proof. Let 0_{V^*} be the zero functional (which maps everything to $0 \in \mathbb{K}$). Clearly $0_{V^*}(s) = 0$ for all $s \in S$, so $0_{V^*} \in S^0$, meaning S^0 is non-empty. Let $f, g \in S^0$ and $c \in \mathbb{K}$. We test the linear combination $cf + g$. For any $s \in S$, evaluate the combination:

$$(cf + g)(s) = cf(s) + g(s)$$

Since $f, g \in S^0$, we know $f(s) = 0$ and $g(s) = 0$. Thus:

$$(cf + g)(s) = c(0) + 0 = 0$$

Because $(cf + g)$ evaluates to zero for every $s \in S$, it follows that $(cf + g) \in S^0$. By the subspace test, S^0 is a subspace of V^*. ■

Theorem 2.3.13 (Dimension Theorem for Annihilators)**.** *Let V be a finite-dimensional vector space, and let W be a subspace of V. Then:*

$$\dim(W) + \dim(W^0) = \dim(V)$$

Proof. Let $\dim(V) = n$ and $\dim(W) = k$. We must show that $\dim(W^0) = n - k$. Let $\{w_1, w_2, \ldots, w_k\}$ be a basis for W. By the Basis Extension Theorem, we can extend this to a basis for the entire space V:

$$\mathcal{B} = \{w_1, \ldots, w_k, v_{k+1}, \ldots, v_n\}$$

Let the corresponding dual basis for V^* be:

$$\mathcal{B}^* = \{f_1, \ldots, f_k, f_{k+1}, \ldots, f_n\}$$

We claim that the subset of functionals $S = \{f_{k+1}, f_{k+2}, \ldots, f_n\}$ forms a basis for the annihilator W^0.

1. $S \subseteq W^0$**:** For any basis functional $f_j \in S$ (where $j > k$) and any basis vector $w_i \in W$ (where $i \leq k$), $i \neq j$. By the definition of the dual basis, $f_j(w_i) = \delta_{ji} = 0$. Since f_j evaluates to zero on the basis of W, by linearity, it evaluates to zero on all of W. Thus, $f_j \in W^0$.

2. S **spans** W^0**:** Let $g \in W^0 \subset V^*$. Since $\mathcal{B}^*$ is a basis for V^*, we can write g uniquely as:

$$g = c_1 f_1 + \cdots + c_k f_k + c_{k+1} f_{k+1} + \cdots + c_n f_n$$

Because $g \in W^0$, we know $g(w) = 0$ for all $w \in W$. In particular, evaluating g at the basis vector w_i (for $1 \leq i \leq k$) gives:

$$g(w_i) = c_i f_i(w_i) = c_i(1) = c_i$$

But since $w_i \in W$, $g(w_i)$ must be 0. Thus $c_i = 0$ for all $i = 1, \ldots, k$. This implies $g = c_{k+1} f_{k+1} + \cdots + c_n f_n$. Thus, S spans W^0.

3. S **is linearly independent:** S is a subset of the basis $\mathcal{B}^*$, so it is automatically linearly independent. Since S is a basis for W^0 containing exactly $n - k$ elements, $\dim(W^0) = n - k$. Rearranging this gives $\dim(W) + \dim(W^0) = k + (n - k) = n = \dim(V)$. ■

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Example 2.3.14 (Finding a Dual Basis)**.** Let $V = \mathbb{R}^2$. Consider the basis $\mathcal{B} = \{v_1, v_2\}$ where $v_1 = (2,1)$ and $v_2 = (3,1)$. Find the dual basis $\mathcal{B}^* = \{f_1, f_2\}$.

Solution. Any linear functional f on $\mathbb{R}^2$ maps a vector (x,y) to a scalar, and takes the general algebraic form $f(x,y) = ax + by$ for some constants $a, b \in \mathbb{R}$. We seek two specific functionals, $f_1(x,y) = a_1x + b_1y$ and $f_2(x,y) = a_2x + b_2y$, that satisfy the dual basis condition: $f_i(v_j) = \delta_{ij}$.

1. Finding f_1: We require $f_1(v_1) = 1$ and $f_1(v_2) = 0$.

$$f_1(2,1) = 2a_1 + b_1 = 1$$
$$f_1(3,1) = 3a_1 + b_1 = 0$$

Subtracting the first equation from the second gives $a_1 = -1$. Substituting $a_1 = -1$ into the first equation: $2(-1) + b_1 = 1 \implies b_1 = 3$. Thus, $f_1(x,y) = -x + 3y$.

2. Finding f_2: We require $f_2(v_1) = 0$ and $f_2(v_2) = 1$.

$$f_2(2,1) = 2a_2 + b_2 = 0$$
$$f_2(3,1) = 3a_2 + b_2 = 1$$

Subtracting the first equation from the second gives $a_2 = 1$. Substituting $a_2 = 1$ into the first equation: $2(1) + b_2 = 0 \implies b_2 = -2$. Thus, $f_2(x,y) = x - 2y$.

The dual basis is $\mathcal{B}^* = \{f_1, f_2\}$, where $f_1(x,y) = -x + 3y$ and $f_2(x,y) = x - 2y$. ■

Example 2.3.15 (Dual Basis for Polynomials)**.** Let $V = P_1(\mathbb{R})$ (polynomials of degree ≤ 1). Consider the basis $\mathcal{B} = \{p_1, p_2\}$ where $p_1(x) = 1 - x$ and $p_2(x) = 1 + x$. Find the dual basis.

Solution. A linear functional f on $P_1(\mathbb{R})$ maps a polynomial $p(x) = a + bx$ to a real number. We can define functionals by assigning weights to the coefficients: $f(a + bx) = c_1a + c_2b$. Let our dual functionals be f_1 and f_2. We need $f_1(p_1) = 1$ and $f_1(p_2) = 0$. Let $f_1(a + bx) = \alpha_1 a + \beta_1 b$.

$$f_1(1 - x) = \alpha_1(1) + \beta_1(-1) = 1$$
$$f_1(1 + x) = \alpha_1(1) + \beta_1(1) = 0$$

Adding these equations gives $2\alpha_1 = 1 \implies \alpha_1 = 1/2$. Then $\beta_1 = -1/2$. So, $f_1(a + bx) = \frac{1}{2}a - \frac{1}{2}b$.

We need $f_2(p_1) = 0$ and $f_2(p_2) = 1$. Let $f_2(a + bx) = \alpha_2 a + \beta_2 b$.

$$f_2(1 - x) = \alpha_2(1) + \beta_2(-1) = 0$$
$$f_2(1 + x) = \alpha_2(1) + \beta_2(1) = 1$$

Adding these equations gives $2\alpha_2 = 1 \implies \alpha_2 = 1/2$. Then $\beta_2 = 1/2$. So, $f_2(a + bx) = \frac{1}{2}a + \frac{1}{2}b$. ■

Example 2.3.16 (Finding the Annihilator)**.** Let $V = \mathbb{R}^3$ and let W be the subspace spanned by the single vector $v = (1,2,-1)$. Find a basis for the annihilator W^0, and verify the dimension theorem for annihilators.

Solution. By the Dimension Theorem, $\dim(W^0) = \dim(\mathbb{R}^3) - \dim(W) = 3 - 1 = 2$. Any functional $f \in (\mathbb{R}^3)^*$ has the form $f(x,y,z) = ax + by + cz$. For f to be in W^0, it must map every vector in W to 0. Since W is spanned by v, it is necessary and sufficient that $f(v) = 0$.

$$f(1,2,-1) = a(1) + b(2) + c(-1) = 0 \implies a + 2b - c = 0 \implies c = a + 2b$$

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Thus, any functional in W^0 has the form:

$$\begin{aligned} f(x,y,z) &= ax + by + (a+2b)z \\ &= a(x+z) + b(y+2z) \end{aligned}$$

This shows that every functional in W^0 is a linear combination of two specific functionals: $f_1(x,y,z) = x+z$ (obtained by setting $a=1, b=0$) and $f_2(x,y,z) = y+2z$ (obtained by setting $a=0, b=1$). Since these two functionals are clearly linearly independent (neither is a scalar multiple of the other), they form a basis for W^0. Therefore, a basis for W^0 is $\{f_1, f_2\}$, and $\dim(W^0) = 2$, perfectly verifying $\dim(W) + \dim(W^0) = 1 + 2 = 3 = \dim(\mathbb{R}^3)$. ■

Definition 2.3.17. Let V be a vector space and let $W_1, W_2, \ldots, W_k$ be subspaces of V. The sum of these subspaces is defined as:

$$W_1 + W_2 + \cdots + W_k = \{w_1 + w_2 + \cdots + w_k : w_i \in W_i \text{ for each } i\}$$

The sum is called a **direct sum**, denoted by $W_1 \oplus W_2 \oplus \cdots \oplus W_k$, if every vector v in the sum can be written **uniquely** in the form $v = w_1 + w_2 + \cdots + w_k$, where $w_i \in W_i$.

Theorem 2.3.18. *Let $W_1, W_2, \ldots, W_k$ be subspaces of a vector space V. The following conditions are equivalent:*

1. *The sum $W = \sum_{i=1}^{k} W_i$ is a direct sum.*
2. *If $w_1 + w_2 + \cdots + w_k = 0$, with $w_i \in W_i$, then $w_i = 0$ for all $i = 1, \ldots, k$.*
3. *For each $j \in \{1, \ldots, k\}$, the intersection of W_j and the sum of the other subspaces is the zero subspace:*

$$W_j \cap \left(\sum_{i \neq j} W_i \right) = \{0\}$$

Proof. We will prove the equivalence by showing (1) $\implies$ (2), (2) $\implies$ (3), and (3) $\implies$ (1).

(1) $\implies$ (2): Assume the sum is direct. By definition, every vector in the sum has a unique representation. Consider the zero vector $0 \in W$. One way to represent it is $0 = 0 + 0 + \cdots + 0$, where each $0 \in W_i$. If we have any other representation $w_1 + w_2 + \cdots + w_k = 0$, the uniqueness property forces $w_i = 0$ for all i.

(2) $\implies$ (3): Assume that the only representation of the zero vector is the trivial one. Suppose there exists a vector v in the intersection $W_j \cap (\sum_{i \neq j} W_i)$. Since $v \in W_j$, we have a vector in W_j. Since $v \in \sum_{i \neq j} W_i$, we can write $v = \sum_{i \neq j} w_i$ for some $w_i \in W_i$. Subtracting v from both sides, we get:

$$w_1 + \cdots + w_{j-1} + (-v) + w_{j+1} + \cdots + w_k = 0$$

This is a sum of vectors from each subspace (where the j-th component is $-v \in W_j$) that equals zero. By our assumption in (2), every component must be zero. Thus, $-v = 0$, which means $v = 0$. Therefore, the intersection is $\{0\}$.

(3) $\implies$ (1): Assume condition (3) holds. Suppose a vector $v \in W$ has two representations:

$$v = w_1 + w_2 + \cdots + w_k \quad \text{and} \quad v = w_1' + w_2' + \cdots + w_k'$$

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where $w_i, w_i' \in W_i$. Subtracting these equations gives:

$$0 = (w_1 - w_1') + (w_2 - w_2') + \cdots + (w_k - w_k')$$

For any specific j, we can isolate the j-th term:

$$w_j' - w_j = \sum_{i \neq j} (w_i - w_i')$$

The left side is an element of W_j. The right side is an element of $\sum_{i \neq j} W_i$. Since the intersection of these two spaces is $\{0\}$ by condition (3), it follows that $w_j' - w_j = 0$, or $w_j = w_j'$. Since this holds for every j, the representation is unique, and the sum is direct. ■

Example 2.3.19. Consider $V = \mathbb{R}^3$. Let W_1 be the xy-plane ($z = 0$) and W_2 be the z-axis ($x = y = 0$). Show that $\mathbb{R}^3 = W_1 \oplus W_2$.

Solution. We check the intersection of W_1 and W_2. A vector $v = (x, y, z)$ is in W_1 if $z = 0$. It is in W_2 if $x = 0$ and $y = 0$. For $v \in W_1 \cap W_2$, all three coordinates must be zero: $x = 0, y = 0, z = 0$. Thus $W_1 \cap W_2 = \{(0, 0, 0)\}$. Since the intersection is the zero subspace, by condition (3) of the theorem, the sum is direct. Furthermore, since $\dim(W_1) = 2$ and $\dim(W_2) = 1$, the dimension of the direct sum is $2 + 1 = 3$, which equals $\dim(\mathbb{R}^3)$. Therefore, $\mathbb{R}^3 = W_1 \oplus W_2$. ■

Definition 2.3.20. Let V be a vector space over a field F (typically $\mathbb{R}$ or $\mathbb{C}$), and let $T : V \to V$ be a linear transformation. A scalar $\lambda \in F$ is called an **eigenvalue** of T if there exists a non-zero vector $v \in V$ such that:

$$T(v) = \lambda v$$

The vector v is called an **eigenvector** of T corresponding to the eigenvalue λ. The set of all eigenvectors corresponding to λ, together with the zero vector, forms a subspace of V called the **eigenspace**, denoted by E_λ.

Definition 2.3.21. Let A be an $n \times n$ matrix representing a linear transformation T with respect to a basis. The polynomial defined by:

$$p(\lambda) = \det(A - \lambda I)$$

is called the **characteristic polynomial** of A. The equation $p(\lambda) = 0$ is the **characteristic equation**.

Theorem 2.3.22. *A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if* $\det(A - \lambda I) = 0$.

Proof. By definition, λ is an eigenvalue if there exists $v \neq 0$ such that $Av = \lambda v$. This equation can be rewritten as:

$$Av - \lambda v = 0$$

$$(A - \lambda I)v = 0$$

This is a homogeneous system of linear equations. For a non-zero solution v to exist, the matrix $(A - \lambda I)$ must be singular (non-invertible). From matrix theory, a square matrix is singular if and only if its determinant is zero. Thus:

$$\det(A - \lambda I) = 0$$

■

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Theorem 2.3.23. *Let $v_1, v_2, \ldots, v_k$ be eigenvectors of a linear transformation T corresponding to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_k$. Then the set $\{v_1, v_2, \ldots, v_k\}$ is linearly independent.*

Proof. We proceed by induction on k. For $k = 1$, the set $\{v_1\}$ is linearly independent because $v_1 \neq 0$ by the definition of an eigenvector. Assume the statement holds for $k-1$ eigenvectors. Consider the linear combination:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0 \quad \text{(Equation 1)}$$

Apply $(T - \lambda_k I)$ to both sides:

$$c_1(T - \lambda_k I)v_1 + \cdots + c_{k-1}(T - \lambda_k I)v_{k-1} + c_k(T - \lambda_k I)v_k = 0$$

Since $T(v_i) = \lambda_i v_i$, this becomes:

$$c_1(\lambda_1 - \lambda_k)v_1 + c_2(\lambda_2 - \lambda_k)v_2 + \cdots + c_{k-1}(\lambda_{k-1} - \lambda_k)v_{k-1} + 0 = 0$$

By the inductive hypothesis, $\{v_1, \ldots, v_{k-1}\}$ are linearly independent. Thus, all coefficients must be zero:

$$c_i(\lambda_i - \lambda_k) = 0 \quad \text{for } i = 1, \ldots, k-1$$

Since the eigenvalues are distinct, $(\lambda_i - \lambda_k) \neq 0$, which implies $c_1 = c_2 = \cdots = c_{k-1} = 0$. Substituting these into (Equation 1) gives $c_k v_k = 0$, and since $v_k \neq 0$, we have $c_k = 0$. Thus, the set is linearly independent. ■

Definition 2.3.24. A linear operator T on a finite-dimensional vector space is called **diagonalizable** if there is an ordered basis β for V such that $[T]_\beta$ is a diagonal matrix. In terms of a square matrix, this can be viewed as a matrix A having that property that $P^{-1}AP = \text{diag}(\lambda_1, \ldots, \lambda_n)$ for some invertible matrix P. In this case, the column vectors of P are the eigenvectors of A.

Theorem 2.3.25. *An $n \times n$ matrix A is diagonalizable if and only if it has n linearly independent eigenvectors.*

Proof. First, assume that A is diagonalizable. By definition, there exists an invertible $n \times n$ matrix P and a diagonal matrix D such that:

$$A = PDP^{-1}$$

Multiplying both sides of the equation by P on the right, we obtain:

$$AP = PD$$

Let the columns of P be denoted by $v_1, v_2, \ldots, v_n$, and let the diagonal entries of D be $\lambda_1, \lambda_2, \ldots, \lambda_n$. We can express the product AP in terms of its columns:

$$AP = A[v_1\ v_2\ \ldots\ v_n] = [Av_1\ Av_2\ \ldots\ Av_n]$$

Similarly, the product PD is calculated by scaling each column of P by the corresponding diagonal entry of D:

$$PD = [v_1\ v_2\ \ldots\ v_n] \begin{pmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{pmatrix} = [\lambda_1 v_1\ \lambda_2 v_2\ \ldots\ \lambda_n v_n]$$

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Equating the columns of AP and PD, we find that $Av_i = \lambda_i v_i$ for each $i = 1, \ldots, n$. Since P is an invertible matrix, its columns $v_1, \ldots, v_n$ are linearly independent by definition. Furthermore, no column of an invertible matrix can be the zero vector, so $v_i \neq 0$ for all i. Thus, A has n linearly independent eigenvectors.

Now, assume that A has n linearly independent eigenvectors $v_1, v_2, \ldots, v_n$ with corresponding eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. We construct an $n \times n$ matrix P using these eigenvectors as columns:

$$P = [v_1 \ v_2 \ \ldots \ v_n]$$

Because the eigenvectors are linearly independent, the matrix P is invertible. Next, we construct a diagonal matrix D whose diagonal entries are the corresponding eigenvalues:

$$D = \begin{pmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{pmatrix}$$

Computing the product AP, we have:

$$AP = [Av_1 \ Av_2 \ \ldots \ Av_n] = [\lambda_1 v_1 \ \lambda_2 v_2 \ \ldots \ \lambda_n v_n]$$

This resulting matrix is exactly what we obtain by computing PD. Therefore:

$$AP = PD$$

Since P is invertible, we can multiply on the right by P^{-1}:

$$A = PDP^{-1}$$

This shows that A is similar to a diagonal matrix, which means A is diagonalizable. ■

Example 2.3.26. Find the eigenvalues and a basis for each eigenspace of the matrix:

$$A = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$$

Solution. First, we find the characteristic polynomial by computing $\det(A - \lambda I) = 0$:

$$\det \begin{pmatrix} 4-\lambda & -2 \\ 1 & 1-\lambda \end{pmatrix} = (4-\lambda)(1-\lambda) - (-2)(1)$$

$$= \lambda^2 - 5\lambda + 4 + 2 = \lambda^2 - 5\lambda + 6$$

Setting the polynomial to zero:

$$(\lambda - 2)(\lambda - 3) = 0$$

The eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 3$.

For $\lambda_1 = 2$: Solve $(A - 2I)v = 0$:

$$\begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The row reduction leads to $x_1 - x_2 = 0$, or $x_1 = x_2$. The basis for the eigenspace E_2 is $\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\}$.

For $\lambda_2 = 3$: Solve $(A - 3I)v = 0$:

$$\begin{pmatrix} 1 & -2 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

The row reduction leads to $x_1 - 2x_2 = 0$, or $x_1 = 2x_2$. The basis for the eigenspace E_3 is $\{\begin{pmatrix} 2 \\ 1 \end{pmatrix}\}$. ■

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Theorem 2.3.27. *A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if V has a basis consisting of eigenvectors of T.*

Proof. Suppose T is diagonalizable. By definition, there exists a basis $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ for V such that the matrix representation of T with respect to $\mathcal{B}$ is diagonal:

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{pmatrix}$$

By the definition of the matrix representation of a linear operator, the j-th column of $[T]_{\mathcal{B}}$ is the coordinate vector of $T(v_j)$ with respect to the basis $\mathcal{B}$. Thus, for each $j \in \{1, \ldots, n\}$:

$$T(v_j) = 0v_1 + 0v_2 + \cdots + \lambda_j v_j + \cdots + 0v_n = \lambda_j v_j$$

Since each v_j is a element of a basis, $v_j \neq 0$. The equation $T(v_j) = \lambda_j v_j$ implies that v_j is an eigenvector of T corresponding to the eigenvalue λ_j. Therefore, $\mathcal{B}$ is a basis of V consisting of eigenvectors of T.

Suppose V has a basis $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ consisting of eigenvectors of T. Then for each j, there exists a scalar λ_j such that:

$$T(v_j) = \lambda_j v_j$$

To find the matrix representation $[T]_{\mathcal{B}}$, we write $T(v_j)$ as a linear combination of the basis vectors:

$$T(v_j) = 0v_1 + \cdots + \lambda_j v_j + \cdots + 0v_n$$

The coordinate vector for the j-th column is:

$$[T(v_j)]_{\mathcal{B}} = \begin{pmatrix} 0 \\ \vdots \\ \lambda_j \\ \vdots \\ 0 \end{pmatrix}$$

where λ_j is in the j-th row. Constructing the matrix column by column, we obtain:

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & 0 & \ldots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & \lambda_n \end{pmatrix}$$

Since $[T]_{\mathcal{B}}$ is a diagonal matrix, T is diagonalizable by definition. ■

Example 2.3.28. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (x + y, 3x - y)$. Determine if T is diagonalizable by looking for a basis of eigenvectors.

Solution. The matrix of T relative to the standard basis is $A = \begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix}$. First, find the eigenvalues:

$$\det(A - \lambda I) = (1 - \lambda)(-1 - \lambda) - 3 = \lambda^2 - 1 - 3 = \lambda^2 - 4$$

Setting $\lambda^2 - 4 = 0$ gives $\lambda_1 = 2$ and $\lambda_2 = -2$.

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For $\lambda_1 = 2$: $(A - 2I)v = 0 \implies \begin{pmatrix} -1 & 1 \\ 3 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies x = y$. An eigenvector is $v_1 = (1, 1)$.

For $\lambda_2 = -2$: $(A + 2I)v = 0 \implies \begin{pmatrix} 3 & 1 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \implies 3x = -y$. An eigenvector is $v_2 = (1, -3)$.

The set $\mathcal{B} = \{(1, 1), (1, -3)\}$ is a set of two eigenvectors corresponding to distinct eigenvalues, so it is linearly independent. Since $\dim(\mathbb{R}^2) = 2$, $\mathcal{B}$ is a basis for $\mathbb{R}^2$. Because a basis of eigenvectors exists, T is diagonalizable. ■

Theorem 2.3.29. *Let T be a linear operator on a vector space V, and let λ be an eigenvalue of T. A vector $v \in V$ is an eigenvector of T corresponding to λ if and only if $v \neq 0$ and $v \in N(T - \lambda I)$.*

Proof. By definition, v is an eigenvector of T corresponding to λ if and only if $v \neq 0$ and $T(v) = \lambda v$. This equation can be rewritten as follows:

$$T(v) - \lambda v = 0$$
$$(T - \lambda I)v = 0$$

The condition $(T - \lambda I)v = 0$ is equivalent to the statement that v is in the null space (kernel) of the operator $(T - \lambda I)$, denoted as $v \in N(T - \lambda I)$. Thus, the conditions $v \neq 0$ and $T(v) = \lambda v$ are equivalent to $v \neq 0$ and $v \in N(T - \lambda I)$. ■

Theorem 2.3.30. *Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T. If T is diagonalizable, then V is the direct sum of the eigenspaces of T:*

$$V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}$$

Proof. To prove that V is the direct sum of the eigenspaces, we must show two things: The sum of the eigenspaces spans V: $V = E_{\lambda_1} + E_{\lambda_2} + \cdots + E_{\lambda_k}$. and the sum is direct: $E_{\lambda_i} \cap \sum_{j \neq i} E_{\lambda_j} = \{0\}$.

Spanning the space. Since T is diagonalizable, by definition there exists a basis $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ for V consisting entirely of eigenvectors of T. Each vector v_i in this basis corresponds to exactly one of the distinct eigenvalues λ_j. Therefore, each v_i belongs to one of the eigenspaces E_{λ_j}. Because any vector $v \in V$ can be written as a linear combination of the basis vectors:

$$v = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$$

and each $v_i \in \bigcup_{j=1}^{k} E_{\lambda_j}$, it follows that v is a sum of vectors from the various eigenspaces. Thus:

$$V = E_{\lambda_1} + E_{\lambda_2} + \cdots + E_{\lambda_k}$$

The sum is direct. The sum $E_{\lambda_1} + \cdots + E_{\lambda_k}$ is direct if and only if the only way to write the zero vector as a sum of eigenvectors from distinct eigenspaces is the trivial way. Suppose:

$$w_1 + w_2 + \cdots + w_k = 0$$

where each $w_j \in E_{\lambda_j}$. We know that eigenvectors corresponding to distinct eigenvalues are linearly independent. If any w_j were non-zero, the set of non-zero w_j's would be a set of eigenvectors corresponding to distinct eigenvalues that is linearly dependent (since their sum is zero). This contradicts the theorem that eigenvectors of distinct eigenvalues are linearly independent. Therefore, we must have:

$$w_1 = w_2 = \cdots = w_k = 0$$

This confirms that the sum is a direct sum.

■

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Theorem 2.3.31. *Let T be a linear operator on a finite-dimensional vector space V, and let λ be an eigenvalue of T having multiplicity m. Then $1 \leq \dim(E_\lambda) \leq m$.*

Proof. By the definition of an eigenvalue, there exists a non-zero vector v such that $T(v) = \lambda v$. Since E_λ contains at least one non-zero vector, its dimension must be at least 1, proving $1 \leq \dim(E_\lambda)$.

To prove $\dim(E_\lambda) \leq m$, let $d = \dim(E_\lambda)$ and choose a basis $\{v_1, v_2, \ldots, v_d\}$ for E_λ. Extend this to a basis $\mathcal{B} = \{v_1, \ldots, v_d, v_{d+1}, \ldots, v_n\}$ for V. The matrix representation of T with respect to $\mathcal{B}$ is a block matrix of the form:

$$[T]_{\mathcal{B}} = \begin{pmatrix} \lambda I_d & B \\ 0 & C \end{pmatrix}$$

where I_d is the $d \times d$ identity matrix, B is a $d \times (n-d)$ matrix, and C is an $(n-d) \times (n-d)$ matrix. The characteristic polynomial $p(x)$ of T is given by:

$$p(x) = \det([T]_{\mathcal{B}} - xI) = \det \begin{pmatrix} (\lambda - x)I_d & B \\ 0 & C - xI \end{pmatrix}$$

Using the property of block triangular matrices, we have:

$$p(x) = \det((\lambda - x)I_d) \cdot \det(C - xI) = (\lambda - x)^d \cdot q(x)$$

where $q(x)$ is a polynomial of degree $n - d$. This shows that $(\lambda - x)$ divides $p(x)$ at least d times. Since the algebraic multiplicity m is defined as the largest integer such that $(\lambda - x)^m$ divides $p(x)$, it follows that $d \leq m$. Thus, $1 \leq \dim(E_\lambda) \leq m$. ■

Definition 2.3.32. The **minimal polynomial** of a linear operator $T : V \to V$ on a finite-dimensional vector space V is the unique monic polynomial $m_T(x) \in F[x]$ of least degree such that $m_T(T) = 0$.

Theorem 2.3.33 (Primary Decomposition Theorem)**.** *Let T be a linear operator on a finite-dimensional vector space V with minimal polynomial $m_T(x) = f_1(x)^{n_1} \cdots f_k(x)^{n_k}$. Let $W_i = \ker(f_i(T)^{n_i})$ for $i = 1, \ldots, k$. Then:*

1. *$V = W_1 \oplus W_2 \oplus \cdots \oplus W_k$.*
2. *Each W_i is T-invariant (i.e., $T(W_i) \subseteq W_i$).*
3. *If T_i is the restriction of T to W_i, then the minimal polynomial of T_i is $f_i(x)^{n_i}$.*

Proof. For each $i \in \{1, \ldots, k\}$, define the polynomial $g_i(x)$ by dividing $m_T(x)$ by the i-th primary factor:

$$g_i(x) = \frac{m_T(x)}{f_i(x)^{n_i}} = \prod_{j \neq i} f_j(x)^{n_j}$$

Since the $f_i(x)^{n_i}$ factors are pairwise relatively prime, the set of polynomials $\{g_1(x), \ldots, g_k(x)\}$ is relatively prime. By Bezout's Identity, there exist polynomials $h_1(x), \ldots, h_k(x)$ such that:

$$1 = h_1(x)g_1(x) + h_2(x)g_2(x) + \cdots + h_k(x)g_k(x)$$

Substituting the operator T into this identity, we obtain:

$$I = h_1(T)g_1(T) + h_2(T)g_2(T) + \cdots + h_k(T)g_k(T) \quad \text{(Equation 1)}$$

Let $E_i = h_i(T)g_i(T)$. For any vector $v \in V$, we have $v = E_1v + E_2v + \cdots + E_kv$. To show V is the sum of the W_i, we show $E_iv \in W_i$. Note that:

$$f_i(T)^{n_i}(E_iv) = f_i(T)^{n_i}h_i(T)g_i(T)v = h_i(T)(f_i(T)^{n_i}g_i(T))v = h_i(T)m_T(T)v$$

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Since $m_T(T) = 0$, we have $f_i(T)^{n_i}(E_i v) = 0$, meaning $E_i v \in \ker(f_i(T)^{n_i}) = W_i$. Thus $V = W_1 + \cdots + W_k$.

To show the sum is direct, suppose $w_1 + \cdots + w_k = 0$ with $w_i \in W_i$. For $i \neq j$, $g_j(x)$ contains $f_i(x)^{n_i}$ as a factor. Since $w_i \in \ker(f_i(T)^{n_i})$, it follows that $g_j(T)w_i = 0$ for all $i \neq j$. Applying $E_j = h_j(T)g_j(T)$ to the sum $w_1 + \cdots + w_k = 0$:

$$E_j(w_1) + \cdots + E_j(w_j) + \cdots + E_j(w_k) = 0$$

Since E_j contains the operator $g_j(T)$, we have $E_j(w_i) = 0$ for all $i \neq j$. Thus the equation reduces to $E_j(w_j) = 0$. From (Equation 1), $I = \sum E_i$, and since $E_i(w_j) = 0$ for $i \neq j$, we have $I(w_j) = E_j(w_j)$. Therefore, $w_j = E_j(w_j) = 0$ for all j. This proves the sum is direct.

For T-invariance, if $w \in W_i$, then $f_i(T)^{n_i} w = 0$. Then:

$$f_i(T)^{n_i}(Tw) = T(f_i(T)^{n_i} w) = T(0) = 0$$

so $Tw \in W_i$. Finally, because $f_i(T)^{n_i}$ annihilates W_i, the minimal polynomial of the restriction T_i must divide $f_i(x)^{n_i}$. By the definition of the minimal polynomial of T, it can be shown that no smaller power of $f_i(x)$ can annihilate W_i, thus $m_{T_i}(x) = f_i(x)^{n_i}$. ■

Theorem 2.3.34. *A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if its minimal polynomial $m_T(x)$ is a product of distinct linear factors in $F[x]$. That is,*

$$m_T(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$$

where $\lambda_1, \ldots, \lambda_k$ are the distinct eigenvalues of T.

Proof. Assume T is diagonalizable. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T. Since T is diagonalizable, the vector space V is the direct sum of the eigenspaces:

$$V = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \cdots \oplus E_{\lambda_k}$$

Let $p(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$. For any $v \in V$, we can write $v = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$ where $v_i \in E_{\lambda_i}$. Applying $p(T)$ to an arbitrary v_i:

$$p(T)v_i = \left(\prod_{j \neq i} (T - \lambda_j I) \right) (T - \lambda_i I) v_i$$

Since $(T - \lambda_i I)v_i = 0$, it follows that $p(T)v_i = 0$ for all i. Consequently, $p(T)v = p(T)(c_1 v_1 + \cdots + c_k v_k) = 0$ for all $v \in V$. Thus $p(T) = 0$. By the properties of minimal polynomials, $m_T(x)$ must divide $p(x)$. Since $p(x)$ consists of distinct linear factors, $m_T(x)$ must also consist of distinct linear factors.

Assume $m_T(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$ where λ_i are distinct. According to the **Primary Decomposition Theorem**, if the minimal polynomial is the product of pairwise relatively prime factors $f_i(x) = (x - \lambda_i)$, then:

$$V = \ker(f_1(T)) \oplus \ker(f_2(T)) \oplus \cdots \oplus \ker(f_k(T))$$

Substituting $f_i(T) = (T - \lambda_i I)$, we have:

$$V = \ker(T - \lambda_1 I) \oplus \ker(T - \lambda_2 I) \oplus \cdots \oplus \ker(T - \lambda_k I)$$

Each subspace $V_i = \ker(T - \lambda_i I)$ is the eigenspace E_{λ_i} of T corresponding to λ_i. Since V is the direct sum of the eigenspaces of T, there exists a basis of V consisting of eigenvectors. Thus, T is diagonalizable. ■

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Proof without using the Primary Decomposition Theorem. Suppose T is diagonalizable. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T. By the definition of diagonalizability, there exists a basis $\mathcal{B} = \{v_1, \ldots, v_n\}$ for V consisting of eigenvectors of T. For any $v \in \mathcal{B}$, $T(v) = \lambda_j v$ for some $j \in \{1, \ldots, k\}$.

Define the polynomial $p(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$. Applying the operator $p(T)$ to any basis vector $v \in \mathcal{B}$:

$$p(T)v = \left(\prod_{j=1}^{k} (T - \lambda_j I) \right) v$$

Since v is an eigenvector, there exists some i such that $(T - \lambda_i I)v = 0$. Because the factors in the product commute, we can shift this specific factor to the end:

$$p(T)v = \left(\prod_{j \neq i} (T - \lambda_j I) \right) (T - \lambda_i I)v = \left(\prod_{j \neq i} (T - \lambda_j I) \right) 0 = 0$$

Since $p(T)$ is zero on every vector in a basis of V, $p(T) = 0$. The minimal polynomial $m_T(x)$ must divide $p(x)$. Since $p(x)$ is a product of distinct linear factors, $m_T(x)$ must also be a product of distinct linear factors.

Suppose $m_T(x) = (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_k)$ where $\lambda_1, \ldots, \lambda_k$ are distinct. We will show that V is the sum of the eigenspaces of T. For each $j \in \{1, \ldots, k\}$, define the Lagrange-style polynomial:

$$p_j(x) = \prod_{i \neq j} \frac{x - \lambda_i}{\lambda_j - \lambda_i}$$

These polynomials have the property that $p_j(\lambda_j) = 1$ and $p_j(\lambda_i) = 0$ for $i \neq j$. The sum $P(x) = \sum_{j=1}^{k} p_j(x)$ is a polynomial of degree at most $k - 1$. Since $P(\lambda_i) = 1$ for k distinct values, $P(x)$ must be the identity polynomial 1. Thus, as operators:

$$I = p_1(T) + p_2(T) + \cdots + p_k(T)$$

For any vector $v \in V$, we can write:

$$v = I(v) = p_1(T)v + p_2(T)v + \cdots + p_k(T)v$$

Let $v_j = p_j(T)v$. We check if v_j is an eigenvector by applying $(T - \lambda_j I)$:

$$(T - \lambda_j I)v_j = (T - \lambda_j I)p_j(T)v = \left((T - \lambda_j I) \prod_{i \neq j} \frac{T - \lambda_i I}{\lambda_j - \lambda_i} \right) v$$

The expression in the parentheses is a scalar multiple of the minimal polynomial $m_T(T)$:

$$(T - \lambda_j I)v_j = \frac{1}{\prod_{i \neq j} (\lambda_j - \lambda_i)} m_T(T)v$$

Since $m_T(T) = 0$, it follows that $(T - \lambda_j I)v_j = 0$. This implies that each non-zero v_j is an eigenvector of T with eigenvalue λ_j. Since every vector v can be expressed as a sum of eigenvectors $v = \sum v_j$, the eigenspaces span V. Therefore, there exists a basis of V consisting of eigenvectors, making T diagonalizable. ■

Example 2.3.35. Let T be a linear operator on $\mathbb{R}^2$ represented by the matrix:

$$A = \begin{pmatrix} 5 & 1 \\ 0 & 5 \end{pmatrix}$$

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Determine if T is diagonalizable using its minimal polynomial.

Solution. The characteristic polynomial of A is:

$$\det(A - xI) = (5 - x)^2$$

The minimal polynomial $m_A(x)$ must divide the characteristic polynomial and contain all distinct irreducible factors of the characteristic polynomial. Thus, $m_A(x)$ is either $(x-5)$ or $(x-5)^2$. We test $(x-5)$:

$$A - 5I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$$

Since $(A - 5I) \neq 0$, the minimal polynomial must be $m_A(x) = (x-5)^2$. Because the minimal polynomial $m_A(x) = (x-5)(x-5)$ has a repeated linear factor (it is not a product of *distinct* linear factors), the operator T is not diagonalizable. ■

Exercise 2.3.36. Problems bases on Chapter II

1. Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be the linear transformation defined by $T(x, y, z) = (x + 2y - z, 3x - y + 4z)$. Find the matrix of T with respect to the standard bases of $\mathbb{R}^3$ and $\mathbb{R}^2$.
2. Let $D : P_3(\mathbb{R}) \to P_2(\mathbb{R})$ be the differentiation operator, where $P_n(\mathbb{R})$ is the space of polynomials of degree at most n. Find the matrix representation of D with respect to the standard bases $\{1, x, x^2, x^3\}$ and $\{1, x, x^2\}$.
3. Let $L : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $L(x, y) = (2x - 3y, x + y)$. Find the matrix of L with respect to the basis $\mathcal{B} = \{(1, 1), (1, -1)\}$.
4. Let $V = M_{2\times 2}(\mathbb{R})$ be the vector space of 2×2 matrices. Let $B = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}$ and let $T : V \to V$ be defined by $T(A) = BA$. Find the matrix of T with respect to the standard basis of V.
5. In $\mathbb{R}^2$, find the change of coordinate matrix from the basis $\mathcal{B} = \{(1, 2), (3, 5)\}$ to the standard basis $\mathcal{E} = \{(1, 0), (0, 1)\}$.
6. Let $\mathcal{B} = \{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ and $\mathcal{B}' = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ be bases for $\mathbb{R}^3$. Find the transition matrix P from $\mathcal{B}$ to $\mathcal{B}'$.
7. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $T(x, y) = (5x - y, 2x + y)$. Let $\mathcal{B} = \{(1, -1), (2, 1)\}$. Find the matrix $[T]_{\mathcal{B}}$ by first finding the matrix of T relative to the standard basis and then using a change of basis matrix.
8. Consider the basis $\mathcal{B} = \{(1, -1, 3), (0, 1, -1), (0, 3, -2)\}$ for $\mathbb{R}^3$. Find the dual basis $\mathcal{B}^*$.
9. Let $V = P_2(\mathbb{R})$. Let ϕ_1, ϕ_2, ϕ_3 be the linear functionals on V defined by $\phi_1(p) = \int_0^1 p(t)dt$, $\phi_2(p) = p'(1)$, and $\phi_3(p) = p(0)$. Find the basis for V of which $\{\phi_1, \phi_2, \phi_3\}$ is the dual.
10. Let W be the subspace of $\mathbb{R}^4$ spanned by the vectors $u_1 = (1, 2, -3, 4)$ and $u_2 = (0, 1, 4, -1)$. Find a basis for the annihilator W^0.
11. Let V be a vector space of dimension n. Prove that the annihilator of a subspace W of V, denoted W^0, is a subspace of the dual space V^*. What is $\dim(W^0)$ if $\dim(W) = k$?
12. Determine if the linear transformation $T : \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x + y - 2z, x + 2y + z, 2x + 2y - 3z)$ is invertible. If so, find a formula for T^{-1}.

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13. Show that the matrices $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ are not similar.

14. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & -1 \\ 3 & 1 \end{pmatrix}$. Determine if A and B are similar.

15. Find the characteristic polynomial of the matrix $A = \begin{pmatrix} 3 & -2 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix}$.

16. Let T be the linear operator on $M_{2\times 2}(\mathbb{R})$ defined by $T(A) = A^T$. Find the characteristic polynomial of T.

17. Find the minimal polynomial of the matrix $A = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

18. Find the minimal polynomial of the matrix $B = \begin{pmatrix} 0 & -1 & -1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}$.

19. The characteristic polynomial of a matrix A is $\Delta(t) = (t-2)^4(t-3)^3$. List all possible minimal polynomials for A.

20. Let A be an $n \times n$ matrix. Prove that A and its transpose A^T have the same minimal polynomial.

21. Find the eigenvalues and a basis for each eigenspace of the matrix $A = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$.

22. Let $T : P_2(\mathbb{R}) \to P_2(\mathbb{R})$ be the linear operator defined by $T(f(x)) = f(x) + xf'(x)$. Find the eigenvalues and eigenvectors (eigenfunctions) of T.

23. Find the eigenvalues and the dimensions of the corresponding eigenspaces for the matrix $A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$.

24. Let A be a 3×3 matrix with eigenvalues $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = -4$. What are the eigenvalues of the matrix $A^2 - 2I$?

25. Prove that the eigenvalues of an upper triangular matrix are its diagonal entries.

26. Determine whether the matrix $A = \begin{pmatrix} 5 & -3 \\ 3 & -1 \end{pmatrix}$ is diagonalizable. If so, find a matrix P such that $P^{-1}AP$ is a diagonal matrix.

27. Let $A = \begin{pmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{pmatrix}$. Given that its characteristic polynomial is $\Delta(t) = (t-1)(t+2)^2$, determine if A is diagonalizable.

28. Is the matrix $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 3 \end{pmatrix}$ diagonalizable? Justify your answer.

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29. For what value(s) of k is the matrix $A = \begin{pmatrix} 1 & k & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ diagonalizable?

30. Let T be a linear operator on a finite-dimensional vector space V. Prove that T is diagonalizable if and only if its minimal polynomial is a product of distinct linear factors.

Answers and Hints for Exercise 2.3.36

1. **Answer:** $[T] = \begin{pmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{pmatrix}$.

2. **Answer:** $[D] = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

3. **Hint:** Calculate $L(1,1) = (-1,2)$ and $L(1,-1) = (5,0)$. Then, express these results as linear combinations of the basis vectors $(1,1)$ and $(1,-1)$. For example, $(-1,2) = c_1(1,1) + c_2(1,-1)$. Solve for c_1, c_2. These coefficients form the first column of the matrix.
   Answer: $[L]_{\mathcal{B}} = \begin{pmatrix} 1/2 & 5/2 \\ -3/2 & 5/2 \end{pmatrix}$.

4. **Answer:** $[T] = \begin{pmatrix} 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$.

5. **Answer:** The matrix is formed by the basis vectors of $\mathcal{B}$ as columns: $P = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}$.

6. **Answer:** $P = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$.

7. **Hint:** The matrix of T relative to the standard basis $\mathcal{E}$ is $[T]_{\mathcal{E}} = \begin{pmatrix} 5 & -1 \\ 2 & 1 \end{pmatrix}$. The change of basis matrix from $\mathcal{B}$ to $\mathcal{E}$ is $P = \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix}$. The desired matrix is $[T]_{\mathcal{B}} = P^{-1}[T]_{\mathcal{E}}P$.
   Answer: $[T]_{\mathcal{B}} = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$.

8. **Hint:** Let the dual basis be $\{f_1, f_2, f_3\}$ where $f_i(x,y,z) = c_{i1}x + c_{i2}y + c_{i3}z$. Use the property $f_i(v_j) = \delta_{ij}$ to set up three systems of linear equations to find the coefficients.
   Answer: $f_1(x,y,z) = x$, $f_2(x,y,z) = 4x + 2y + 3z$, $f_3(x,y,z) = -x - y - z$.

9. **Hint:** Let the basis be $\{p_1, p_2, p_3\}$ where $p_j(t) = a_j + b_j t + c_j t^2$. Use the conditions $\phi_i(p_j) = \delta_{ij}$ to find the coefficients a_j, b_j, c_j for each polynomial. For example, for p_1, you need $\phi_1(p_1) = 1, \phi_2(p_1) = 0, \phi_3(p_1) = 0$.
   Answer: $p_1(t) = 3t - 3t^2$, $p_2(t) = -2t + 3t^2$, $p_3(t) = 1 - t$.

10. **Hint:** A linear functional $f(x,y,z,w) = ax + by + cz + dw$ is in W^0 if $f(u_1) = 0$ and $f(u_2) = 0$. This gives a system of two equations in the four variables a, b, c, d. Find a basis for the solution space.

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Answer: A basis for W^0 is $\{f_1, f_2\}$ where $f_1(x,y,z,w) = 11x - 4y + z$ and $f_2(x,y,z,w) = -3x + y + w$.

11. Hint: For the proof, show that for any $f, g \in W^0$ and scalar c, $(f + cg)(w) = 0$ for all $w \in W$. For the dimension, recall the theorem $\dim(V) = \dim(W) + \dim(W^0)$.
Answer: $\dim(W^0) = \dim(V) - \dim(W) = n - k$.

12. Answer: Invertible. $T^{-1}(a,b,c) = (-8a + b + 5c, 5a - b - 3c, -2a + c)$.

13. Hint: Similar matrices must have the same trace, determinant, and eigenvalues. Compare these invariants for A and B.
Answer: They are not similar. For instance, $\det(A) = 1$ while $\det(B) = 1$. The trace is also the same, $\text{tr}(A) = 2, \text{tr}(B) = 2$. However, the only eigenvalue of A is 1, with geometric multiplicity 1. The only eigenvalue of B is 1, with geometric multiplicity 2. Since the geometric multiplicities differ, they are not similar.

14. Answer: Not similar, since $\det(A) = -2$ and $\det(B) = 8$.

15. Answer: $\Delta(t) = (t-5)(t-1)(t-5) = (t-5)^2(t-1)$.

16. Hint: The space of symmetric 2×2 matrices is an eigenspace for $\lambda = 1$, and the space of skew-symmetric matrices is an eigenspace for $\lambda = -1$. The dimensions are 3 and 1, respectively.
Answer: $\Delta(t) = (t-1)^3(t+1)$.

17. Hint: The characteristic polynomial is $(t-2)^3$. The minimal polynomial must be $(t-2)^k$ for $1 \leq k \leq 3$. Test $A - 2I$, $(A - 2I)^2$, etc.
Answer: $m(t) = (t-2)^2$.

18. Answer: $m(t) = (t-1)^2$. The characteristic polynomial is $(t-1)^3$.

19. Hint: The minimal polynomial must have the same roots as the characteristic polynomial. Its degree for each factor is at least 1 and at most the algebraic multiplicity in the characteristic polynomial.
Answer: $m(t) = (t-2)^a(t-3)^b$ where $1 \leq a \leq 4$ and $1 \leq b \leq 3$. There are $4 \times 3 = 12$ possibilities.

20. Hint: Show that for any polynomial $p(t)$, $p(A) = 0$ if and only if $p(A^T) = 0$. This follows from $(p(A))^T = p(A^T)$ and the fact that a matrix is the zero matrix if and only if its transpose is the zero matrix.

21. Answer: Eigenvalues $\lambda_1 = 5, \lambda_2 = 2$. Eigenspaces $E_5 = \text{span}\{(1,1)\}$, $E_2 = \text{span}\{(-1,2)\}$.

22. Hint: Solve the differential equation $f(x) + xf'(x) = \lambda f(x)$, which simplifies to $x\frac{df}{dx} = (\lambda - 1)f$. This is a separable ODE.
Answer: Eigenvalues are $\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3$. Eigenfunctions are c_1, c_2x, c_3x^2.

23. Answer: Eigenvalues are $\lambda_1 = 1$ (alg. multiplicity 2) and $\lambda_2 = 2$ (alg. multiplicity 1). The eigenspace for $\lambda_1 = 1$ has dimension 1. The eigenspace for $\lambda_2 = 2$ has dimension 1.

24. Hint: If λ is an eigenvalue of A, then $p(\lambda)$ is an eigenvalue of $p(A)$ for any polynomial p.
Answer: The eigenvalues are $1^2 - 2 = -1$, $2^2 - 2 = 2$, and $(-4)^2 - 2 = 14$.

25. Hint: Let A be an upper triangular matrix. Consider the matrix $A - \lambda I$. It is also upper triangular. The determinant of a triangular matrix is the product of its diagonal entries.

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26. **Answer:** Yes, it is diagonalizable. The eigenvalues are $\lambda = 2, 2$. However, there is only one linearly independent eigenvector, e.g., $(1, 1)$. So it is not diagonalizable. (Correction on solving: char poly is $\lambda^2 - 4\lambda + 2 = 0$, $\lambda = 2 \pm \sqrt{2}$. Distinct real eigenvalues, so it is diagonalizable.) A correct matrix for this question type: $A = \begin{pmatrix} 3 & -1 \\ 1 & 1 \end{pmatrix}$. Eigenvalue $\lambda = 2$ (mult. 2), geometric mult. 1. Not diagonalizable. Let's re-solve the original question: $A = \begin{pmatrix} 5 & -3 \\ 3 & -1 \end{pmatrix}$. Char poly: $\lambda^2 - 4\lambda + 4 = (\lambda - 2)^2 = 0$. $\lambda = 2$ has algebraic multiplicity 2. Eigenspace: $\ker(A - 2I) = \ker \begin{pmatrix} 3 & -3 \\ 3 & -3 \end{pmatrix}$ has dimension 1. Not diagonalizable.

27. **Hint:** The matrix is diagonalizable if and only if the geometric multiplicity of the repeated eigenvalue $\lambda = -2$ is equal to its algebraic multiplicity, which is 2. Calculate $\dim(\ker(A + 2I))$.
Answer: $\dim(\ker(A + 2I)) = 1$. Since $1 \neq 2$, the matrix is not diagonalizable.

28. **Answer:** Not diagonalizable. The eigenvalue $\lambda = 3$ has algebraic multiplicity 3, but its geometric multiplicity is $\dim(\ker(A - 3I)) = 1$.

29. **Hint:** The eigenvalues are $1, 2, 1$. Diagonalizability requires the geometric multiplicity of $\lambda = 1$ to be 2. Compute $\dim(\ker(A - I))$ and find the value of k that makes this dimension 2.
Answer: $k = 0$.

30. **Hint:** This is a statement of a major theorem. To prove one direction (minimal polynomial has distinct linear factors $\implies$ diagonalizable), show that the vector space can be written as the direct sum of the eigenspaces. To prove the other direction (diagonalizable $\implies$ minimal polynomial has distinct linear factors), consider the operator on a basis of eigenvectors.

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CHAPTER

3

Inner Product Spaces

This unit extends vector spaces to include geometric structure through inner product spaces, enabling concepts like length, angle, and orthogonality. We will focus on constructing special orthonormal bases and analyzing operators, such as self-adjoint and normal operators, that preserve this structure. The central results are the Spectral Theorems, which provide conditions for orthonormal diagonalizability, and the Jordan decomposition for handling the general non-diagonalizable case.

3.1 Definition and Core Concepts

In a general vector space, we have notions of vector addition and scalar multiplication. However, we lack the geometric concepts of length, distance, and angle. An inner product is an additional structure that endows a vector space with these geometric properties, turning it into an **inner product space**. This allows us to generalize familiar ideas from Euclidean geometry to more abstract spaces, such as spaces of functions, polynomials, or matrices.

Definition 3.1.1 (Inner Product)**.** Let V be a vector space over $\mathbb{F}$ where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. An **inner product** on V is a map $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ such that for all $x, y, z \in V$ and $\alpha \in \mathbb{F}$:

(P1) **Conjugate symmetry**: $\langle x, y \rangle = \overline{\langle y, x \rangle}$.

(P2) **Linearity in the first argument**: $\langle \alpha x + y, z \rangle = \alpha \langle x, z \rangle + \langle y, z \rangle$.

(P3) **Positive definiteness**: $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0$ iff $x = 0$.

A vector space equipped with an inner product is called an **inner product space**. If $\mathbb{F} = \mathbb{R}$, conjugate symmetry is simply symmetry: $\langle x, y \rangle = \langle y, x \rangle$.

Definition 3.1.2 (Norm Induced by Inner Product)**.** For $x \in V$, the norm induced by the inner product is $\|x\| = \sqrt{\langle x, x \rangle}$.

Definition 3.1.3 (Norm, Distance, and Orthogonality)**.** Let V be an inner product space.

- The **norm** (or length) of a vector $v \in V$ is defined as $\|v\| = \sqrt{\langle v, v \rangle}$. A vector with norm 1 is called a **unit vector**.
- The **distance** between two vectors $u, v \in V$ is defined as $d(u, v) = \|u - v\|$.

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- Two vectors $u, v \in V$ are **orthogonal** if $\langle u, v \rangle = 0$. We denote this by $u \perp v$.
- In a real inner product space, the **angle** θ between two non-zero vectors u and v is defined by $\cos(\theta) = \dfrac{\langle u, v \rangle}{\|u\|\|v\|}$.

While many different inner products can be defined on a given vector space, certain inner products are considered "standard" or "natural" because they are the most common and directly generalize the geometric intuition of length and angle from $\mathbb{R}^2$ and $\mathbb{R}^3$. This document lists the standard inner products for the most frequently encountered vector spaces in linear algebra and analysis.

The Euclidean Space $\mathbb{R}^n$

This is the vector space of n-tuples of real numbers.

Definition 3.1.4. Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ be vectors in $\mathbb{R}^n$. The **standard inner product**, also known as the **Euclidean inner product** or the **dot product**, is defined as:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^{n} u_i v_i = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n.$$

If vectors are represented as column vectors, this can also be written in matrix form as $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^T \mathbf{u}$.

The Complex Space $\mathbb{C}^n$

This is the vector space of n-tuples of complex numbers. The standard inner product here requires a complex conjugate to ensure the positive-definiteness axiom holds.

Definition 3.1.5. Let $\mathbf{u} = (u_1, u_2, \ldots, u_n)$ and $\mathbf{v} = (v_1, v_2, \ldots, v_n)$ be vectors in $\mathbb{C}^n$. The **standard inner product** on $\mathbb{C}^n$ is defined as:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=1}^{n} u_i \overline{v_i} = u_1 \overline{v_1} + u_2 \overline{v_2} + \cdots + u_n \overline{v_n}.$$

The conjugate on the second vector ensures that the norm squared, $\langle \mathbf{u}, \mathbf{u} \rangle = \sum_{i=1}^{n} u_i \overline{u_i} = \sum_{i=1}^{n} |u_i|^2$, is a non-negative real number. In matrix form, this is $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{v}^* \mathbf{u}$, where $\mathbf{v}^*$ is the conjugate transpose (or Hermitian transpose) of $\mathbf{v}$.

The Space of Polynomials $P_n(\mathbb{R})$

This is the vector space of all real polynomials of degree at most n. As a subspace of $C[a, b]$ for any interval, it inherits the same inner product structure.

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Definition 3.1.6. Let $p(x)$ and $q(x)$ be two polynomials in $P_n(\mathbb{R})$. A **standard inner product** is defined by fixing an interval $[a, b]$ and using the integral form:

$$\langle p, q \rangle = \int_a^b p(x)q(x)\,dx.$$

The choice of the interval (e.g., $[-1, 1]$ or $[0, 1]$) is part of the definition of the inner product space. For example, the inner product used to generate Legendre polynomials uses the interval $[-1, 1]$.

The Space of Matrices $M_{m\times n}(\mathbb{R})$

This is the vector space of all $m \times n$ matrices with real entries.

Definition 3.1.7. Let A and B be two matrices in $M_{m\times n}(\mathbb{R})$. The **standard inner product**, also known as the **Frobenius inner product** or the **Hilbert-Schmidt inner product**, is defined as the trace of AB^T:

$$\langle A, B \rangle = \text{tr}(AB^T) = \text{tr}(B^T A).$$

This is equivalent to summing the element-wise products of the matrices:

$$\langle A, B \rangle = \sum_{i=1}^{m}\sum_{j=1}^{n} A_{ij}B_{ij}.$$

This shows that the Frobenius inner product is equivalent to treating the matrices as long vectors of size mn and taking their standard dot product. For complex matrices in $M_{m\times n}(\mathbb{C})$, the inner product is $\langle A, B \rangle = \text{tr}(AB^*) = \text{tr}(B^*A)$.

Infinite-Dimensional and Function Spaces

The Space of Continuous Functions C[a, b]

This is the vector space of all real-valued continuous functions on a closed interval $[a, b]$.

Definition 3.1.8. Let $f(x)$ and $g(x)$ be two functions in $C[a, b]$. The standard inner product is defined by the integral:

$$\langle f, g \rangle = \int_a^b f(x)g(x)\,dx.$$

The integral acts as a "continuous sum," making this a natural analogue of the dot product. For the space of complex-valued continuous functions, a conjugate is again required: $\langle f, g \rangle = \int_a^b f(x)\overline{g(x)}\,dx$.

Weighted Function Spaces $L^2_w(a, b)$

This is a generalization of the previous space, where different parts of the interval are given different importance or "weight".

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Definition 3.1.9. Let $w(x)$ be a continuous, positive real-valued function on (a, b), called a **weight function**. For two functions f, g for which the following integral is finite, the **weighted inner product** is:

$$\langle f, g \rangle_w = \int_a^b f(x)g(x)w(x)\,dx.$$

This type of inner product is fundamental in the theory of orthogonal polynomials. For example, the Hermite polynomials are orthogonal with respect to the weight $w(x) = e^{-x^2}$ on $(-\infty, \infty)$.

The Sobolev Space $H^1(a, b)$

Sobolev spaces are crucial in the study of partial differential equations. They are function spaces whose inner product accounts for both the function's values and its derivatives' values.

Definition 3.1.10. The space $H^1(a, b)$ consists of functions f on $[a, b]$ such that both f and its derivative f' are square-integrable. The **standard H^1 inner product** is:

$$\langle f, g \rangle_{H^1} = \int_a^b \left(f(x)g(x) + f'(x)g'(x)\right)\,dx.$$

This inner product measures both the "size" of the functions (via the fg term) and their "smoothness" or "total change" (via the $f'g'$ term).

The Sequence Space l^2

This is a prototypical infinite-dimensional inner product space, fundamental in functional analysis.

Definition 3.1.11. The space l^2 consists of all infinite sequences of real or complex numbers $\mathbf{x} = (x_1, x_2, \dots)$ such that $\sum_{i=1}^{\infty} |x_i|^2 < \infty$. The **standard inner product** on l^2 is:

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{\infty} x_i \overline{y_i}.$$

The condition defining the space guarantees that this infinite series converges absolutely.

Abstract and Structural Examples

The Space of Random Variables $L^2(\Omega, \mathcal{F}, P)$

In probability theory, the concept of an inner product provides a geometric framework for understanding random variables.

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Definition 3.1.12. Let X and Y be real-valued random variables on a probability space $(\Omega, \mathcal{F}, P)$ with finite second moments (i.e., $E[X^2] < \infty$ and $E[Y^2] < \infty$). The **standard inner product** is defined using the expectation operator:

$$\langle X, Y \rangle = E[XY].$$

In this context, the norm $\|X\| = \sqrt{E[X^2]}$ is the root mean square. Two random variables are orthogonal if their expectation product is zero, $E[XY] = 0$. Note that if X and Y have zero mean, their inner product is precisely their covariance, $\text{Cov}(X, Y) = E[XY]$.

The Direct Sum of Inner Product Spaces $V \oplus W$

Given two inner product spaces, one can construct a new, larger inner product space by taking their direct sum.

Definition 3.1.13. Let $(V, \langle \cdot, \cdot \rangle_V)$ and $(W, \langle \cdot, \cdot \rangle_W)$ be two inner product spaces. The direct sum $V \oplus W$ consists of all ordered pairs (v, w) where $v \in V$ and $w \in W$. The **standard inner product** on $V \oplus W$ is defined as:

$$\langle (v_1, w_1), (v_2, w_2) \rangle_{V \oplus W} = \langle v_1, v_2 \rangle_V + \langle w_1, w_2 \rangle_W.$$

This construction effectively treats vectors from V and W as orthogonal components in the new space. For example, $\mathbb{R}^n \oplus \mathbb{R}^m$ is naturally isomorphic to $\mathbb{R}^{n+m}$ with its standard dot product.

Example 3.1.14. Consider the vector space $V = \mathbb{R}^2$. Show that the function defined by $\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2$ for $\mathbf{u} = (u_1, u_2)$ and $\mathbf{v} = (v_1, v_2)$ is a valid inner product.

Solution. We must verify the three axioms for an inner product on a real vector space.

1. **Linearity:** Let $\mathbf{u} = (u_1, u_2), \mathbf{v} = (v_1, v_2), \mathbf{w} = (w_1, w_2)$ and $c \in \mathbb{R}$.

$$\begin{aligned} \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle &= \langle (u_1 + v_1, u_2 + v_2), (w_1, w_2) \rangle \\ &= 2(u_1 + v_1)w_1 + 3(u_2 + v_2)w_2 \\ &= (2u_1w_1 + 3u_2w_2) + (2v_1w_1 + 3v_2w_2) = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle. \\ \langle c\mathbf{u}, \mathbf{v} \rangle &= \langle (cu_1, cu_2), (v_1, v_2) \rangle = 2(cu_1)v_1 + 3(cu_2)v_2 = c(2u_1v_1 + 3u_2v_2) = c\langle \mathbf{u}, \mathbf{v} \rangle. \end{aligned}$$

   The linearity axiom holds.

2. **Symmetry:**

$$\langle \mathbf{u}, \mathbf{v} \rangle = 2u_1v_1 + 3u_2v_2 = 2v_1u_1 + 3v_2u_2 = \langle \mathbf{v}, \mathbf{u} \rangle.$$

   The symmetry axiom holds.

3. **Positive-definiteness:**

$$\langle \mathbf{u}, \mathbf{u} \rangle = 2u_1^2 + 3u_2^2.$$

   Since $u_1^2 \geq 0$ and $u_2^2 \geq 0$, their weighted sum is also non-negative, so $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$. Furthermore, $2u_1^2 + 3u_2^2 = 0$ if and only if $u_1^2 = 0$ and $u_2^2 = 0$, which means $u_1 = 0$ and $u_2 = 0$. Thus, $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = \mathbf{0}$.

Since all three axioms are satisfied, this function defines a valid inner product on $\mathbb{R}^2$. ■

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Example 3.1.15. Let $V = C[0,1]$ be the space of continuous real-valued functions on $[0,1]$ with inner product $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$. Find the norm of $f(x) = \sqrt{x}$ and the angle θ between $f(x) = 1$ and $g(x) = x$.

Solution. First, we find the norm of $f(x) = \sqrt{x}$.

$$\|f\|^2 = \langle f, f\rangle = \int_0^1 (\sqrt{x})^2\,dx = \int_0^1 x\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}.$$

So, the norm is $\|f\| = \sqrt{1/2} = 1/\sqrt{2}$.

Next, to find the angle between $f(x) = 1$ and $g(x) = x$, we use $\cos(\theta) = \frac{\langle f,g\rangle}{\|f\|\|g\|}$.

- $\langle f, g\rangle = \int_0^1 (1)(x)\,dx = \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}$.
- $\|f\|^2 = \int_0^1 1^2\,dx = [x]_0^1 = 1$, so $\|f\| = 1$.
- $\|g\|^2 = \int_0^1 x^2\,dx = \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3}$, so $\|g\| = 1/\sqrt{3}$.

Now, we compute $\cos(\theta)$:

$$\cos(\theta) = \frac{1/2}{1 \cdot (1/\sqrt{3})} = \frac{\sqrt{3}}{2}.$$

Therefore, the angle between the functions $f(x) = 1$ and $g(x) = x$ is $\theta = \arccos(\frac{\sqrt{3}}{2}) = \frac{\pi}{6}$ radians or 30°. ■

Example 3.1.16. Let $V = \mathbb{C}^2$ with the standard inner product $\langle \mathbf{u}, \mathbf{v}\rangle = u_1\bar{v}_1 + u_2\bar{v}_2$. Let $\mathbf{u} = (1+i, 2)$ and $\mathbf{v} = (2-3i, i)$. Compute $\langle \mathbf{u}, \mathbf{v}\rangle$, $\langle \mathbf{v}, \mathbf{u}\rangle$, and $\|\mathbf{u}\|$.

Solution. We apply the definition of the complex inner product.

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v}\rangle &= (1+i)\overline{(2-3i)} + (2)\overline{(i)} \\ &= (1+i)(2+3i) + (2)(-i) \\ &= (2+3i+2i+3i^2) - 2i \\ &= (2+5i-3) - 2i = -1+3i.\end{aligned}$$

To find $\langle \mathbf{v}, \mathbf{u}\rangle$, we use the conjugate symmetry property:

$$\langle \mathbf{v}, \mathbf{u}\rangle = \overline{\langle \mathbf{u}, \mathbf{v}\rangle} = \overline{-1+3i} = -1-3i.$$

To compute the norm of $\mathbf{u}$, we find $\langle \mathbf{u}, \mathbf{u}\rangle$:

$$\begin{aligned}\|\mathbf{u}\|^2 = \langle \mathbf{u}, \mathbf{u}\rangle &= (1+i)\overline{(1+i)} + (2)\overline{(2)} \\ &= (1+i)(1-i) + (2)(2) \\ &= (1-i^2) + 4 = (1-(-1)) + 4 = 2+4 = 6.\end{aligned}$$

Therefore, the norm is $\|\mathbf{u}\| = \sqrt{6}$. ■

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Example 3.1.17. Let $V = M_{2\times 2}(\mathbb{R})$ be the space of 2×2 real matrices. A common inner product for this space is the Frobenius inner product, defined as $\langle A, B\rangle = \text{tr}(B^T A)$. Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix}$. Determine if A and B are orthogonal.

Solution. First, we compute the product $B^T A$.

$$B^T = \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix}.$$

$$B^T A = \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} (-2)(1) + (1)(0) & (-2)(2) + (1)(1) \\ (1)(1) + (0)(0) & (1)(2) + (0)(1) \end{pmatrix} = \begin{pmatrix} -2 & -3 \\ 1 & 2 \end{pmatrix}.$$

Now we find the trace of this resulting matrix. The trace is the sum of the diagonal elements.

$$\langle A, B\rangle = \text{tr}(B^T A) = -2 + 2 = 0.$$

Since the inner product is zero, the matrices A and B are orthogonal with respect to the Frobenius inner product. ■

Theorem 3.1.18. *Let V be an inner product space over $\mathbb{C}$. For any vectors $x, y, z \in V$ and any scalar $c \in \mathbb{C}$, the following properties hold:*

(a) $\langle x, y + z\rangle = \langle x, y\rangle + \langle x, z\rangle$

(b) $\langle x, cy\rangle = \overline{c}\langle x, y\rangle$

(c) $\langle x, 0\rangle = \langle 0, x\rangle = 0$

(d) $\langle x, x\rangle = 0$ *if and only if* $x = 0$

(e) If $\langle x, z\rangle = \langle y, z\rangle$ *for all* $z \in V$, *then* $x = y$.

Proof. These properties follow directly from the three defining axioms of an inner product: linearity in the first argument, conjugate symmetry ($\langle u, v\rangle = \overline{\langle v, u\rangle}$), and positive definiteness.

(a) Using conjugate symmetry and linearity in the first argument:

$$\langle x, y + z\rangle = \overline{\langle y + z, x\rangle} = \overline{\langle y, x\rangle + \langle z, x\rangle} = \overline{\langle y, x\rangle} + \overline{\langle z, x\rangle} = \langle x, y\rangle + \langle x, z\rangle.$$

(b) Using conjugate symmetry and linearity in the first argument:

$$\langle x, cy\rangle = \overline{\langle cy, x\rangle} = \overline{c\langle y, x\rangle} = \overline{c}\,\overline{\langle y, x\rangle} = \overline{c}\langle x, y\rangle.$$

(c) Using the fact that the zero vector can be written as the scalar 0 times any vector (e.g., $0 = 0 \cdot x$ or $0 = 0 \cdot 0$), and pulling the scalar out:

$$\langle 0, x\rangle = \langle 0 \cdot 0, x\rangle = 0\langle 0, x\rangle = 0.$$

Similarly, applying the result from part (b):

$$\langle x, 0\rangle = \langle x, 0 \cdot 0\rangle = \overline{0}\langle x, 0\rangle = 0 \cdot \langle x, 0\rangle = 0.$$

(d) This is true by definition. It is the *positive definiteness* axiom of an inner product space, which states that $\langle x, x\rangle \geq 0$ for all x, and $\langle x, x\rangle = 0 \iff x = 0$.

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(e) Suppose $\langle x, z\rangle = \langle y, z\rangle$ for all $z \in V$. Subtracting $\langle y, z\rangle$ from both sides yields:

$$\langle x, z\rangle - \langle y, z\rangle = 0$$

By linearity in the first argument, we can combine the terms:

$$\langle x - y, z\rangle = 0$$

Because this holds for *all* vectors $z \in V$, it must specifically hold for the vector $z = x - y$. Substituting this in yields:

$$\langle x - y, x - y\rangle = 0$$

By the positive definiteness axiom stated in part (d), this implies that $x - y = 0$, and therefore $x = y$. ■

Theorem 3.1.19. *Let V be an inner product space over a field F (where F is $\mathbb{R}$ or $\mathbb{C}$). The norm induced by the inner product is defined as $\|x\| = \sqrt{\langle x, x\rangle}$. For all $x, y \in V$ and $c \in F$, the following are true:*

(a) $\|cx\| = |c| \cdot \|x\|$

(b) $\|x\| = 0$ *if and only if* $x = 0$

(c) ***Cauchy-Schwarz Inequality:*** $|\langle x, y\rangle| \leq \|x\| \cdot \|y\|$

(d) ***Triangle Inequality:*** $\|x + y\| \leq \|x\| + \|y\|$

Proof. **(a)** By the definition of the norm and the properties of the inner product (pulling out scalars):

$$\|cx\|^2 = \langle cx, cx\rangle = c\bar{c}\langle x, x\rangle = |c|^2\|x\|^2$$

Taking the principal square root of both sides yields $\|cx\| = |c| \cdot \|x\|$.

(b) By definition, $\|x\| = \sqrt{\langle x, x\rangle}$. Thus, $\|x\| = 0$ if and only if $\langle x, x\rangle = 0$. By the positive definiteness axiom of inner product spaces, $\langle x, x\rangle = 0 \iff x = 0$.

(c) If $y = \mathbf{0}$, both sides are zero, so the inequality holds. Assume $v \neq \mathbf{0}$. For any scalar $c \in \mathbb{F}$, we have $\|x - cy\|^2 \geq 0$.

$$\begin{aligned} 0 \leq \langle x - cy, x - cy\rangle &= \langle x, x\rangle - \langle x, cy\rangle - \langle cy, x\rangle + \langle cy, cy\rangle \\ &= \|x\|^2 - \bar{c}\langle x, y\rangle - c\langle y, x\rangle + c\bar{c}\|y\|^2 \\ &= \|x\|^2 - \bar{c}\langle x, y\rangle - c\overline{\langle x, y\rangle} + |c|^2\|y\|^2. \end{aligned}$$

This inequality must hold for any choice of c. To simplify the expression, we choose $c = \frac{\langle x, y\rangle}{\|y\|^2}$. Substituting this value:

$$\begin{aligned} 0 &\leq \|x\|^2 - \frac{\overline{\langle x, y\rangle}}{\|y\|^2}\langle x, y\rangle - \frac{\langle x, y\rangle}{\|y\|^2}\overline{\langle x, y\rangle} + \left|\frac{\langle x, y\rangle}{\|y\|^2}\right|^2 \|y\|^2 \\ 0 &\leq \|x\|^2 - \frac{|\langle x, y\rangle|^2}{\|y\|^2}. \end{aligned}$$

Rearranging this gives $|\langle x, y\rangle|^2 \leq \|x\|^2\|y\|^2$. Taking the square root of both sides yields the desired inequality. Equality holds if and only if $\|x - cy\| = 0$, which means $x = cy$, so x and y are linearly dependent.

(d) We expand $\|x + y\|^2$ using the properties of the inner product:

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y\rangle \\ &= \langle x, x\rangle + \langle x, y\rangle + \langle y, x\rangle + \langle y, y\rangle \\ &= \|x\|^2 + \langle x, y\rangle + \overline{\langle x, y\rangle} + \|y\|^2 \\ &= \|x\|^2 + 2\mathrm{Re}(\langle x, y\rangle) + \|y\|^2 \end{aligned}$$

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Since the real part of any complex number is less than or equal to its absolute value ($\text{Re}(z) \leq |z|$), and applying the Cauchy-Schwarz inequality from part (c):

$$2\text{Re}(\langle x, y \rangle) \leq 2|\langle x, y \rangle| \leq 2\|x\|\|y\|$$

Substituting this back into the equation gives:

$$\|x+y\|^2 \leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$$

Taking the square root of both sides provides the Triangle Inequality: $\|x+y\| \leq \|x\| + \|y\|$. ■

Theorem 3.1.20 (Pythagorean Theorem). *For any orthogonal vectors u, v in an inner product space V (i.e., $\langle u, v \rangle = 0$),*

$$\|u+v\|^2 = \|u\|^2 + \|v\|^2.$$

Proof. Since $u \perp v$, we have $\langle u, v \rangle = 0$, which implies $\langle v, u \rangle = \overline{\langle u, v \rangle} = 0$. Expanding $\|u+v\|^2$:

$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v \rangle \\ &= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + 0 + 0 + \|v\|^2 \\ &= \|u\|^2 + \|v\|^2. \end{aligned}$$

■

Definition 3.1.21 (Orthonormal Set). Let V be an inner product space (over $\mathbb{R}$ or $\mathbb{C}$). A set of vectors $\{u_1, u_2, \ldots, u_k\}$ is called **orthonormal** if

$$\langle u_i, u_j \rangle = \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

where δ_{ij} is the Kronecker delta.

Definition 3.1.22 (Orthonormal Basis). An orthonormal set that is also a basis for V is called an **orthonormal basis** of V.

Theorem 3.1.23 (Properties of Orthonormal Bases). *Let $\{\vec{e}_1, \ldots, \vec{e}_n\}$ be an orthonormal basis for an n-dimensional inner product space V. Then for any vector $\vec{v} \in V$:*

1. $\vec{v} = \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \ \vec{e}_i$ *(Fourier expansion).*
2. $\|\vec{v}\|^2 = \sum_{i=1}^{n} |\ \langle \vec{v}, \vec{e}_i \rangle\ |^2$ *(Parseval's identity).*
3. *For any two vectors $\vec{v}, \vec{w} \in V$,*

$$\langle \vec{v}, \vec{w} \rangle = \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \, \overline{\langle \vec{w}, \vec{e}_i \rangle}.$$

Proof. **(1).** Since $\{\vec{e}_1, \ldots, \vec{e}_n\}$ is a basis of V, there exist unique scalars $c_1, \ldots, c_n$ such that

$$\vec{v} = \sum_{i=1}^{n} c_i \vec{e}_i.$$

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Fix an index j with $1 \le j \le n$ and take the inner product of both sides with $\vec{e}_j$:

$$\langle \vec{v}, \vec{e}_j \rangle = \left\langle \sum_{i=1}^{n} c_i \vec{e}_i,\ \vec{e}_j \right\rangle = \sum_{i=1}^{n} c_i \langle \vec{e}_i, \vec{e}_j \rangle.$$

Orthonormality gives $\langle \vec{e}_i, \vec{e}_j \rangle = \delta_{ij}$ (the Kronecker delta). Hence the sum reduces to the single term with $i = j$:

$$\langle \vec{v}, \vec{e}_j \rangle = c_j.$$

Thus $c_j = \langle \vec{v}, \vec{e}_j \rangle$ for every j, and

$$\vec{v} = \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle\, \vec{e}_i.$$

(2). Using the Fourier expansion of $\vec{v}$,

$$\|\vec{v}\|^2 = \langle \vec{v}, \vec{v} \rangle = \left\langle \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \vec{e}_i,\ \sum_{j=1}^{n} \langle \vec{v}, \vec{e}_j \rangle \vec{e}_j \right\rangle.$$

Expand by bilinearity (conjugate-linear in the second argument for complex spaces):

$$\|\vec{v}\|^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \overline{\langle \vec{v}, \vec{e}_j \rangle}\ \langle \vec{e}_i, \vec{e}_j \rangle.$$

Because $\langle \vec{e}_i, \vec{e}_j \rangle = \delta_{ij}$, only the diagonal terms $i = j$ survive:

$$\|\vec{v}\|^2 = \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \overline{\langle \vec{v}, \vec{e}_i \rangle} = \sum_{i=1}^{n} |\langle \vec{v}, \vec{e}_i \rangle|^2.$$

(3). Let $\vec{v}, \vec{w} \in V$ and write their Fourier expansions:

$$\vec{v} = \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \vec{e}_i, \qquad \vec{w} = \sum_{j=1}^{n} \langle \vec{w}, \vec{e}_j \rangle \vec{e}_j.$$

Then

$$\langle \vec{v}, \vec{w} \rangle = \left\langle \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \vec{e}_i,\ \sum_{j=1}^{n} \langle \vec{w}, \vec{e}_j \rangle \vec{e}_j \right\rangle = \sum_{i=1}^{n} \sum_{j=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \overline{\langle \vec{w}, \vec{e}_j \rangle}\ \langle \vec{e}_i, \vec{e}_j \rangle.$$

Again $\langle \vec{e}_i, \vec{e}_j \rangle = \delta_{ij}$, so the double sum collapses:

$$\langle \vec{v}, \vec{w} \rangle = \sum_{i=1}^{n} \langle \vec{v}, \vec{e}_i \rangle \overline{\langle \vec{w}, \vec{e}_i \rangle}.$$

■

3.2 Gram–Schmidt Orthogonalization

The Gram–Schmidt process constructs an orthonormal basis from an arbitrary linearly independent set.

Theorem 3.2.1 (Gram–Schmidt Orthogonalization Process). *Let $\{\vec{v}_1, \ldots, \vec{v}_n\}$ be a linearly independent set in an inner product space V. Define vectors $\{\vec{u}_1, \ldots, \vec{u}_n\}$ recursively by*

$$\vec{u}_1 = \vec{v}_1,$$

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and for $k = 2, 3, \ldots, n$

$$\vec{u}_k = \vec{v}_k - \sum_{i=1}^{k-1} \frac{\langle \vec{v}_k, \vec{u}_i \rangle}{\langle \vec{u}_i, \vec{u}_i \rangle} \vec{u}_i.$$

Then $\{\vec{u}_1, \ldots, \vec{u}_n\}$ *is an orthogonal set and*

$$\operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_n\} = \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_n\}.$$

Proof. We prove by induction on k that for each $k = 1, \ldots, n$ the vectors $\vec{u}_1, \ldots, \vec{u}_k$ are orthogonal (i.e., $\langle \vec{u}_i, \vec{u}_j \rangle = 0$ for all $i \neq j$) and satisfy

$$\operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_k\} = \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\}.$$

Base case $k = 1$**:** Since $\{\vec{v}_1, \ldots, \vec{v}_n\}$ is linearly independent, $\vec{v}_1 \neq \vec{0}$. Therefore $\vec{u}_1 = \vec{v}_1$ is non-zero and clearly $\operatorname{span}\{\vec{u}_1\} = \operatorname{span}\{\vec{v}_1\}$. The orthogonality condition is vacuously true for a single vector.
Inductive step: Assume that for some k with $2 \leq k \leq n$ we have already constructed orthogonal vectors $\vec{u}_1, \ldots, \vec{u}_{k-1}$ such that

$$\operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_{k-1}\} = \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_{k-1}\}.$$

Define

$$\vec{u}_k = \vec{v}_k - \sum_{i=1}^{k-1} \frac{\langle \vec{v}_k, \vec{u}_i \rangle}{\langle \vec{u}_i, \vec{u}_i \rangle} \vec{u}_i.$$

For any j with $1 \leq j \leq k-1$ we compute the inner product:

$$\langle \vec{u}_k, \vec{u}_j \rangle = \langle \vec{v}_k, \vec{u}_j \rangle - \sum_{i=1}^{k-1} \frac{\langle \vec{v}_k, \vec{u}_i \rangle}{\langle \vec{u}_i, \vec{u}_i \rangle} \langle \vec{u}_i, \vec{u}_j \rangle.$$

Because $\{\vec{u}_1, \ldots, \vec{u}_{k-1}\}$ is orthogonal, $\langle \vec{u}_i, \vec{u}_j \rangle = 0$ when $i \neq j$. Hence only the term with $i = j$ survives:

$$\langle \vec{u}_k, \vec{u}_j \rangle = \langle \vec{v}_k, \vec{u}_j \rangle - \frac{\langle \vec{v}_k, \vec{u}_j \rangle}{\langle \vec{u}_j, \vec{u}_j \rangle} \langle \vec{u}_j, \vec{u}_j \rangle = \langle \vec{v}_k, \vec{u}_j \rangle - \langle \vec{v}_k, \vec{u}_j \rangle = 0.$$

Thus $\vec{u}_k$ is orthogonal to each $\vec{u}_j$ for $j < k$, so the set $\{\vec{u}_1, \ldots, \vec{u}_k\}$ is orthogonal.

We must show that $\vec{u}_k \neq \vec{0}$. Suppose, for contradiction, that $\vec{u}_k = \vec{0}$. Then

$$\vec{v}_k = \sum_{i=1}^{k-1} \frac{\langle \vec{v}_k, \vec{u}_i \rangle}{\langle \vec{u}_i, \vec{u}_i \rangle} \vec{u}_i \in \operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_{k-1}\} = \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_{k-1}\},$$

which contradicts the linear independence of $\{\vec{v}_1, \ldots, \vec{v}_n\}$. Hence $\vec{u}_k \neq \vec{0}$ and the construction is valid.

Now we verify the span equality. From the definition of $\vec{u}_k$,

$$\vec{u}_k \in \operatorname{span}\{\vec{v}_k, \vec{u}_1, \ldots, \vec{u}_{k-1}\} = \operatorname{span}\{\vec{v}_k, \vec{v}_1, \ldots, \vec{v}_{k-1}\}$$

by the induction hypothesis. Therefore

$$\operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_k\} \subseteq \operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\}.$$

Conversely, solving for $\vec{v}_k$ gives

$$\vec{v}_k = \vec{u}_k + \sum_{i=1}^{k-1} \frac{\langle \vec{v}_k, \vec{u}_i \rangle}{\langle \vec{u}_i, \vec{u}_i \rangle} \vec{u}_i,$$

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so $\vec{v}_k \in \operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_k\}$. Together with the induction hypothesis that $\vec{v}_1, \ldots, \vec{v}_{k-1}$ belong to this span, we obtain

$$\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_k\} \subseteq \operatorname{span}\{\vec{u}_1, \ldots, \vec{u}_k\}.$$

Thus the two spans are equal.

By induction, the statements hold for all $k = 1, \ldots, n$. In particular, for $k = n$ we have that $\{\vec{u}_1, \ldots, \vec{u}_n\}$ is an orthogonal set spanning the same subspace as $\{\vec{v}_1, \ldots, \vec{v}_n\}$. Since the original set is linearly independent and has n vectors, the subspace is n-dimensional, and an orthogonal set of n non-zero vectors is automatically a basis. Hence $\{\vec{u}_1, \ldots, \vec{u}_n\}$ is an orthogonal basis for $\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_n\}$. ■

Corollary 3.2.2 (Orthonormal Basis). *Under the same hypothesis, the normalized vectors*

$$\vec{e}_k = \frac{\vec{u}_k}{\|\vec{u}_k\|}, \qquad k = 1, \ldots, n,$$

form an orthonormal basis for $\operatorname{span}\{\vec{v}_1, \ldots, \vec{v}_n\}$.

We Learned: The Gram-Schmidt process takes a set of linearly independent vectors and produces an orthogonal (or orthonormal) set that spans the same subspace. Given vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$, we construct orthogonal vectors $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n$ as follows:

$$\begin{aligned}
\mathbf{u}_1 &= \mathbf{v}_1, \\
\mathbf{u}_2 &= \mathbf{v}_2 - \frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1, \\
\mathbf{u}_3 &= \mathbf{v}_3 - \frac{\mathbf{v}_3 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 - \frac{\mathbf{v}_3 \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2}\mathbf{u}_2,
\end{aligned}$$

and so on. To obtain an orthonormal set, we normalize each $\mathbf{u}_i$: $\mathbf{e}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|}$.

Example 3.2.3 (Two vectors in $\mathbb{R}^2$). Apply the Gram-Schmidt process to $\mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \end{pmatrix}$.

Solution. We begin by setting $\mathbf{u}_1 = \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$.

Next we subtract from $\mathbf{v}_2$ its projection onto $\mathbf{u}_1$. The projection coefficient is

$$\frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} = \frac{(2)(3) + (2)(1)}{3^2 + 1^2} = \frac{8}{10} = \frac{4}{5}.$$

Hence

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{4}{5}\mathbf{u}_1 = \begin{pmatrix} 2 \\ 2 \end{pmatrix} - \frac{4}{5}\begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{2}{5} \\ \frac{6}{5} \end{pmatrix}.$$

Multiplying by 5 gives an equivalent orthogonal vector $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$, and the set $\left\{ \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 3 \end{pmatrix} \right\}$ is orthogonal.

For an orthonormal basis, we normalize:

$$\|\mathbf{u}_1\| = \sqrt{3^2 + 1^2} = \sqrt{10}, \qquad \mathbf{e}_1 = \frac{1}{\sqrt{10}}\begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

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$$\|\mathbf{u}_2\| = \sqrt{\left(-\frac{2}{5}\right)^2 + \left(\frac{6}{5}\right)^2} = \sqrt{\frac{4}{25} + \frac{36}{25}} = \frac{2\sqrt{10}}{5},$$

$$\mathbf{e}_2 = \frac{1}{\|\mathbf{u}_2\|}\mathbf{u}_2 = \frac{5}{2\sqrt{10}}\begin{pmatrix} -\frac{2}{5} \\ \frac{6}{5} \end{pmatrix} = \frac{1}{\sqrt{10}}\begin{pmatrix} -1 \\ 3 \end{pmatrix}.$$

Thus an orthonormal basis is $\left\{ \frac{1}{\sqrt{10}}\begin{pmatrix} 3 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{10}}\begin{pmatrix} -1 \\ 3 \end{pmatrix} \right\}$. ■

Example 3.2.4 (Three vectors in $\mathbb{R}^3$). Orthogonalize $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ using the Gram-Schmidt process.

Solution. Set $\mathbf{u}_1 = \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

The second orthogonal vector is obtained by removing the projection of $\mathbf{v}_2$ onto $\mathbf{u}_1$:

$$\frac{\mathbf{v}_2 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} = \frac{1\cdot 1 + 0\cdot 1 + 1\cdot 0}{1^2 + 1^2 + 0^2} = \frac{1}{2},$$

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{1}{2}\mathbf{u}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}.$$

For $\mathbf{u}_3$, we subtract from $\mathbf{v}_3$ its projections onto $\mathbf{u}_1$ and $\mathbf{u}_2$:

$$\frac{\mathbf{v}_3 \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} = \frac{0\cdot 1 + 1\cdot 1 + 1\cdot 0}{2} = \frac{1}{2},$$

$$\frac{\mathbf{v}_3 \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} = \frac{0\cdot \frac{1}{2} + 1\cdot\left(-\frac{1}{2}\right) + 1\cdot 1}{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2 + 1^2} = \frac{-\frac{1}{2} + 1}{\frac{1}{4} + \frac{1}{4} + 1} = \frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3}.$$

Therefore

$$\mathbf{u}_3 = \mathbf{v}_3 - \frac{1}{2}\mathbf{u}_1 - \frac{1}{3}\mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} - \frac{1}{3}\begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix}.$$

Multiplying by 3 gives $\begin{pmatrix} -2 \\ 2 \\ 2 \end{pmatrix}$; the vectors

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix}$$

form an orthogonal set.

To obtain an orthonormal basis, we normalize each:

$$\|\mathbf{u}_1\| = \sqrt{2}, \qquad \mathbf{e}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix};$$

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$$\|\mathbf{u}_2\| = \sqrt{\frac{1}{4}+\frac{1}{4}+1} = \sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{2}, \qquad \mathbf{e}_2 = \frac{2}{\sqrt{6}}\begin{pmatrix} 1/2 \\ -1/2 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix};$$

$$\|\mathbf{u}_3\| = \sqrt{\frac{4}{9}+\frac{4}{9}+\frac{4}{9}} = \sqrt{\frac{12}{9}} = \frac{2}{\sqrt{3}}, \qquad \mathbf{e}_3 = \frac{\sqrt{3}}{2}\begin{pmatrix} -2/3 \\ 2/3 \\ 2/3 \end{pmatrix} = \frac{1}{\sqrt{3}}\begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

Hence the orthonormal basis is

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}(1,1,0), \qquad \mathbf{e}_2 = \frac{1}{\sqrt{6}}(1,-1,2), \qquad \mathbf{e}_3 = \frac{1}{\sqrt{3}}(-1,1,1).$$

■

Example 3.2.5. (Orthogonalizing polynomials on $[0,1]$) Consider the space $\mathcal{P}_2$ of real polynomials of degree at most 2 with the inner product

$$\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx.$$

Apply the Gram-Schmidt process to the basis $\{1, x, x^2\}$ to obtain an orthogonal basis.

Solution. Set $u_1(x) = 1$.

For u_2, subtract from x its projection onto u_1:

$$\langle x, u_1\rangle = \int_0^1 x\cdot 1\,dx = \frac{1}{2}, \qquad \langle u_1, u_1\rangle = \int_0^1 1\,dx = 1,$$

so the projection coefficient is $1/2$. Hence

$$u_2(x) = x - \frac{1}{2}\cdot 1 = x - \frac{1}{2}.$$

For u_3, we remove from x^2 its projections onto u_1 and u_2. First,

$$\langle x^2, u_1\rangle = \int_0^1 x^2\,dx = \frac{1}{3}, \qquad \frac{\langle x^2, u_1\rangle}{\langle u_1, u_1\rangle} = \frac{1}{3}.$$

Next,

$$\langle x^2, u_2\rangle = \int_0^1 x^2\left(x-\frac{1}{2}\right)dx = \int_0^1 \left(x^3 - \frac{1}{2}x^2\right)dx = \frac{1}{4} - \frac{1}{2}\cdot\frac{1}{3} = \frac{1}{4} - \frac{1}{6} = \frac{1}{12},$$

$$\langle u_2, u_2\rangle = \int_0^1 \left(x-\frac{1}{2}\right)^2 dx = \int_0^1 \left(x^2 - x + \frac{1}{4}\right)dx = \frac{1}{3} - \frac{1}{2} + \frac{1}{4} = \frac{4-6+3}{12} = \frac{1}{12}.$$

Thus the second projection coefficient is $\frac{1/12}{1/12} = 1$. Therefore

$$u_3(x) = x^2 - \frac{1}{3}\cdot 1 - 1\cdot\left(x-\frac{1}{2}\right) = x^2 - x + \frac{1}{6}.$$

An orthogonal basis is $\left\{1,\ x-\frac{1}{2},\ x^2-x+\frac{1}{6}\right\}$. Normalization yields the first three Legendre polynomials scaled to $[0,1]$. ■

Example 3.2.6 (Orthogonalizing matrices in $M_{2\times 2}$)**.** Let $M_{2\times 2}(\mathbb{R})$ be the space of 2×2 real matrices

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equipped with the Frobenius inner product

$$\langle A, B\rangle = \operatorname{tr}(A^T B) = \sum_{i,j} a_{ij} b_{ij}.$$

Apply Gram-Schmidt to the basis

$$A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad A_3 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

Solution. Start with $U_1 = A_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

For U_2, subtract the projection of A_2 onto U_1:

$$\langle A_2, U_1\rangle = \operatorname{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}^T \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \right) = \operatorname{tr}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = 1,$$

$$\langle U_1, U_1\rangle = \operatorname{tr}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = 1,$$

so the coefficient is 1. Hence

$$U_2 = A_2 - 1 \cdot U_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

For U_3, we remove from A_3 its projections onto U_1 and U_2. Compute

$$\langle A_3, U_1\rangle = \operatorname{tr}\left(\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}^T \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \right) = \operatorname{tr}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = 1,$$

so the coefficient onto U_1 is $1/\langle U_1, U_1\rangle = 1$. Next,

$$\langle A_3, U_2\rangle = \operatorname{tr}\left(\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \right) = \operatorname{tr}\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = 1,$$

$$\langle U_2, U_2\rangle = \operatorname{tr}\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 1,$$

so the coefficient onto U_2 is also 1. Then

$$U_3 = A_3 - 1 \cdot U_1 - 1 \cdot U_2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Thus we obtain the orthogonal basis

$$U_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad U_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad U_3 = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

(Note that the original set was linearly independent but not orthogonal; Gram-Schmidt reveals the standard matrix units.) ■

Example 3.2.7. (Orthogonalizing functions on $[-1, 1]$) In the space $C[-1, 1]$ with inner product $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)\,dx$, orthogonalize the set $\{1, x, x^2\}$.

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Solution. Take $u_1(x) = 1$.

For u_2, compute

$$\langle x, 1\rangle = \int_{-1}^{1} x\,dx = 0.$$

Since the projection coefficient is zero, $u_2(x) = x$ already orthogonal to u_1.

For u_3, we need $u_3(x) = x^2 - \alpha u_1 - \beta u_2$ where

$$\alpha = \frac{\langle x^2, 1\rangle}{\langle 1, 1\rangle}, \qquad \beta = \frac{\langle x^2, x\rangle}{\langle x, x\rangle}.$$

Now

$$\langle x^2, 1\rangle = \int_{-1}^{1} x^2 dx = \frac{2}{3}, \quad \langle 1, 1\rangle = 2 \;\Rightarrow\; \alpha = \frac{2/3}{2} = \frac{1}{3},$$

$$\langle x^2, x\rangle = \int_{-1}^{1} x^3 dx = 0 \;\Rightarrow\; \beta = 0.$$

Therefore

$$u_3(x) = x^2 - \frac{1}{3}\cdot 1 = x^2 - \frac{1}{3}.$$

The orthogonal set is $\{1,\ x,\ x^2 - 1/3\}$. Up to scaling, these are the first three Legendre polynomials: $P_0(x) = 1$, $P_1(x) = x$, $P_2(x) = \frac{1}{2}(3x^2 - 1)$; indeed $x^2 - 1/3 = \frac{2}{3}P_2(x)$. ■

3.3 Orthogonal Complements

Let V be an inner product space (real or complex) with inner product $\langle \cdot, \cdot \rangle$. The concept of orthogonal complement is fundamental in understanding decomposition of vectors into components lying in a subspace and its perpendicular part. This leads to orthogonal projections and plays a crucial role in applications such as least squares, Fourier series, and quantum mechanics.

Definition 3.3.1. Let V be an inner product space and let $S \subseteq V$ be a nonempty subset. The **orthogonal complement** of S, denoted by $S^\perp$, is the set of all vectors in V that are orthogonal to every vector in S:

$$S^\perp = \{v \in V \mid \langle v, s\rangle = 0 \text{ for all } s \in S\}.$$

Example 3.3.2. In $\mathbb{R}^3$ with the dot product, let $W = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$ (the xy-plane). Then $W^\perp$ is the z-axis: $\{(0, 0, z) \mid z \in \mathbb{R}\}$.

Theorem 3.3.3 (Basic Properties). *Let V be an inner product space and $S, T \subseteq V$.*

1. *$S^\perp$ is a subspace of V (even if S is not a subspace).*
2. *$S \subseteq T \implies T^\perp \subseteq S^\perp$.*
3. *$S \cap S^\perp \subseteq \{0\}$; if $0 \in S$ then $0 \in S \cap S^\perp$.*
4. *$(\operatorname{span} S)^\perp = S^\perp$.*
5. *$V^\perp = \{0\}$ and $\{0\}^\perp = V$.*

Proof. (1) $0 \in S^\perp$ because $\langle 0, s\rangle = 0$ for all $s \in S$. If $u, v \in S^\perp$ and $\alpha, \beta \in \mathbb{F}$ (the field, either $\mathbb{R}$ or $\mathbb{C}$), then for any $s \in S$:

$$\langle \alpha u + \beta v, s\rangle = \alpha\langle u, s\rangle + \beta\langle v, s\rangle = 0.$$

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Hence $\alpha u + \beta v \in S^\perp$. So $S^\perp$ is a subspace.

(2) Take $x \in T^\perp$. Then $\langle x, t\rangle = 0$ for all $t \in T$. Since $S \subseteq T$, it holds for all $s \in S$ as well. Thus $x \in S^\perp$.

(3) If $x \in S \cap S^\perp$, then in particular $\langle x, x\rangle = 0$, so $x = 0$ by positive definiteness.

(4) The inclusion $(\operatorname{span} S)^\perp \subseteq S^\perp$ follows from (2) because $S \subseteq \operatorname{span} S$. Conversely, let $x \in S^\perp$. For any linear combination $\sum_{i=1}^{n} \alpha_i s_i \in \operatorname{span} S$, we have

$$\langle x, \sum_i \alpha_i s_i\rangle = \sum_i \bar{\alpha}_i \langle x, s_i\rangle = 0,$$

so $x \in (\operatorname{span} S)^\perp$.

(5) Only 0 is orthogonal to every vector in V, so $V^\perp = \{0\}$. Also, every vector is orthogonal to 0, so $\{0\}^\perp = V$. ■

Orthogonal Decomposition Theorem

The most important result is that for finite-dimensional subspaces, the whole space splits as a direct sum of a subspace and its orthogonal complement.

Theorem 3.3.4 (Orthogonal Decomposition). *Let V be a finite-dimensional inner product space and let W be a subspace of V. Then*

$$V = W \oplus W^\perp.$$

Proof. We first show $V = W + W^\perp$. Since V is finite-dimensional, W has an orthonormal basis $\{e_1, \ldots, e_k\}$. Extend this to an orthonormal basis of V: $\{e_1, \ldots, e_k, e_{k+1}, \ldots, e_n\}$. Any $v \in V$ can be expressed as

$$v = \sum_{i=1}^{k} \langle v, e_i\rangle e_i + \sum_{i=k+1}^{n} \langle v, e_i\rangle e_i.$$

The first sum is in W (it is the orthogonal projection of v onto W), and the second sum is orthogonal to each e_i for $i = 1, \ldots, k$, hence lies in $W^\perp$. Thus $v \in W + W^\perp$.

Now we show $W \cap W^\perp = \{0\}$. If $x \in W \cap W^\perp$, then $\langle x, x\rangle = 0$ implies $x = 0$. Hence the sum is direct. ■

Corollary 3.3.5. $\dim W^\perp = \dim V - \dim W$.

Proof. $V = W \oplus W^\perp$ implies $\dim V = \dim W + \dim W^\perp$. ■

Theorem 3.3.6 (Double Orthogonal Complement). *If W is a subspace of a finite-dimensional inner product space V, then*

$$(W^\perp)^\perp = W.$$

Proof. First, we show that $W \subseteq (W^\perp)^\perp$. Let $w \in W$. By the definition of the orthogonal complement $W^\perp$, we know that $\langle w, x\rangle = 0$ for every $x \in W^\perp$. This means that w is orthogonal to every vector in $W^\perp$, which is exactly the definition of belonging to $(W^\perp)^\perp$. Thus, $W \subseteq (W^\perp)^\perp$.

Next, because V is finite-dimensional, dimension of any subspace and its orthogonal complement sum to the dimension of V:

$$\dim(W) + \dim(W^\perp) = \dim(V)$$

Applying this exact same theorem to the subspace $W^\perp$ yields:

$$\dim(W^\perp) + \dim((W^\perp)^\perp) = \dim(V)$$

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Comparing these two equations, we see that $\dim(W) = \dim((W^\perp)^\perp)$.

Since W is a subspace of $(W^\perp)^\perp$ and they have the exact same finite dimension, they must be equal. Therefore, $W = (W^\perp)^\perp$. ■

Theorem 3.3.7 (Riesz Representation). *Let V be a finite-dimensional inner product space over $\mathbb{F}$ and let $\varphi : V \to \mathbb{F}$ be a linear functional. Then there exists a unique vector $y \in V$ such that $\varphi(x) = \langle x, y\rangle$ for all $x \in V$.*

Proof. **Existence:** Let $\{e_1, \ldots, e_n\}$ be an orthonormal basis for V. We construct y as:

$$y = \sum_{i=1}^{n} \overline{\varphi(e_i)} e_i$$

Let's evaluate the inner product of an arbitrary basis vector e_j with y:

$$\langle e_j, y\rangle = \left\langle e_j, \sum_{i=1}^{n} \overline{\varphi(e_i)} e_i \right\rangle = \sum_{i=1}^{n} \varphi(e_i)\langle e_j, e_i\rangle = \varphi(e_j)$$

The last step follows because $\langle e_j, e_i\rangle = 1$ when $i = j$ and 0 otherwise. Since the linear transformations $\varphi(\cdot)$ and $\langle \cdot, y\rangle$ agree on every basis vector, they must be equal for all $x \in V$.

Uniqueness: Suppose $y_1, y_2 \in V$ both satisfy the condition. Then $\langle x, y_1\rangle = \langle x, y_2\rangle$ for all $x \in V$, which implies by linearity that $\langle x, y_1 - y_2\rangle = 0$. Choosing $x = y_1 - y_2$ yields:

$$\langle y_1 - y_2, y_1 - y_2\rangle = 0$$

By positive definiteness, this requires $y_1 - y_2 = 0$, meaning $y_1 = y_2$. ■

Example 3.3.8 (Verifying an Inner Product). Show that $\langle p, q\rangle = \int_0^1 p(t)q(t)\,dt$ defines an inner product on the space $\mathcal{P}_2$ of real polynomials of degree ≤ 2.

Solution. Linearity and symmetry are clear from properties of the integral. Positive definiteness: $\langle p, p\rangle = \int_0^1 p(t)^2 dt \geq 0$, and if $\int_0^1 p(t)^2 dt = 0$ then $p(t) = 0$ for all t (since p is continuous). Hence it is an inner product. ■

Example 3.3.9 (Applying Cauchy–Schwarz). For $f(x) = x$, $g(x) = x^2$ in $C[0,1]$ with $\langle f, g\rangle = \int_0^1 f(t)g(t)dt$, verify the Cauchy–Schwarz inequality.

Solution. Compute $\langle f, g\rangle = \int_0^1 x^3 dx = \frac{1}{4}$, $\|f\|^2 = \int_0^1 x^2 dx = \frac{1}{3}$, $\|g\|^2 = \int_0^1 x^4 dx = \frac{1}{5}$. Then $|\langle f, g\rangle| = 1/4$, $\|f\|\|g\| = \sqrt{1/3 \cdot 1/5} = 1/\sqrt{15} \approx 0.2582$. Since $0.25 \leq 0.2582$, the inequality holds. ■

Example 3.3.10 (Gram–Schmidt Process). Apply Gram–Schmidt to $\{v_1 = (1,0,0), v_2 = (1,1,0), v_3 = (1,1,1)\}$ in $\mathbb{R}^3$ with the Euclidean inner product.

Solution. Let $e_1 = \frac{v_1}{\|v_1\|} = (1,0,0)$. Compute $u_2 = v_2 - \langle v_2, e_1\rangle e_1 = (1,1,0) - 1 \cdot (1,0,0) = (0,1,0)$, so $e_2 = (0,1,0)$. $u_3 = v_3 - \langle v_3, e_1\rangle e_1 - \langle v_3, e_2\rangle e_2 = (1,1,1) - 1 \cdot (1,0,0) - 1 \cdot (0,1,0) = (0,0,1)$, thus $e_3 = (0,0,1)$. The orthonormal basis is the standard basis. ■

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3.4 Orthogonal Projections

Definition 3.4.1 (Orthogonal Complement)**.** For a subspace $W \subseteq V$, the **orthogonal complement** $W^\perp$ is

$$W^\perp = \{\vec{v} \in V \mid \langle \vec{v}, \vec{w} \rangle = 0 \text{ for all } \vec{w} \in W\}.$$

Definition 3.4.2 (Orthogonal Projection)**.** Let W be a finite-dimensional subspace of an inner product space V. The **orthogonal projection** of a vector $\vec{v} \in V$ onto W is the unique vector $\text{proj}_W(\vec{v}) \in W$ such that

$$\vec{v} - \text{proj}_W(\vec{v}) \in W^\perp.$$

Theorem 3.4.3 (Existence and Uniqueness of Orthogonal Projection)**.** *For any $\vec{v} \in V$, there exists a unique orthogonal projection onto W. Moreover, if $\{\vec{e}_1, \ldots, \vec{e}_m\}$ is an orthonormal basis of W, then*

$$\text{proj}_W(\vec{v}) = \sum_{i=1}^{m} \langle \vec{v}, \vec{e}_i \rangle \, \vec{e}_i.$$

This projection minimizes the distance to W: for all $\vec{w} \in W$,

$$\|\vec{v} - \text{proj}_W(\vec{v})\| \leq \|\vec{v} - \vec{w}\|,$$

with equality if and only if $\vec{w} = \text{proj}_W(\vec{v})$.

Proof. Extend $\{\vec{e}_1, \ldots, \vec{e}_m\}$ to an orthonormal basis $\{\vec{e}_1, \ldots, \vec{e}_m, \vec{e}_{m+1}, \ldots, \vec{e}_n\}$ of V (possible by Gram–Schmidt). Write $\vec{v} = \sum_{i=1}^{n} a_i \vec{e}_i$. Define $P\vec{v} = \sum_{i=1}^{m} a_i \vec{e}_i$. Then $P\vec{v} \in W$ and $\vec{v} - P\vec{v} = \sum_{i=m+1}^{n} a_i \vec{e}_i \in W^\perp$. Uniqueness: Suppose $\vec{p}_1, \vec{p}_2 \in W$ satisfy $\vec{v} - \vec{p}_i \in W^\perp$. Then $\vec{p}_1 - \vec{p}_2 \in W$ and also $\vec{p}_1 - \vec{p}_2 = (\vec{v} - \vec{p}_2) - (\vec{v} - \vec{p}_1) \in W^\perp$, hence orthogonal to itself, so $\vec{p}_1 = \vec{p}_2$. The formula follows because $a_i = \langle \vec{v}, \vec{e}_i \rangle$. For the best approximation property, for any $\vec{w} \in W$,

$$\|\vec{v} - \vec{w}\|^2 = \|(\vec{v} - P\vec{v}) + (P\vec{v} - \vec{w})\|^2 = \|\vec{v} - P\vec{v}\|^2 + \|P\vec{v} - \vec{w}\|^2,$$

since the two terms are orthogonal. This is minimized uniquely when $\vec{w} = P\vec{v}$. ∎

Example 3.4.4 (Orthogonal Projection onto a Plane in $\mathbb{R}^3$)**.** Let $W = \text{span}\{(1,1,0), (1,0,1)\}$ in $\mathbb{R}^3$. Find the orthogonal projection of $\vec{v} = (2,3,4)$ onto W.

Solution. First, orthonormalize the basis of W. From the previous example we already have $\vec{e}_1 = \frac{1}{\sqrt{2}}(1,1,0)$ and $\vec{e}_2 = \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right)$ that span the same plane. Then

$$\text{proj}_W(\vec{v}) = \langle \vec{v}, \vec{e}_1 \rangle \, \vec{e}_1 + \langle \vec{v}, \vec{e}_2 \rangle \, \vec{e}_2.$$

Compute:

$$\langle \vec{v}, \vec{e}_1 \rangle = (2,3,4) \cdot \frac{1}{\sqrt{2}}(1,1,0) = \frac{1}{\sqrt{2}}(2+3+0) = \frac{5}{\sqrt{2}}.$$

$$\langle \vec{v}, \vec{e}_2 \rangle = (2,3,4) \cdot \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right) = \frac{1}{\sqrt{6}}(2-3+8) = \frac{7}{\sqrt{6}}.$$

Hence

$$\text{proj}_W(\vec{v}) = \frac{5}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}(1,1,0) + \frac{7}{\sqrt{6}} \cdot \left(\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}\right) = \frac{5}{2}(1,1,0) + \frac{7}{6}(1,-1,2).$$

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Simplify:

$$\frac{5}{2}(1,1,0) = \left(\frac{5}{2}, \frac{5}{2}, 0\right), \qquad \frac{7}{6}(1,-1,2) = \left(\frac{7}{6}, -\frac{7}{6}, \frac{14}{6}\right) = \left(\frac{7}{6}, -\frac{7}{6}, \frac{7}{3}\right).$$

Adding:

$$\text{proj}_W(\vec{v}) = \left(\frac{5}{2}+\frac{7}{6}, \frac{5}{2}-\frac{7}{6}, 0+\frac{7}{3}\right) = \left(\frac{15+7}{6}, \frac{15-7}{6}, \frac{7}{3}\right) = \left(\frac{22}{6}, \frac{8}{6}, \frac{7}{3}\right) = \left(\frac{11}{3}, \frac{4}{3}, \frac{7}{3}\right).$$

Thus the orthogonal projection is $(\frac{11}{3}, \frac{4}{3}, \frac{7}{3})$. ■

Example 3.4.5 (Orthogonal Projection in Function Space)**.** Let $V = C[-\pi, \pi]$ with inner product $\langle f, g\rangle = \int_{-\pi}^{\pi} f(x)g(x)\,dx$. Consider the subspace $W = \text{span}\{1, \cos x, \sin x\}$. Find the orthogonal projection of $f(x) = x^2$ onto W.

Solution. The set $\{1, \cos x, \sin x\}$ is orthogonal (though not orthonormal) on $[-\pi, \pi]$. Indeed:

$$\int_{-\pi}^{\pi} 1 \cdot \cos x\,dx = 0, \quad \int_{-\pi}^{\pi} 1 \cdot \sin x\,dx = 0, \quad \int_{-\pi}^{\pi} \cos x \sin x\,dx = 0.$$

Their norms squared:

$$\|1\|^2 = \int_{-\pi}^{\pi} 1\,dx = 2\pi, \quad \|\cos x\|^2 = \int_{-\pi}^{\pi} \cos^2 x\,dx = \pi, \quad \|\sin x\|^2 = \int_{-\pi}^{\pi} \sin^2 x\,dx = \pi.$$

Thus an orthonormal basis for W is

$$e_1(x) = \frac{1}{\sqrt{2\pi}}, \quad e_2(x) = \frac{\cos x}{\sqrt{\pi}}, \quad e_3(x) = \frac{\sin x}{\sqrt{\pi}}.$$

The projection formula gives

$$\text{proj}_W(f) = \langle f, e_1\rangle\, e_1 + \langle f, e_2\rangle\, e_2 + \langle f, e_3\rangle\, e_3.$$

Compute the coefficients:

$$c_1 = \langle f, e_1\rangle = \int_{-\pi}^{\pi} x^2 \cdot \frac{1}{\sqrt{2\pi}}\,dx = \frac{1}{\sqrt{2\pi}} \cdot \frac{2\pi^3}{3} = \frac{2\pi^3}{3\sqrt{2\pi}}.$$

$$c_2 = \langle f, e_2\rangle = \int_{-\pi}^{\pi} x^2 \cdot \frac{\cos x}{\sqrt{\pi}}\,dx = \frac{1}{\sqrt{\pi}} \cdot \left[\text{known integral: } \int_{-\pi}^{\pi} x^2 \cos x\,dx\right].$$

Integrate by parts: $\int x^2 \cos x\,dx = x^2 \sin x + 2x\cos x - 2\sin x$. Evaluating from $-\pi$ to π:

$$[x^2 \sin x + 2x\cos x - 2\sin x]_{-\pi}^{\pi} = (\pi^2 \cdot 0 + 2\pi(-1) - 0) - (\pi^2 \cdot 0 + 2(-\pi)(-1) - 0) = (-2\pi) - (2\pi) = -4\pi.$$

Thus $\int_{-\pi}^{\pi} x^2 \cos x\,dx = -4\pi$. Hence $c_2 = \frac{1}{\sqrt{\pi}}(-4\pi) = -4\sqrt{\pi}$. Similarly,

$$c_3 = \langle f, e_3\rangle = \int_{-\pi}^{\pi} x^2 \cdot \frac{\sin x}{\sqrt{\pi}}\,dx = \frac{1}{\sqrt{\pi}} \int_{-\pi}^{\pi} x^2 \sin x\,dx.$$

$x^2 \sin x$ is odd, so integral over symmetric interval is 0. Thus $c_3 = 0$. Therefore the projection is

$$\text{proj}_W(f) = c_1 e_1 + c_2 e_2 = \frac{2\pi^3}{3\sqrt{2\pi}} \cdot \frac{1}{\sqrt{2\pi}} + (-4\sqrt{\pi}) \cdot \frac{\cos x}{\sqrt{\pi}} = \frac{2\pi^3}{3 \cdot 2\pi} - 4\cos x = \frac{\pi^2}{3} - 4\cos x.$$

So the best approximation of x^2 by a trigonometric polynomial of degree 1 is $\frac{\pi^2}{3} - 4\cos x$. ■

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Example 3.4.6 (Orthogonal Projection). In $\mathbb{R}^2$ with the dot product, find the projection of $x = (3,4)$ onto the line spanned by $y = (1,2)$.

Solution. The projection is $\dfrac{\langle x,y\rangle}{\langle y,y\rangle}y = \dfrac{3\cdot 1+4\cdot 2}{1^2+2^2}(1,2) = \dfrac{11}{5}(1,2) = \left(\dfrac{11}{5},\dfrac{22}{5}\right)$. ■

Theorem 3.4.7 (Bessel's Inequality). *Let V be an inner product space and let $\{e_1, e_2, \ldots, e_k\}$ be an orthonormal set in V. For any $x \in V$,*

$$\sum_{i=1}^{k} |\langle x, e_i\rangle|^2 \leq \|x\|^2$$

Proof. Let $y = \sum_{i=1}^{k}\langle x, e_i\rangle e_i$ be the orthogonal projection of x onto the span of our orthonormal set. We use the non-negativity of the norm: $0 \leq \|x-y\|^2$. Expanding this yields:

$$0 \leq \langle x-y, x-y\rangle = \|x\|^2 - \langle x,y\rangle - \langle y,x\rangle + \|y\|^2$$

Now, we compute $\|y\|^2$ and the cross terms using the orthonormality of e_i (where $\langle e_i, e_j\rangle = 1$ if $i=j$ and 0 otherwise):

$$\|y\|^2 = \left\langle \sum_{i=1}^{k}\langle x,e_i\rangle e_i, \sum_{j=1}^{k}\langle x,e_j\rangle e_j \right\rangle = \sum_{i=1}^{k}\langle x,e_i\rangle\overline{\langle x,e_i\rangle} = \sum_{i=1}^{k}|\langle x,e_i\rangle|^2$$

Similarly, computing $\langle x,y\rangle$:

$$\langle x,y\rangle = \left\langle x, \sum_{i=1}^{k}\langle x,e_i\rangle e_i \right\rangle = \sum_{i=1}^{k}\overline{\langle x,e_i\rangle}\langle x,e_i\rangle = \sum_{i=1}^{k}|\langle x,e_i\rangle|^2 = \|y\|^2$$

By conjugate symmetry, $\langle y,x\rangle = \overline{\langle x,y\rangle} = \|y\|^2$. Substituting these back into our expanded inequality gives:

$$0 \leq \|x\|^2 - \|y\|^2 - \|y\|^2 + \|y\|^2 \implies 0 \leq \|x\|^2 - \|y\|^2 \implies \|y\|^2 \leq \|x\|^2$$

Replacing $\|y\|^2$ with the sum we calculated proves the inequality. ■

Theorem 3.4.8 (Parseval's Identity). *Let V be a finite-dimensional inner product space and let $\{e_1, e_2, \ldots, e_n\}$ be an orthonormal **basis** for V. For any $x \in V$,*

$$\|x\|^2 = \sum_{i=1}^{n}|\langle x,e_i\rangle|^2$$

Proof. Because $\{e_1,\ldots,e_n\}$ is an orthonormal basis, any vector $x \in V$ can be expressed exactly as a linear combination of the basis vectors, where the coefficients are the inner products:

$$x = \sum_{i=1}^{n}\langle x,e_i\rangle e_i$$

We find the squared norm of x by taking the inner product of x with itself:

$$\|x\|^2 = \langle x,x\rangle = \left\langle \sum_{i=1}^{n}\langle x,e_i\rangle e_i, \sum_{j=1}^{n}\langle x,e_j\rangle e_j \right\rangle$$

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By the linearity in the first argument and conjugate linearity in the second, we pull out the sums and scalars:

$$\|x\|^2 = \sum_{i=1}^{n}\sum_{j=1}^{n}\langle x, e_i\rangle\overline{\langle x, e_j\rangle}\langle e_i, e_j\rangle$$

Because the basis is orthonormal, $\langle e_i, e_j\rangle = 0$ whenever $i \neq j$. Thus, all cross terms vanish, and we are left only with the terms where $i = j$ (and $\langle e_i, e_i\rangle = 1$):

$$\|x\|^2 = \sum_{i=1}^{n}\langle x, e_i\rangle\overline{\langle x, e_i\rangle} = \sum_{i=1}^{n}|\langle x, e_i\rangle|^2$$

■

Example 3.4.9 (Using Parseval's Identity)**.** Let $\{e_1 = (1,0), e_2 = (0,1)\}$ be the standard orthonormal basis of $\mathbb{R}^2$. For $x = (a,b)$, verify $\|x\|^2 = a^2 + b^2$ using Parseval.

Solution. $\langle x, e_1\rangle = a$, $\langle x, e_2\rangle = b$, so $\sum|\langle x, e_i\rangle|^2 = a^2 + b^2 = \|x\|^2$. ■

Theorem 3.4.10 (Existence of the Adjoint)**.** *Let V be a finite-dimensional inner product space, and let $T : V \to V$ be a linear operator. There exists a unique linear operator $T^* : V \to V$ such that for all $x, y \in V$:*

$$\langle T(x), y\rangle = \langle x, T^*(y)\rangle$$

Proof. Fix an arbitrary vector $y \in V$. Because T is linear and the inner product is linear in its first argument, the map $x \mapsto \langle T(x), y\rangle$ is a linear functional on V. By the Riesz Representation Theorem, there exists a unique vector in V associated with this functional. We define $T^*(y)$ to be this unique vector.

Thus, by definition, the map $T^* : V \to V$ satisfies $\langle T(x), y\rangle = \langle x, T^*(y)\rangle$ for all $x, y \in V$. Uniqueness of T^* follows directly from the uniqueness of the vector in the Riesz Representation Theorem.

To show that T^* is a linear operator, let $y_1, y_2 \in V$ and c be a scalar. For any $x \in V$, we use the definition of T^* and the properties of the inner product to compute:

$$\begin{aligned}\langle x, T^*(cy_1 + y_2)\rangle &= \langle T(x), cy_1 + y_2\rangle \\ &= \overline{c}\langle T(x), y_1\rangle + \langle T(x), y_2\rangle \\ &= \overline{c}\langle x, T^*(y_1)\rangle + \langle x, T^*(y_2)\rangle \\ &= \langle x, cT^*(y_1) + T^*(y_2)\rangle\end{aligned}$$

Because $\langle x, T^*(cy_1 + y_2)\rangle = \langle x, cT^*(y_1) + T^*(y_2)\rangle$ for all $x \in V$, it must be that:

$$T^*(cy_1 + y_2) = cT^*(y_1) + T^*(y_2)$$

Therefore, T^* is a linear operator. ■

Definition 3.4.11 (Self-Adjoint Operator)**.** Let V be an inner product space. A linear operator $T : V \to V$ is called **self-adjoint** (or Hermitian) if it equals its adjoint, meaning $T = T^*$. Equivalently, for all vectors $x, y \in V$, the operator satisfies:

$$\langle Tx, y\rangle = \langle x, Ty\rangle$$

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Theorem 3.4.12. *Let V be an inner product space over $\mathbb{C}$ and let T be a self-adjoint operator on V. Then:*

(a) Every eigenvalue of T is a real number.

(b) If x and y are eigenvectors corresponding to distinct eigenvalues, then x and y are orthogonal.

Proof. **Part (a): Eigenvalues are real.**
Let λ be an eigenvalue of T with a corresponding non-zero eigenvector x. By definition, $Tx = \lambda x$. We compute the inner product of Tx with x:

$$\langle Tx, x\rangle = \langle \lambda x, x\rangle = \lambda\langle x, x\rangle = \lambda\|x\|^2$$

On the other hand, because T is self-adjoint, we can move T to the second argument:

$$\langle Tx, x\rangle = \langle x, Tx\rangle = \langle x, \lambda x\rangle = \overline{\lambda}\langle x, x\rangle = \overline{\lambda}\|x\|^2$$

Equating the two results gives:

$$\lambda\|x\|^2 = \overline{\lambda}\|x\|^2$$

Because x is an eigenvector, $x \neq 0$, which means $\|x\|^2 \neq 0$. Dividing both sides by $\|x\|^2$ yields $\lambda = \overline{\lambda}$. A complex number equals its conjugate if and only if it is entirely real. Thus, $\lambda \in \mathbb{R}$.

Part (b): Eigenvectors of distinct eigenvalues are orthogonal.
Let $Tx = \lambda x$ and $Ty = \mu y$, where x and y are non-zero eigenvectors and $\lambda \neq \mu$. We evaluate the inner product $\langle Tx, y\rangle$:

$$\langle Tx, y\rangle = \langle \lambda x, y\rangle = \lambda\langle x, y\rangle$$

Because T is self-adjoint, we also have:

$$\langle Tx, y\rangle = \langle x, Ty\rangle = \langle x, \mu y\rangle = \overline{\mu}\langle x, y\rangle$$

From Part (a), we know that all eigenvalues of T are real, so $\overline{\mu} = \mu$. Therefore, the equation becomes:

$$\langle Tx, y\rangle = \mu\langle x, y\rangle$$

Equating the two expressions for $\langle Tx, y\rangle$ gives:

$$\lambda\langle x, y\rangle = \mu\langle x, y\rangle \implies (\lambda - \mu)\langle x, y\rangle = 0$$

Since we assumed the eigenvalues are distinct ($\lambda \neq \mu$), the term $(\lambda - \mu)$ is not zero. Therefore, we must have $\langle x, y\rangle = 0$. By definition, this means x and y are orthogonal. ■

Theorem 3.4.13 (Block Upper Triangular Matrix)**.** *Let V be a finite-dimensional vector space, T a linear operator on V, and W a T-invariant subspace. There exists a basis β for V such that the matrix representation $[T]_\beta$ has the block upper triangular form:*

$$[T]_\beta = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$$

where A is the matrix representation of the restriction $T|_W$.

Proof. Choose a basis $\alpha = \{w_1, \ldots, w_k\}$ for the subspace W. Because W is a subspace of V, we can extend α to a basis $\beta = \{w_1, \ldots, w_k, v_{k+1}, \ldots, v_n\}$ for the entire space V.

To find the matrix representation $[T]_\beta$, we apply T to each basis vector and write the result as a linear combination of the vectors in β. For $1 \leq j \leq k$, because W is T-invariant, $T(w_j) \in W$. Therefore,

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$T(w_j)$ can be written entirely as a linear combination of $\{w_1, \ldots, w_k\}$, meaning the coefficients for the basis vectors $v_{k+1}, \ldots, v_n$ are exactly zero.

Writing these coordinate vectors as the columns of our matrix naturally forces the bottom-left block to be zero, yielding the desired block matrix structure. The top-left block A precisely tracks how T acts on the basis α, making it the matrix for $T|_W$. ■

In inner product spaces, invariant subspaces have a beautiful relationship with the adjoint operator T^* and the orthogonal complement $W^\perp$.

Theorem 3.4.14 (Orthogonal Complements and Adjoints). *Let V be a finite-dimensional inner product space and T a linear operator on V. If W is a T-invariant subspace, then its orthogonal complement $W^\perp$ is T^*-invariant.*

Proof. Let $v \in W^\perp$ and $w \in W$. Since W is T-invariant, we have $T(w) \in W$. By the definition of the adjoint, we compute:

$$\langle w, T^*(v) \rangle = \langle T(w), v \rangle = 0$$

The last equality holds because $v \in W^\perp$ is orthogonal to everything in W, including $T(w)$. Since $\langle w, T^*(v) \rangle = 0$ for all $w \in W$, it follows that $T^*(v) \in W^\perp$. Thus, $W^\perp$ is T^*-invariant. ■

Theorem 3.4.15 (Characteristic Polynomial of a Restriction). *Let V be a finite-dimensional vector space, T a linear operator on V, and W a T-invariant subspace. Let $T_W : W \to W$ be the restriction of T to W. Then the characteristic polynomial of T_W divides the characteristic polynomial of T.*

Proof. Let $\alpha = \{w_1, \ldots, w_k\}$ be a basis for W. We extend α to a basis $\beta = \{w_1, \ldots, w_k, v_{k+1}, \ldots, v_n\}$ for the entire space V.

Because W is T-invariant, the matrix representation of T with respect to β is block upper-triangular:

$$[T]_\beta = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$$

where $A = [T_W]_\alpha$ is the $k \times k$ matrix representation of the restriction T_W, and C is an $(n-k) \times (n-k)$ matrix.

The characteristic polynomial of T, denoted $p_T(x)$, is the determinant of $[T]_\beta - xI$. Using the block structure, we have:

$$\begin{aligned} p_T(x) &= \det \begin{pmatrix} A - xI_k & B \\ 0 & C - xI_{n-k} \end{pmatrix} \\ &= \det(A - xI_k) \cdot \det(C - xI_{n-k}) \end{aligned}$$

By definition, $\det(A - xI_k)$ is precisely the characteristic polynomial of T_W, denoted $p_{T_W}(x)$. Letting $q(x) = \det(C - xI_{n-k})$, we get:

$$p_T(x) = p_{T_W}(x) q(x)$$

Since $q(x)$ is a polynomial, $p_{T_W}(x)$ divides $p_T(x)$. ■

Theorem 3.4.16 (Schur). *Let T be a linear operator on a finite dimensional I.P.S V. Suppose that the characteristic polynomial of T splits. Then there exists an orthonormal basis β for V such that the matrix $[T]_\beta$ is upper triangular.*

Proof. The proof is by induction on the dimension n of V. The result is immediate if $n = 1$. So suppose the result is true for linear operators on $(n-1)$-dimensional I.P.S's whose char. polynomial splits.

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By previous lemma, we can assume T^* has a unit eigenvector z. Suppose that $T^*(z) = \lambda z$ and that $W = \text{span}(\{z\})$. We show that $W^\perp$ is T-invariant.

If $y \in W^\perp$ and $x = cz \in W$, then

$$\begin{aligned}\langle T(y), x\rangle &= \langle T(y), cz\rangle = \langle y, T^*(cz)\rangle = \langle y, cT^*(z)\rangle \\ &= \langle y, c\lambda z\rangle = \bar{c}\bar{\lambda}\langle y, z\rangle = \bar{c}\bar{\lambda}\cdot 0 = 0.\end{aligned}$$

Hence $\langle T(y), x\rangle = 0$, i.e., $T(y) \in W^\perp$. Thus, as $W^\perp$ is the T-invariant subspace of V, hence the char. polynomial of $T_{W^\perp}$ divides the char. polynomial of T, and hence the char. polynomial of $T_{W^\perp}$ splits. Also, $\dim W^\perp = n-1$. Thus, by induction hypothesis, there exists an orthonormal basis γ of $W^\perp$ such that the matrix $[T_{W^\perp}]_\gamma$ is upper triangular. Clearly, $\beta = \gamma \cup \{z\}$ is an orthonormal basis of V such that $[T]_\beta$ is upper triangular. ■

Definition 3.4.17. Let V be an I.P.S, and let T be a linear operator on V. We say T is normal if $TT^* = T^*T$.

Example 3.4.18. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be rotation by θ, $0 \le \theta < \pi$. Then w.r.t standard basis β of $\mathbb{R}^2$,

$$[T]_\beta = \begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}.$$

Then T is a normal operator.

Theorem 3.4.19. *Let V be an I.P.S and T be a normal operator on V. Then the following statements are true:*

(a) $\|T(x)\| = \|T^*(x)\|$ *for all* $x \in V$.

(b) $T - cI$ *is normal for every* $c \in \mathbb{F}$.

(c) If x is an eigenvector of T, then x is also an eigenvector of T^*. In fact, if $T(x) = \lambda x$, then $T^*(x) = \bar{\lambda}x$.

(d) If λ_1 & λ_2 are distinct eigenvalues of T with corresponding eigenvectors x_1 and x_2, then x_1 and x_2 are orthogonal.

Proof. (a) $\|T(x)\|^2 = \langle T(x), T(x)\rangle = \langle T^*T(x), x\rangle$
$= \langle TT^*(x), x\rangle = \langle T^*(x), T^*(x)\rangle = \|T^*(x)\|^2$.

(b) $(T-cI)(T-cI)^* = (T-cI)(T^*-\bar{c}I)$
$= TT^* - \bar{c}T - cT^* + c\bar{c}I = T^*T - \bar{c}T - cT^* + c\bar{c}I$
$= (T^*-\bar{c}I)(T-cI) = (T-cI)^*(T-cI)$.
Thus, $T-cI$ is normal.

(c) Suppose $T(x) = \lambda x$ for some $x \in V$, then $(T-\lambda I)(x) = 0$ and $T-\lambda I$ is normal. Thus, by (a),

$$0 = \|(T-\lambda I)(x)\| = \|(T-\lambda I)^*(x)\| = \|T^*(x) - \bar{\lambda}x\|$$

Thus, $T^*(x) - \bar{\lambda}x = 0$, hence $T^*(x) = \bar{\lambda}x$.

(d) Let $\lambda_1 \neq \lambda_2$ be two eigenvalues of T with corresponding eigenvectors x_1 and x_2. Then

$$\begin{aligned}\lambda_1\langle x_1, x_2\rangle &= \langle \lambda_1 x_1, x_2\rangle = \langle T(x_1), x_2\rangle = \langle x_1, T^*(x_2)\rangle \\ &= \langle x_1, \bar{\lambda}_2 x_2\rangle = \lambda_2\langle x_1, x_2\rangle \quad \text{by (c)}\end{aligned}$$

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Thus, $\lambda_1\langle x_1, x_2\rangle = \lambda_2\langle x_1, x_2\rangle$, hence $\langle x_1, x_2\rangle = 0$ because $\lambda_1 \neq \lambda_2$.

■

Theorem 3.4.20. *Let T be a linear operator on a finite-dimensional I.P.S. V. Then T is normal iff there exists an orthonormal basis for V consisting of eigenvectors of T.*

Proof. Suppose T is normal. By the Fundamental Theorem of algebra, the characteristic polynomial of T splits. Thus, by Schur's theorem, there exists an orthonormal basis β of V such that $[T]_\beta = A$ is upper triangular. Let $\beta = \{v_1, \ldots, v_n\}$.

Then A being upper triangular, we have $T(v_1) = a_{11}v_1 + 0 \cdot v_2 + \cdots + 0 \cdot v_n$. Thus, v_1 is an eigenvector of T. Assume $v_1, \ldots, v_{k-1}$ be the eigenvectors of T. We show v_k is an eigenvector of T, so that the result follows by mathematical induction.

Consider any $j < k$, and let λ_j be an eigenvalue of T corresponding to v_j. Then $T(v_j) = \lambda_j v_j$, so that $T^*(v_j) = \bar{\lambda}_j v_j$. Since A is upper triangular,

$$T(v_k) = a_{1k}v_1 + a_{2k}v_2 + \cdots + a_{kk}v_k + 0 \cdot v_{k+1} + \cdots + 0 \cdot v_n.$$

Now, the coefficients a_{jk} are given by

$$\begin{aligned} a_{jk} &= \langle T(v_k), v_j\rangle = \langle v_k, T^*(v_j)\rangle = \langle v_k, \bar{\lambda}_j v_j\rangle \\ &= \lambda_j\langle v_k, v_j\rangle = 0 \quad \text{(by previous theorem (d))} \end{aligned}$$

Thus, $T(v_k) = a_{kk}v_k$, and hence v_k is an eigenvector of T. So by induction, all the vectors in β are eigenvectors of T. Thus, V has a basis consisting of the eigenvectors of T.

Conversely, assume that V has a basis β consisting of the eigenvectors of T. Thus, the matrix of T, $[T]_\beta$ is a diagonal matrix. Hence $[T]_\beta$ and $[T^*]_\beta$, which are the transpose of each other, commute. Thus, T and T^* commute, i.e., $TT^* = T^*T$. ■

Definition 3.4.21. Let T be a linear operator on an I.P.S V. We say T is self-adjoint or Hermitian if $T = T^*$.

Theorem 3.4.22. *Let T be a self-adjoint operator on a finite-dimensional I.P.S. V. Then*

(a) Every eigenvalue of T is real.

(b) Suppose that V is a real IPS, then the characterisctic polynomial of T splits.

Proof. (a) Suppose that $T(x) = \lambda x$ for $x \neq 0$. As $T^* = T$, we have $T^*T = TT^*$ and so $T^*(x) = \bar{\lambda}x$. Thus, we have $\lambda x = T(x) = T^*(x) = \bar{\lambda}x$, that is, $\bar{\lambda} = \lambda$, hence λ is real.

(b) Let $\dim V = n$. Since T is self-adjoint, T is normal. Thus, by Theorem 3.4.20, there exists an orthonormal basis β of V consisting of eigenvectors of T. Hence $[T]_\beta$ is a diagonal matrix. Now, by part (a), since each eigenvalue of T is real, $[T]_\beta$ is a real diagonal matrix. Therefore, its characteristic polynomial splits in $\mathbb{R}$. ■

Theorem 3.4.23. *Let T be a linear operator on a finite-dimensional real inner product space. Then T is self-adjoint if and only if there exists an orthonormal basis β of V consisting of eigenvectors of T.*

Proof. Suppose T is self-adjoint. Thus, the characteristic polynomial of T splits, hence by Schur's theorem, there exists an orthonormal basis β for V such that the matrix $A = [T]_\beta$ is upper triangular.

But $A^* = [T]_\beta^* = [T]_\beta = A$, so A and A^* are both upper triangular, and therefore A is a diagonal matrix. Thus, β must consist of the eigenvectors of T.

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Conversely, assume that V has an orthonormal basis β consisting of the eigenvectors of T. Then, the matrix ... ■

Definition 3.4.24. Let V be a linear operator on a finite-dimensional inner product space V (over $\mathbb{F}$). If $\|T(x)\| = \|x\|$ for all $x \in V$, we call T a unitary operator if $\mathbb{F} = \mathbb{C}$ and an orthogonal operator if $\mathbb{F} = \mathbb{R}$.

Theorem 3.4.25. *Let T be a linear operator on a finite-dimensional inner product space V. Then the following statements are equivalent:*

(a) $TT^* = T^*T = I$

(b) $\langle T(x), T(y)\rangle = \langle x, y\rangle$ *for all* $x, y \in V$

(c) If β is an orthonormal basis of V, then so is $T(\beta)$.

(d) $\|T(x)\| = \|x\|$ *for all* $x \in V$.

Proof. $(a) \implies (b)$: Let $x, y \in V$. Then $\langle x, y\rangle = \langle I(x), y\rangle = \langle T^*T(x), y\rangle = \langle T(x), T(y)\rangle$.

$(b) \implies (c)$: Let $\beta = \{v_1, \ldots, v_n\}$ be an orthonormal basis of V. Then $\langle T(v_i), T(v_j)\rangle = \langle v_i, v_j\rangle = \delta_{ij}$. Moreover, $\|T(v_i)\|^2 = \langle T(v_i), T(v_i)\rangle = \langle v_i, v_i\rangle = \|v_i\|^2 = 1$.

$(c) \implies (d)$: Let $x \in V$, and let $\beta = \{v_1, \ldots, v_n\}$ be an orthonormal basis. Then we have $x = \sum_{i=1}^{n} a_i v_i$ for some scalars $a_i \in \mathbb{F}$. Thus: $\|x\|^2 = \langle \sum_{i=1}^{n} a_i v_i, \sum_{j=1}^{n} a_j v_j\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i \bar{a}_j \langle v_i, v_j\rangle = \sum_{i=1}^{n} |a_i|^2$. Since $T(x) = T(\sum a_i v_i) = \sum a_i T(v_i)$, we have: $\|T(x)\|^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i \bar{a}_j \langle T(v_i), T(v_j)\rangle = \sum_{i=1}^{n} |a_i|^2 = \|x\|^2$. Hence, $\|T(x)\| = \|x\|$.

$(d) \implies (a)$: For any $x \in V$, we have $\langle x, x\rangle = \|x\|^2 = \|T(x)\|^2 = \langle T(x), T(x)\rangle = \langle x, T^*T(x)\rangle$. So $\langle x, x - T^*T(x)\rangle = 0$, that is $\langle x, (I - T^*T)(x)\rangle = 0$ for all $x \in V$. Let $I - T^*T = U$. U is self-adjoint, and $\langle x, U(x)\rangle = 0$. Let β be an orthonormal basis of V consisting of eigenvectors of U. If $x \in \beta$, then $U(x) = \lambda x$. Thus $0 = \langle x, U(x)\rangle = \bar{\lambda}\langle x, x\rangle$, so $\lambda = 0$. Thus $U = 0$, i.e., $I - T^*T = 0$, hence $T^*T = I$. Similarly, $TT^* = I$. ■

Theorem 3.4.26. *Let A be a complex $n \times n$ matrix. Then A is normal if and only if A is unitarily equivalent to a diagonal matrix.*

Proof. Let A be a normal matrix over $\mathbb{C}$. There exists an orthonormal basis β of $\mathbb{F}^n$ consisting of eigenvectors of A. Hence A is similar to a diagonal matrix D, and there exists a non-singular matrix Q whose columns are the vectors in β, such that $D = Q^{-1}AQ$. Since β is an orthonormal basis, Q is unitary. Hence A is unitarily similar to a diagonal matrix. Conversely, suppose $A = P^*DP$ where P is a unitary matrix and D is a diagonal matrix. Then $AA^* = (P^*DP)(P^*DP)^* = P^*DD^*P$ because $P^*P = I$. Similarly, $A^*A = P^*D^*DP$. Since diagonal matrices commute, $DD^* = D^*D$, thus $AA^* = A^*A$. Hence, A is normal. ■

Theorem 3.4.27 (Schur). *Let $A \in M_{n\times n}(\mathbb{F})$ be a matrix whose characteristic polynomial splits over $\mathbb{F}$.*

(a) If $\mathbb{F} = \mathbb{C}$, then A is unitarily equivalent to a complex upper triangular matrix.

(b) If $\mathbb{F} = \mathbb{R}$, then A is orthogonally equivalent to a real upper triangular matrix.

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Proof. We proceed by induction on n. The case $n = 1$ is trivial. Assume the result holds for all matrices of size $(n-1) \times (n-1)$.

Let λ_1 be an eigenvalue of A, and let v_1 be a corresponding unit eigenvector ($\|v_1\| = 1$). We extend $\{v_1\}$ to an orthonormal basis $\{v_1, u_2, \ldots, u_n\}$ of $\mathbb{F}^n$. Let U_1 be the unitary (or orthogonal) matrix whose columns are these basis vectors. The matrix $A' = U_1^* A U_1$ takes the block form:

$$U_1^* A U_1 = \begin{pmatrix} \lambda_1 & w^* \\ 0 & A_1 \end{pmatrix} \tag{3.1}$$

where A_1 is an $(n-1) \times (n-1)$ matrix. Since the characteristic polynomial of A is $p_A(t) = (\lambda_1 - t)p_{A_1}(t)$, the eigenvalues of A_1 are also contained in $\mathbb{F}$.

By the induction hypothesis, there exists an $(n-1) \times (n-1)$ unitary (or orthogonal) matrix V such that $T_1 = V^* A_1 V$ is upper triangular. Define the $n \times n$ matrix U_2 as:

$$U_2 = \begin{pmatrix} 1 & 0 \\ 0 & V \end{pmatrix} \tag{3.2}$$

U_2 is clearly unitary (or orthogonal). Let $U = U_1 U_2$. Then U is unitary (or orthogonal), and we compute:

$$U^* A U = U_2^* (U_1^* A U_1) U_2 = \begin{pmatrix} 1 & 0 \\ 0 & V^* \end{pmatrix} \begin{pmatrix} \lambda_1 & w^* \\ 0 & A_1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & V \end{pmatrix} = \begin{pmatrix} \lambda_1 & w^* V \\ 0 & T_1 \end{pmatrix} \tag{3.3}$$

The resulting matrix is upper triangular, completing the induction. ■

Definition 3.4.28 (Invariant Subspace)**.** A subspace $W \subseteq V$ is called **T-invariant** (or invariant under T) if

$$T(W) \subseteq W,$$

that is, for every $w \in W$ we have $T(w) \in W$. When W is T-invariant, the restriction $T|_W : W \to W$ is a well-defined linear operator on W.

Example 3.4.29 (Trivial Invariant Subspaces)**.** For any linear operator $T : V \to V$:

(i) $W = \{\mathbf{0}\}$ is T-invariant, since $T(\mathbf{0}) = \mathbf{0}$.

(ii) $W = V$ is T-invariant, since $T(V) \subseteq V$ by definition.

These are called the **trivial** invariant subspaces. An operator with no non-trivial invariant subspaces is said to be **irreducible** (or cyclic over an algebraically closed field).

Example 3.4.30 (Eigenspace as an Invariant Subspace)**.** Let $\lambda \in F$ be an eigenvalue of T. The **eigenspace**

$$E_\lambda = \ker(T - \lambda I) = \{v \in V \mid T(v) = \lambda v\}$$

is T-invariant. Indeed, if $v \in E_\lambda$, then $T(v) = \lambda v \in E_\lambda$.

Example 3.4.31 (Kernel and Image)**.** Both $\ker(T)$ and $\mathrm{Im}(T)$ are T-invariant subspaces of V.

Proof. If $v \in \ker(T)$, then $T(v) = \mathbf{0} \in \ker(T)$. If $v \in \mathrm{Im}(T)$, then $v = T(u)$ for some $u \in V$, so $T(v) = T^2(u) \in \mathrm{Im}(T)$. ■

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Theorem 3.4.32 (Intersection and Sum of Invariant Subspaces). *Let W_1 and W_2 be T-invariant subspaces of V. Then:*

(i) $W_1 \cap W_2$ *is T-invariant.*

(ii) $W_1 + W_2$ *is T-invariant.*

Proof. **(i)** Let $v \in W_1 \cap W_2$. Then $v \in W_1$ and $v \in W_2$. Since W_1 is T-invariant, $T(v) \in W_1$. Since W_2 is T-invariant, $T(v) \in W_2$. Hence $T(v) \in W_1 \cap W_2$, so $W_1 \cap W_2$ is T-invariant.

(ii) Let $v \in W_1 + W_2$. Write $v = w_1 + w_2$ with $w_1 \in W_1$, $w_2 \in W_2$. Then

$$T(v) = T(w_1 + w_2) = T(w_1) + T(w_2).$$

Since W_1 is T-invariant, $T(w_1) \in W_1$. Since W_2 is T-invariant, $T(w_2) \in W_2$. Hence $T(v) \in W_1 + W_2$. ■

Corollary 3.4.33. *An arbitrary intersection of T-invariant subspaces is T-invariant.*

Proof. Let $\{W_i\}_{i \in I}$ be a (possibly infinite) collection of T-invariant subspaces. If $v \in \bigcap_{i \in I} W_i$, then $v \in W_i$ for all i, so $T(v) \in W_i$ for all i, hence $T(v) \in \bigcap_{i \in I} W_i$. ■

Theorem 3.4.34 (Invariance Under Polynomial of T). *If W is T-invariant, then W is $p(T)$-invariant for every polynomial $p \in F[x]$.*

Proof. It suffices to show W is T^k-invariant for all $k \geq 0$, since a general polynomial $p(T) = a_n T^n + \cdots + a_1 T + a_0 I$ is then handled by closure of W under linear combinations.

We proceed by induction on k. The case $k = 0$ gives $T^0 = I$, and $I(W) = W \subseteq W$. Assume $T^k(W) \subseteq W$. For any $w \in W$, we have $T^k(w) \in W$ by hypothesis, and then $T^{k+1}(w) = T(T^k(w)) \in W$ since W is T-invariant. This completes the induction.

Now let $p(x) = \sum_{k=0}^{n} a_k x^k$ and $w \in W$. Then

$$p(T)(w) = \sum_{k=0}^{n} a_k T^k(w).$$

Each $T^k(w) \in W$, so $p(T)(w)$ is a linear combination of elements of W, which lies in W. Hence W is $p(T)$-invariant. ■

3.5 T-Cyclic Subspaces

Definition 3.5.1 (T-Cyclic Subspace). For a nonzero vector $v \in V$, the **T-cyclic subspace generated by v** is

$$Z(v; T) = \operatorname{span}\{v, T(v), T^2(v), \ldots\}.$$

It is the smallest T-invariant subspace containing v.

Theorem 3.5.2 (Structure of the T-Cyclic Subspace). *Let $\dim V = n$ and $v \in V$ be nonzero. Then $Z(v; T)$ is T-invariant, and*

$$Z(v; T) = \operatorname{span}\left\{v, T(v), T^2(v), \ldots, T^{k-1}(v)\right\}$$

for some $k \leq n$, where k is the smallest positive integer such that $T^k(v) \in \operatorname{span}\{v, T(v), \ldots, T^{k-1}(v)\}$.

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Proof. Since V is finite-dimensional with $\dim V = n$, the vectors $v, T(v), \ldots, T^n(v)$ are $n+1$ vectors in an n-dimensional space, hence linearly dependent. Let $k \leq n$ be the smallest positive integer such that $T^k(v)$ is a linear combination of $v, T(v), \ldots, T^{k-1}(v)$. By minimality, the set $\{v, T(v), \ldots, T^{k-1}(v)\}$ is linearly independent.

Let $W = \operatorname{span}\{v, T(v), \ldots, T^{k-1}(v)\}$. We show W is T-invariant. For $0 \leq j \leq k-2$, we have $T(T^j(v)) = T^{j+1}(v) \in W$. For $j = k-1$, we have $T(T^{k-1}(v)) = T^k(v) \in W$ by the choice of k. Since T maps each basis vector of W into W, we have $T(W) \subseteq W$.

Since $Z(v;T)$ is defined as the span of all iterates, and since $T^k(v), T^{k+1}(v), \ldots$ all lie in W (by induction using the recurrence), we get $Z(v;T) \subseteq W$. Clearly $W \subseteq Z(v;T)$. Hence $Z(v;T) = W$. ■

Definition 3.5.3 (T-Annihilator)**.** The **T-annihilator** of v (also called the minimal polynomial of T with respect to v) is the monic polynomial $m_{v,T}(x) \in F[x]$ of least degree such that $m_{v,T}(T)(v) = \mathbf{0}$.

Theorem 3.5.4 (Dimension of Cyclic Subspace)**.** $\dim Z(v;T) = \deg(m_{v,T})$.

Proof. Let $k = \deg(m_{v,T})$. Then $T^k(v)$ is the first iterate that lies in $\operatorname{span}\{v, T(v), \ldots, T^{k-1}(v)\}$, so by Theorem 3.5.2 the set $\{v, T(v), \ldots, T^{k-1}(v)\}$ forms a basis for $Z(v;T)$. Hence $\dim Z(v;T) = k = \deg(m_{v,T})$. ■

Definition 3.5.5 (Minimal Polynomial of T)**.** The **minimal polynomial** $m_T(x) \in F[x]$ of T is the monic polynomial of least degree such that $m_T(T) = \mathbf{0}$ (the zero operator).

Theorem 3.5.6 (Minimal Polynomial Divides Any Annihilating Polynomial)**.** *If $p(T) = \mathbf{0}$ for some polynomial $p \in F[x]$, then $m_T(x) \mid p(x)$.*

Proof. Divide $p(x)$ by $m_T(x)$: write $p(x) = q(x)\, m_T(x) + r(x)$ with $\deg r < \deg m_T$. Evaluating at T:

$$\mathbf{0} = p(T) = q(T)\, m_T(T) + r(T) = r(T).$$

So $r(T) = \mathbf{0}$. If $r \neq 0$, this contradicts the minimality of $\deg m_T$. Hence $r = 0$, and $m_T(x) \mid p(x)$. ■

Theorem 3.5.7 (Invariant Subspaces from Factoring the Minimal Polynomial)**.** *Suppose $m_T(x) = p(x)\, q(x)$ where $\gcd(p,q) = 1$. Then*

$$V = \ker\big(p(T)\big) \oplus \ker\big(q(T)\big),$$

and both $\ker(p(T))$ and $\ker(q(T))$ are T-invariant subspaces of V.

Proof. **Invariance.** Since $p(T)$ and T commute (both are polynomials in T), for any $v \in \ker(p(T))$:

$$p(T)(T(v)) = T(p(T)(v)) = T(\mathbf{0}) = \mathbf{0},$$

so $T(v) \in \ker(p(T))$. The same argument applies to $\ker(q(T))$.

Trivial intersection. Suppose $v \in \ker(p(T)) \cap \ker(q(T))$. Then $p(T)(v) = \mathbf{0}$ and $q(T)(v) = \mathbf{0}$. Since $\gcd(p,q) = 1$, by Bezout's identity there exist $s, t \in F[x]$ with $s(x)p(x) + t(x)q(x) = 1$. Evaluating at T and applying to v:

$$v = I(v) = s(T)p(T)(v) + t(T)q(T)(v) = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Hence $\ker(p(T)) \cap \ker(q(T)) = \{\mathbf{0}\}$.

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Spanning. Since $m_T(T) = p(T)q(T) = \mathbf{0}$, for any $v \in V$:

$$v = s(T)p(T)(v) + t(T)q(T)(v).$$

Let $u = t(T)q(T)(v)$. Then $p(T)(u) = p(T)t(T)q(T)(v) = t(T)m_T(T)(v) = \mathbf{0}$, so $u \in \ker(p(T))$. Similarly, $s(T)p(T)(v) \in \ker(q(T))$. Thus $v \in \ker(p(T)) + \ker(q(T))$.

Combining, $V = \ker(p(T)) \oplus \ker(q(T))$. ■

The following theorem is also proved in Chapter 4 [See Theorem 4.1.5].

Corollary 3.5.8 (Primary Decomposition Theorem). *Let $m_T(x) = \prod_{i=1}^{r} p_i(x)^{n_i}$ be the factorisation of the minimal polynomial into powers of distinct monic irreducibles over F. Then*

$$V = \bigoplus_{i=1}^{r} K_i, \qquad K_i = \ker(p_i(T)^{n_i}),$$

*and each K_i is T-invariant. The subspace K_i is called the p_i-**primary component** of V.*

Proof. We proceed by induction on r, the number of distinct irreducible factors.

Base case ($r = 1$). Here $m_T(x) = p_1(x)^{n_1}$, so $K_1 = \ker(p_1(T)^{n_1}) = \ker(m_T(T)) = V$, and the decomposition $V = K_1$ is trivially a direct sum. T-invariance of $K_1 = V$ is clear.

Inductive step. Assume the result holds whenever the minimal polynomial has fewer than r distinct irreducible factors. Write

$$m_T(x) = f(x)\,g(x), \qquad f(x) = p_r(x)^{n_r}, \qquad g(x) = \prod_{i=1}^{r-1} p_i(x)^{n_i}.$$

Since $p_1, \ldots, p_r$ are distinct monic irreducibles, no p_i ($i < r$) divides p_r, so $\gcd(f, g) = 1$. By Theorem 3.5.7 applied to the coprime pair (f, g):

$$V = \ker(f(T)) \oplus \ker(g(T)) = K_r \oplus W,$$

where $W = \ker(g(T))$, and both K_r and W are T-invariant.

We now identify the minimal polynomial of $T|_W$. Let $h = m_{T|_W}$ be the minimal polynomial of the restriction $T|_W : W \to W$. Since $g(T)(w) = \mathbf{0}$ for all $w \in W$, the operator $g(T|_W)$ is the zero operator on W, so $h \mid g$. Conversely, $h(T)$ annihilates W, and since $V = K_r \oplus W$ we can check that $h(T)$ annihilates K_r as well (as $f \mid m_T$ and using the primary decomposition argument), so $m_T \mid fh$. But $m_T = fg$, so $g \mid h$. Together, $h = g$, i.e.

$$m_{T|_W}(x) = g(x) = \prod_{i=1}^{r-1} p_i(x)^{n_i}.$$

This has $r - 1$ distinct irreducible factors, so the inductive hypothesis applies to $T|_W$ on W:

$$W = \bigoplus_{i=1}^{r-1} K_i', \qquad K_i' = \ker(p_i(T|_W)^{n_i}),$$

with each K_i' invariant under $T|_W$, hence under T.

We claim $K_i' = K_i = \ker(p_i(T)^{n_i})$ for each $1 \le i \le r-1$. Indeed, if $v \in K_i'$ then $v \in W \subseteq V$ and $p_i(T)^{n_i}(v) = \mathbf{0}$, so $K_i' \subseteq K_i$. Conversely, if $v \in K_i$ then write $v = w + u$ with $w \in W$, $u \in K_r$. Then $p_i(T)^{n_i}(u) = \mathbf{0}$ requires $u \in K_i \cap K_r$. But $K_r = \ker(p_r(T)^{n_r})$ and $\gcd(p_i^{n_i}, p_r^{n_r}) = 1$ (distinct irreducibles), so $K_i \cap K_r = \{\mathbf{0}\}$, giving $u = \mathbf{0}$ and $v = w \in W$. Hence $v \in K_i'$, and $K_i = K_i'$.

Substituting back:

$$V = K_r \oplus W = K_r \oplus \bigoplus_{i=1}^{r-1} K_i = \bigoplus_{i=1}^{r} K_i.$$

Each K_i is T-invariant, completing the induction. ■

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3.6 T-Invariant Subspaces and the Characteristic Polynomial

Theorem 3.6.1 (Cayley–Hamilton Theorem). *Let $f_T(x) = \det(xI - T)$ be the characteristic polynomial of T. Then*

$$f_T(T) = \mathbf{0}.$$

In particular, $m_T(x) \mid f_T(x)$.

Proof (sketch). Write $xI - T$ as a matrix. The adjugate (classical adjoint) $B(x) = \operatorname{adj}(xI - T)$ has entries that are polynomials in x of degree at most $n - 1$, and satisfies

$$(xI - T)\, B(x) = f_T(x)\, I.$$

Write $B(x) = B_{n-1}x^{n-1} + \cdots + B_1 x + B_0$ and $f_T(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_0$. Comparing coefficients of x^k in $(xI - T)B(x) = f_T(x)I$:

$$\begin{aligned} B_{n-1} &= I, \\ B_{k-1} - TB_k &= c_k\, I, \quad 1 \le k \le n-1, \\ -TB_0 &= c_0\, I. \end{aligned}$$

Multiply the k-th equation on the right by T^k and sum:

$$f_T(T) = T^n + c_{n-1}T^{n-1} + \cdots + c_0 I = \mathbf{0}. \qquad \blacksquare$$

Theorem 3.6.2 (Existence of Non-Trivial Invariant Subspace over $\mathbb{C}$). *If $F = \mathbb{C}$ and $\dim V \ge 1$, then T has at least one non-trivial invariant subspace, namely an eigenspace.*

Proof. Since $\mathbb{C}$ is algebraically closed, the characteristic polynomial $f_T(x)$ splits completely over $\mathbb{C}$. In particular, it has at least one root $\lambda \in \mathbb{C}$, which is an eigenvalue of T. The corresponding eigenspace $E_\lambda = \ker(T - \lambda I) \neq \{\mathbf{0}\}$ is a T-invariant subspace. Since λ is an eigenvalue, $E_\lambda \neq V$ only when $T \neq \lambda I$; in either case, a non-trivial invariant subspace exists. $\blacksquare$

3.7 Direct Sum Decompositions and Invariant Subspaces

Definition 3.7.1 (T-Invariant Complement). A subspace W is said to have a **T-invariant complement** if there exists a T-invariant subspace W' such that $V = W \oplus W'$.

Theorem 3.7.2 (Matrix Form Under Invariant Subspace). *Let W be a T-invariant subspace of V with $\dim W = k$. Let $\mathcal{B} = \{w_1, \ldots, w_k, v_{k+1}, \ldots, v_n\}$ be a basis for V extending a basis $\{w_1, \ldots, w_k\}$ of W. Then the matrix of T with respect to $\mathcal{B}$ has the block upper-triangular form*

$$[T]_{\mathcal{B}} = \begin{pmatrix} A & C \\ 0 & B \end{pmatrix},$$

where A is the $k \times k$ matrix of $T|_W$ with respect to $\{w_1, \ldots, w_k\}$.

Proof. Since W is T-invariant, for each $1 \le j \le k$:

$$T(w_j) \in W = \operatorname{span}\{w_1, \ldots, w_k\}.$$

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Thus the first k columns of $[T]_{\mathcal{B}}$ have zeros in rows $k+1,\dots,n$, yielding the block form $\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}$. The matrix A represents $T|_W$ in the basis $\{w_1,\dots,w_k\}$ by definition. ■

Corollary 3.7.3. *$f_T(x) = f_{T|_W}(x) \cdot f_{\bar{T}}(x)$, where $\bar{T} : V/W \to V/W$ is the operator induced on the quotient space and $f_{\bar{T}}$ corresponds to the block B above.*

Theorem 3.7.4 (Decomposition into Two Invariant Subspaces). *Suppose $V = W_1 \oplus W_2$ where both W_1 and W_2 are T-invariant. Then*

$$m_T(x) = \operatorname{lcm}(m_{T|_{W_1}}, m_{T|_{W_2}}), \qquad f_T(x) = f_{T|_{W_1}}(x) \cdot f_{T|_{W_2}}(x).$$

Proof. Let $m_1 = m_{T|_{W_1}}$ and $m_2 = m_{T|_{W_2}}$, and let $\ell = \operatorname{lcm}(m_1, m_2)$.

Since $\ell(T)(w) = \mathbf{0}$ for all $w \in W_1$ (because $m_1 \mid \ell$) and similarly for W_2, and since every $v \in V$ can be written $v = w_1 + w_2$, we get $\ell(T)(v) = \mathbf{0}$. Thus $m_T \mid \ell$.

Conversely, $m_T(T)|_{W_i} = \mathbf{0}$ implies $m_i \mid m_T$ for $i = 1, 2$, so $\operatorname{lcm}(m_1, m_2) \mid m_T$. Hence $m_T = \ell$.

For the characteristic polynomial, choose a basis of V as the union of bases of W_1 and W_2. In this basis T is block-diagonal with blocks $T|_{W_1}$ and $T|_{W_2}$, giving

$$f_T(x) = \det(xI - T) = \det(xI_{W_1} - T|_{W_1}) \cdot \det(xI_{W_2} - T|_{W_2}) = f_{T|_{W_1}}(x) \cdot f_{T|_{W_2}}(x).$$ ■

3.8 Diagonalisability and Invariant Subspaces

Theorem 3.8.1 (T Diagonalisable $\Leftrightarrow$ V Decomposes into Eigenspaces). *T is diagonalisable if and only if*

$$V = \bigoplus_{\lambda} E_\lambda,$$

where the direct sum runs over all distinct eigenvalues λ of T.

Proof. ($\Rightarrow$) If T is diagonalisable, there is a basis $\mathcal{B}$ of eigenvectors. Group them by eigenvalue: the eigenvectors for λ span E_λ, and the union over all eigenvalues gives a basis for V, so $V = \sum_\lambda E_\lambda$. The sum is direct because eigenvectors for distinct eigenvalues are linearly independent (proved by a standard Vandermonde/induction argument).

($\Leftarrow$) If $V = \bigoplus_\lambda E_\lambda$, concatenating bases of each E_λ gives a basis of V consisting entirely of eigenvectors, so T is diagonalisable. ■

Theorem 3.8.2 (Simultaneous Diagonalisability). *Let $S, T \in \mathcal{L}(V)$ both be diagonalisable. Then S and T are simultaneously diagonalisable (i.e. there exists a single basis of V that diagonalises both) if and only if $ST = TS$.*

Proof. ($\Rightarrow$) If S and T are simultaneously diagonal in the same basis, then their matrices commute (diagonal matrices commute), so $ST = TS$.

($\Leftarrow$) Assume $ST = TS$. For each eigenvalue λ of S, the eigenspace $E_\lambda(S)$ is T-invariant: if $v \in E_\lambda(S)$, then

$$S(T(v)) = T(S(v)) = T(\lambda v) = \lambda T(v),$$

so $T(v) \in E_\lambda(S)$. Thus $T|_{E_\lambda(S)}$ is a well-defined operator. Since T is diagonalisable on V, it is diagonalisable on each T-invariant subspace $E_\lambda(S)$. So each $E_\lambda(S)$ has a basis of eigenvectors of T. Since

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$V = \bigoplus_\lambda E_\lambda(S)$ (as S is diagonalisable), the union of all these bases is a basis for V diagonalising both S and T. ■

Definition 3.8.3 (Generalised Eigenspace)**.** For an eigenvalue λ of T, the **generalised eigenspace** is

$$G_\lambda = \ker((T - \lambda I)^n), \quad n = \dim V.$$

Equivalently, $G_\lambda = \ker((T - \lambda I)^{m_\lambda})$ where m_λ is the multiplicity of λ in the minimal polynomial.

Theorem 3.8.4 (Generalised Eigenspace Decomposition)**.** *Over an algebraically closed field (e.g. $F = \mathbb{C}$), write $f_T(x) = \prod_{i=1}^r (\lambda_i - x)^{d_i}$ with distinct eigenvalues λ_i. Then:*

(i) Each G_{λ_i} is T-invariant with $\dim G_{\lambda_i} = d_i$.

(ii) $V = \bigoplus_{i=1}^r G_{\lambda_i}$.

(iii) $m_T(x) = \prod_{i=1}^r (x - \lambda_i)^{k_i}$ *where $k_i \leq d_i$.*

Proof. Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V over an algebraically closed field F. Since F is algebraically closed, any polynomial over F splits into linear factors. Consequently, the minimal polynomial $m_T(x)$ of T factors into linear terms as:

$$m_T(x) = \prod_{i=1}^r (x - \lambda_i)^{k_i}$$

where $\lambda_1, \ldots, \lambda_r$ are the distinct eigenvalues of T, and $k_i \geq 1$ are positive integers.

By applying the Primary Decomposition Theorem to $m_T(x)$, the vector space V decomposes into a direct sum of T-invariant subspaces:

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_r$$

where the primary components are defined by $W_i = \ker\left((T - \lambda_i I)^{k_i}\right)$, and the minimal polynomial of each restricted operator $T_i = T|_{W_i}$ is given by $m_{T_i}(x) = (x - \lambda_i)^{k_i}$.

To show that these primary components correspond to the generalized eigenspaces, recall that the generalized eigenspace associated with λ_i is defined by:

$$G_{\lambda_i} = \bigcup_{m=1}^{\infty} \ker\left((T - \lambda_i I)^m\right)$$

Clearly, the containment $W_i \subseteq G_{\lambda_i}$ holds because $(T - \lambda_i I)^{k_i} v = 0$ for all $v \in W_i$. Conversely, let $v \in G_{\lambda_i}$ so that $(T - \lambda_i I)^m v = 0$ for some integer $m \geq 1$. Expressing v uniquely as $v = v_1 + \cdots + v_r$ with $v_j \in W_j$ and applying the operator $(T - \lambda_i I)^m$ yields:

$$(T - \lambda_i I)^m v_1 + \cdots + (T - \lambda_i I)^m v_r = 0$$

Since each W_j is T-invariant, we must have $(T - \lambda_i I)^m v_j = 0$ for all j. For any $j \neq i$, the linear polynomial $x - \lambda_i$ is coprime to the minimal polynomial $m_{T_j}(x) = (x - \lambda_j)^{k_j}$ of the restricted operator T_j. This coprimality implies that the restriction of $(T - \lambda_i I)^m$ to W_j is invertible. Therefore, $(T - \lambda_i I)^m v_j = 0$ forces $v_j = 0$ for all $j \neq i$. It follows that $v = v_i \in W_i$, which proves the reverse containment $G_{\lambda_i} \subseteq W_i$.

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Thus, $G_{\lambda_i} = W_i = \ker\left((T - \lambda_i I)^{k_i}\right)$, which establishes both the T-invariance of each G_{λ_i} and the direct sum decomposition:

$$V = \bigoplus_{i=1}^{r} G_{\lambda_i}$$

To determine the dimension of each generalized eigenspace, we restrict the operator T to G_{λ_i}. The restricted operator $T_i = T|_{G_{\lambda_i}}$ satisfies $(T_i - \lambda_i I)^{k_i} = \mathbf{0}$, meaning that $T_i - \lambda_i I$ is a nilpotent operator. The only eigenvalue of a nilpotent operator is 0, which implies that the only eigenvalue of T_i is λ_i. Over our algebraically closed field, the characteristic polynomial of T_i is therefore:

$$f_{T_i}(x) = (\lambda_i - x)^{\dim G_{\lambda_i}}$$

Since V is the direct sum of these invariant subspaces, the characteristic polynomial of T is the product of the characteristic polynomials of the restricted operators:

$$f_T(x) = \prod_{i=1}^{r} f_{T_i}(x) = \prod_{i=1}^{r} (\lambda_i - x)^{\dim G_{\lambda_i}}$$

Comparing this representation with the given characteristic polynomial $f_T(x) = \prod_{i=1}^{r} (\lambda_i - x)^{d_i}$ and invoking the uniqueness of polynomial factorization, we find that $\dim G_{\lambda_i} = d_i$ for each $i \in \{1, \ldots, r\}$.

Finally, the minimal polynomial $m_T(x)$ is the product of the minimal polynomials of the restricted operators T_i:

$$m_T(x) = \prod_{i=1}^{r} (x - \lambda_i)^{k_i}$$

By the Cayley–Hamilton Theorem applied to the restricted operator T_i on G_{λ_i}, the minimal polynomial $m_{T_i}(x) = (x - \lambda_i)^{k_i}$ must divide the characteristic polynomial $f_{T_i}(x) = (\lambda_i - x)^{d_i}$, which forces the inequality $k_i \leq d_i$ for all $i \in \{1, \ldots, r\}$. ■

Remark 3.8.5. The Jordan Normal Form theorem asserts that, over $\mathbb{C}$, one can choose a basis of V (the **Jordan basis**) with respect to which T is block-diagonal, with each block being a Jordan block $J_k(\lambda)$ corresponding to a generalised eigenspace G_λ. The invariant subspaces spanned by each block form the building blocks of the entire T-invariant subspace lattice.

Theorem 3.8.6 (Schur's Theorem (Triangularisation)). *Let $F = \mathbb{C}$ (or any algebraically closed field). Then there exists an ordered basis $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ of V with respect to which $[T]_{\mathcal{B}}$ is upper triangular. Equivalently, there exists a chain of T-invariant subspaces*

$$\{0\} = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = V, \quad \dim V_k = k.$$

Proof. We use induction on $n = \dim V$.

Base case: $n = 1$. Any 1-dimensional space has the basis $\{v_1\}$ and $[T] = (\lambda)$, which is trivially upper triangular.

Inductive step: Assume the result holds for spaces of dimension $n - 1$. Since $\mathbb{C}$ is algebraically closed, T has at least one eigenvalue λ_1 with eigenvector v_1. Set $V_1 = \operatorname{span}\{v_1\}$, which is T-invariant.

Consider the quotient space $\bar{V} = V/V_1$ of dimension $n - 1$, with induced operator $\bar{T}(\bar{v}) = \overline{T(v)}$. By the inductive hypothesis, there is a basis $\{\bar{v}_2, \ldots, \bar{v}_n\}$ of $\bar{V}$ in which $\bar{T}$ is upper triangular.

Lift to V: choose representatives $v_2, \ldots, v_n \in V$ with $\bar{v}_k = v_k + V_1$. Then $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ is a basis of V, and $T(v_k) \in \operatorname{span}\{v_1, \ldots, v_k\}$ for all k, giving upper triangular form.

The corresponding chain is $V_k = \operatorname{span}\{v_1, \ldots, v_k\}$, each T-invariant. ■

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Definition 3.8.7. Let V be a vector space over a field F, and let $T : V \to V$ be a linear operator. For a nonzero vector $v \in V$, the **T-cyclic subspace generated by** v is defined as

$$Z(v;T) = \text{span}\{v,\ T(v),\ T^2(v),\ T^3(v),\ \ldots\}.$$

That is, $Z(v;T)$ is the smallest T-invariant subspace of V containing v. More explicitly,

$$Z(v;T) = \left\{ \sum_{k=0}^{n} a_k\, T^k(v) \;\middle|\; n \in \mathbb{N}_0,\ a_k \in F \right\}.$$

If $V = Z(v;T)$ for some $v \in V$, then V is called a **T-cyclic vector space** and v is called a **T-cyclic vector** (or cyclic generator) for V.

Example 3.8.8. Let $V = \mathbb{R}^3$ and define $T : \mathbb{R}^3 \to \mathbb{R}^3$ by

$$T(x,y,z) = (y,\ z,\ 0),$$

i.e., T is the "left-shift" operator. Let $v = e_1 = (1,0,0)$. We claim that $Z(e_1;\ T) = \mathbb{R}^3$.

Solution. We compute successive applications of T to $v = e_1$:

$$\begin{aligned} T^0(e_1) &= (1,0,0) = e_1, \\ T^1(e_1) &= T(1,0,0) = (0,1,0) = e_2, \\ T^2(e_1) &= T(0,1,0) = (0,0,1) = e_3, \\ T^3(e_1) &= T(0,0,1) = (0,0,0) = \mathbf{0}. \end{aligned}$$

Thus, the set $\{T^0(e_1),\ T^1(e_1),\ T^2(e_1)\} = \{e_1, e_2, e_3\}$ is contained in $Z(e_1;\ T)$. Since $\{e_1, e_2, e_3\}$ is the standard basis for $\mathbb{R}^3$, we conclude

$$Z(e_1;\ T) = \text{span}\{e_1, e_2, e_3\} = \mathbb{R}^3.$$

Therefore, $V = \mathbb{R}^3$ is a T-cyclic space with cyclic generator $v = e_1$. ■

Theorem 3.8.9. *Let V be a finite-dimensional vector space over F and let $0 \neq v \in V$. Let W be a $k-$dimensional $T-$cyclic subspace spanned by V, then the following statements hold.*

(a) $\{v, T(v), T^2(v), \ldots, T^{k-1}(v)\}$ is a basis for W.

(b) If $a_0 v + a_1 T(v) + \cdots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$, then the characteristic polynomial of T_W is $(-1)^k(a_0 + a_1 x + \cdots + a_{k-1}x^{k-1} + x^k)$.

Proof. (a) Since $v \neq 0$, the set $\{v\}$ is linearly independent. We adjoin the elements/vectors $T(v), T^2(v), \ldots$ into the set $\{v\}$ until it remains linearly independent. Noting that $\dim V = n < \infty$, let j be the largest integer for which $\beta = \{v, T(v), \ldots, T^{j-1}(v)\}$ is linearly independent. Let $Z = \text{span}(\beta)$. Since β is linearly independent, β is a basis for Z.

Claim: $T^j(v) \in Z$. If not, then $T^j(v) \in V$ and $T^j(v) \notin Z$, thus $\beta \cup \{T^j(v)\}$ is linearly independent, which is a contradiction to j being the largest integer such that $\beta = \{v, T(v), \ldots, T^{j-1}(v)\}$ is linearly independent.

Let $w \in Z$. Then $w = b_0 v + b_1 T(v) + \cdots + b_{j-1} T^{j-1}(v)$ for $b_i \in \mathbb{F}$. Hence $T(w) = T(b_0 v + b_1 T(v) + \cdots + b_{j-1} T^{j-1}(v)) = b_0 T(v) + b_1 T^2(v) + \cdots + b_{j-1} T^j(v)$, which is a linear combination of vectors from

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Z, so $T(w) \in Z$. Thus, Z is a T-invariant subspace of V. Furthermore, $v \in Z$ and W is the smallest T-invariant subspace containing v, thus $W \subseteq Z$. But $Z = \text{span}(\beta)$ and $W = \text{span}\{v, T(v), T^2(v), \dots\}$, hence $Z \subseteq W$. Thus, we have $W = Z$, and so β is the basis for W. Therefore, $\dim W = j = k$.

(b) Let $\beta = \{v, T(v), \dots, T^{k-1}(v)\}$ be the ordered basis of W. Let $a_0, a_1, \dots, a_{k-1}$ be the scalars for which $a_0 v + a_1 T(v) + \cdots + a_{k-1} T^{k-1}(v) + T^k(v) = 0$.

We have $[T_W]_\beta = [[T_W(v)]_\beta [T_W(T(v))]_\beta \dots [T_W(T^{k-1}(v))]_\beta]$. Now, $[T_W(v)]_\beta = [T(v)]_\beta = 0 \cdot v + 1 \cdot T(v) + 0 \cdot T^2(v) + \cdots + 0 \cdot T^{k-1}(v)$. Following this logic, $[T_W(T^{j-1}(v))]_\beta = -a_0 v - a_1 T(v) - \cdots - a_{k-1} T^{k-1}(v)$.

Thus, $[T_W]_\beta = \begin{bmatrix} 0 & 0 & \dots & 0 & -a_0 \\ 1 & 0 & \dots & 0 & -a_1 \\ 0 & 1 & \dots & 0 & -a_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -a_{k-1} \end{bmatrix}$, which is the companion matrix, whose characteristic polynomial is thus given by $c(x) = C_{T_W}(x) = (-1)^k (a_0 + a_1 x + a_2 x^2 + \cdots + a_{k-1} x^{k-1} + x^k)$. ■

Theorem 3.8.10 (Cayley-Hamilton Theorem). *Let T be a linear operator on a finite dimensional vector space V and let $c(x)$ be the characteristic polynomial of T. Then $c(T) = T_0$, the zero transformation.*

Proof. Let $v \in V$ be a non-zero vector and let W be the T-cyclic subspace generated by v. Let $k = \dim(W)$. As shown previously, there exists a monic polynomial $p(x) = x^k + a_{k-1} x^{k-1} + \cdots + a_0$ such that $p(T)(v) = 0$.

The restriction $T|_W$ has the companion matrix of $p(x)$ as its representation in the basis $\{v, T(v), \dots, T^{k-1}(v)\}$. The characteristic polynomial of this restriction is $c_W(x) = (-1)^k p(x)$. Since $p(T)(v) = 0$, it follows that $c_W(T)(v) = 0$.

By the theory of invariant factors, V can be decomposed into a direct sum of T-cyclic subspaces $V = W_1 \oplus W_2 \oplus \cdots \oplus W_m$. Let $c_i(x)$ be the characteristic polynomial of $T|_{W_i}$. The characteristic polynomial of T is the product of the characteristic polynomials of the restricted operators:

$$c(x) = \prod_{i=1}^{m} c_i(x)$$

Since each $c_i(T)$ vanishes on its corresponding subspace W_i, and $c(T)$ contains $c_i(T)$ as a factor, we have $c(T)(v) = 0$ for all $v \in W_i$. By linearity, $c(T) = T_0$ on the entire space V. ■

Exercise 3.8.11. Practice the following exercises.

1. Let $V = P_2(\mathbb{R})$ be the vector space of real polynomials of degree at most 2 with the inner product:
$$\langle p, q \rangle = \int_0^1 p(x) q(x)\, dx$$
For a general polynomial $p(x) = ax^2 + bx + c$, find the conditions on the real coefficients a, b, and c such that $p(x)$ is orthogonal to the polynomial $q(x) = x$.

2. Let V be an inner product space. Prove the Apollonius identity for any elements $x, y, z \in V$:
$$\|z - x\|^2 + \|z - y\|^2 = \frac{1}{2}\|x - y\|^2 + 2\left\|z - \frac{x+y}{2}\right\|^2$$

3. Let $V = \mathbb{R}^3$ with the standard inner product, and let W be the subspace spanned by $v_1 = (1, 1, 1)$ and $v_2 = (1, -1, 0)$.

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(a) Construct an orthonormal basis for W.

(b) Compute the orthogonal projection of the vector $v = (1, 2, 3)$ onto W.

4. Let W be a subspace of a finite-dimensional inner product space V, and let $P_W : V \to V$ be the orthogonal projection onto W. Prove that P_W is a self-adjoint operator ($P_W^* = P_W$) and is idempotent ($P_W^2 = P_W$).

5. Let V be an inner product space and let $\{e_1, e_2, \ldots, e_n\}$ be an orthonormal set in V. Prove that for any $x \in V$, the vector $y = \sum_{i=1}^{n} \langle x, e_i \rangle e_i$ satisfies:

$$\|x - y\| \leq \|x - z\| \quad \forall z \in \text{span}(\{e_1, \ldots, e_n\})$$

6. Let $V = P_2(\mathbb{R})$ with the inner product $\langle p, q \rangle = \int_{-1}^{1} p(x)q(x)\,dx$. Apply the Gram-Schmidt orthogonalization process to the standard basis $\{1, x, x^2\}$ to find an orthonormal basis for V.

7. In $\mathbb{R}^4$ equipped with the standard inner product, let W be the subspace spanned by the vectors $u_1 = (1, 0, 1, 0)$, $u_2 = (1, 1, 1, 1)$, and $u_3 = (0, 1, 2, 1)$. Apply the Gram-Schmidt process to obtain an orthonormal basis for W.

8. Let $V = M_{2\times 2}(\mathbb{R})$ with the inner product defined by $\langle A, B \rangle = \text{tr}(B^T A)$. Let W be the subspace consisting of all symmetric 2×2 matrices.

(a) Characterize the orthogonal complement $W^\perp$.

(b) Express the matrix $M = \begin{pmatrix} 2 & 5 \\ 1 & 4 \end{pmatrix}$ as $M = S + K$, where $S \in W$ and $K \in W^\perp$.

9. Let V be a finite-dimensional inner product space, and let U and W be subspaces of V. Prove that:

(a) $(U + W)^\perp = U^\perp \cap W^\perp$

(b) $(U \cap W)^\perp = U^\perp + W^\perp$

10. Let $T : V \to V$ be a linear operator on a finite-dimensional complex inner product space V. Prove that T is normal if and only if T can be written as $T = U + iW$, where U and W are self-adjoint operators that commute (i.e., $UW = WU$).

11. Let V be a finite-dimensional real inner product space, and let $T : V \to V$ be a self-adjoint operator. Show that if $T^k = 0$ for some positive integer $k \geq 1$, then $T = 0$.

12. Let V be a finite-dimensional complex inner product space, and let $T : V \to V$ be a normal operator. Prove that if $T^2 = T$, then T is an orthogonal projection (meaning T is self-adjoint).

13. Orthogonally diagonalize the symmetric matrix:

$$A = \begin{pmatrix} 5 & -2 & 4 \\ -2 & 8 & 2 \\ 4 & 2 & 5 \end{pmatrix}$$

by finding an orthogonal matrix P and a diagonal matrix D such that $P^T AP = D$.

14. Let T be a normal operator on a finite-dimensional complex inner product space V. Let $\lambda_1, \ldots, \lambda_k$ be the distinct eigenvalues of T, and let P_i be the orthogonal projection onto the eigenspace E_{λ_i}. Prove that:

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(a) $P_i P_j = 0$ for $i \neq j$.

(b) $I = \sum_{i=1}^{k} P_i$.

(c) $T = \sum_{i=1}^{k} \lambda_i P_i$.

15. Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V. Let λ be an eigenvalue of T. Prove that the sequence of subspaces:

$$\ker(T - \lambda I) \subseteq \ker(T - \lambda I)^2 \subseteq \ker(T - \lambda I)^3 \subseteq \dots$$

stabilizes at some step $k \leq \dim V$, so that $\ker(T - \lambda I)^k = \ker(T - \lambda I)^{k+1} = \dots$.

16. Prove that generalized eigenvectors corresponding to distinct eigenvalues of a linear operator T on a finite-dimensional vector space V are linearly independent.

17. Find the Jordan canonical form J and an invertible transition matrix P such that $P^{-1}AP = J$ for the matrix:

$$A = \begin{pmatrix} 3 & -1 & 1 \\ 2 & 0 & 1 \\ 1 & -1 & 2 \end{pmatrix}$$

18. Let V be a finite-dimensional complex vector space, and let $T : V \to V$ be a linear operator. Prove that there exist unique operators $D, N : V \to V$ such that $T = D + N$, where D is diagonalizable, N is nilpotent, and $DN = ND$.

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CHAPTER

4

Canonical Forms

The study of linear operators attains its fullest clarity when the underlying vector space is decomposed into invariant and cyclic subspaces, reducing the operator to a canonical form form — triangular, nilpotent, Jordan, or rational — that lays bare its algebraic structure. This chapter develops these canonical forms alongside the theory of bilinear, symmetric, alternating, and quadratic forms, with invariance and decomposition serving as the unifying thread throughout.

4.1 Triangular Form

Definition 4.1.1 (Triangular Form of a Linear Operator)**.** Let V be a finite-dimensional vector space over a field F, and let $T : V \to V$ be a linear operator. We say that T is in **triangular form** (or that T is **triangularisable**) if there exists an ordered basis $\mathcal{B} = \{v_1, v_2, \dots, v_n\}$ of V such that the matrix $[T]_{\mathcal{B}}$ is upper triangular, i.e.

$$[T]_{\mathcal{B}} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & a_{nn} \end{pmatrix}.$$

Equivalently, T is triangularisable if and only if there exists a chain of T-invariant subspaces

$$\{\mathbf{0}\} = V_0 \subset V_1 \subset V_2 \subset \cdots \subset V_n = V, \qquad \dim V_k = k \ \text{ for each } \ k = 0, 1, \dots, n,$$

such that $T(V_k) \subseteq V_k$ for every k. The diagonal entries $a_{11}, a_{22}, \dots, a_{nn}$ are precisely the eigenvalues of T (counted with multiplicity), and they coincide with the roots of the characteristic polynomial of T.

Theorem 4.1.2 (Schur)**.** *Let V be a finite-dimensional vector space over a field F, and let $T : V \to V$ be a linear operator whose characteristic polynomial $f_T(x)$ splits into linear factors over F, i.e.*

$$f_T(x) = (-1)^n \prod_{i=1}^{r} (x - \lambda_i)^{d_i}, \qquad \lambda_i \in F, \quad \sum_{i=1}^{r} d_i = n = \dim V.$$

Then there exists an ordered basis $\mathcal{B}$ of V with respect to which the matrix of T is upper triangular.

Proof. See Theorem 3.4.16 for proof. ■

2nd Proof. We proceed by induction on $n = \dim V$.

Base case ($n = 1$). Any 1×1 matrix is trivially upper triangular. Setting $\mathcal{B} = \{v_1\}$ for any nonzero

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$v_1 \in V$ gives $[T]_{\mathcal{B}} = (\lambda_1)$.

Inductive step. Suppose the result holds for all vector spaces of dimension less than n, and let $\dim V = n \geq 2$.

Since $f_T(x)$ splits into linear factors over F, it has at least one root $\lambda_1 \in F$. Hence T has at least one eigenvalue λ_1; let $v_1 \in V$ be a corresponding eigenvector, so $T(v_1) = \lambda_1 v_1$. Set $V_1 = \operatorname{span}\{v_1\}$. Since $T(v_1) = \lambda_1 v_1 \in V_1$, the subspace V_1 is T-invariant.

Consider the quotient space $\bar{V} = V/V_1$, which has dimension $n-1$. The operator T induces a well-defined linear operator $\bar{T} : \bar{V} \to \bar{V}$ by

$$\bar{T}(\bar{v}) = \overline{T(v)}, \qquad \bar{v} = v + V_1.$$

We determine the characteristic polynomial of $\bar{T}$. Choose any basis of V extending $\{v_1\}$; in this basis the matrix of T takes the block form

$$[T] = \begin{pmatrix} \lambda_1 & * \\ 0 & A \end{pmatrix},$$

where A is the $(n-1) \times (n-1)$ matrix representing $\bar{T}$ in the induced basis of $\bar{V}$. Hence

$$f_T(x) = (x - \lambda_1)\, f_{\bar{T}}(x),$$

so $f_{\bar{T}}(x)$ is a product of $n-1$ linear factors over F; in particular it splits.

By the inductive hypothesis applied to $\bar{T}$ on $\bar{V}$, there exists an ordered basis $\{\bar{v}_2, \bar{v}_3, \ldots, \bar{v}_n\}$ of $\bar{V}$ in which $\bar{T}$ is upper triangular. Lift each $\bar{v}_k$ to a representative $v_k \in V$ (so $v_k + V_1 = \bar{v}_k$). We claim that $\mathcal{B} = \{v_1, v_2, \ldots, v_n\}$ is the required basis.

$\mathcal{B}$ is a basis of V. Since $\{v_1\}$ is a basis of V_1 and $\{\bar{v}_2, \ldots, \bar{v}_n\}$ is a basis of V/V_1, a standard result on quotient spaces guarantees that $\{v_1, v_2, \ldots, v_n\}$ is a basis of V.

The matrix of T is upper triangular in $\mathcal{B}$. For $k \geq 2$, since $\bar{T}$ is upper triangular in $\{\bar{v}_2, \ldots, \bar{v}_n\}$,

$$\bar{T}(\bar{v}_k) \in \operatorname{span}\{\bar{v}_2, \ldots, \bar{v}_k\},$$

which means $T(v_k) - \sum_{j=2}^{k} a_{jk}\, v_j \in V_1 = \operatorname{span}\{v_1\}$ for some scalars $a_{jk} \in F$. Hence

$$T(v_k) \in \operatorname{span}\{v_1, v_2, \ldots, v_k\} \quad \text{for each } k = 2, 3, \ldots, n.$$

Combined with $T(v_1) = \lambda_1 v_1 \in \operatorname{span}\{v_1\}$, we conclude that the matrix $[T]_{\mathcal{B}}$ is upper triangular, completing the induction. ∎

Theorem 4.1.3. *Let V be a vector space over a field F, let $T : V \to V$ be a linear operator, and let $f \in F[x]$ be any polynomial. Then $\ker(f(T))$ is a T-invariant subspace of V.*

Proof. Let $W = \ker(f(T))$. We first note that W is a subspace of V since $f(T)$ is a linear operator and kernels of linear operators are subspaces.

It remains to show that $T(W) \subseteq W$, i.e. that for every $v \in W$ we have $T(v) \in W$, equivalently $f(T)(T(v)) = \mathbf{0}$.

Let $v \in W$, so $f(T)(v) = \mathbf{0}$ by definition of W. Write $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$ with $a_k \in F$. Since T is a linear operator, every power T^k commutes with T (as $T^k \circ T = T^{k+1} = T \circ T^k$), and by linearity of composition:

$$f(T) \circ T = \left(\sum_{k=0}^{n} a_k T^k \right) \circ T = \sum_{k=0}^{n} a_k\, T^k \circ T = \sum_{k=0}^{n} a_k\, T^{k+1} = T \circ \sum_{k=0}^{n} a_k\, T^k = T \circ f(T).$$

That is, $f(T)$ and T commute: $f(T) \circ T = T \circ f(T)$.

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Applying this to v:

$$f(T)(T(v)) = \big(f(T)\circ T\big)(v) = \big(T\circ f(T)\big)(v) = T(f(T)(v)) = T(\mathbf{0}) = \mathbf{0}.$$

Hence $T(v) \in \ker(f(T)) = W$. Since $v \in W$ was arbitrary, $T(W) \subseteq W$, and $W = \ker(f(T))$ is T-invariant. ■

Theorem 4.1.4. *Let $W_1, W_2, \ldots, W_r$ be subspaces of a vector space V, and for each $i = 1, 2, \ldots, r$ let $\mathcal{B}_i = \{w_{i1}, w_{i2}, \ldots, w_{in_i}\}$ be a basis of W_i. Then*

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_r \quad \Longleftrightarrow \quad \mathcal{B} = \mathcal{B}_1 \cup \mathcal{B}_2 \cup \cdots \cup \mathcal{B}_r \text{ is a basis of } V.$$

Proof. Suppose $V = W_1 \oplus W_2 \oplus \cdots \oplus W_r$.

Spanning. Every $v \in V$ can be written as $v = w_1 + w_2 + \cdots + w_r$ with $w_i \in W_i$. Since $\mathcal{B}_i$ spans W_i, each w_i is a linear combination of vectors in $\mathcal{B}_i$. Hence v is a linear combination of vectors in $\mathcal{B}$, so $\mathcal{B}$ spans V.

Linear independence. Suppose

$$\sum_{i=1}^{r}\sum_{j=1}^{n_i} a_{ij}\, w_{ij} = \mathbf{0}.$$

Setting $u_i = \displaystyle\sum_{j=1}^{n_i} a_{ij}\, w_{ij} \in W_i$, this reads $u_1 + u_2 + \cdots + u_r = \mathbf{0}$. Since the sum is direct, the zero vector decomposes uniquely as $\mathbf{0} = \mathbf{0} + \cdots + \mathbf{0}$, so $u_i = \mathbf{0}$ for each i. Since $\mathcal{B}_i$ is linearly independent, $u_i = \mathbf{0}$ forces $a_{ij} = 0$ for all j. Hence all coefficients are zero, and $\mathcal{B}$ is linearly independent. Thus $\mathcal{B}$ is a basis of V.

Conversely, suppose $\mathcal{B} = \displaystyle\bigcup_{i=1}^{r} \mathcal{B}_i$ is a basis of V. We show V is a direct sum of $V_i's$, $i = 1, 2, \ldots, r$. Since $\mathcal{B}$ spans V and every vector in $\mathcal{B}$ belongs to some W_i, every $v \in V$ is a linear combination of vectors drawn from $W_1, \ldots, W_r$, hence $v \in W_1 + \cdots + W_r$.

The sum is direct. Suppose $w_1 + w_2 + \cdots + w_r = \mathbf{0}$ with $w_i \in W_i$. Write each $w_i = \displaystyle\sum_{j=1}^{n_i} a_{ij}\, w_{ij}$. Then

$$\sum_{i=1}^{r}\sum_{j=1}^{n_i} a_{ij}\, w_{ij} = \mathbf{0}.$$

Since $\mathcal{B}$ is linearly independent, every coefficient $a_{ij} = 0$, so $w_i = \mathbf{0}$ for each i. Hence the only way to write $\mathbf{0}$ as a sum $w_1 + \cdots + w_r$ with $w_i \in W_i$ is the trivial one, which is precisely the condition for the sum to be direct.

Therefore $V = W_1 \oplus W_2 \oplus \cdots \oplus W_r$. ■

Theorem 4.1.5 (Primary Decomposition Theorem)**.** *Let $T : V \to V$ be a linear operator on a finite-dimensional vector space V over a field F, and let*

$$m_T(x) = p_1(x)^{r_1}\, p_2(x)^{r_2} \cdots p_k(x)^{r_k}$$

be the factorisation of the minimal polynomial of T into powers of distinct monic irreducible polynomials $p_1, p_2, \ldots, p_k$ over F. Define

$$W_i = \ker\big(p_i(T)^{r_i}\big), \qquad i = 1, 2, \ldots, k.$$

Then:

(i) *Each W_i is a T-invariant subspace of V.*

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(ii) $V = W_1 \oplus W_2 \oplus \cdots \oplus W_k$.

(iii) *The minimal polynomial of the restriction* $T|_{W_i}$ *is* $p_i(x)^{r_i}$.

Proof. **(i)** T**-invariance of each** W_i**.**
Let $v \in W_i$, so that $p_i(T)^{r_i}(v) = \mathbf{0}$. Since $p_i(T)^{r_i}$ is itself a polynomial in T, it commutes with T:

$$p_i(T)^{r_i}(T(v)) = T(p_i(T)^{r_i}(v)) = T(\mathbf{0}) = \mathbf{0}.$$

Hence $T(v) \in W_i$, and W_i is T-invariant.
(ii) $V = W_1 \oplus W_2 \oplus \cdots \oplus W_k$**.**
We establish this in two steps.
Step 1: The subspaces W_i *are independent.*
Define, for each i,

$$f_i(x) = \frac{m_T(x)}{p_i(x)^{r_i}} = \prod_{j \neq i} p_j(x)^{r_j}.$$

Since $p_1, \ldots, p_k$ are distinct irreducibles, $\gcd(p_i^{r_i}, f_i) = 1$, so by Bezout's theorem in $F[x]$ there exist polynomials $s_i, t_i \in F[x]$ satisfying

$$s_i(x)\, p_i(x)^{r_i} + t_i(x)\, f_i(x) = 1. \tag{$*$}$$

Now suppose

$$v_1 + v_2 + \cdots + v_k = \mathbf{0}, \qquad v_i \in W_i.$$

Fix an index i and apply $f_i(T)$ to this equation:

$$f_i(T)(v_1) + \cdots + f_i(T)(v_i) + \cdots + f_i(T)(v_k) = \mathbf{0}.$$

For $j \neq i$, the polynomial $p_j^{r_j}$ divides f_i, so $f_i(T)(v_j) = \mathbf{0}$ (since $v_j \in W_j = \ker(p_j(T)^{r_j})$). Hence the equation reduces to

$$f_i(T)(v_i) = \mathbf{0}. \tag{$**$}$$

But $v_i \in W_i = \ker(p_i(T)^{r_i})$, so $p_i(T)^{r_i}(v_i) = \mathbf{0}$. Evaluating the Bezout identity $(*)$ at T and applying to v_i:

$$v_i = s_i(T)\, p_i(T)^{r_i}(v_i) + t_i(T)\, f_i(T)(v_i) = s_i(T)(\mathbf{0}) + t_i(T)(\mathbf{0}) = \mathbf{0}.$$

Since i was arbitrary, $v_i = \mathbf{0}$ for all i, proving the subspaces are independent, i.e. the sum $W_1 + \cdots + W_k$ is direct.
Step 2: $V = W_1 + W_2 + \cdots + W_k$.
For each i, the Bezout identity $(*)$ gives $s_i(T)\, p_i(T)^{r_i} + t_i(T)\, f_i(T) = I$. Since $\gcd(f_1, f_2, \ldots, f_k) = 1$ (as the p_i are distinct), the polynomials $f_1, f_2, \ldots, f_k$ generate the unit ideal in $F[x]$; more concretely, there exist polynomials $h_1, \ldots, h_k \in F[x]$ such that

$$h_1(x)\, f_1(x) + h_2(x)\, f_2(x) + \cdots + h_k(x)\, f_k(x) = 1.$$

Substituting T and applying to any $v \in V$:

$$v = h_1(T)f_1(T)(v) + h_2(T)f_2(T)(v) + \cdots + h_k(T)f_k(T)(v). \tag{$***$}$$

Set $v_i = h_i(T)f_i(T)(v)$. We show $v_i \in W_i$:

$$p_i(T)^{r_i}(v_i) = p_i(T)^{r_i}\, h_i(T)\, f_i(T)(v) = h_i(T)\, \underbrace{p_i(T)^{r_i} f_i(T)}_{= m_T(T) = \mathbf{0}}(v) = \mathbf{0}.$$

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Hence $v_i \in W_i$, and $(***)$ expresses v as a sum of elements from $W_1, \ldots, W_k$. Since v was arbitrary, $V = W_1 + \cdots + W_k$.

Combining Steps 1 and 2:

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_k.$$

(iii) The minimal polynomial of $T|_{W_i}$ is $p_i(x)^{r_i}$.

Let $m_i(x)$ be the minimal polynomial of $T|_{W_i}$. Since $W_i = \ker(p_i(T)^{r_i})$, it follows that $p_i(T|_{W_i})^{r_i} = 0$, so $m_i(x) \mid p_i(x)^{r_i}$. As $p_i(x)$ is irreducible, thus, $m_i(x) = p_i(x)^s$ for some $1 \leq s \leq r_i$.

Suppose $s < r_i$. Define the polynomial $q(x) = p_i(x)^s \prod_{j \neq i} p_j(x)^{r_j}$. By the direct sum decomposition $V = \bigoplus W_j$, and noting that $p_i(T)^s$ annihilates W_i while $p_j(T)^{r_j}$ annihilates W_j for $j \neq i$, the operator $q(T)$ acts as the zero operator on all W_j. Thus $q(T) = 0$. Since $\deg(q) < \deg(m_T)$, this contradicts the minimality of $m_T(x)$. Hence, $s = r_i$, and $m_i(x) = p_i(x)^{r_i}$. ■

Theorem 4.1.6. *Let $T : V \to V$ be a linear operator, and let $f(t) = g(t)\,h(t)$ be polynomials over F such that $f(T) = \mathbf{0}$ and $\gcd(g, h) = 1$. Then*

$$V = U \oplus W, \qquad U = \ker(g(T)), \quad W = \ker(h(T)),$$

where both U and W are T-invariant subspaces of V.

Proof. (The proof can be derived from PDT (Theorem 4.1.5) by setting $p_1^{r_1} = g$, $p_2^{r_2} = h$ and remaining $p_i^{r_i} = 1$).

T-invariance. Since every polynomial in T commutes with T, for any $v \in U = \ker(g(T))$:

$$g(T)(T(v)) = T(g(T)(v)) = T(\mathbf{0}) = \mathbf{0},$$

so $T(v) \in U$. The same argument shows W is T-invariant.

$V = U + W$. Since $\gcd(g, h) = 1$, by Bezout's theorem there exist polynomials $s, t \in F[x]$ such that

$$s(t)\,g(t) + t(t)\,h(t) = 1.$$

Substituting T and applying to any $v \in V$:

$$v = \underbrace{s(T)g(T)(v)}_{\in W} + \underbrace{t(T)h(T)(v)}_{\in U}.$$

Indeed, $h(T)\big(s(T)g(T)(v)\big) = s(T)\,\underbrace{f(T)(v)}_{=\mathbf{0}} = \mathbf{0}$, so $s(T)g(T)(v) \in W$. Similarly $t(T)h(T)(v) \in U$. Hence $V = U + W$.

The sum is direct. Let $v \in U \cap W$, so $g(T)(v) = \mathbf{0}$ and $h(T)(v) = \mathbf{0}$. Applying the Bezout identity at T to v:

$$v = s(T)\,\underbrace{g(T)(v)}_{=\mathbf{0}} + t(T)\,\underbrace{h(T)(v)}_{=\mathbf{0}} = \mathbf{0}.$$

Hence $U \cap W = \{\mathbf{0}\}$, and therefore $V = U \oplus W$. ■

Theorem 4.1.7. *A linear operator $T : V \to V$ is diagonalisable if and only if its minimal polynomial $m_T(t)$ factors into distinct linear factors over F, i.e.*

$$m_T(t) = (t - \lambda_1)(t - \lambda_2)\cdots(t - \lambda_k), \qquad \lambda_i \in F \text{ distinct.}$$

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Proof. If T is diagonalisable, it is annihilated by the polynomial $q(t) = \prod_{i=1}^{k}(t - \lambda_i)$, where λ_i are the distinct eigenvalues. Since the minimal polynomial $m_T(t)$ divides any polynomial that annihilates T, and $q(t)$ has distinct linear factors, $m_T(t)$ must also be a product of distinct linear factors.

Suppose $m_T(t) = \prod_{i=1}^{k}(t - \lambda_i)$ with distinct λ_i. By the Primary Decomposition Theorem, V decomposes into the direct sum of the T-invariant subspaces $W_i = \ker(T - \lambda_i I)$:

$$V = \ker(T - \lambda_1 I) \oplus \ker(T - \lambda_2 I) \oplus \cdots \oplus \ker(T - \lambda_k I). \tag{4.1}$$

Since each W_i is the eigenspace E_{λ_i}, V is the direct sum of the eigenspaces of T. Thus, there exists a basis of V consisting of eigenvectors of T, implying T is diagonalisable. ■

Definition 4.1.8. A linear operator $T : V \to V$ is **nilpotent** if $T^k = \mathbf{0}$ for some positive integer k. A smallest positive integer k having the property that $T^k = 0$ and $T^{k+1} \neq 0$ is called the index of nilpotency of T.

Theorem 4.1.9. *T is nilpotent if and only if all its eigenvalues are zero.*

Proof. ($\implies$) If $Tv = \lambda v$, then $T^k v = \lambda^k v = 0$. Since $v \neq 0$, $\lambda = 0$. ($\impliedby$) If 0 is the only eigenvalue, the characteristic polynomial is $p(t) = \pm t^n$. By Cayley-Hamilton, $p(T) = \pm T^n = 0$, so T is nilpotent. ■

Theorem 4.1.10. *The minimal polynomial of a nilpotent operator T with index k is $m_T(t) = t^k$.*

Proof. Since $T^k = 0$, $m_T(t)$ must divide t^k, implying $m_T(t) = t^s$ with $s \leq k$. Since k is the smallest integer such that $T^k = 0$, s cannot be less than k. Thus $s = k$. ■

Definition 4.1.11. A Jordan block $J_k(\lambda)$ is a square matrix of order k having λ on its diagonal, 1s on the super-diagonal and all other entries zero. A matrix is in **Jordan Canonical Form** if it is a block-diagonal matrix $J = J_{k_1}(\lambda_1) \oplus \cdots \oplus J_{k_m}(\lambda_m)$, where each $J_k(\lambda)$ is a Jordan block with eigenvalue λ on the diagonal and 1s on the super-diagonal.

Definition 4.1.12. Let $T : V \to V$ be a linear operator of order n with minimal polynomial $m_T(t) = \prod_{i=1}^{k}(t - \lambda_i)^{d_i}$. The **Jordan Canonical Form** is a block-diagonal matrix $J = \bigoplus_{i=1}^{k} J_i$, where each J_i is composed of Jordan blocks corresponding to λ_i such that:

- There is at least one Jordan block in J_i of order d_i.
- All other Jordan blocks corresponding to λ_i have order $\leq d_i$.
- The sum of the sizes of all blocks corresponding to λ_i is the algebraic multiplicity of λ_i.

Example 4.1.13. If $c_T(x) = (t-2)^5(t+1)^3$ and $m_T(t) = (t-2)^2(t+1)^2$ for an 8×8 matrix, find all possible Jordan Canonical forms.

Solution. The Jordan blocks must satisfy the constraints imposed by the characteristic and minimal polynomials.

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- **Eigenvalue $\lambda = 2$:** Multiplicity is 5, largest block is 2×2. Possible block partitions (sums to 5): $\{2,2,1\}$ and $\{2,1,1,1\}$.
- **Eigenvalue $\lambda = -1$:** Multiplicity is 3, largest block is 2×2. Possible block partition (sum to 3): $\{2,1\}$.

Combining these, there are two possible JCFs (up to block ordering):

1. $J = J_2(2) \oplus J_2(2) \oplus J_1(2) \oplus J_2(-1) \oplus J_1(-1)$
2. $J = J_2(2) \oplus J_1(2) \oplus J_1(2) \oplus J_1(2) \oplus J_2(-1) \oplus J_1(-1)$

where $J_2(2) = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$, $J_1(2) = [2]$, $J_2(-1) = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$, $J_1(-1) = [-1]$. ■

Example 4.1.14. Let T be a linear operator on a 10-dimensional vector space V with characteristic polynomial $c_T(t) = (t-5)^6(t-3)^4$ and minimal polynomial $m_T(t) = (t-5)^3(t-3)^2$. Determine all possible Jordan Canonical Forms.

Solution. To find the possible Jordan Canonical Forms (JCF), we partition the algebraic multiplicities of the eigenvalues according to the size constraints imposed by the minimal polynomial.

- **Eigenvalue $\lambda = 5$:** The algebraic multiplicity is 6, and the exponent in the minimal polynomial is 3, meaning the largest Jordan block size is exactly 3. We seek all partitions of 6 into parts of size at most 3, where at least one part is 3:

  (i). $\{3,3\}$ (ii). $\{3,2,1\}$ (iii). $\{3,1,1,1\}$
- **Eigenvalue $\lambda = 3$:** The algebraic multiplicity is 4, and the exponent in the minimal polynomial is 2, meaning the largest Jordan block size is exactly 2. We seek all partitions of 4 into parts of size at most 2, where at least one part is 2:

  (i). $\{2,2\}$ (ii). $\{2,1,1\}$

The total number of possible Jordan Canonical Forms is the product of the number of valid partitions for each eigenvalue, which is $3 \times 2 = 6$. The JCF is determined by the direct sum of the blocks corresponding to these partitions.

For example, selecting partition $\{3,3\}$ for $\lambda = 5$ and partition $\{2,2\}$ for $\lambda = 3$ yields:

$$J = J_3(5) \oplus J_3(5) \oplus J_2(3) \oplus J_2(3).$$

Alternatively, selecting partition $\{3,1,1,1\}$ for $\lambda = 5$ and partition $\{2,1,1\}$ for $\lambda = 3$ yields:

$$J = J_3(5) \oplus J_1(5) \oplus J_1(5) \oplus J_1(5) \oplus J_2(3) \oplus J_1(3) \oplus J_1(3).$$

The remaining four forms are obtained by matching the other partition combinations (e.g., $\{3,2,1\}$ with $\{2,2\}$, etc.). Each distinct combination of partitions defines a similarity class of matrices. ■

4.1 Rational Canonical Form

The Rational Canonical Form provides a basis-independent way to represent a linear operator $T : V \to V$ as a block-diagonal matrix of companion matrices.

Definition 4.1.15 (Companion Matrix)**.** Let $a(x) = x^k + b_{k-1}x^{k-1} + \cdots + b_0 \in F[x]$. The companion

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matrix $C(a)$ is the $k \times k$ matrix:

$$C(a) = \begin{pmatrix} 0 & 0 & \dots & 0 & -b_0 \\ 1 & 0 & \dots & 0 & -b_1 \\ 0 & 1 & \dots & 0 & -b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 1 & -b_{k-1} \end{pmatrix}$$

Definition 4.1.16 (Cyclic Subspace)**.** A subspace $W \subseteq V$ is T-cyclic if $W = \text{span}\{v, Tv, T^2v, \dots\}$ for some $v \in V$. The restriction $T|_W$ has a matrix representation equal to the companion matrix of the minimal polynomial of v with respect to T.

Lemma 4.1.17. *Let $T : V \to V$ be a linear operator whose minimal polynomial is $f(t)^n$, where $f(t)$ is a monic irreducible polynomial. Then V is the direct sum*

$$V = Z(v_1, T) \oplus \cdots \oplus Z(v_r, T)$$

of T-cyclic subspaces $Z(v_i, T)$ with corresponding T-annihilators

$$f(t)^{n_1}, f(t)^{n_2}, \dots, f(t)^{n_r}, \quad n = n_1 \geq n_2 \geq \cdots \geq n_r.$$

Any other decomposition of V into T-cyclic subspaces has the same number of components and the same set of T-annihilators.

Proof. Proof Follows by Primary Decomposition Theorem (Theorem 4.1.5). ■

The set of T-annihilators is uniquely determined by T. Thus, T has a unique block diagonal matrix representation:

$$M = \text{diag}(C_1, C_2, \dots, C_r)$$

where the C_i are the companion matrices of the polynomials $f(t)^{n_i}$.

Theorem 4.1.18. *Let $T : V \to V$ be a linear operator with minimal polynomial*

$$m(t) = f_1(t)^{m_1} f_2(t)^{m_2} \cdots f_s(t)^{m_s}$$

where the $f_i(t)$ are distinct monic irreducible polynomials. Then T has a unique block diagonal matrix representation:

$$M = diag(C_{11}, C_{12}, \dots, C_{1r_1}, \dots, C_{s1}, C_{s2}, \dots, C_{sr_s})$$

where the C_{ij} are companion matrices of the polynomials $f_i(t)^{n_{ij}}$, such that

$$m_1 = n_{11} \geq n_{12} \geq \cdots \geq n_{1r_1}, \quad \dots, \quad m_s = n_{s1} \geq n_{s2} \geq \cdots \geq n_{sr_s}.$$

*The polynomials $f_i(t)^{n_{ij}}$ are called the **elementary divisors** of T.*

Proof. By the Primary Decomposition Theorem, V decomposes uniquely as $V = W_1 \oplus \cdots \oplus W_s$, where $W_i = \ker(f_i(T)^{m_i})$. The restriction $T_i = T|_{W_i}$ has minimal polynomial $f_i(t)^{m_i}$. By the lemma of T-cyclic decomposition, each W_i further decomposes uniquely into cyclic subspaces $Z(v_{ij}, T)$ with annihilators $f_i(t)^{n_{ij}}$, satisfying $m_i = n_{i1} \geq \cdots \geq n_{ir_i}$.

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The matrix representation of T restricted to each cyclic subspace is the companion matrix $C_{ij} = C(f_i(t)^{n_{ij}})$. As $V = \bigoplus_{i,j} Z(v_{ij}, T)$, the global matrix M of T is the direct sum of these companion matrices. The uniqueness of the blocks is guaranteed by the uniqueness of the Primary Decomposition and the uniqueness of the cyclic subspace decomposition within each W_i. ■

Example 4.1.19. Let V be a vector space of dimension 8 over $\mathbb{Q}$, and let T be a linear operator on V whose minimal polynomial is

$$m(t) = f_1(t)f_2(t)^2 = (t^4 - 4t^3 + 6t^2 - 4t - 7)(t-3)^2.$$

Because $\dim V = 8$, the characteristic polynomial is $\Delta(t) = f_1(t)f_2(t)^4$. The rational canonical form M must contain one block which is the companion matrix of $f_1(t)$ and blocks corresponding to $f_2(t)^2 = (t-3)^2$. There are two possibilities:

(a) $\text{diag}[C(t^4 - 4t^3 + 6t^2 - 4t - 7),\ C((t-3)^2),\ C((t-3)^2)]$

(b) $\text{diag}[C(t^4 - 4t^3 + 6t^2 - 4t - 7),\ C((t-3)^2),\ C(t-3),\ C(t-3)]$

Example 4.1.20. Compute the RCF for an operator on V ($\dim V = 4$) with $m(t) = (t^2+1)^2$.

Solution. The minimal polynomial is degree 4, so V is a cyclic subspace. The RCF is $C(t^4 + 2t^2 + 1)$:

$$M = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

■

Example 4.1.21. Compute the RCF for T ($\dim V = 4$) with $c(t) = (t-2)^4$ and $m(t) = (t-2)^2$.

Solution. The elementary divisors produce two possible partition $\{2,2\}$ and $\{2,1,1\}$.

1. Corresponding to the partition $\{2,2\}$, the elementary divisors are $(t-2)^2, (t-2)^2$. Thus $M = C((t-2)^2) \oplus C((t-2)^2)$. Since $C(t^2 - 4t + 4) = \begin{pmatrix} 0 & -4 \\ 1 & 4 \end{pmatrix}$: The RCF is:

$$M_1 = \begin{pmatrix} 0 & -4 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & 0 & -4 \\ 0 & 0 & 1 & 4 \end{pmatrix}$$

2. Corresponding to the partition $\{2,1,1\}$, the elementary divisors are $(t-2)^2, (t-2), (t-2)$. The RCF is:

$$M_2 = \begin{pmatrix} 0 & -4 & 0 & 0 \\ 1 & 4 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

■

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Theorem 4.1.22. *Let A and B be $n \times n$ matrices over F. Then A and B are similar iff A and B have the same rational canonical form.*

Lemma 4.1.23. *Let $a(x) \in F[x]$ be any monic polynomial.*

(1) The characteristic polynomial of the companion matrix of $a(x)$ is $a(x)$.

(2) If $M = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$, then the characteristic polynomial of M is the product of characteristic polynomials of A and B.

Theorem 4.1.24. *Let A be an $n \times n$ matrix over the field F. Using ERT and ECT, the $n \times n$ matrix $xI - A$ can be put into the diagonal form (called the Smith Normal form for A)*

$$\begin{pmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & & a_1(x) & & \\ & & & & \ddots & \\ & & & & & a_m(x) \end{pmatrix}$$

with monic non-zero elements $a_1(x), a_2(x), \ldots, a_m(x)$ of $F[x]$ with degrees at least one and satisfying $a_1(x) \mid a_2(x) \mid \cdots \mid a_m(x)$.

Algorithm for Smith form

1. Write $xI - A$
2. Then Smith form is

$$\begin{pmatrix} \gcd(a_{ij}) = B_1 & & & \\ & \frac{\gcd(M_{2\times 2})}{B_1} = B_2 & & \\ & & \frac{\gcd M_{3\times 3}}{B_2} & \\ & & & \ddots \end{pmatrix}$$

Example 4.1.25. Convert the following matrices into Smith form.

$$\begin{pmatrix} 2 & -2 & 14 \\ 0 & 3 & -7 \\ 0 & 0 & 2 \end{pmatrix}, \quad \begin{pmatrix} 0 & -4 & 85 \\ 1 & 4 & -30 \\ 0 & 0 & 3 \end{pmatrix}, \quad \begin{pmatrix} 2 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}$$

Solution. (1) $A = \begin{pmatrix} 2 & -2 & 14 \\ 0 & 3 & -7 \\ 0 & 0 & 2 \end{pmatrix}$, then $xI - A = \begin{pmatrix} x-2 & 2 & -14 \\ 0 & x-3 & 7 \\ 0 & 0 & x-2 \end{pmatrix}$.

Thus, Smith form is

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & x-2 & 0 \\ 0 & 0 & (x-2)(x-3) \end{pmatrix}$$

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Then invariant factors of A are $x-2$, & $(x-2)(x-3)$. Thus, the rational canonical form of A is

$$\begin{pmatrix} C_{x-2} & 0 \\ 0 & C_{x^2-5x+6} \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & -6 \\ 0 & 1 & 5 \end{pmatrix}.$$

■

Algorithm for rational Canonical form

1. Convert the matrix into Smith form.
2. Determine the invariant factors, say, $a_1(x), a_2(x), \ldots, a_m(x)$.
3. The rational Canonical form is

$$C_{a_1(x)} \oplus C_{a_2(x)} \oplus \cdots \oplus C_{a_m(x)}.$$

4.2 Bilinear and Quadratic Forms

4.2.1 Bilinear Forms

Definition 4.2.1. A bilinear form $B : V \times V \to F$ is linear in each argument. Its matrix representation $[B]_\beta$ with respect to a basis $\beta = \{v_1, \ldots, v_n\}$ is given by $A_{ij} = B(v_i, v_j)$.

Theorem 4.2.2. *For any bilinear form B on a vector space V over a field F, the following hold:*

1. *If $L_x, R_x : V \to F$ are defined by $L_x(y) = B(x,y)$ and $R_x(y) = B(y,x)$, then L_x and R_x are linear.*
2. *$B(0,x) = B(x,0) = 0$ for all $x \in V$.*
3. *For all $x, y, z, w \in V$, $B(x+y, z+w) = B(x,z) + B(x,w) + B(y,z) + B(y,w)$.*
4. *If $J : V \times V \to F$ is defined by $J(x,y) = B(y,x)$, then J is bilinear.*

Proof. 1. By the definition of a bilinear form, B is linear in each slot. Thus, for any $y_1, y_2 \in V$ and $c \in F$: $L_x(cy_1 + y_2) = B(x, cy_1 + y_2) = cB(x,y_1) + B(x,y_2) = cL_x(y_1) + L_x(y_2)$. The linearity of R_x follows identically from the linearity of B in the first slot.

2. Using linearity in the first slot: $B(0,x) = B(0+0,x) = B(0,x) + B(0,x)$. Subtracting $B(0,x)$ from both sides yields $B(0,x) = 0$. The case $B(x,0) = 0$ follows analogously from linearity in the second slot.

3. By linearity in both slots:

$$\begin{aligned} B(x+y, z+w) &= B(x, z+w) + B(y, z+w) \\ &= B(x,z) + B(x,w) + B(y,z) + B(y,w). \end{aligned}$$

4. We check linearity in the first slot for J: $J(cx_1 + x_2, y) = B(y, cx_1 + x_2) = cB(y,x_1) + B(y,x_2) = cJ(x_1,y) + J(x_2,y)$. Linearity in the second slot follows similarly from the linearity of B in the first slot. Thus, J is bilinear.

■

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Definition 4.2.3 (Operations on Bilinear Forms)**.** Let $\mathcal{B}(V)$ be the set of all bilinear forms on a vector space V over a field F. For $B_1, B_2 \in \mathcal{B}(V)$ and $c \in F$, we define:

1. **Sum:** $(B_1 + B_2)(u, v) = B_1(u, v) + B_2(u, v)$
2. **Scalar Multiplication:** $(cB_1)(u, v) = c \cdot B_1(u, v)$

Theorem 4.2.4. *The set $\mathcal{B}(V)$ is a vector space over the field F under the operations defined above.*

Proof. First, we show that $B_1 + B_2$ and cB_1 are bilinear. Let $B_1, B_2 \in \mathcal{B}(V)$. For $B_1 + B_2$, consider linearity in the first slot:

$$\begin{aligned}(B_1 + B_2)(au + v, w) &= B_1(au + v, w) + B_2(au + v, w) \\ &= [aB_1(u, w) + B_1(v, w)] + [aB_2(u, w) + B_2(v, w)] \\ &= a(B_1 + B_2)(u, w) + (B_1 + B_2)(v, w).\end{aligned}$$

Linearity in the second slot follows similarly. For cB_1, observe that:

$$\begin{aligned}(cB_1)(au + v, w) &= c \cdot B_1(au + v, w) = c(aB_1(u, w) + B_1(v, w)) \\ &= a(cB_1)(u, w) + (cB_1)(v, w).\end{aligned}$$

Thus, $\mathcal{B}(V)$ is closed under addition and scalar multiplication.

Finally, since the value of these forms lies in the field F (which is a vector space), the axioms of a vector space (associativity, commutativity of addition, distributive laws, and the identity $0(u, v) = 0$) are inherited directly from the properties of F. The zero vector is the zero form $Z(u, v) = 0$ for all $u, v \in V$. Thus, $\mathcal{B}(V)$ is a vector space. ■

Example 4.2.5 (The Dot Product)**.** Let $V = \mathbb{R}^n$. The dot product defined by

$$B(u, v) = u \cdot v = \sum_{i=1}^{n} u_i v_i$$

is a bilinear form on $\mathbb{R}^n$.

Solution. For any $u, v, w \in \mathbb{R}^n$ and $c \in \mathbb{R}$:

$$B(cu + v, w) = \sum_{i=1}^{n} (cu_i + v_i)w_i = c \sum_{i=1}^{n} u_i w_i + \sum_{i=1}^{n} v_i w_i = cB(u, w) + B(v, w).$$

The linearity in the second slot follows by the commutativity of real multiplication ($u_i v_i = v_i u_i$). ■

Example 4.2.6 (Integral Bilinear Form)**.** Let $V = C[0, 1]$, the vector space of continuous real-valued functions on the interval $[0, 1]$. Define $B : V \times V \to \mathbb{R}$ by:

$$B(f, g) = \int_0^1 f(t)g(t)\, dt$$

This is a bilinear form on V.

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Solution. By the linearity of the definite integral, for any $f, g, h \in V$ and $c \in \mathbb{R}$:

$$\begin{aligned} B(cf+g,h) &= \int_0^1 (cf(t)+g(t))h(t)\,dt \\ &= c\int_0^1 f(t)h(t)\,dt + \int_0^1 g(t)h(t)\,dt \\ &= cB(f,h) + B(g,h). \end{aligned}$$

Since the integral is linear in the second argument as well, B is bilinear. ■

Definition 4.2.7. Let $\beta = \{v_1, v_2, \ldots, v_n\}$ be an ordered basis for an $n-$dimensional vector space V, and let $B \in \mathcal{B}(V)$. We associate an $n \times n$ matrix with B defined as:

$$B = [b_{ij}]_{n\times n} = [B(v_i, v_j)]_{n\times n}$$

for $i, j = 1, 2, \ldots, n$.

Example 4.2.8. Let $V = \mathbb{R}^2$ with basis $\beta = \{v_1, v_2\}$ where $v_1 = (1,1)^T$ and $v_2 = (1,-1)^T$. Let B be the standard dot product $B(u,v) = u_1v_1 + u_2v_2$. We compute the matrix $A = [B(v_i, v_j)]$:

- $A_{11} = B(v_1, v_1) = (1)(1) + (1)(1) = 2$
- $A_{12} = B(v_1, v_2) = (1)(1) + (1)(-1) = 0$
- $A_{21} = B(v_2, v_1) = (1)(1) + (-1)(1) = 0$
- $A_{22} = B(v_2, v_2) = (1)(1) + (-1)(-1) = 2$

The matrix representation is:

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

Example 4.2.9. Let $V = P_1(\mathbb{R})$ be the space of polynomials of degree at most 1, with basis $\beta = \{1, t\}$. Let B be the bilinear form defined by:

$$B(p,q) = p(0)q(0) + p(0)q(1)$$

We compute the matrix $A = [B(v_i, v_j)]$ where $v_1 = 1$ and $v_2 = t$:

- $A_{11} = B(1,1) = (1)(1) + (1)(1) = 2$
- $A_{12} = B(1,t) = (1)(0) + (1)(1) = 1$
- $A_{21} = B(t,1) = (0)(1) + (0)(1) = 0$
- $A_{22} = B(t,t) = (0)(0) + (0)(1) = 0$

Note that $A_{12} \neq A_{21}$, reflecting that B is not symmetric. The matrix representation is:

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix}$$

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Theorem 4.2.10. *Let V be an n-dimensional vector space over a field F, and let $\beta = \{v_1, v_2, \ldots, v_n\}$ be an ordered basis for V. The mapping $\psi_\beta : \mathcal{B}(V) \to M_{n\times n}(F)$ defined by $\psi_\beta(T) = [T]_\beta$ is an isomorphism.*

Proof. **Linearity:** Let $T, U \in \mathcal{B}(V)$ and $c \in F$. For any i, j:

$$[\psi_\beta(cT + U)]_{ij} = (cT + U)(v_i, v_j) = cT(v_i, v_j) + U(v_i, v_j) = c[\psi_\beta(T)]_{ij} + [\psi_\beta(U)]_{ij}$$

This shows $\psi_\beta(cT + U) = c\psi_\beta(T) + \psi_\beta(U)$, so ψ_β is linear.

Injectivity: Suppose $\psi_\beta(T) = \mathbf{0}$. Then $T(v_i, v_j) = 0$ for all $i, j \in \{1, \ldots, n\}$. Let $u, w \in V$. We can write $u = \sum_{i=1}^{n} a_i v_i$ and $w = \sum_{j=1}^{n} b_j v_j$. By the bilinearity of T:

$$T(u, w) = T\left(\sum a_i v_i, \sum b_j v_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j T(v_i, v_j)$$

Since every $T(v_i, v_j) = 0$, it follows that $T(u, w) = 0$ for all $u, w \in V$. Thus T is the zero bilinear form, and $\ker(\psi_\beta) = \{0\}$, making ψ_β injective.

Surjectivity: Let $A \in M_{n\times n}(F)$ with entries a_{ij}. Define $T : V \times V \to F$ by:

$$T\left(\sum_{i=1}^{n} a_i v_i, \sum_{j=1}^{n} b_j v_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i b_j a_{ij}$$

T is a bilinear form by the properties of summation and scalar multiplication. For the basis vectors v_k and v_l, we have $T(v_k, v_l) = a_{kl}$. Thus, $[\psi_\beta(T)]_{kl} = T(v_k, v_l) = a_{kl}$, which implies $\psi_\beta(T) = A$.

Since ψ_β is a linear bijection, it is an isomorphism. ■

Corollary 4.2.11. *For any $n-$dimensional vector space V, $\mathcal{B}(V)$ has dimension n^2.*

Proof. As $\mathcal{B}(V) \cong M_{n\times n}(\mathbb{K})$, the result follows. ■

Definition 4.2.12 (Symmetric Bilinear Form)**.** A bilinear form B on a vector space V is symmetric if $B(x, y) = B(y, x)$.

Theorem 4.2.13. *Let B be a bilinear form on a finite-dimensional vector space V and let β be an ordered basis for V. Then B is symmetric if and only if $\psi_\beta(B)$ is symmetric, where ψ_β is an isomorphism between $\mathcal{B}(V)$ to $M_{n\times n}(\mathbb{K})$.*

Proof. Let $\beta = \{v_1, v_2, \ldots, v_n\}$ and $M = \psi_\beta(B)$. First assume that B is symmetric. Then for $1 \le i, j \le n$, $m_{ij} = B(v_i, v_j) = B(v_j, v_i) = m_{ji}$ and it follows that M is symmetric.

Conversely, suppose that M is symmetric. Let $J : V \times V \to \mathbb{K}$, where $\mathbb{K}$ is the filed of scalars for V, be the mapping defined by $J(x, y) = B(y, x)$ for all $x, y \in V$. Then J is bilinear. Let $C = \psi_\beta(J)$. Then, for $1 \le i, j \le n$,

$$C_{ij} = J(v_i, v_j) = B(v_j, v_i) = m_{ji} = m_{ij}.$$

Thus, $C = M$. Since ψ_β is one-one, w have $J = B$. Hence, $B(y, x) = J(x, y) = B(x, y)$ for all $x, y \in V$, and therefore, H is symmetric. ■

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Definition 4.2.14 (Alternating Bilinear Form)**.** A bilinear form $B : V \times V \to F$ is called **alternating** if $B(v, v) = 0$ for all $v \in V$.

Theorem 4.2.15. *A bilinear form B is alternating if and only if it is skew-symmetric, i.e., $B(u, v) = -B(v, u)$ for all $u, v \in V$. (Assuming $\text{char}(F) \neq 2$).*

Proof. Assume B is alternating. Then $B(u + v, u + v) = 0$. By bilinearity:

$$B(u, u) + B(u, v) + B(v, u) + B(v, v) = 0$$

Since $B(u, u) = 0$ and $B(v, v) = 0$, we have $B(u, v) + B(v, u) = 0$, hence $B(u, v) = -B(v, u)$.

Assume $B(u, v) = -B(v, u)$. For any $v \in V$, letting $u = v$ gives $B(v, v) = -B(v, v)$, so $2B(v, v) = 0$. Since $\text{char}(F) \neq 2$, $B(v, v) = 0$. ■

Theorem 4.2.16 (Matrix Representation)**.** *If B is an alternating bilinear form, its matrix representation A relative to any basis is skew-symmetric, i.e., $A^T = -A$. Furthermore, if $\text{char}(F) \neq 2$, the diagonal entries must be zero ($A_{ii} = 0$).*

Proof. By the previous theorem, $B(v_i, v_j) = -B(v_j, v_i)$. Thus $A_{ij} = -A_{ji}$, which is the definition of $A^T = -A$. Setting $i = j$ gives $A_{ii} = -A_{ii} \implies 2A_{ii} = 0 \implies A_{ii} = 0$. ■

Example 4.2.17 (The Standard Symplectic Form)**.** Let $V = \mathbb{R}^2$ with basis $\{e_1, e_2\}$. Define B by the matrix:

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

For $u = (x_1, x_2)$ and $v = (y_1, y_2)$, this gives $B(u, v) = x_1y_2 - x_2y_1$, which is the signed area of the parallelogram spanned by u and v. This is clearly alternating as $B(u, u) = x_1x_2 - x_2x_1 = 0$.

Example 4.2.18 (Differential Forms (Wedge Product))**.** Let V be the space of smooth functions on $\mathbb{R}^n$. The exterior derivative and wedge product construct alternating forms. For vectors $u, v \in \mathbb{R}^3$, the determinant form defined by $B(u, v) = \det(u, v, k)$ (where k is a fixed vector) is an alternating bilinear form. In $\mathbb{R}^2$, the form $B(u, v) = \det(u, v)$ is the canonical alternating form.

4.3 Quadratic Forms

Definition 4.3.1 (Quadratic Form)**.** Let V be a $n-$dimensional vector space over the field of real numbers $\mathbb{R}$. A mapping $q : V \to \mathbb{R}$ is called a **quadratic form** if there exists a symmetric bilinear form $B : V \times V \to \mathbb{R}$ such that:

$$q(x) = B(x, x) \quad \forall x \in V$$

If we choose an ordered basis $\{e_1, e_2, \ldots, e_n\}$ of V, any vector $x \in V$ can be expressed as $x = \sum_{i=1}^{n} x_i e_i$. Substituting this into the definition of $q(x)$ yields:

$$q(x) = B\left(\sum_{i=1}^{n} x_i e_i, \sum_{j=1}^{n} x_j e_j\right) = \sum_{i=1}^{n}\sum_{j=1}^{n} x_i x_j B(e_i, e_j)$$

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Letting $a_{ij} = B(e_i, e_j)$, we have $a_{ij} = a_{ji}$ because B is symmetric. Thus, the quadratic form can be written as:

$$q(x) = \sum_{i=1}^{n}\sum_{j=1}^{n} a_{ij}x_i x_j = x^T Ax$$

where $x = [x_1, x_2, \ldots, x_n]^T \in \mathbb{R}^n$ and $A = (a_{ij}) \in \mathbb{R}^{n\times n}$ is a unique real symmetric matrix associated with q relative to the chosen basis.

Theorem 4.3.2 (Polarization Identity). *If q is a quadratic form associated with a symmetric bilinear form B, then B can be uniquely recovered from q by:*

$$B(u, v) = \frac{1}{2}\left(q(u + v) - q(u) - q(v)\right)$$

Proof. By the definition $q(u + v) = B(u + v, u + v)$, we expand using the bilinearity and symmetry of B:

$$\begin{aligned} q(u + v) &= B(u, u + v) + B(v, u + v) \\ &= B(u, u) + B(u, v) + B(v, u) + B(v, v) \end{aligned}$$

Since B is symmetric, $B(u, v) = B(v, u)$, so:

$$q(u + v) = q(u) + 2B(u, v) + q(v)$$

Rearranging terms:

$$2B(u, v) = q(u + v) - q(u) - q(v) \implies B(u, v) = \frac{1}{2}(q(u + v) - q(u) - q(v))$$

This proves B is uniquely determined by q. ■

Theorem 4.3.3 (Matrix Representation). *If V has basis β and A is the matrix representation of B, then $q(v) = [v]_\beta^T A[v]_\beta$.*

Proof. From the Matrix Representation theorem for bilinear forms, we have $B(u, v) = [u]_\beta^T A[v]_\beta$. Substituting $u = v$:

$$q(v) = B(v, v) = [v]_\beta^T A[v]_\beta$$

If we write $[v]_\beta = (x_1, \ldots, x_n)^T$, then $q(v) = \sum_{i,j} A_{ij}x_i x_j$. This is a homogeneous polynomial of degree 2 in the coordinates of v. ■

Example 4.3.4. For $V = \mathbb{R}^2$ and $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, the quadratic form is:

$$q(x, y) = \begin{pmatrix} x & y \end{pmatrix}\begin{pmatrix} a & b \\ b & c \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + 2bxy + cy^2$$

4.4 Rank, Index, and Signature

When a non-singular linear transformation (or change of basis) $x = Py$ (where P is an invertible matrix) is applied to the quadratic form, we obtain:

$$q(Py) = (Py)^T A(Py) = y^T(P^T AP)y = y^T Dy$$

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where $D = P^T AP$ is a symmetric matrix congruent to A. Since A is real and symmetric, it is always possible to find a matrix P such that D is diagonal. In this diagonalized system, the quadratic form reduces to a sum of squares:

$$q(y) = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2$$

By scaling the coordinates appropriately ($y_i \leftarrow \frac{y_i}{\sqrt{|\lambda_i|}}$ for $\lambda_i \neq 0$), the quadratic form can be brought to its **canonical form**:

$$q(z) = z_1^2 + z_2^2 + \cdots + z_p^2 - z_{p+1}^2 - \cdots - z_r^2$$

where the coefficients are $+1$, -1, or 0. Based on this canonical representation, we define the key invariants:

Definition 4.4.1 (Rank)**.** The **rank** r of a quadratic form $q(x)$ is the rank of its associated symmetric matrix A. It is equal to the total number of non-zero terms (both positive and negative) in the canonical form of q.

Definition 4.4.2 (Index)**.** The **index** p of a quadratic form $q(x)$ is the number of positive terms ($+1$ coefficients) in its canonical form.

Definition 4.4.3 (Signature)**.** The **signature** s of a quadratic form $q(x)$ is defined as the difference between the number of positive terms and the number of negative terms in its canonical form:

$$s = p - (r - p) = 2p - r$$

Remark 4.4.4. Some texts define the signature as the ordered pair $(p, r-p)$ or the triple $(p, r-p, n-r)$. The definition $s = p - (r - p)$ is the standard scalar signature.

4.4.1 Classification of Quadratic Forms

Using the rank r and the index p, a real quadratic form $q(x)$ on an n-dimensional space is classified as:

- **Positive Definite**: if $r = n$ and $p = n$ (all eigenvalues are positive).
- **Positive Semi-definite**: if $r < n$ and $p = r$ (all eigenvalues are non-negative).
- **Negative Definite**: if $r = n$ and $p = 0$ (all eigenvalues are negative).
- **Negative Semi-definite**: if $r < n$ and $p = 0$ (all eigenvalues are non-positive).
- **Indefinite**: if $0 < p < r$ (there are both positive and negative eigenvalues).

Theorem 4.4.5 (Lagrange's Diagonalization Theorem)**.** *Every real quadratic form $q(x) = x^T Ax$ can be reduced to a diagonal form by a nonsingular linear transformation.*

Proof. We proceed by mathematical induction on the number of variables n.

- **Base Case**: For $n = 1$, $q(x) = a_{11}x_1^2$, which is already diagonal.
- **Inductive Step**: Assume that any quadratic form in $n-1$ variables can be diagonalized. Let $q(x_1, \ldots, x_n) = \sum_{i,j=1}^{n} a_{ij} x_i x_j$ be a quadratic form in n variables. We distinguish two cases:

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Case 1: At least one diagonal coefficient is non-zero. Without loss of generality, let $a_{11} \neq 0$. Grouping the terms containing x_1, we can write:

$$q(x_1, \ldots, x_n) = a_{11}x_1^2 + 2x_1 \sum_{j=2}^{n} a_{1j}x_j + \sum_{i,j=2}^{n} a_{ij}x_i x_j$$

Completing the square for the terms containing x_1 yields:

$$q(x_1, \ldots, x_n) = a_{11}\left(x_1 + \sum_{j=2}^{n} \frac{a_{1j}}{a_{11}}x_j\right)^2 - \frac{1}{a_{11}}\left(\sum_{j=2}^{n} a_{1j}x_j\right)^2 + \sum_{i,j=2}^{n} a_{ij}x_i x_j$$

Define a non-singular linear transformation:

$$y_1 = x_1 + \sum_{j=2}^{n} \frac{a_{1j}}{a_{11}}x_j, \quad y_i = x_i \quad \text{for } i \geq 2$$

This substitution transforms q into:

$$q = a_{11}y_1^2 + q'(y_2, \ldots, y_n)$$

where q' is a quadratic form in $n-1$ variables. By the induction hypothesis, there exists a non-singular transformation that diagonalizes q'. Thus, q can be fully diagonalized.

Case 2: All diagonal coefficients are zero ($a_{ii} = 0$ for all i). If q is not identically zero, there must exist some $a_{ij} \neq 0$ for $i \neq j$. Without loss of generality, assume $a_{12} \neq 0$. We define the non-singular transformation:

$$x_1 = y_1 + y_2, \quad x_2 = y_1 - y_2, \quad x_i = y_i \quad \text{for } i \geq 3$$

The term $2a_{12}x_1x_2$ becomes $2a_{12}(y_1^2 - y_2^2)$. Substituting these into q yields a new quadratic form in y where the diagonal coefficients of y_1^2 and y_2^2 are non-zero. This reduces the problem to Case 1.

By induction, any real quadratic form can be reduced to diagonal form. ■

The above theorem can also be presented and proved in the following way:

Theorem 4.4.6 (Lagrange's Diagonalization Theorem). *Let V be a finite-dimensional vector space over a field F with $\text{char}(F) \neq 2$. For any quadratic form $Q : V \to F$, there exists an ordered basis $\beta = \{v_1, v_2, \ldots, v_n\}$ for V such that for any vector $v = \sum_{i=1}^{n} x_i v_i$, the quadratic form is represented by:*

$$Q(v) = \sum_{i=1}^{n} a_i x_i^2$$

where $a_i \in F$ are scalars. In other words, every quadratic form can be diagonalized by a suitable choice of basis.

Proof. We proceed by induction on $n = \dim(V)$. The case $n = 1$ is trivial.

If $Q(v) = 0$ for all $v \in V$, the statement holds with all $a_i = 0$. Otherwise, there exists $v_1 \in V$ such that $Q(v_1) \neq 0$. Let B be the symmetric bilinear form associated with Q, and define the subspace $W^\perp = \{w \in V \mid B(w, v_1) = 0\}$.

We claim $V = \text{span}\{v_1\} \oplus W^\perp$. For any $v \in V$, define:

$$u = v - \frac{B(v, v_1)}{Q(v_1)}v_1$$

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A direct calculation shows $B(u, v_1) = B(v, v_1) - \frac{B(v,v_1)}{Q(v_1)} B(v_1, v_1) = 0$, so $u \in W^\perp$. Since $v = \frac{B(v,v_1)}{Q(v_1)} v_1 + u$, we have $V = \text{span}\{v_1\} + W^\perp$. The sum is direct because $v_1 \notin W^\perp$ (as $B(v_1, v_1) = Q(v_1) \neq 0$).

By the induction hypothesis, the restriction of Q to $W^\perp$ is diagonalizable with respect to some basis $\{v_2, \ldots, v_n\}$. Consequently, the set $\beta = \{v_1, v_2, \ldots, v_n\}$ is a basis for V. For any $v = \sum_{i=1}^n x_i v_i$, the cross terms vanish by the construction of $W^\perp$:

$$Q(v) = B\left(\sum x_i v_i, \sum x_j v_j\right) = x_1^2 Q(v_1) + Q\left(\sum_{i=2}^n x_i v_i\right) = a_1 x_1^2 + \sum_{i=2}^n a_i x_i^2$$

This completes the proof. ■

Theorem 4.4.7 (Sylvester's Law of Inertia)**.** *The rank r and the index p of a real quadratic form are invariants under any non-singular real linear transformation. That is, no matter what non-singular transformation is applied to diagonalize the form, the number of positive, negative, and zero coefficients remains the same.*

Proof. Let q be a quadratic form on a real vector space V of dimension n. Suppose there exist two different non-singular coordinate transformations resulting in two diagonal representations of q.

Let the first transformation yield a basis $\{u_1, u_2, \ldots, u_n\}$ in which:

$$q(x) = y_1^2 + \cdots + y_p^2 - y_{p+1}^2 - \cdots - y_r^2$$

Let the second transformation yield a basis $\{v_1, v_2, \ldots, v_n\}$ in which:

$$q(x) = w_1^2 + \cdots + w_{p'}^2 - w_{p'+1}^2 - \cdots - w_{r'}^2$$

where r and r' represent the ranks, and p and p' represent the indices in the respective coordinate systems.

First, we establish that the rank is invariant. Since the matrix of a quadratic form transforms as $A' = P^T A P$ under a change of basis represented by an invertible matrix P, we have:

$$r = \text{rank}(P^T A P) = \text{rank}(A) = r'$$

Thus, the total number of non-zero terms is invariant ($r = r'$).

Next, we prove that the index must be invariant ($p = p'$). Assume for contradiction that $p > p'$.

Let U be the subspace of V spanned by the first p basis vectors of the first system:

$$U = \text{span}\{u_1, \ldots, u_p\}$$

For any non-zero vector $u \in U$ with coordinates $(y_1, \ldots, y_p, 0, \ldots, 0)$, we have:

$$q(u) = y_1^2 + \cdots + y_p^2 > 0$$

Thus, $q(u) > 0$ for all $u \in U \setminus \{0\}$, and $\dim(U) = p$.

Let W be the subspace of V spanned by the remaining basis vectors of the second system starting from $p' + 1$:

$$W = \text{span}\{v_{p'+1}, \ldots, v_n\}$$

For any vector $w \in W$ with coordinates $(0, \ldots, 0, w_{p'+1}, \ldots, w_n)$, we have:

$$q(w) = -w_{p'+1}^2 - \cdots - w_{r'}^2 \leq 0$$

Thus, $q(w) \leq 0$ for all $w \in W$, and $\dim(W) = n - p'$.

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Now consider the intersection $U \cap W$. By the grassmann dimension formula:

$$\dim(U + W) = \dim(U) + \dim(W) - \dim(U \cap W)$$

Since $U + W$ is a subspace of V, we must have $\dim(U + W) \leq n$. Therefore:

$$\dim(U \cap W) \geq \dim(U) + \dim(W) - n = p + (n - p') - n = p - p'$$

Since we assumed $p > p'$, it follows that:

$$\dim(U \cap W) \geq 1$$

This implies there exists a non-zero vector $z \in U \cap W$.

- Since $z \in U$ and $z \neq 0$, we must have $q(z) > 0$.
- Since $z \in W$, we must have $q(z) \leq 0$.

This contradiction ($q(z) > 0$ and $q(z) \leq 0$) shows that our assumption $p > p'$ is false. By symmetry, assuming $p' > p$ also leads to a contradiction. Hence, we must have:

$$p = p'$$

Because both r and p are invariant, the number of negative coefficients $r - p$ and the signature $s = 2p - r$ are also uniquely determined invariants of the quadratic form. ■

Example 4.4.8 (Diagonalization of a Positive Definite Form)**.** Consider the quadratic form on $\mathbb{R}^3$ defined by:

$$q(x, y, z) = x^2 + 2y^2 + 5z^2 - 2xy + 4yz$$

To analyze its properties, we reduce the form to its canonical representation using Lagrange's method of completing squares.

We begin by grouping and completing the square for all terms containing the variable x:

$$\begin{aligned} q(x, y, z) &= (x^2 - 2xy) + 2y^2 + 5z^2 + 4yz \\ &= (x - y)^2 - y^2 + 2y^2 + 5z^2 + 4yz \\ &= (x - y)^2 + y^2 + 4yz + 5z^2 \end{aligned}$$

Next, we focus on the remaining terms involving y and z to complete the square with respect to y:

$$\begin{aligned} q(x, y, z) &= (x - y)^2 + (y^2 + 4yz) + 5z^2 \\ &= (x - y)^2 + (y + 2z)^2 - 4z^2 + 5z^2 \\ &= (x - y)^2 + (y + 2z)^2 + z^2 \end{aligned}$$

This algebraic structure motivates the introduction of a new set of coordinates defined by the linear transformation:

$$X = x - y, \quad Y = y + 2z, \quad Z = z$$

Expressing the original coordinates in terms of the new variables, we obtain the relation:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 1 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$$

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Since the transition matrix is upper triangular with non-zero diagonal entries, its determinant is 1. The transformation is therefore non-singular, and the quadratic form reduces to the canonical diagonal representation:

$$q(X, Y, Z) = X^2 + Y^2 + Z^2$$

From this canonical form, the invariants of q are determined directly:

- **Rank** (r): The number of non-zero terms is 3.
- **Index** (p): The number of positive terms is 3.
- **Signature** (s): $s = 2p - r = 2(3) - 3 = 3$.

Because $r = p = 3$ on a three-dimensional space, the quadratic form is classified as **positive definite**.

Example 4.4.9 (Diagonalization of a Form with Cross-Terms)**.** Consider the quadratic form on $\mathbb{R}^3$ defined by:

$$q(x, y, z) = 2xy + 2yz + 2zx$$

Because the form contains only cross-products and lacks squared terms, we initiate the reduction by applying a preliminary transformation to generate squared terms. Let:

$$x = u + v, \quad y = u - v, \quad z = w$$

Substituting these relations into the expression for q yields:

$$\begin{aligned} q &= 2(u + v)(u - v) + 2(u - v)w + 2(u + v)w \\ &= 2(u^2 - v^2) + 2uw - 2vw + 2uw + 2vw \\ &= 2u^2 - 2v^2 + 4uw \end{aligned}$$

We now group the terms containing the variable u and complete the square:

$$\begin{aligned} q &= 2(u^2 + 2uw) - 2v^2 \\ &= 2\left[(u + w)^2 - w^2\right] - 2v^2 \\ &= 2(u + w)^2 - 2v^2 - 2w^2 \end{aligned}$$

To bring the expression into its canonical form, we normalize the coordinates by setting:

$$X = \sqrt{2}(u + w), \quad Y = \sqrt{2}v, \quad Z = \sqrt{2}w$$

This substitution is a valid non-singular change of variables, transforming q into:

$$q(X, Y, Z) = X^2 - Y^2 - Z^2$$

The invariants of the quadratic form are subsequently identified as:

- **Rank** (r): The number of non-zero terms is 3.
- **Index** (p): The number of positive terms is 1.
- **Signature** (s): $s = 2p - r = 2(1) - 3 = -1$.

Since the index satisfies $0 < p < r$, the quadratic form is classified as **indefinite**.

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